FICTION

Jane's All the World's Aircraft. 1913

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Several illustrations are either blank or have only text in them. Those were rendered as fully as possible in plain text. Other illustrations are noted with an [Illustration] tag and the caption, if there is one.

The advertisements which were originally at the front of the book have been moved to the back.]

JANE'S ALL THE WORLD'S AIRCRAFT 1913

A Reprint of the 1913 Edition of All The World's Air-craft Edited by FRED T. JANE

ARCO PUBLISHING COMPANY, INC. New York

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FOUNDED AND EDITED BY FRED T. JANE, Founder and Editor of "FIGHTING SHIPS" (Naval Annual), Etc.

PART A.--AEROPLANES AND DIRIGIBLES OF THE WORLD. PART B.--HISTORICAL AEROPLANES OF THE LAST SIX YEARS. PART C.--THE WORLD'S AERIAL ENGINES. PART D.--AERIAL "WHO'S WHO" AND DIRECTORY.

FIFTH YEAR OF ISSUE. (Founded 1909.)

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PREFACE.

As conjectured last year, considerable further changes have been produced in this edition.

When, some five years ago, work on this annual was first commenced, the military aviator was an idle dream. Fighting men in dirigibles were a bare possibility; but nothing more than that. Every amateur building an aeroplane (or even merely intending to build one) in his back garden was a possible "conqueror of the air." The aeroplane was going to oust the motor car as a sporting vehicle--everyone was quite certain about that! Beyond that, nothing!

To-day everything is completely changed and except as a war machine the aeroplane is of little interest or use to anyone. A few civilian aviators are still flying, but in practically every case they are doing so in connection with the business aspect of the question. There is no "sport of aviation" such as the prophets foretold a few years ago.

An increasing number of people obtain their pilot certificates and lists of these are still given, although the title of "aviator" is in the bulk of cases somewhat of a courtesy one, since so few keep on flying once they have secured their brevets.

It is as a _war machine_ that the aeroplane has come into its own. The Italian aeroplanes over and over again proved their utility in Tripoli. Although in the Balkan

War aircraft were less in evidence than many expected, this may be attributed to the peculiar circumstances of the campaign and also to the scarcity of available machines.

Every country is now engaged in forming its aerial fleets. How far the naval and military branches will coalesce, or how far they will differentiate remains to be seen. The probabilities, at present, all point in the latter direction, and that just as an army is made up of cavalry, infantry, artillery, etc., and a navy of battleships, cruisers, torpedo craft and submarines, so the sky fleets seem destined to consist of groups of different types of machines, each type designed for some special purpose.

The increased war utility of aircraft has necessitated an extension of the pages devoted to organisation of military aviation, etc. The details given are by no means as full as I could wish; but all organisations are being so continually changed owing to increased experience that satisfactory data are not very easy to come by.

During the past twelve months or so we have learned at least one or two important things. The mere fact of the possession of aeroplanes by a nation is a military factor of comparatively little importance. A nation possessing next to no aeroplanes can easily acquire a few hundreds in case of emergency _if she has the people to build them_. The real problem is two-fold. First, of course, is the possession of trained and efficient aviators to fly the machines. Naval and military officers who have merely secured their brevets at a flying school are of no immediate value; civilians of the same kind are of still less utility.

Second to this is the productive capacity of any country; which may roughly be gauged from the number and importance of its firms engaged in construction.

These points cannot too strongly be enforced. The air strength of any nation in case of war resides in its efficient flying men and in its own productive capacity. The next war will see aircraft quite as much "contraband" as warships, and the nation which relies upon aerial imports will be foredoomed at once. One month is probably the utmost effective life of an aeroplane on hard active service and it may well be a good deal less. And firms capable of building efficient machines cannot be improvised.

A remarkable feature of the last twelve months has been the recrudescence of the dirigible, which is now in far greater esteem than it was a year ago, or for that matter, ever before. In the past there is no doubt that progress was hampered by arguments between the advocates of "heavier than air" and "lighter than air," and a curious notion that the one could only exist at the expense of the other.

Such ideas are now dead, and it is recognised that for war purposes both have their uses and that both are interdependent. It is not quite yet realised how intense this interdependence is likely to be.

Briefly the present situation may be summed up as follows: the dirigible has enormous potentialities for attack on fortified bases and the like, but its powers of defence, guns or no guns, are very slight. A single aeroplane should be able to disable or destroy without very great difficulty the finest dirigible yet built (supposing it able

to find the airship in the vastness of the air). The damage that a single aeroplane can do to land defences or ships is, however, entirely trivial--at any rate at present.

Hence the aerial war unit already formed in Germany, and likely to be in existence everywhere else ere another year or so has passed. This unit is a dirigible of great offensive powers, associated with a number of aeroplanes presumably intended to defend it and ward off and defeat attack by hostile aeroplanes.

This is merely the crude beginning, it seems reasonably safe to prophecy that in the early future the aerial war-unit will be made up somewhat as follows:--

(_a_) An offensive dirigible, carrying the maximum of bombs, etc.

(_b_) One or two dirigibles carrying oil and petrol for the aeroplanes--possibly capable of dealing with all minor repairs and of carrying a certain number of aeroplanes on board.

(_c_) A number of war aeroplanes specially designed for fighting other aeroplanes and attacking hostile dirigibles as chances may occur.

(_d_) A few very swift one man aeroplanes which will be the eyes of the unit.

This seems an early certainty. After all it merely reproduces for the air what centuries of experience have shewn to be essential for fleets and armies.

The matter is a fascinating subject for speculation; but in connection with a work that exists merely to deal with things as they are at present, is perhaps, better not now pursued further. One point, however, may perhaps be mentioned, and that is that victory or defeat in aerial warfare seems likely to depend upon which side can first destroy the other's bases. A base-less dirigible will not live long. This is likely to lead to very great attention being paid at an early date to anti-aircraft guns and other devices for the defence of aerial bases.

Reverting to the arrangement of the present edition, a few words may be said about some of the changes. As stated last year the clumsy old system of grouping monoplanes, biplanes, etc., separately has been abolished. So many firms specialise in both that any such grouping could only lead to confusion.

A tabular system has been generally adopted for most new matter. This will be found far more convenient for reference, and of course, saves a great deal of space.

The effective age of aeroplanes is somewhat of a vexed question, for while one year probably represents the really effective war utility endurance, even in peace time, school life is more or less indefinite and so is ordinary private life. Consequently--although "dead machines" are excluded it has not been possible to draw an exactly uniform age limit line beyond that. Speaking generally modern machines represent as a rule detail improvements rather than the complete changes of the past. For example, the gap between 1911 and 1913, is far less than the gap between 1909 and 1911. This fact is beginning to make itself felt in war machines.

In Part B an attempt has been made to collect illustrations of aeroplanes of the past which for one reason or another possess an historical interest. This section is

remarkable for two totally different things (1) the early anticipations of some modern practice, and (2) the past prevalence of certain other ideas which are now totally extinct.

Part C deals with aero-engines. It is mainly remarkable--in comparison with past issues--for the large number of engines which have ceased to exist. It is probably still too ample; as a year hence quite half the makers still recorded are likely to disappear. The mere ability to construct motor car engines is no longer of value. The aeroplane engine designer needs to be a specialist. The absolutely ideal aero-engine no doubt yet remains to be produced; but meanwhile the tendency of users to concentrate upon fewer makes is increasingly evident, despite the fact that the best engine for one particular type of machine is not necessarily the best for some other type.

In conclusion I tender my most grateful thanks to all those who have so kindly collaborated with or for me in the various sections. The book is still some way from being near my ideal, but I have every hope that this edition will be generally considered a very considerable improvement upon previous issues.

FRED T. JANE.

Bedhampton, _Hants.,_ _England._

GLOSSARY OF TECHNICAL TERMS, Etc.

ENGLISH. | DUTCH. | FRENCH. | GERMAN. | ITALIAN. ---------- Abaft | Achterste deel | Arriere | Hinter | A poppa Accessories | Onderdeelen | Accessoires | Zubehoer | Accessori Accumulator | Accumulator | Accumulateur | Akkumulator | Accumulatore ~AEROPLANE~ | Dekvlieger | Aeroplane | Drachenflieger | Aereoplano Aeronaut | Luchtvaarder | Aeronaute | Luftschiffer | Aereonauta | | Aviateur | | Aerostat | Luchtbal | Aerostat | Freiballon | Aereostato Aft | Achterdeel | Arriere | Hinten | Addietro After (rear) | Achter | Arriere | Hinterer | Poppa Air-cooled | Luchtgekoeld | Refroidit par | Luftgekuhlt | Raffredda ad | | Pair | | aria Angleiron | Hoekyzer | Corniere | Eck Schiene | Ferro ad angolo Anti-friction | Wit metaal | Metal | Lagermetall | Metallo beanco metal | | anti friction | | (anti frizione) | | on regule | | Aviation | Vliegtechniek | Aviation | Flugtechnik | Aviazione Babbit Metal | Babbits metaal | Metal Babbitt | Lagermetall | Metallo Babbitt | | on regule | | Balance | Evenwicht | Equilibre | Gleichgewicht | Equilibrio Ball bearings | Kogellagers | Coussinets a | Kugel Lager | Cuscinetti a | | billes | | sfere Ballonet | Luchtzak | Ballonet | Ballonet | Palloncino | | | | compensatore Battery | Battery | Batterie | Batterie | Pila a secco Bearing metal | Kussenmetaal | Metal pour les | Lager metall | Metallo par | | coussinets on | | cuscinette | | regule | | Behind | Achter | Derriere | Hinter | Di dietro Bevel geared | Kegelraderwerk | Engrenage | Konischer | Ingranaggio | | Conique | Antrieb | conico Biplane | Tweedekker | Biplan | Zwei decker | Biplano Blades | Bladen | Pales | Fluegel | Pale (of propeller)| (der schroef) | | | delt'elica Body | Romp | Fuselage | Koerper | Telaio o | | | | chassis Bolt | Bout | Bonlon | Bolzen | Bollone Box-kite | Kabel-vlieger | Cerf-volant | Drachen | Aquilone a celle Bracket | Klamp | Tasseau | Stuetze | Sostegno Brake | Rem | Frein | Bremse | Freno Breadth | Breedte | Largeur | Breite | Larghezza Canvas | Doek | Toile | Leinwand |

Tela Car | Gondel | Nacelle | Gondel | Navicella Carburetter | Vergasser | Carburateur | Vergaser | Carburatore Casting | Gietstuk | Moulage | Guss Stueck | Getto Centre of | Zwaartepunt | Centre de | Schwerpunkt | Centro di Gravity | | Gravite | | gravita Chain driven | Door ketting | Transmission | Ketten antrieb | Trasmissione a | gedreven | par chaine | | catena Chassis | Gestel | Chassis | Motor Rahmen | Chassis Circumference | Omtrek | Circonferance | Umfang | Cuconferenza Clutch | Haak | Embrayage | Kupplung | Innesto Connection | Schakeling | Couplage | Kupplung | Connessione Control | Stuurinrichting | Direction | Lenk | Meccanismo di | | | Ubersetsung | direzione Coupled | Gekoppeld | Jumele | Paarweise | Accoppiato Crank shaft | Krukas | Arbre a | Kurbelwelle | Albero delle | | manivelle | | manovelle Cylinder | Cylinder | Cylindre | Zylinder | Cilindro Die cast | Ondermetaallager | Coussinets | Schalenguss | Cuscinette fusi Bearings | | moutes | Lager | in conchiglia | | encogiulles | | ~DIRIGIBLE~ | Motorballon | Dirigeable: | Motorluftschiff | Dirigibile | | Aeronat | | Diameter | Middellyn | Diametre | Durchmesser | Diametro Direct driven | Direct | Prise directe | Direkter | Presa diretta | gekoppeld | | Antrieb | Electric | Electrische | Soudure | Elektrisches | Soldatura welding | Lassching | electrique | Schweissen | elettrica Elevator | Hoogtestuur | Gouvernail de | Hohensteuer | Timone (horizontal | | profondeur | | orizzontale rudder) | | | | Engine | Motor | Moteur | Motor | Motore Fan | Ventilator | Ventilateur | Ventilator | Ventilatore Fittings | Fittings | Garniture | Garnitur | Armamento Flight | Vlucht | Vol | Flug | Volo Flown | Gevlogen | Vole | Geflogen | Volato Fore | Voor | Avant | Vorderer | Ouvanti Forward | Van Voren | En avant | Vor | Davanti (in front) | | | | Frame | Romp | Fuselage | Rahm | Telais Framework | Geraamte | Fuselage | Gerueste | Intelaiatura Gas bag | Gaszak | Enveloppe | Luftballon (Huelle) | Involucro Geared to | Vertand | Multiplie a | Uebersetst auf | Moltiplicato a Gear driven | Met tandrad- | | durch Zahnraedern | Trasmissione | overbrenging | | getrieben | a ingranaggi Girder | Balk | Poutre | Balken | Longarin Glider | Glydvlieger | Planeur | Gleitflieger | Apparecehio a | | | | planare Gondola | Gondel | Nacelle | Gondel | Navicella Helices | Schroeven | Helices | Schranben | Eliche Helicopter | Schroefvlieger | Helicoptere | Schraubenflieger | Elicoplano | | | | Elicottero Horizontal | Horizontaalvlak | Plan horizontal| Horizontal flaeche | Piano plane (in a) | (in een) | | | orizzontale Horse power | Paardekracht | Puissance en | Pferdekraft | Forza cavalli | | chevaux | | Hydrogen | Waterstof | Hydrogene | Wasserstoff | Idrogens Ignition | Ontsteking | Allumage | Zuendung | Accensione Inch | Duim | 25.39 m/m. | 25.39 m/m. | Pollice = | | | | 25.39 m/m. Inclination | Helling | Inclination | Schraegstellung | Inclinazione Keel | Kiel | Carene | Kiel | Chiglia K.P.M. | K.P.U. (kilom. | Kilometres par | Kilometre pro | Chilometre (kilometres | per uur) | heure | Stunde | all'ora per hour) | | | | Kite | Vlieger | Cerf volant | Drachen | Aquilone Length | Lengte | Longueur | Laenge | Lunghezza Lining metal | Lagermetaal | Metal pour | Lagermetall | Metallo per | | garnir less | | bronzine dei | | coussinets ou | | cuscinetti | | regule | | Lower (planes)| Onder (vlakken) | Inferieur | Untere Flaechen | Piani inferiori | | (plans) | | Magneto | Magneet | Magneto | Magnet | Magneto ~Maximum~ | Maximum | Maximum | Maximum | Massimo Middle (plane)| Midden (vlak) | (Plan)au | Mittel Deck | Piano medio | | milieu | | Mile | Myl | Mile | Meile | Miglio Military | Militair | Militaire | Militaerische | Militare Miscellaneous | Verschillend | General | Verschiedenes | Diversi | (allerlei) | | | ~Monoplane~ | Eendekker | Monoplan | Ein decker |

Monoplano ~Motor~ | Motor | Moteur | Motor | Motore M.P.H. | M.P.U. | Vitesse | M.P.S. | Miglia all'ora (miles per | (myl per uur) | | | hour) | | | | Multiplane | Veeldekker | Multiplan | Vieldecker | Multiplano Nacelle | Schuitje | Nacelle | Gondel | Navicella ~Non-rigid~ | Slap | Souple | Unstarr | Non-rigido-- | | | | flessibile Petrol | Benzine | Essence | Benzin | Benzina gasoline) | | | | Pilot (driver)| Bestuurder | Flyer: Aviateur| Fuehrer | Aviatore Pivot | Tap | Pivot | Gewinde Zapfen | Perno Planes | Vlakken | Plans | Flaechen | Piani Plug | Kaars, stop | Bougie | Zuend Kerze | Candela Pound (lb.) | Eng pond = | 0.453 kg. | 0.453 kg. | Libbra = | 0,453 K.G. | | | 0.453 kg. Pressure | Druk | Pression | Druck | Pressione Propeller | Schroef | Helice | Schraube | Eliche Quadruplane | Vierdekker | Quadruplan | Vier decker | Qudruplani Quintuplane | Vyfdekker | Quintuplan | Fuenf decker | Quintuplani Radiator | Koeler | Radiateur | Kuehler | Radiatore Rear (in) | Achterkant | En arriere | Hinten | Indictro | (aan de) | | | Reduction | Reductie- |Engrenage de | Ubersetzung | Ingranaggi di gearing | overbrenging |demultiplication| | ridugione R.P.M. | Omw. per minuut | Tours | Umlauf | Giri al minuto (revolutions | | | | per minute) | | | | ~Rigid~ | Styf | Rigide | Starr | Rigido Rises | Stygt | S'eleve | Hebt sich | Si eleva Rubber | Gummi | Caoutchouc | Gummi | Gomma Rudder | Roer, Stuur | Gouvernail | Steuer | Timone Section | Doorsnede | Section | Durchschnitt | Regione ~Semi-rigid~ | Halfstyf | Demi-rigide | Halb Starr | Semi-rigido Span | Spanwydte | Envergure | Spanweite | Apertura ~Speed~ | Snelheid | Vitesse | Geschwindigkeit | Velocita Stability | Evenwicht | Stabilite | Gleichgewicht | Stabilita Stabilising | Evenwichtsvlakken| Ailerons | Gleichgwichtsflaechen| Piani fins | | | | stabilizzaton Steel | Staal | Acier | Stahl | Acciaio ~Steering | Stuurtoestel | Direction | Steuerung | Meccanismo Gear~ | | | | | | | di direzione Steering Wheel| Stuurwiel | Volant | Steuerrad | Volante di | | | | direzione ~Supporting~ | Draagvlak | Surface | Tragflaeche | Superficio ~surface~ | | | | di sostegno Surfaces | Oppervlakken | Surfaces | Flaechen | Superfici Suspension | Ophanging | Suspension | Aufhaengung | Sospensioni Switch | Omschakelaar | Interrupteur | Schalter | Interruttore | | | | Tail | Staart | Queu | Schwanz | Coda ~Total weight~| Totaal gewicht | Poids totale | Gesamtlast | Peso totale Transmission | Overbrengingsas | Arbre de | Transmissions Welle | Albero di Shaft | | transmission | | trasmissione Trial | Proef | Essai | Probe | Prova ~Triplane~ | Driedekker | Triplan | Drei decker | Triplano | | | | Universal | Kogelgewricht | Joint | Kardan | Guinta Joint | | | | | | universel | | universale Unladen | Onbelast, leeg | a vide | Leerlaufend | Upper (planes)| Boven (vlakken) | Superior | Ob ere | Piani Superior ~Useful lift~ | Nuttier last | Poids utile | Outlast | Forza utile di | | | | elevation Valve | Kelp | Soup ape | Lentil | Valvular Vertical plane| Vertical vlak | Plan vertical | (in der) | Neal piano (in the) | (in heat) | | Vertikalflaeche | verticale Vertical | Zystuur | Gouvernail | Seitensteuer | Timone rudder | | | | | verticale | | verticale ~Volume~ | Inhoud | Volume | Inhalt | Volume | | | | Water-cooled | Watergekoeld | Refroidissement| Wasserkuehlung | Raffreddata | | par eau | | ad acqua ~Weight~ | Gewicht | Poids | Gewicht | Peso Wheels | Wielen | Roues | Raeder | Ruote ~Wings~ | Vleugels | Ailes | Fluegel | Ali Wood | Hout | Bois | Holz | Legno | | | | Yard (measure)| Yard (maat)= | 0.914 metres | 0.914 meter | Jarda=0.914 m. | 0,914 M | | |

Part A.

AEROPLANES & DIRIGIBLES.

ARRANGED BY NATIONALITIES IN ALPHABETICAL ORDER.

Note.--Every nation is given in the following fixed order:--

List of Aerial Societies and Clubs, with addresses and Secretaries where possible. List of Aerial Journals, with addresses, price, and dates of publication.

List of Flying Grounds for aeroplanes, and hangars for dirigibles (if any).

List of Military and Naval Machines and aviators.

List of Private Aviators, total of machines, etc.

AEROPLANES in alphabetical order, _uniform scale_ plans, and particulars.

DIRIGIBLES: Military and private _uniform scale_ plans, and particulars.

Note.--The uniform scale of dirigible plans is a smaller scale than that used for aeroplanes.

ARGENTINE.

(Revised by J. SCHIERE, Aeronautical Engineer and Librarian, Dutch R. Ae. C.)

~Aerial Societies~:--

Ae.C., Argentino, 561, S. Martin, Buenos Ayres.

~Aerial Journals~:--

Boletin del Ae.C., Argentino (Monthly).

~Flying Grounds~:--

~Aerodromo Villa Lugano~. (P. Castabert, Director.)

~Aerodromo del Palomar~. (Military).

~Military Aviation.~

At the end of 1912 there were 6 military aeroplanes (3 _Bleriot_, 1 _Castaibert_, 1 _Nieuport_, and 1 _H Farman_)--all 1912 models.

Marcel Paillette is director of the military flying ground at the Palomar Aerodrome.

More machines will be added and by the end of the present year it is probable that a very considerable air force will exist.

~Private Aviators.~

Bregi, Henri (A.C.F. 26) de Bruyn, A. (3) Castaibert, B. (1) Fels, T. (9) England, Gordon F.C. (British p.) Goffre, C.A. (4) Hentsch, H. (5) Mascias, A.R. (8) Melchior, E. (11) Newbury, G. (6) Origone, M.F. (10) Paillette, Marcel (French p.) Parravicini, F. (7) Roth, J.A. (2) Valleton A. (French p.)

PABLO CASTAIBERT. Monoplane.

| ~1911.~ | ~1912.~ | Type | _Bleriot-Hanriot_ type. | _Bleriot_ type. | _____|_____|_____| | | | Length | 26-1/2 feet (8.15 m.) | 28 feet (8.47 m.) | Span | 29 feet (8.80 m.) | 30 feet (9.35 m.) | Area | 206 sq. ft. (19-20 m squared.) | 194 sq. ft. (18 m squared.) | Weight (total) | 705 lbs. (320 kgs.) | 617 lbs. (280 kgs.) | Motor (h.p.) | 25 Anzani | 50 Gnome | Speed (p.h.) | 46-1/2 m. (75 km.) | 50 m. (80 km.) |

Note. Both fly well. Description in _Boletin de Ae.C. Argentino_.

AUSTRO-HUNGARIAN.

(By our Austrian Editor.)

~Aerial Societies:--~

Cesky Club Automobilistn. Aviatische Sektion. (Prague.)

Deutscher Luftfahrt-Verein in Boehmen (Teplitz-Schoenau).

Flugtechnischer Verein fuer Maehren (Brunn.).

Flugtechnischer Verein in Schlesien (Troppau).

Flugtechnischer Verein in Steiermark (Graz, Schmiedgasse 31).

K.k. Oesterreichischer Flugtechnischer Verein (Wienstrasse 31, Vienna).

Kaertner Automobil Club (Klagenfuert.)

Klub Awiata (Obertynska Str. 8, Lemburg, Galicia).

Magyar Automobil Club (Budapest).

Magyar Athletikai Club (Abt. fur Aviatik) Budapest.

Oberoesterreichischer Verein f.L. in Linz (Landstr. 119, Linz).

Oesterreichischer Aero Club (St. Annahoff, Vienna) (_formerly Wiener Aero Club_).

Oesterreichischer Flugsport Club (Breitegasse 7, Vienna VII).

K.k. Oesterreichischer Flugtechnischer Verein (Weinstrasse 31, Vienna).

Oesterreichischer Luftflotten-Verein (Vienna).

Oesterreichischer Wintersportklub (Vienna). _Glider club._

Verein fuer L. in Tirol (Innsbruck).

~Aerial Journals:--~

Allgemeine Automobil Zeitung. (Fleischmarkt 5, Vienna) weekly.

Allgemeine Sport Zeitung. (St. Annhoff, Vienna) weekly.

H.P. Fachzeitung fuer Automobilismus und Flugtechnik. (Vienna) weekly.

Oesterreichische Flugzeitschrift. (Aspernplatz, Wien I) fortnightly.

Wiener Luftschiffer-Zeitung. (St. Annahof, Vienna) fortnightly.

~Flying Grounds:--~

Military.

~Fischamend~ (Principal Army).

~Goerz.~

~Zaule b. Triest.~

Naval.

~Pola.~

Private.

~Aspern bei Wien~, Vienna.

~Rakos bei Budapest.~

~Wiener Neustadt.~

AUSTRO-HUNGARIAN AEROPLANES.

~Military Aviation: General.~

In June, 1912, a central aeronautical committee was created, under the presidency of Prince Fuerstenberg, to deal with the creation of a national aerial fleet. One of the objects is the perfection of the Austrian machines and factories.

About the same date, Pola was selected as a naval aviation school, and two _Paulhan-Curtiss_ hydro-avions purchased.

In August the record making _Lohner_ was purchased for the Army.

During September Captain Odolek tested before the military authorities a parachute of his invention; and a number were ordered.

In October very strict regulations were issued as to aircraft flying over prohibited areas, a rule that any offenders would be shot at was subsequently modified.

In November a _Donnet-Lereque_ was purchased for the Navy and another ordered at the Whitehead Works, Fiume.

~Army Section.~

At the end of 1911 the Army possessed 4 monoplanes and one biplane (a _Lohner_), now available for school work.

During 1912 there were acquired:--

~20 monoplanes.~--1 _Bleriot_, 2 _Nieuport_, 15 _Etrich-Taube_, 1 _Etrich_ limousine, 1 _Deperdussin_.

~6 biplanes.~--4 _Lohner-Daimler_, 1 _Mars_, 1 _Klobucar_, (of the above the 2 _Nieuports_, 1 _Etrich_ limousine and the 4 _Lohners_ were the only ones built in Austria).

~Naval Section.~

~4 hydro-avions~ were acquired during 1912; 2 _Donnet-Leveque_ and 2 _Paulhan-Curtiss_.

~Military Pilots.~

Banfield, Ob. Leut. Blaschke, v. Ob. Leut. Eyb, Ob. Leut. Flassig, Leut. Holeka, Ob. Leut. Kenese, Ob. Leut. Klobucar, Ob. Leut. Miller, Ob. Leut. (5) Oelwein, Ob. Leut. Perini, Leut. Petroczy, Haupt. von. Riedlinger, Ob. Leut. von. Schindler, Leut. Schuenzel, Leut. Stohanzl, Ob. Leut. K (14) Umlauff, Major von. (10) Uzelac, Ob. Leut. Venczel, Leut. Welhelm, Ob. Leut. von

The military centre is at Goerz, the naval one at Pola.

Flying officers receive each a grant of 1,600 crowns; also 15 crowns a month for upkeep. Special certificate brings 2,000 crowns extra grant.

~Private Aviators.~

* = Superior brevet. + = Killed.

Auer, J. (6) Baar, R. Baboncse, K. Banfield, K. Bauer, Dr. V.R. von. Bernat, M. *Brier, H. (18) Blaschke von. Z.R. Booms, W. (9) Bratmann, J. Buchstaetter, A. Cejnek, J. Cihak, E. Ciszek, J. Czermak, J. Dworak, W. *Economo, C.F. von. (7) Fiedler, P. (19) *Flesch, J. (11) Friedmann, W. Haner, E. Hesse, M. Heyrowski, A. Hieronimus, O. Hinter, K. Hold, Hermann Huss, H. *Illner, K. Javor, J. Kaiserfeld, R. von. Kasulakow, W. Keck, Z. Kenese, W. Kiraly, K. Klobucar, V. Knirsch, A. Kolowrat, A. Graf. (15) Kreiner, E. Lagler, B.V. Latzel, J. Lettis, A. Libowitzky, A. Mandl. Mazuranic, B. +Mosen. Nemec, H.E. von. Nittner, E. Ockermueller, H. +Petrovics, A. von. (13) Pischof. A.R. von. (2) Rabis, M. Reisner, H. Riedlinger, E. v. Kastrenberg. Rosenthal, F. +Russjan. Sablatnig, J. (12) Schartner, H. Schindler, A. Schonowsky, B. Schoenpflug, F. Seidl, Franz. Simon, R. (4) Stanger, R. Steiner-Goeltl, E. v. A. Stiploschek, M. *Szekely, M. Tauszig, A. Teufl von. Ferland, R. Umlauff von F. Vlaicu, A. *Warchalowski, A. (1) Warchalowski, K. (8) Weiner, T. Widmer, J. +Wiesenbach, V. Wosecek, W.

~Private Aeroplanes.~

At the end of March, 1913, the total number of private aeroplanes in the country was about twenty.

ETRICH Monoplanes. Etrich Flieger Werke, Wiener Neustadt. Igo Etrich was a very early experimenter in conjunction with Wels. In 1909 he produced on his own account the first _Etrich_ monoplane, a characteristic machine, which except for detail improvement, varying dimensions, etc., has not been appreciably altered since. (See Historical Section.)

[Illustration: _Photo, C. Maleuit._]

----------+----------+----------+----------+ | | ~VIII 1911-12.~ | ~1912-13.~ | Model and date. | ~VII 1911.~ | 2-seater | Limousine 2-seater. | ----------+----------+----------+----------+ ~Length~ feet (m.) | 37 (11.30) | 30-3/4 (9.30) | 26-1/4 (8) | ~Span~ feet (m.) | 48 (14.60) | 42 (12.80) | 31-1/4

(9.50) | ~Area~ sq. feet (m squared.) | 380 (35) | 323 (30) | 280 (26) | {total lbs. (kgs.)| ... | ... | ... | ~Weight~ { | | | | {useful lbs. (kgs.)| ... | ... | ... | ~Motor~ h.p. | 120 Daimler | 100 | 60 Daimler | ~Speed~ m.p.h. (km.) | ... | ... | ... | Number built during 1912 | 5 | 2 | 2 | ----------+----------+----------+----------+

Remarks.--A number of _VII_ & _VIII_ have been sold for military purposes to the Austrian, Russian, German, and other governments.

[Illustration: Etrich VIII. UAS.]

[Illustration: Etrich. Limousine. _Photo, Guld._]

LOHNER-DAIMLER. This firm is now amalgamated with Etrich.

[Illustration]

----------+----------+---------- | | ~1912-13.~ | ~1911.~ | Lohner Daimler | | Pfeilflieger. ----------+----------+---------- ~Length~ feet (m.) | ... | 32 (9.70) ~Span~ feet (m.) | ... | 44-1/4 (13.50) ~Area~ sq. feet (m squared.) | ... | 450 (42) {total lbs. (kgs.) | ... | 926 (420) ~Weight~ { | | {useful lbs.(kgs.) | ... | ... ~Motor~ h.p. | 60 Aust. Daimler | 125 Aust. Daimler ~Speed~ m.p.h. (km.) | 50 (80) | 62 (100) Number built during 1912 | ? | 4 ----------+----------+----------

Remarks.--Staggered and ~V~ shape. Late in 1911 one was purchased for the Austrian Army. In 1912 made a world's altitude. Passenger record, 4,530 metres (14,862 feet.)

MERCEP Monoplanes. Mihalis Mercep, Aeroplanwerkstatte, Agram, Hungary. Russjan was connected with this firm, which built 2 biplanes to his designs in 1909. Russjan was killed in the second of these. In 1911, a _Mercep_ was built.

----------+----------+---------- | ~1911.~ | ~1912-13.~ ----------+----------+---------- ~Length~ feet (m.) | 29-1/2 (9) | 23 (7) ~Span~ feet (m.) | 34-1/3 (10.50) | 32-1/2 (10) ~Area~ sq. feet (m squared) | ... | 204 (19) {machine, etc. lbs. (kgs.) | ... | 617 (280) ~Weight~ { | | {useful lbs. (kgs.) | ... | 661 (300) ~Motor~ h.p. | ... | 50 Gnome Number built | 1 | 1 ----------+----------+----------

[Illustration: Mercep. 1912-13.]

WARCHALOWSKI, Biplane. Karl Warchalowski, Autoplan Werke, Odoakergasse 35, Vienna XVI.

[Illustration]

A machine generally on _M. Farman_ lines, but with different shaped ailerons and corners of the leading edge rounded.

WHITEHEAD. Whitehead & Co., Fiume, Austria. The Whitehead Torpedo Co. has laid down plant for the production of hydro-aeroplanes.

ZIEGLER Monoplane. Flugzengwerke Johann Ziegler, Vienna.

---------+---------- | ~1912-13.~ ---------+---------- ~Length~ feet (m.) | 59 (18) ~Span~ feet (m.) | 42-3/4 (13) ~Area~ sq. feet (m squared.) | 586 (55) { total | 1656 (750) ~Weight~ { | { useful | ... ~Motor~ h.p. | 100 Mercedes ~Speed~ m.p.h. (k.m.) | 50 (80) Number built during 1912 | 2 ---------+----------

AUSTRO-HUNGARIAN DIRIGIBLES.

~Military.~

-------+-----------+----------+-------+---------+-------+----------+---------- Date | | | Capacity | | Speed. | of | Name. | Make. | Type. | in m cubed. | H.P. | m.p.h. (k.p.h.) | Remarks. order. | | | | | | -------+-----------+----------+-------+---------+-------+----------+---------- | | | | | | 1909 | ~M 1~ | Parseval P.L. 4 | n.r. | 2300 | 70 | 27 (45) | 1909 | ~M 2~ | Lebaudy-Juillot 6 | s.r. | 4800 | 100 | 23 (37) | Wrecked 1911 | | | | | | but repaired 1910 | ~M 3~ | Koerting (K.W. 1) | n.r. | 3600 | 150 | 30-1/2 (49) | 1912 | ~M 4~ | Zeppelin | | 22,000 | 450 | 47 (75) | _Building._ -------+-----------+----------+-------+---------+-------+----------+----------

Military Dirigible Pilots.

Cajanek, V. Grebenz, K. Hauswirth, J. Heller, S. Hofstaetter, E. Macher, M. Tauber, F. Tepser, G.E. von. Weiss, H.

~Private.~

----------+----------+----------+-------+---------+------+----------+---------- Date of | | | Capacity | | Speed. | commencement. | Name. | Make. | Type. | in m cubed. | H.P. | m.p.h. (k.p.h.) | Remarks. ----------+----------+----------+-------+---------+------+----------+---------- 1910 | ~MANNSBARTH-STAGL~ | Mann-Sl. | n.r. | 8200 | 300 | 40 (65) | 1912 | ~BOEMCHER II~ | Boemcher II | | 2750 | | 25 (40) | ----------+----------+----------+-------+---------+------+----------+----------

Two _Renners_ and a _Boemcher I_ have ceased to exist.

Private Dirigible Pilots.

Adrario, K. Baumann, F. Becker, T. Berlepsch, F.F. von. Cassinone, A. Fuerst, A. Hoffory, W. Hinterstoisser, F. Kaiser, K. Mannsbarth, F. Nowy, V. Richter, von. B. Stagel, H. Stratmann, W. Wagner, E. von. F. Zborowski, J.

BOEMCHER II. (1912-13.) Non-rigid.

+----------+ | | | _Building._ | | | +----------+

~Length~, ? feet (? m.) ~diameter~, ? feet (? m.) ~volume~, 77,000 c. feet (2,750 m cubed.)

~Gas bags.~--

~Motor.~--

~Speed.~--25 m.p.h. (40 k.p.h.)

~Propeller.~--

LEBAUDY-JUILLOT 6=Military M II. (1910.) Semi-rigid.

[Illustration]

~Length,~ 229-3/4 feet (70 m.) ~diameter,~ 36 feet (11 m.) ~volume,~ 170,000 c. feet (4,800 m cubed.)

~Gas bags.~--Austro-American Rubber Co.

~Motor.~--100 h.p. Mercedes.

~Speed.~--27 m.p.h. (45 km.)

~Propellers.~--Two 2-bladed.

Remarks.--Built by the Austrian Daimler works to _Lebaudy-J._ designs. Sister to the Russian _Lebed_.

KOeRTING-WIMPASSING (K-WI). Non-rigid=Military. M III. (1911.)

[Illustration]

~Maximum length~, 213-1/4 feet (65 m.) ~maximum diameter,~ 34-1/2 feet (10.50 m.) ~volume,~ 127,150 c. feet (3,600 m cubed.)

~Total lift.~-- lbs. (kgs.) ~Useful lift,~ lbs. (kgs.)

~Gas bags.~--2 ballonets of 15,900 c. feet (450 m cubed.)

~Motors.~--2 Koerting, of 75 h.p. each (= 150 h.p.)

~Speed.~--30-1/2 m.p.h. (49 km.) Made on trials March, 1911.

~Propellers.~--Two 4-bladed. Diameter, 9-3/4 feet (3 m.)

~Steering.~--_Parseval_ style. This ship is generally an adaptation of the _Parseval_ type. Accommodates 8 people. Completed 1911. Military airship.

Remarks.--

[Illustration: UDS.]

MANNSBARTH. Non-rigid (1911). _(Alias ~STAGL MANNSBARTH.~)_

[Illustration]

~Maximum length,~ ? feet (? m.) ~maximum diameter,~ ? feet (? m.) ~volume,~ 289,600 c. feet (8,200 m cubed.)

~Total lift.~-- ? lbs. (? kgs.) ~Useful lift,~ ? lbs. (? kgs.)

~Gas bags.~--Divided into 4 compartments. Ballonet in each.

~Motors.~--2 of 150 h.p. each (= 300 h.p.)

~Speed.~--40 m.p.h. (65 k.p.h.)

~Propellers.~--3. Diameter, 13 feet (4 m.) Also 1 helice.

~Steering.~--Helice used as elevator. Rudder aft. Forward and after ballonets also used as elevators, _Parseval_ style. Completed 1911.

Remarks.--Built for Government, but not taken over.

PARSEVAL P.L. 4. Non-rigid. = MILITARY M I. (1909.)

[Illustration]

~Maximum length,~ 164 feet (50 m.) ~approx. diameter,~ 28-1/4 feet (8.60 m.) ~volume,~ ? c. feet (2,300 m cubed.)

~Total lift.~--5,730 lbs. (2,600 kgs.)

~Gas bags.~--Rubbered fabric by Austrian American Rubber Co.

~Motor.~--70-100 h.p. Mercedes Daimler. 1,200 r.p.m. at 70 h.p.

~Speed.~--27 m.p.h. (45 km.) made on trials.

~Propellers.~--_Parseval_ type, semi-rigid, chain driven, 3-bladed. Diameter, 11-1/2 feet (3.50 m.)

~Steering.~--_Parseval_ system.

Remarks.--Built by the Austrian Motorluftschiff Gesellschaft to the _Parseval_ type C design (see Germany). Taken over by the Austro-Hungarian Army, December, 1909. Has flown 6-1/2 hours. Has risen to 1,150 metres and subsequently flown 1-1/2 hours. Carries 4 persons, _about_ 400 lbs. (180 kgs.) ballast, and fuel for 12 hours. _Station_: Fischamend.

BELGIAN.

(Revised by M. BRACKE, Aeronautical Engineer & Editor of "L'Aero Mecanique.")

~Aerial Societies:~--

Aero Club Belgique.

Aero Club of Hainault.

La Ligue Nationale Belgique.

Delta Club (Kites).

Ae. C. de Flanders.

Ae. C. du Littoral.

Ae. C. de Liege-Spa.

~Aerial Journals:~--

La Conquete de l'Air. (214 Rue Royale, Brussels) bimonthly, 5 francs p.a.

L'Aero Mecanique (Brussels edition). Chemin de St. Denis, Casteau, Mons., Belgium, 2.50 f.

L'Aviation Industrielle et Commerciale (monthly), Chemin de St. Denis, Casteau, Mons., 1.50 f.

~Flying Grounds:~--

Berchem.

Brasschaet (Military).

Camp de Casteau. (Aviation Industrielle & Commerciale).

Etterbeek, near Brussels.

Kiewit.

St. Job (private property Baron de Caters.).

BELGIAN AEROPLANES.

~Military Aeroplanes.~

At the end of 1912 the military air force consisted of three 50 h.p. Gnome _H. Farman_ 1911 military, used for instructional purposes, and twenty-four 70 h.p. Gnome _H. Farmans_ (model 1912 military), for war work:

The military school is at Brasschaet, near Antwerp. Major Campion in command.

The course is as follows:

1. _Theoretical course._--Lectures on meteorology, structure of aeroplanes, aviation motors, etc.

2. _Practical._--This, in addition to flight, consists of dismounting and replacing parts of aeroplanes and aerial motors, all general repairs, erecting hangars and aerial photography.

The school possesses nine hangars, of which three are Bessonneau type, three wooden, and three metal.

For 1913 the sum of L20,000 is to be expended for purchase of aeroplanes and the establishment of aerial squadrons at Antwerp, Liege, and Brasschaet.

These are organised into six squadrons of four units each. The full complement of each squadron is eight aviators, fifteen to 20 mechanics, etc., and six citizen soldiers.

The question of hydro-avions for the Congo is under consideration.

~AVIATORS.~

(The number against any name is, unless otherwise stated, the Ae. C. Belgique pilot certificate number.)

Military.

Broune, Lieut. (37) Cozic, R. (23) Dhanis, Lieut. (35) Heinter Poorten (47) Lebon, Lieut. (36) Moulin, E. (45) Movtens, Lieut. (19) Nelis, Lieut, (in command) (28) Robert, V. (47) Sarteel, Lieut. (26) Sournoy, J. (46) Tocy, Lieut.

Private.

Allard, E. (4) Armand, C. (22) Baugniet, Edmond (18) Boel Bracke, A. Camille, Amand (22) Christiaens, Joseph (7) Crombez (25) De Caters, Baron (1) De Heel, Emile (24) D'Hespel, Comte Joseph (15) De la Hault, Adhemar De Laet, E. (31) De Laminne, Chev (9) De Jonckeer (44) Depireux, Isidore (20) De Petrovsky, Alexandre (11) De Ridder, Alphonse (13) De Roy, W. (41) Descommines Deudeuner, A. (43) Dolphyn (40) Doneryos, J. (33) Duray, A. (3) Dutrieu, Mdlle. H. (27) Fischer, Jules (12) Frenay, Fernand (21) Hanciau, P. (34) Hanouilleo, P. (42) Hasen Lamblotte, F. (29) Lanser, Alfred (16) Lescart, F. (30) Mestagh, G. (39) Michez, S.R. (32) Olieslagers, Jan (5) Olieslagers, Max Orta, Jose Peeters Pickard Stellingwerff, J. (49) Tyck, Jules (8) Van den Born, Chas. (6) Verschaeve, Fernand (17) Verstraeten, Leon (14)

The following Belgian aviators have been killed:--

+----------+ | Kinet, Daniel (2) | | Kinet, Nicolas (10) | | Verrept, John (38) | +----------+

BEHUEGHE (Bron), in Herseun. Built in 1912. A monoplane that flew very well at camp of Casteau Aerodrome during May--October. ~Motor.~--25. h.p. type Morane. New designs in wing construction, landing chassis, etc.

A. BRACKE (formerly Bracke, Missyon & Co.), Casteau, Mons. In 1910, constructed the first aeroplane built by a Belgium firm--a monoplane with planes at 120. This machine has not been duplicated: but the firm have since built machines to private specifications. The only firm which has in Belgium the speciality of aeronautical patents.

DE BROUCKERE, 23 rue Joardens, Brussels. Biplane. H. Farman. type. Built in 1911, modified in 1912.

DE LA HAULT Adhemar de la Hault, 214 rue Royale, Brussels. In 1906, built a flapper of novel design. This was followed in 1910, by a machine on monoplane lines with one fixed plane and two flapping wings. This failed to fly, and in August, 1911, was altered into a biplane. It did not succeed, however. M. Hault is still pursuing the ornithopter question.

HAREL I. Biplane. ~Length.~--49-1/4 feet (15 m.) ~Surface.~--344-1/2 sq. feet (32 m squared.) ~Weight.~--771 lbs. (350 kgs.), flying order. Warping wings. Monoplane tail. ~Motor.~--50 h.p. Gnome, mounted just under and forward of the upper wing. ~Tractor.~--1 Chauviere. Elevator placed 1 in front and 1 in rear, _H. Farman_ style. Rudders, 2 in rear. Completed May, 1911. For further details see _Conquete de l'Air_, July 1st, 1911. Property of M. Van der Stegen.

WILLIAMS. Biplane. ~Motor.~--70 h.p. E.N.V. Generally of headless _Voisin_ type on a _Farman_ body. Completed 1911. Has flown fairly well.

BELGIAN DIRIGIBLES.

Military.

~1910.~

1. LA BELGIQUE II~ (late ~I~)_ 4,000 m cubed.

~1911.~

2. LA BELGIQUE III

Note.--_La Belgique I_ was built in 1909 and re-built 1910.

Private.

VILLE DE BRUXELLES 6,000 m cubed.

LA BELGIQUE II. (No. 1 rebuilt.) Military.

[Illustration]

~Length,~ 226 feet (64.8 m.) ~maximum diameter,~ 35 feet (10.75 m.) ~capacity,~ 141,300 cubic feet (4,000 m cubed.) ~total lift,~ 9,921 lbs. (4,500 kgs.)

~Gas bag.~--Rubber proofed Continental fabric. Ballonet, filled by a separate motor giving 7.5 inches of water pressure. Warm air can be pumped in if required. Ballonet, 28,250 c. feet (800 m cubed.)

~Motors.~--2, each of 60 h.p., Vivinus, 4-cylinder, 112x130.

~Propellers.~--1 in front of the car. 285 revolutions per minute. Wood construction.

~Speed.~--25 miles per hour. 40 km. per hour.

~Planes.~--Horizontal: a gas tube bent horizontally round the tail. Vertical: vertical fins on the tail, and a long vertical keel under the gas bag.

~Car.~--A girder, square in section, tapered at both ends. Built of tubular steel. Length, 82 feet (25 m.)

~Miscellaneous.~--Built by L. Godard, France, 1909. Crew, 3 men. Accommodation for 1 passenger. Fuel for 10 hours. Greatest height attainable, 3,280 feet (1,000 m.)

~Table of weights.~-- Gas bag, complete with ballonet, valves, planes, lbs. kgs. suspension, etc 1,951 885 Propellers (2) 275-1/2 125 Blower 33 15 3 h.p. motor for blower 33 15 Motors (2) complete with gearing and shafting 1,410 640 Car 992 450 Fuel for 10 hours 738-1/2 335 Ballast 826-3/4 375 1 passenger (or ballast) 154 70 Crew (3) 463 210 Guide ropes, etc 220 100 Miscellaneous 88 40 ----- ----- _About_ 7,165 3,250

Remarks.--Reconstructed in the winter of 1909. There are two noteworthy innovations in connection with the ballonet. (1) The ballonet can be warmed by the motor. (2) In case of real emergency air can be pumped direct into the gas bag. Experiments of the utmost importance to all airships are in progress with a view to ridding the gas of this air cheaply and quickly.

LA BELGIQUE III. Military.

Presented 1910 to the Belgian Government by H.M. the King of the Belgians. 4,500 m cubed. Practically same as II, but has 3 propellers. ~Motors.~--Two 100 h.p. Germain.

VILLE DE BRUXELLES.

(Formerly known as LA FLANDRE.)

(Astra type.)

[Illustration]

~Maximum length,~ 256 feet (78 m.) ~maximum diameter,~ 41 feet (12.4 m.) ~volume,~ 212,000 c. feet (6,000 m cubed.)

~Total lift.~--15,763 lbs. (7,150 kgs.) ~Useful lift,~ lbs. (kgs.)

~Gas bags.~--Continental rubbered fabric, yellow. Ballonet, 16,146 c. feet (1,500 m cubed.)

~Motors.~--2 Pipe motors of 100 h.p. each, placed in line with each other in the fore and aft line, and with clutches and the necessary gearing in between them.

~Speed.~--35 m.p.h.

~Propellers.~--3, namely: one at the fore end, driven by the two motors when coupled together, and two placed above and on either side of the centre of the car, for use when only one motor is running. Chauviere propellers.

~Steering.~--Vertical steering by means of a large double aeroplane fixed above the car, about a third from the front. Horizontal steering by means of a double vertical rudder above the rear end of the car. Stability is secured by the usual Astra pear shaped stabilising gas bags, with fins of rubbered cloth spread between the inner edges of these shapes.

Remarks.--The distinctive feature of this ship is the arrangement of the propellers. Both motors can be coupled either on to the front propeller or on to the two rear propellers, or on to all three together, but they are actually intended only to drive the front one. On stopping either motor the other is connected to the two rear propellers, which are designed for a slower speed of translation than the front one, with the result that the running motor does not find itself overloaded as it would if the same propeller had to serve both for one and for two motors.

BRAZILIAN.

~AVIATORS.~

Garos, Queiroz, Robert, Henri, Santos-Dumont, Versepuiz.

There are possibly one to two aeroplanes in Brazil, but the well-known aviators live in France. Little or nothing seems doing in Brazil as yet.

BRITISH.

~Aerial Societies:~--

Royal Aero Club.

Aerial League.

Aeronautical Society. (Premier Society, founded 1866.)

Brooklands Aero Club.

There were once a great many local aero clubs, but the majority of these have ceased to exist and with one or two possible exceptions all the rest are moribund.

~Aerial Journals, etc.:~--

Aeronautical Journal. Quarterly. 53, Victoria St., London, S.W.

Aeronautics. 3d. monthly. 27, Chancery Lane, London, W.C.

The Aero. 6d. monthly. 20, Tudor St., London, E.C.

Flight. 3d. every Saturday. 44, St. Martins Lane, London, W.C. (Official organ of the R. Ae. C.)

The Aeroplane. 1d. weekly, 166, Piccadilly, London.

All the World's Aircraft. 21/-. Annual. 100, Southwark Street, London, S.E. and 5, Queen Victoria Street, London, E.C.

In addition, the _Car Illustrated_ and the _Motor_ devote considerable space to aerial matters.

~Principal Flying Grounds:~--

~Aldershot.~--Army school.

~Brighton,~ Shoreham Aerodrome. Aero school.

~Brooklands.~--Bristol school.

~Camber Sands,~ Rye, Sussex.--At low tide moderately hard sand and soft places. Area two miles by one mile.

~Dagenham~ (Aeronautical Society).

~Dartford Marsh.~--Vickers school.

~Dunstall Park,~ Wolverhampton.

~Eastbourne.~ Aerodrome School.

~Eastchurch,~ Sheppey.--(R. Ae. C.) 350 acres. Sheds. Members only. R. Naval school.

~Filey.~--Blackburn school.

~Hendon.~--Grahame-White, Blackburn, Bleriot, Deperdussin, Temple and Ewen schools.

~Lanark.~--Deperdussin school.

~Liverpool~ (Melly school).

~Llandudno & North Wales.~--Aerodrome.

~Mapplin Sands,~ Essex.--(Foulness). Very hard sand at low tide. Area ten miles by four miles. Property of War Office. Flying forbidden in winter.

~Salisbury Plain.~--Bristol school. Vast space available. Plenty of fairly smooth ground. Army school.

~Shoreham.~--(See Brighton).

~Upavon.~ Central flying school (R. Flying Corps.)

~BRITISH MILITARY AVIATION.~

~Royal Flying Corps.~

In 1912 the Royal Flying Corps was instituted. It consists of two wings, navy and army, with a central flying school at Upavon, Salisbury Plain.

The staff is as follows:--

Commandant: Paine, Capt. G.M., M.V.O., R.N.

Secretary: Lidderdale, Asst. Paymaster J.H., R.N.

Medical Officer: Lithgow, Capt. E.G.R., R.A.M.C.

Quarter-Master: Kirby, Hon. Lieut. (Qr.-Mr.), V.C.

Instructor in Theory and Construction: Cook, Lieut.-Col. H. R., R.A.

Instructor in Meteorology: Dobson, G., Esq.

Instructors in Flying:

Fulton, Capt. J. D. B., R.A.

Gerrard, Capt. E. L., R.M.

Shepherd, Lieut. P. A., R.N.

Trenchard, Mt. Maj. H. M., D.S.O., R. Sc. Fus.

Salmond, Capt. J. M., R. Lanc. R.

Inspector of Engines: Randall, Eng.-Lieut. C. R. J., R.N.

~Royal Aircraft Factory.~

This is situated at Farnborough. Mervyn O'Gorman is superintendent. There are large sheds. Some _B E_ biplanes have been built here, but the principal object of the factory is understood to be repairs and maintenance.

~Naval Wing Royal Flying Corps, Aeroplane Section.~

There is a special Air Department at the Admiralty with Captain M. F. Sueter, as Director, Commander O. Schwann and Lieut. C. L'Estrange-Malone, as Assistants,

Eng. Lieut. G. W. S. Aldwell, as Eng. Inspector.

Officers are graded Flying Officers, then Flight Commanders, thence to Squadron Commanders.

The flying school is at Eastchurch, Sheppey. Commander Sampson, S.C., in command. There are at present four air stations: (1) Isle of Grain, (2) Calshot, (3) Harwich, (4) Yarmouth.

At the end of March, 1913, the total number of aeroplanes including those on order, school machines, etc., was about 32; of which about 16 were effective for war purposes or available at short notice.

These machines were as follows:--

7 monoplanes (= 1 Bleriot, 2 Deperdussin, 1 Etrich, 1 Nieuport, 2 Short).

15 biplanes (= 1 Avro, 2 Bristol, 1 Breguet, 1 Caudron, 2 H. Farman, 1 M. Farman, 5 Short, 2 Sopwith).

10 hydro-avions (= 1 Astra, 1 Avro, 2 Borel, 1 Donnet-Leveque, 1 H. Farman, 1 M. Farman, 3 Short).

The _personnel_ is as follows (number after names is the R. Ae. C. brevet number):--

~Squadron Commanders.~

Gerrard, Capt. F. L., R. M. (76) Gordon, Capt., R. M. (161) Gregory, Lieut. (75) L'Estrange-Malone, Lieut. C. (195) Longmore, Lieut. Sampson, Com. C. R. (71) Shepherd, Lieut. P. A. (215)

~Flight Commanders.~

Courtney, Lieut. I. T. (R. M.) Grey, Lieut. Spencer (117) Risk, Capt. C. E., R. M. (303) Seddon, Lieut. J. W. (296)

~Flying Officers.~

Those marked * are under instruction, not yet graded.

*Agar, Lieut. A.W.S. Babington, Lieut. J.T. (408) Bigsworth, Lieut. A.W. (390) *Bobbett, Boatswain H.C. (334) Bowhill, Lieut. F.W. (397) *Brodribb, Lieut. F.G. (481) Courtney, Lieut. I.T., R.M. Courtney, Lieut. C.L. (328) *Davies, Lieut. R.B. (90) *Edmonds, Lieut. G.H.K. *Fawcett, Capt. H., R.M. *Gaskell, Lieut. A.B. *Hathorn, Lieut. G.H.V., R.M. Hewlett, Sub. Lieut., F.E.J. Kennedy, Lieut. J.B. *Maude, Lieut. C.E. *Noyes, Asst. Paymaster, C.R.F. Oliver, Lieut. E.A. (425) *Parker, Asst. Paymaster E.B. (415) Rathbone, Lieut., C.E., R.M. Ross, Lieut. R.P. (422) *Sitwell, Lieut. W.G. Travers, Lieut. J.L. Vernon, Lieut. H.D. (404) Wildman-Lushington, Lt. G.V., R.M.A.

The following R.N. officers and men are aviators employed in various duties at the Admiralty, at the Central Flying School or at Eastchurch:--

Aldwell, Eng. Lieut. G.W.S. Andrews, J.C. (372) Ashton, Ldg. Seaman Batemad, Able Seaman P.E. (446) Briggs, Eng. Lieut. E.F. Brownridge, Carp. Collins, Art. Eng. J.V. Cresswell, Lieut. T.S., R.M. (420) Deakin, A. (333) Gerrard, Capt., R.M. (76) L'Estrange-Malone, Lieut. C. (195) Lidderdale, Asst. Paymaster H.J. (402) O'Connor, Art. Eng. T. (280) Paine, Capt. G.M. (217) Randall, Eng. Lieut. (81) Schwann, Com. O. (203) Scarff, Art. Eng. F.W. Shaw, Shipwright D. (465) Shepherd, Lieut. P. (288) Susans, F. (380) Wells, Staff. Surg. H.V.

The following have privately secured pilot certificates in the years mentioned but are not employed in the R.F.C. for aeroplane work. Some of them, however (D), are employed in the airship section:--

~1911.~

Bower, Lieut. J.A. (161) Clark-Hall, Lieut. (127) Leveson-Gower, Com. Williamson, Lieut. (150) Williamson, Lieut. H.A. (160)

~Naval. 1912.~

Blatherwick, Lieut. G. (450) Brown, Com. A M.T. (345) Edwards, Lieut. C.H.H. D Freeman Williams, Lt. F.A.P.(202) Head, Lieut, G.G.W. (191) Hooper, Sub. Lt. C.W.W. (382) Johnson, Capt. C.D. D Masterman, Com. E.A.D. (Ae.C.F.) Prickett, Lieut. C.B. (381) Trewin, Asst. Paymaster (294) D Usborne, Lieut. N.F. (449) Wheeler, Mid. N.F. (370)

~Naval. 1913.~

D Boothby, Lieut. F.L.M. (Ae.C.F.) Brady, B.J.W. (394) Brown, Lieut. A.C.G. (398) Dobie, Lieut. W.F.R. (448) Fitzmaurice, Lieut. R. (447) Freeman, S.T. (393) Littleton, Sub. Lieut. H.A. (405) Picton-Warlow, Lieut. W. (451) Ross, Lieut. R.P. (422)

~Army Wing Royal Flying Corps, Aeroplane Section.~

The Army wing has its headquarters at S. Farnborough, its constitution being as follows:--

1st squadron (airships or kites) see Dirigible Section. 2nd " (aeroplanes) base at Montrose. 3rd " " " " Salisbury Plain. 4th " " " " S. Farnborough. (Four more aeroplane squadrons _pro._)

An aeroplane squadron nominally consists of 18 aeroplanes (9 in service, 9 remounts). At the end of March, 1913, the total number of aeroplanes, including those on order, school machines, etc., was about 110, of which about 50 (including some monoplanes not in use) were effective for war purposes or available at short notice.

The total of 110 was thus made up:--

22 monoplanes (= 2 Bleriot, 4 Bristol, 5 Deperdussin, 4 Howard-Flanders, 1 Martinsyde, 6 Nieuport).

86 biplanes (= 4 Avro, 22 B.E. type,[A] various makers), 2 Breguet, 2 Caudrons, 30 Farman (various types), 6 Short--and about 20 Avro or Farman or Short not delivered.

~Squadron Commanders.~

Brooke-Popham, Capt. H.R.M. (108) Burke, Capt, C.J. (46) (Ae.C.F. 260) Carden, Lt. A. D. (239) Cook, Lt.-Col. H. R. (42) Fulton, Major J. D. B. (27) Raleigh, Capt. G. H. (196) Trenchard, Major H. M. (270)

~Flight Commanders.~

Allen, Capt. C. R. W. (159) Beor, Lt. B. R. W. (R.A.) (185) Becke, Capt. J. H. W. (236) Connor, Lt. D. G. (54) Fox, Lt. A. G. (176) Higgins, Major J. F. A. (R.A.) (264) Longcroft, Lt. C. A. H. (192) Reynolds, Lt. H. R. P. (R.E.) Salmond, Capt. J. M. Webb-Bowen, Capt. T. I. (242)

~Flying Officers.~

Abercromby, 2nd Lt. R. O. (134) Allen, Lt. D. L. (318) Anderson, Lt. E. V. (247) Atkinson, Lt. K. P. (267) Barrington-Kennett, Lt. B. H. (Adjutant) (43) Beatty, Capt. W. D. (89) *Birch, Lt. W. C. K. (375) Board, Capt. A. G. S. (36) Boyle, Lt. the Hon. D. G. Burchardt-Ashton, Lt. A. E. Burroughs, Lt. J. E. G. Carmichael, Lt. G. I. (316) *Chinnery, Lt. E. F. (211) Cholmondeley, Lt. R. (271) *Christie, Lt. A. (R.A.) (245) Conran, Lt. E. L. (342) *Corbalis, Lt. E. R. L. Darbyshire, Capt. C. (257) Dawes, Lt. L. (228) Dawes, Capt. G. W. P. (17) *Gill, Lt. N. J. (174) Glanville, Lt. H. F. (307) Gould, 2nd Lt. C. G. S. (282) Harvey, Lt. E. G. *Harvey-Kelley, Lt. H. D. Herbert, Capt. P. L. W. (244) Holt, Lt. A. V. (312) Hubbard, 2nd Lt. T. O. B. (202) Hynes, Lt. G. B. (R.A.) (40) James, Lt. B. T. Joubert, de la F. Lt. P. B. (280) Lawrence, Lt. W. MacDonnell, Capt. H. C. (273) MacClean, Lt. A. C. H. *Mapplebeck, Lt. G. W. C. (386) Martyn, Lt. R. B. Mead, Sergt. J. (475) Mellor, Capt. C. (155) *Mills, Lt. R. P. (377) Moss, Bt.-Major L. B. (241) *Musgrave, Capt. H. (R.E.) *Mulcahy-Morgan, Lt. T. W. *Noel, Lt. M. W. (416) Pepper, Lt. J. W. (98) *Picton-Warlow, Lt. W. (451) Playfair, 2nd Lt. P. H. L. (283) *Pretyman, Lt. G. F. (341) Porter, Lt. G. T. (R.A.) (169) Pryce, Hon. Lt. W. J. D. (Qr.-mr.) *Read, Lt. A. M. (336) *Rodwell, Lt. R. M. Roupell, 2nd Lt N. S. (237) Shepherd, Capt. G. S. (215) Soames, Lt. A. H. L. Small, Lt. F. G. D. (429) *Small, Lt. R. G. (343) Smith-Barry, 2nd Lt. R. R. (161) Stopford. Lt. G. B. *Todd, Lt. E. (185) Thompson, Lt. A. B. Tucker, Capt. F. St. G. *Vaughan, 2nd Lt. R. M. Wadham, 2nd Lt. V. H. N. (243) Waldron, Lt. F. F. (260) Wanklyn, Lt. F. A. (284)

~Reserve.~

Ashmore, Major E. B. (281) Bell, 2nd Lt. C. G. (100) De Havilland, 2nd Lt. G. (53) Hartree. 2nd Lt. A. (214) Henderson, Col. D. (118) Marks, Lt. C. H. (83) Pizey, 2nd Lt. C. P. (61) Salmond, Capt. W. G. H. Smith, Lt. S. C. W. Unwin, Lt. E. F. Warter, 2nd Lt. H. de V. (107)

~Special Reserve.~ (_2nd Lieuts. on probation._)

Biard, H. C., de la F. (218) Busteed, H. R. (194) Charteris, R. L. (197) Cutler, H. D. (189) Davies, E. K. (22) *Fuller, E. N. (325) Fuller, H. C. (Ae. C. F.) Gibson. W. E. (129) Hammond, J. J. (32) Humphreys, G. N. (390) Lerwill, F. W. H. Metford, L. S. (146) Perry, E. W. C. (130) Rickards, G. B. (400) Sippe, S. V. (172) Spratt, N. C. (339) Ware, D. C. Wilson, C. D. (Ae. C. F. 136) *Wilson, C. W. (329) Young, D. G. (207)

The following have qualified privately, R. Ae. C. brevets, but are not at present employed in the Aeroplane Section:--

~1910.~

Gibb, Lt. (10) Snowden Smith, Lt. (29) Watkins, Lt. H. E. (25) Wood, Capt. H. F. (37)

~1911.~

Blacker, Lt. (12) Cross, Lt. (151) Dickson, Capt. (Ae. C. F. 260) Harford, Lt. (152) Harrison, Capt. (158) Hoare, Capt. (126) Hooper, Lt. (149) Hutchinson, Capt. Steele (143) Manisty, Lt. G. (135) Pitcher, Capt. (125) Sebag-Montefiore, Lt. (93) Smeaton, Lt.-Col. (115) Strover, Lt. E. J. (145)

~1912.~

Agnew, Capt. C. H. (240) Alston, Capt. R. C. W. (255) Ashton, Lt. A. E. B. (201) Bannerman, Major Sir A. (213) Boger, Capt. R. (335) Borton, Lt. A. E. (170) Boyle, Capt. M. (241) Brodigan, Lt. F. J. (200) Broke-Smith, Capt. D. W. (204) Bulkeley, Lt. H. T. (246) Carfrae, Lt. G. T. (188) Chamier, Capt. J. A. (340) Cordner, Capt. R. H. L. (277) Ellington, Capt. E. L. (305) Empsom, Lt. J. (387) Fielding, L. H. C. (212) Fletcher, Lt. (229) Hanlon, Lt. D. R. (311) Jones, Lt. B. T. (230) Lewis, Lt. D. (216) Mackay, Lt. M. E. (177) Mackworth, Lt. J. D. (209) Martin-Barry, Lt. (Ae. C. F.) McCudden, Capt. J. H. (269) Miller, Capt. G. R. (313) Murray, Lt. R. G. H. (320) Nicholas, Capt. C. P. (266) Penn-Gaskell, Lt. L. de C. (308) Percival, Lt. D. (226) Pollok, Lt. R. V. (379) Powell, Capt. D. W. (389) Price, Capt. C. L. (299) Rawson, Lt. K. (249) Reilly, Lt. H. L. (252) Ridd, Corporal F. (227) Roger, Capt. R. (335) Stott, Capt. J. N. J. (373) Styles, Lt. F. E. (338) Thomas, Staff-Sergt. (276) Trevenon, Lt. B. J. (230) Weeding, Capt. (182) Winfield-Smith, Lt. S. G. (187) Worthington-Wilmer, Lt. F. M. (254)

~1913.~

Archer, Lt. R. H. (434) Bayly, Lt. C. G. G. (441) Bruce, Sergt. W. R. (467) Bourke, Lt. U. J. D. (479) Cameron, Major N. J. (478) Chidson, Lt. M. R. (471) Crogan, Lt. F. J. L. (460) Harrison, Lt. Hawker, Lt. L. G. (435) Hordern, Lt. L. C. (440) Hosking, Lt. C. G. (472) Hunter, Sergt. Kemper, Sergt. K. (444) Lee, Lt. C. F. (431) Maclean, Lt. L. L. (427) Marshall, Lt. R. (470) McMullern, Lt. J. D. (436) Merrick, Major G. C. (484) Mitchell, Lt. W. G. S. (483) Read, Lt. W. R. (463) Rees, Lt. Col. W. B. (392) Stafford, Sergt. W. G. (438) Street, Sergt. E. J. (439) Thomas, Sergt. Major Vagg, Sergt. H. R. (443)

The above figures are mainly taken from _The Aeroplane,_ 1st May, 1913.

* = under instruction; not yet graded.

PRIVATE AVIATORS.

(The number against any name is, unless otherwise stated, the R. Ae. C. pilot certificate number).

To end of ~1911.~

Abbott, C. R. (101) Aitken, A. H. (56) Anderson, J. A. (164) Archer, Ernest (Ae. C. F. 214) Ballard, F. M. (151) Barber, H. (30) Barnes, G. A. (16) Blackburn, H. (79) Bowens, R. G. (39) Boyle, Hon. Alan (13) Bretherton, John (136) Breton, J. (136) Brown, H. B. (109) Chataway, J. D. (167) Challenger, G. H. (58) Chambers, C. F. M. (168) Cockburn, G. B. (5) Cockerell, P. (132) Cody, S. F. (9) Conway-Jenkins, F. (74) Crawshay, R. (133) Colmore, G. C. (15) Dacre, G. B. (162) Darroch, G. R. S. (59) Dolphin, W. H. (82) Dunkinfield-Jones (138) Ducroq, M. (23) Dyott, G. M. (114) Driver, E. F. (110) Egerton, M. Hon. (11) England, Gordon (68) Esterre, C. R. (Ae. C. F. 259) Ewen, W. H. (63) Fleming, H. R. (69) George, A. E. (19) Graham-White, Claud (6) (Ae. C. F. 30) Gresswell, C. H. (26) Grey, W. H. de (107) Halse, E. (131) Hamel, Gustav (64) (Ae. C. F. 358) Harding, Howard (Ae. C. F. 213) Harrison, Eric (131) Hewlett, Mrs. (122) Higginbotham, Gerald (96) Hilliard, W. M. (102) Hubert, Charles (57) Hotchkiss, E. (87) Houdini, Harry Hucks, B. G. (91) Hunter, A. (137) Johnston, St. Croix, P. G. (41) Johnstone, W. Barnley (103) Kemp, R. C. (80) Keith-Davies, E. King Knight, Archibald (60) Lawrence, W. (113) Longstaffe, J. L. (140) Loraine, Robert (Ae. C. F. 126) Low, A. R. (34) Macdonald, L. F. (28) Maron, Louis (62) Martin, J. V. Mrs. (55) Macfie, R. (49) McArdle, W. E. (Ae. C. F.) M'Clean, F. K. (21) Mellersh, O. S. (155) Melly, H. G. (Ae. C. F.) Moorhouse, W. B. R. (147) Morrison, O. C. (46) Moore-Brabazon, J. (1) Noel, Louis (116) Ogilvie, A. (7) Pashley, Cecil L. (106) Pashley, E. C. (139) Paterson, C. E. (38) Paul, E. A. (Ae. C. F.) Percival, N. S. (111) Petre, H. A. (128) Philpott, R. W. (81) Pixton, H. (50) Prentice, W. R. (67) Radley, J. (12) Rawlinson, A. (3) Raynham, F. P. (85) Roe, A. V. (18) Salmet, H. (99) Sassoon, E. V. (52) Santoni, L. Singer, A. M. (8) (Ae. C. F. 24) Slack, R. B. (157) Smith, S. E. (33) Smith, W. W. (Ae. C. F.) Spencer, H. (124) Somers-Somerset (Ae. C. F. 151) Sopwith, T. (31) Stanley-Adams, H. (97) Stark (Ae. C. F. 110) Stocks, Mrs. C. de B. (153) Thomas, J. H. (51) Travers, J. L. (86) Turner, C. C. (70) Turner, L. W. F. (66) Valentine, J. (47) Watt, W. O. (112) Weir, J. D. (24) Weston, John (Ae. C. F.) Wickham, R. F. (20) Woodward, G. A. T. (A

To end of ~1912.~

Barnwell, R. H. (278) Beech, A. C. (Ae. C. F.) Bendall, W. (180) Bettington, A. V. (326) Birch, E. (322) Brock, W. L. (285) Cheeseman, W. E. (293) Featherstone, W. (384) Fowler, F. H. (221) Gates, R. T. (225) Garne, T. (173) Geere, A. E. (310) Gill, R. W. R. (258) Hall, H. W. (332) Hall, J. L. (291) Hardman, W. L. (323) Harrison, W. J. (275) Hawker, H. G. (297) Hedley, W. S. (274) Hewitt, V. (302) Higginbotham, V. C. (317) Holyoake, R. G. (268) James, J. H. (315) James, H. H. (344) Kershaw, R. H. (248) Lister, R. A. (250) Nesham, H. P. (219) Nevill, M. R. (223) Manton, M. D. (231) Meredith, C. W. (193) Merriam, F. W. (179) Parr, S. (184) Payze, Arthur (337) Potet, A. (224) Prensiel, G. (198) Simms, R. H. (261) Stodart, Dr. D. E. (321) Summerfield, S. (292) Sutton, E. F. (295) Sweetman-Powell, H. (251) Taylor, V. P. (376) Tremlett, L. A. (208) Wood, V. G. (171) Wynne, A. M. (314) Wright, H. S. (331) Yates, V. (306)

~1913~ (Brevets from 400 onward).

Andreas, F. G. (477) Barron, J. C. (480) Hodgson, W. P. (433) Kehrmann, J. C. (420) King, R. A. (482) Lane, H. T. G. (418) Lawford, E. H. (442) Macandrew, H. E. W. (401) Macneill, W. (Ae. C. F.) McNamara, J. C. (445) Minchin, F. R. (419) Muller, P. M.

(432) Temple, G. L. (424) Thompson, A. B. A. (452) Tower, H. C. (466) Rainey, T. H. (474) Russell, A. L. (406) Stewart, H. (473) Strain, L. H. (476)

The following British aviators have been killed:

+-------------------------------------+ | 1910. | | Rolls, Hon. C. (2) | | | | 1911. | | Benson, R. | | Cammell, Lieut. (45) | | Grace, Cecil (4) | | Napier (104) | | Oxley, H. (78) | | Ridge, T. (119) | | Smith, V.[B] | | | | 1912. | | Allen, D. L. (183) | | Astley, J. H. D. (48) | | Bettington, Lt. C. A. (256) | | Campbell, Lindsay (220) | | Clark, Miss J. | | Fenwick, R. C. (35) | | Fisher, E. V. B. (77) | | Gilmour, Graham (Ae. C. F.) | | Hardwick, A. | | Hamilton, Capt. P. (194) | | Hotchkiss, Lieut. | | Loraine, Capt. (154) | | Petre, Edward (259) | | Parke, Lieut. W. (73) | | Wilson, St. Serg. (232) | | Wyness-Stuart, Lt. A. | | | | 1913. | | Arthur, Lt. Desmond (233) | | Berne, Paym'st'r (R.N.) | | England, G. (301) | | Macdonald, L. F. | | Rogers-Harrison, Lieut. L. C. (205) | +-------------------------------------+

BRITISH AEROPLANES

~A~

AIRCRAFT FACTORY. Royal Aircraft Factory, Farnborough, near Aldershot. For a long time this establishment had been engaged in dirigible construction and repairs. In 1911 it was decided to expand it in connection with the Royal Flying Corps. Its precise functions are somewhat uncertain. Its nominal main purpose is the repair, etc., of Service Aircraft. During 1912, however, it turned out several machines to a design of its own, known as the _"B.E."_ This design was at one time regarded as confidential; but subsequently duplicates were built by private contractors, and the design illustrated below, published by the Advisory Committee for Aeronautics.

[Illustration: B.E. type. R.A.F. UAS.]

~Length,~ 29-1/2 feet (9 m.)

~Span.~--36-3/4 feet (11.20 m.)

~Area.~--374 sq. feet (34-3/4 m squared.)

~Weight.~--

~Motor.~--75 h.p. Renault and others.

~Speed.~--

AERO'S Ltd. St. James' Street, Norwich Union Buildings, Piccadilly, London, S.W. Established 1912 for the sale of all parts and accessories; also for the sale of second hand aeroplanes and motors of all makes. Does not construct at present.

AIRCRAFT MANUFACTURING Co., Ltd. 47, Victoria Street, London, S.W. Works: Hendon, London, N.W. This company established in 1912, holds all the British rights for the _H. & M. Farman_ types. It constructs in England all _Farman_ types at its own works. (See _Farman_, French).

AVRO. Aeroplanes. A. V. Roe & Co., Clifton Street, Miles Platting, Manchester; also Shoreham, Sussex. A. V. Roe designed his first machine, a biplane, in 1906. It was the

first British machine to leave the ground. He then experimented with triplanes in Lea Marshes, where he managed to fly with only 9 h.p. in 1908-9. In August, 1910, built _Roe III_, and in September, _Roe IV_, also triplanes (see 1911 edition for full details). In 1911 he abandoned triplanes for the _Avro_ biplane. School: Shoreham.

[Illustration: Type D (1911). _Photo, Alan H. Burgoyne, Esq., M.P._]

----------+----------+----------+----------+----------+---------- | ~D 1911-12.~ | ~E 1912.~ | ~F 1912.~ | ~G 1912-13.~ | ~E 1912-13.~ Model. | 2-seater | 2-seater | Totally | Totally | Hydro-biplane. | biplane. | biplane. | enclosed | enclosed | | | | mono. | biplane. | ----------+----------+----------+----------+----------+---------- ~Length~ feet (m.) | 31 (9.45) | 29 (8.84) | 23 (7) | 29 (8.84) | 33 (10) ~Span~ feet (m.) | 31 (9.45) | 36 (11) | 28 (8.50) | 36 (11) | 47-1/2 (14.50) ~Area~ sq. ft. (m squared.) | 279 (26) | 335 (32) | 158 (14-1/2) | 335 (32) | 478 (34-1/2) {empty lbs. (kgs.) | 800 (363) | 900 (482) | 550 (249) | 1191 (540) | 1740 (789) ~Weight~ { | | | | | {fully loaded, lbs. (kgs.) | ... | 1300 (589) | 800 (363) | 1700 (771) | 2700 (1224) ~Motor~ h.p. | 35, any make | 50 Gnome | 40 Viale | 60 Green | 100 Gnome ~Speed~ m.p.h. (km.) | 48 (78) | 61 (97) | 65 (105) | 61.8 (100) | 55 (90) Number built during 1912 | several | 6 | 1 | 1 | 1 ----------+----------+----------+----------+----------+----------

Remarks.--Of the above, 4 of the 50 Gnome E type were purchased by the British Royal Flying Corps, and one by the Portuguese Government; the other went to Windermere on January, 1913, for hydro experiments. Climbing speed of this type is 440 feet per min. (134 m.) Dual control fitted. D type are no longer being built. Climbing speed of F type, 300 feet per min. (91.5 m.) Gliding angle, 1 in 6. G has a gliding angle 1 in 6.5. On October 24th, 1912, made British record to date, 7'31-1/2" (=450 miles). The hydro. was delivered to the British R.F.C. naval wing early in 1913.

[Illustration: Avro. Type D (1911-12). U.A.S.]

[Illustration: E type Standard 50 h.p. Avro Biplane.]

+----------+ | | | _No suitable photo available._ | | The machine is on usual lines. The first had a single float, but now | | two floats are used. | | | +----------+ E type 100 h.p. Avro Hydro-biplane.

[Illustration: F type Enclosed Avro Mono.]

[Illustration: G type Enclosed Avro Biplane.]

~B~

BLACKBURN Aeroplanes. Blackburn Aeroplane Co., Balm Road, Leeds. Blackburn produced his first machine early in 1910 (see 1911 edition for details). In the latter part of that year he designed the machine which ultimately developed into the _Blackburn_ military. In 1911 other types were produced, all being fitted with the patent Blackburn triple control. School at Filey Hucks has been the principal _Blackburn_ flyer. The type has also been very successfully flown by naval officers. Capacity of works: about 24 a year.

----------+----------+----------+---------- | ~1912-13.~ | ~1912-13.~ | ~1913.~ | Military. 2-seater. | Military. 1-seater | Hydro-biplane. | | | 2-seater ----------+----------+----------+---------- ~Length~ | 32 feet (9.75 m.) | 25 feet (7.60 m.) | 33 feet (10 km.) ~Span~ | 40 feet (12.20 m.) | 32 feet (9.75 m.) | 44 & 36 ft. (13.40 & 11 km.) ~Area~ | 276 sq. ft. (26 m squared.) | 195 sq. ft. (18 m squared.) | 410 sq. ft. (38 m squared.) ~Weight~ (total) | ... | 750 lbs. (340 kgs.) | 1250 lbs. (507 kgs.) ~Motor~ h.p.| ... | 50 Gnome. | 80 Gnome or 100 Anzani ~Speed~ | 55-65 m. (90-105 km.) | 60 m. (97 km.) | 65 m. (105 km.) ----------+----------+----------+----------

Notes.--Petrol for 5 hours (higher endurances can be fitted). Specially designed for military work--all steel construction. All parts unwelded to admit of rapid displacement. Clear observation provided for.

~Fuselage.~--The fuselage is ~V~ shaped and constructed of weldless steel tubing in the form of a lattice girder. The main longitudinals are of round section; cross members, oval section. Connections are not welded but made with strong steel clips so that should any member become damaged a new one can be readily arranged. The front portion is covered with sheet metal giving additional strength and reducing the head resistance. Stream line form tapering towards the rear which is covered with fabric.

~Chassis.~--Two long skids connected up to fuselage by metal struts. Each skid borne by a pair of wheels, axle held down by elastic shock absorbers. On the axle of the wheels are fitted steel springs which take side thrust. Each pair of wheels held by radius rods forming a bogie.

~Control.~--Patent Blackburn triple, independent or simultaneous on hand wheel, but special foot control for rudder is fitted if desired.

In 1912, five machines were built, of which two were of the mil. model. Others, non-military models (see last edition.)

[Illustration: Military monoplane.]

[Illustration: BLACKBURN. Military Type. Two-seater. UAS]

[Illustration: BLACKBURN. Naval Type.]

BRISTOL. The British & Colonial Aeroplane Co., Ltd., Filton House, Bristol. Founded 1910. Capital (1913), ?. Have very extensive works (area. ? sq. feet) on the outskirts of Bristol, employing over 300 men, where they manufacture to their own designs practically every type of flying machine. Flying grounds: Salisbury Plain, Brooklands. 105 Royal Aero Club certificates won on _Bristol_ machines during 1912 (of which 86 were officers of His Majesty's Forces).

----------+----------+----------+----------+---------- | ~Military~ | ~Military~ | | | ~mono.~ | ~mono.~ | ~Tractor~ | ~School~ | 2-seater. | 2-seater. | ~biplane~ | ~mono.~ | 80 h.p. | 50 h.p. | ~1913.~ | Side by side. | ~1912-13.~ | ~1912-13.~ | |

---------- ~Length~ feet (m.) | 28-1/4 (8.60) | 23-2/3 (7.20) | 27-3/4 (8.47) | ~Span~ feet (m.) | 42-1/3 (12.90) | 39-1/3 (12) | 34-1/3 (10.44) | ~Area~ sq. feet (m squared.) | 221 (20.6) | 226 (22) | 370 (34.4) | ~Total~ {machine, lbs. (kgs.) | 1719 (771) | 1323 (600) | 1764 (800) | ~weight~ {useful lbs. (kgs.) | 710 (322) | 551 (250) | 1200 (544) | ~Motor~ h.p. | 80 Gnome | 50 Gnome | 70 Renault | 50 Gnome ~Speed~ {max. m.p.h. (km.) | 73 (118) | 62 (100) | 70 (112) | {min. m.p.h. (km.) | ... | ... | ... | ~Endurance~ hrs. | 4 | 3-4 | ... | Number built during 1912 | ... | ... | ... |

Notes.--~Monoplane:~ Box section fuselage convex on bottom side to minimise resistance. Mounted on 2 wheels and 2 skids with smaller wheels attached at the forward end. Bristol tractor. ~Biplane:~ Box section fuselage, convex on top and bottom sides. Mounted as monoplane. Bristol tractor. This machine is the latest production of the Bristol Co., and has proved an exceptionally successful flyer. Designed by M. Coanda.

[Illustration: 80 h.p. monoplane.]

[Illustration: 70 h.p. biplane. UAS.]

BLERIOT Aeronautics. Belfast Chambers, 156, Regent Street, London, W. School: Hendon. British office of the _Bleriot_ firm (see France).

BRITISH BREGUET CO., 1, Albemarle Street, Piccadilly, London, W. Works and offices: 5, Hythe Road, Cumberland Park, Willesden, London, N.W. Established 1912. Constructs in England _Breguet_ models, some of which are beginning to vary in detail from the originals (see France).

BRITISH CAUDRON. (See _Ewen_.)

BRITISH DEPERDUSSINS. British Deperdussin Aeroplane Co., Ltd., 39, Victoria Street, Westminster, London, S.W. School: Hendon.

Chairman: Admiral The Hon. Sir E. R. Freemantle, G.C.B., C.M.G.

Managing Directors: Lieut. J. C. Porte, R.N., D. Laurence Santoni.

Secretary: N. D. Thompson.

This firm handles the French models of _Deperdussins_, but has in addition a special hydro-aeroplane of its own, of which one was built in 1912. Details of this special machine are:--~Length,~ 27 feet 10 inches (8.50 m.) ~Span,~ 42 feet (12.80 m.) ~Area,~ 290 sq. feet (27 m squared.) ~Weight,~ total, 1,800 lbs. (816 kg.); useful, 1,250 lbs. (566 kg.) ~Motor,~ 100 h.p. Anzani. ~Speed,~ 67 m.p.h. (110 k.m.) Other models sold by the firm are of French type exactly (see France).

BRITISH DONNET-LEVEQUE. Handled by Aeros, Ltd., 39, St. James' Street, Piccadilly, London, S.W. Company forming March, 1913 (see France). Works and school at Shoreham.

BRITISH FARMANS. (See _Aircraft Co._)

BRITISH HANRIOTS. Hewlett & Blondeau, Omnia Works, Vardens Road, Clapham Junction, London, S.W. Construct all types of _Hanriot_ machines (see France), also build to private specifications, and deal in accessories generally.

BRITISH NIEUPORTS. Company forming 1913. Representative: M. Bonnier, 2, Goulders Green Crescent, London, N.W.

~C~

CODY. Cody flying school, Farnborough. Cody commenced experiments with kites in very early days on behalf of the British Admiralty. Subsequently built the first British Army dirigible, and an experimental Army aeroplane. In 1909, his direct connection with the Army ceased. A _Cody I_ was built in 1908. A _Cody II_ was completed June 1910. The _special features_ of both were: very strong construction, great size (_II_ had area of 857 sq. feet), ailerons. Later types, except that warping is substituted for ailerons, do not differ very materially except in minor details. All wood construction.

----------+----------+---------- | ~1911.~ | ~1913.~ | Model. | 4-seater | 4-seater | May, ~1912.~ | biplane. | biplane. | Monoplane. ----------+----------+----------+---------- ~Length~ feet (m.)| 38 (11.60) | 38 (11.60) | 38 (11.60) ~Span~ feet (m.)| 43 (13) | 43 (13) | 43-1/2 (13.25) ~Area~ sq. feet (m squared.)| 484 (44-3/4) | 483 (44-3/4) | 260 (19) {total lbs. (kgs.)| 1900 (862) | 1900 (862) | 2400 (1088) ~Weight~ { | | | {useful lbs. (kgs.)| 1000 (453) | 1000 (453) | 700 ~Motor~ | 60 Green, later | 120 Aust. Daimler | 120 Aust. Daimler | a 100 Green | | {max m.p.h. (km.)| 70 (115) | 75 (120) | 83 (135) ~Speed~ { | | | {min m.p.h. (km.)| 47 (75) | 47 (75) | 58 (95) Number built to end of last year | 1 | 1 | 1 ----------+----------+----------+----------

Remarks.--The 1911 is the famous _Cody_, which, as a 60 h.p., won both Michelin 1911 prizes, and completed the _Daily Mail_ circuit. As a 100 h.p. it won the 1912 Michelin cross-country. By the end of 1912 it is said to have flown a total of 7000 miles. The 1913 is practically a duplicate with a more powerful engine. _Special features_ of the biplanes, maximum camber to lower plane. Both planes equal span. Very strong landing gear. Propeller chain driven: 1-3/4 to 1 gearing. In February, 1913, four biplanes were ordered for the British Army.

Cody lists a mono. for 1913 a trifle longer than the above; also five variations on the biplane of from 35 to 160 h.p., which can be built if required.

[Illustration: Biplane.]

COVENTRY ORDNANCE. The Coventry Ordnance Works, Ltd., Coventry. London office: 28, Broadway, Westminster, S.W. Established 1912. Capacity: 50 machines a year without difficulty.

----------+---------- | ~1912.~ | | Model 10. | ----------+---------- ~Length~ feet (m.)| 29 (8.80) | ~Span~ feet (m.)| 56 (17) | ~Area~ sq. feet (m squared.)| 630 (58) | {total lbs. (kgs.)| 1900 (861) | ~Weight~ { | | {useful lbs. (kgs.)| 800 (362) | ~Motor~ h.p.| 100 Gnome | {max.

m.p.h. (km.)| 60 (97) | ~Speed~ { | | {min. m.p.h. (km.)| ... | ~Endurance~ hrs.| ... | Number Built during 1912 | 2 | ----------+----------+

Remarks.--Experimental machines.

[Illustration]

~D~

DUNNE. The Blair Atholl Aeroplane Syndicate, Ltd., 1, Queen Victoria Street, London, E.C. School: Eastchurch. In 1906 Lieut. Dunne was employed by the British Army authorities for secret aeroplane experiments. He had at that time patented a monoplane of < type. In 1907 _Dunne I_ was tried on the Duke of Atholl's estate in Scotland, but failed to fly, being smashed on the starting apparatus. _Dunne III_, a glider, 1908, was experimented with successfully by Lieut. Gibbs. In the same year _Dunne IV_, a larger power driven edition made hops of 50 yards or so. Early in 1910 the War Office abandoned the experiments. _Dunne II_, a triplane of 1906 design, was, by consent of the War Office, assigned to Prof. Huntingdon, who made one or two short flights with it at Eastchurch in 1910. At the same time the above syndicate was formed, and _Dunne V_, built by Short Bros., was completed in June, 1910. In 1912-13 the Huntingdon, modified, was flying well.

[Illustration]

----------+----------+----------+----------+---------- | 50 Gnome. | | | | ~1912-13~ | ~1912-13~ | ~1912-13~ | ~1912-13~ Model and Date. | single-seat | 2-seater | biplane. | biplane. | mono. | mono. | ~D 8.~ | ~D 9.~ | ~D 7.~ | ~D 7~ _bis._ | | ----------+----------+----------+----------+---------- ~Length~ feet (m.)| _not given_ | ... | ... | ... ~Span~ feet (m.)| 35 (10.66) | 35 (10.66) | 46 (14) | 45 (13.70) ~Area~ sq. feet (m squared.)| 200 (18.5) | 200 (18.5) | 552 (51) | 448 (42) {total lbs. (kgs.)|1050 (476) | 1200 (544) | 1700 (771) | 1693 (768) ~Weight~ { | | | | {useful lbs. (kgs.)| 359 (161) | 528 (239) | 414 (187) | 509 (231) ~Motor~ h.p.| 50 Gnome | 70 Gnome | 60 Green | 80 Gnome ~Speed~ m.p.h. (km.)| 60 (95) | 60 (95) | 45 (70) | 50 (80) Number built during 1912 | 1 | 1 | 1 | 5 b'lding ('13) ----------+----------+----------+----------+----------

Notes.--Biplane _D 3_ is identical with the original pattern _Dunne V_, except that it has only one propeller instead of two. It has been flown completely uncontrolled in a 20 m.p.h. wind, carrying a R. Ae. C. observer as passenger.

[Illustration: DUNNE. Original Dunne biplane D5.]

~E~

EWEN. The W. H. Ewen Aviation Co., London Aerodrome: Hendon. Also works at Lanark, Scotland (opened February, 1913). Hold the British rights for and construct at their works _Caudron_ aeroplanes (see France).

~F~

FERGUSON. J. B. Ferguson, Ltd., Belfast.

[Illustration]

This machine first appeared in 1910. Owing to an accident to Mr. Ferguson it was laid up for a long time. About the end of 1912 it re-appeared. Principal details:--

~Span.~--40 feet (12.20 m.) ~Area.~--230 sq. feet (21 m squared.) H.P. 40.

~G~

GRAHAME-WHITE. The Grahame-White Aviation Co., Ltd., 166 Piccadilly, London, W. Works and Flying Ground: Hendon. Founded by C. Grahame-White, the well-known aviator, who in 1909 commenced operations with a school at Pau. Later this was removed to England, and a general agency for the sale of aeroplanes, etc., established. This developed, and early in 1911 the firm was handling a special British agency for the U.S. _Burgess_ type known as "The Baby." The Hendon Aerodrome was acquired, and a factory established, which has grown continually ever since. In April, 1912, a monoplane to special design was completed. By the close of the same year biplanes of advanced design were constructed. Capacity of the works, March, 1913, was equal to 150 machines a year if necessary.

[Illustration: Military Type VI. UAS.]

[Illustration: "Popular" biplane. Type VII. UAS]

[Illustration: Hydro-biplane. Type VIII. UAS.]

~H~

HOWARD-FLANDERS. L. Howard-Flanders, Ltd., 31, Townsend Terrace, Richmond, Surrey. School: Brooklands. Established February, 1912, by Howard-Flanders, whose connection with aviation dates from the pioneer days. Richmond Works opened April, 1912. Capacity of the works at end of 1912 was sufficient to turn out from 25 to 35 machines a year.

----------+----------+----------+----------+----------+---------- ------- | ~F 4 1912.~ | ~B 2 1912.~ | ~S 2 1913.~ | ~F 5 1913.~ | ~B 3 1913.~ | 2-seater | 2-seater | single-seat | 2-seater | 2-seater | military | biplane. | monoplane. | monoplane. | biplane. | monoplane. | | | | ----------+----------+----------+----------+----------+---------- ~Length~ feet (m.)| 31-1/2 (9.50) | 31-1/2 (9.50) | 28 (8.50) | 31 (9.45) | 31 (9.45) ~Span~ feet (m.)| 40 (12) | 40 (12) | 35 (10.70) | 39 (11.90) | 40 (12) ~Area~ sq. feet (m squared)| 240 (22) | 390 (36) | 190 (17-3/4) | 250 (23) | 390 (36) {total lbs. (kgs.)| 1850 (839) | 1500 (680) | 1180 (535) | 1600 (726) | 1650 (748) ~Weight~ { | | | | | {useful lbs. (kgs.)| 500 (227) | 450 (204) | 350 (159) | 600 (272) | 600 (272) ~Motor~ h.p.| 70 Renault | 40 A.B.C. | 80 Gnome | 80 Gnome | 80 Gnome {max m.p.h. (km.)| 67 (108) | 56 (90) | 82 (132) | 70 (115) | 68 (110) ~Speed~ { | | | | | {min m.p.h. (km.)| 41 (66) | 38 (61) | 45 (73) | 42 (68) | 40 (65) Number built during 1912 | 4 | 1 | | | ----------+----------+----------+----------+----------+----------

Remarks.--_F 4_ climbing speed 1000 feet (305 m.) in 3-1/2 minutes, 1500 in 5-1/2 mins., 2000 in 8 mins. _B 2_ climbing speed 200 feet (61 m.) per minute. The four _F 4_ type were bought by the British Army during 1912.

[Illustration: Monoplane.]

[Illustration: Biplane.]

[Illustration: FLANDERS. UAS.]

[Illustration: FLANDERS. UAS.]

HANDLEY-PAGE Monoplanes. Handley Page, 72, Victoria Street, S.W. Works: 110, Cricklewood Lane, N.W. Flying ground: Hendon. Established at the end of 1908. In June, 1909, it was turned into a Limited Liability Co. Since then it has been busily employed in producing its own machines, also others to inventor's specifications. About the end of 1911 the firm bought up and sold all the machines of the Aeronautical Syndicate--_Valkyrie_ and _Viking_ types. It is doubtful whether any of these V type still exist--in any case it does not matter. Four were presented to the R. Flying Corps. Of these one was smashed up, the others, one army and two navy, were used to teach mechanics to take down and re-assemble engines, etc. Handley-Page also bought up the _Radley-Moorhouse_ machines (Bleriot copies), and disposed of them.

The 1912-13 _Handley-Page_ type is as follows--a development along regular lines of the original H.P. machine:--

[Illustration: Handley-Page V.]

~Length,~ 27-1/2 feet (8.40 m.) ~span,~ 42-1/2 feet (12.95 m.) ~area,~ 240 sq. feet. (22-1/4 m squared.)

~Weight.~--Total, 1300 lbs. (590 kgs.) Empty, 800 lbs. (363 kgs.)

~Motor.~--50 h.p. Gnome. ~Speed.~ 55 m.p.h. (90 km.)

Remarks.--The fixed tail area is 32 sq. feet. Body is entirely enclosed, stream line form. The passenger sits behind the pilot. Mounted on wheels and one long skid forward. Full description and details, _Flight_, 26th October, 1912.

Principal pilots have been the late E. Petre (who made in it the only flight through London), the late Lieut. Parke, R.N., S. Pickles, and L. R. Whitehouse. The machine has been flown with two passengers, in addition to the pilot.

~Military work.~--During 1912 five biplanes of the _B.E._ type were ordered by the British War Office. Several monoplanes were ordered by foreign governments.

[Illustration: HANDLEY PAGE. UAS.]

~L~

LAKE FLYING Co. Windermere. Established 1911, by E. W. Wakefield, with a view to hydro-aeroplane experiments. The first machine was a _Curtiss_ type built by A. V. Roe, which flew in November, 1911. In 1912, a special biplane generally of _Farman_ type but with more camber to the planes, was built.

[Illustration: WATER HEN.]

~Length.~--36-1/2 feet (11 m.) ~Span.~--42 feet (12.80 m.) ~Area.~--270 sq. feet (25 m. squared) ~Motor.~--Gnome. ~Speed.~--45.33 m.p.h. (72.54 k.p.h.)

The single float is 6 feet wide, flexibly connected. Balancers mounted on a spring board. Water rudders for steering at slow speed. Fuller details see _Flight_, December 7th, 1912. Early in 1913, an _Avro_ was purchased for further experiments.

~M~

MARTINSYDE. Messrs. Martin & Handasyde, Brooklands, Weybridge, Surrey. Output capacity: about 20 per annum.

----------+----------+----------+ | ~1912.~ | ~1913.~ | Model and date. | Mono. 2-seater. | Mono. 2-seater. | ----------+----------+----------+ ~Length~ feet. (m.)| 35-1/2 (10.75) | 35 (10.65) | ~Span~ feet. (m.)| 42-1/2 (12.95) | 42-3/4 (13) | ~Area~ sq. feet (m squared.)| 290 (27) | 285 (26-1/2) | {total lbs. (kgs.)| ... | 1212 (550) | ~Weight~ { | | | {useful lbs. (kgs.)| ... | 551 (250) | ~Motor~ h.p.| 65 Antoinette | 80 Laviator | {max m.p.h. (km.)| 63 (102) | 78 (125) | ~Speed~ { | | | {min m.p.h. (km.)| ... | ... | Number built during 1912 | ... | ... | ----------+----------+----------+

Notes.--Wood construction. Landing: wheels and one skid. _Controls_: warping wings and rear elevator. Triangular body. The two models are very nearly identical.

[Illustration: MARTIN-HANDASYDE. UAS.]

~P~

PIGGOTT. Piggott Bros. & Co., Ltd., 220, 222 & 224, Bishopsgate, London, E.C. This well-known firm of shed makers built a novel biplane in May, 1910 (details _Flight_, May 21st, 1910), and in 1911 a monoplane with enclosed body (_Flight_, April 1st, 1911). In 1912, both were disposed of, and the firm is not proceeding with its experiments. It has, however, a staff of skilled mechanics and a great deal of floor space for the construction of aeroplanes to specifications.

PLANES. Planes, Ltd., 6, Lord Street, Liverpool. Works: Duke Street & Cleveland Street, Birkenhead. Not building at present. In October, 1910, the firm produced a biplane, designed by W. P. Thompson, fitted with a special pendulum stabilising device. This was followed a year or so later by a monoplane.

~R~

RADLEY-ENGLAND. This is not an aeroplane firm, but a special hydro built by two well-known aviators for the _Daily Mail_ competition. ~Length,~ 22 feet. ~Span,~ 50 feet. 2 floats, 15 feet long by 1 foot 5 inches wide. Pilot in starboard float. ~Weight,~ with petrol for 12 hours, 1,380 lbs. ~Motor,~ 150 h.p., made up of 3--50 h.p. Gnomes, but two Greens to be fitted for competition. One 4-bladed propeller in rear. ~Speed,~ 60 m.p.h., with 100 h.p.

~S~

SANDERS. This firm appears to have ceased to exist.

SHORT BROS. Works and flying grounds: Eastchurch, Isle of Sheppey, Kent. London office: Queen's Circus, Battersea Park. Took up construction at a very early date. _Wright_ agents in 1909. Have built numerous biplanes and monoplanes to specifications. Produced their own first machine (see 1911 edition) in 1910.

----------+----------+----------+

~Weight~ { | | | | | | | | | | | | {Useful lbs. (kg.)| 771 (350) | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... ~Motor~ h.p.| 80 Gnome | 100 Gnome | 160 Gnome | 70 Gnome | 80 Gnome | 160 Gnome | 50 Gnome | 80 Gnome | 50 Gnome | 70 Gnome | 50 Gnome. | 70 Gnome. {max (m.p.h.)| 65 (105) | 60 (97) | 74 (120) | 60 (97) | 70 (113) | 74 (120) | 42 (68) | 58 (94) | 39 (63) | 48 (78) | ... | 58 (94) ~Speed~ { | | | | | | | | | | | | {min (m.p.h.)| 50 (80) | 50 (80) | 56 (90) | 50 (80) | 50 (80) | 56 (90) | 35 (57) | 39 (63) | 34 (55) | 38 (61) | ... | ... ~Endurance~ hrs.| 4 | 5 | 6 | 5 | 5 | 6 | 4 | 5 | 4 | 5 | 5 | 5 Number built during 1912 | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | | | | | | | | | | | | | ----------+----------+----------+

| ... | ...
---|---|---|---|---

Notes.--Wood construction. Carriage wheels and skids. _Control:_ balanced ailerons.

[Illustration: Sopwith. Flying boat.]

[Illustration: 1913. Tractor biplane.]

~V~

VICKERS. Vickers, Ltd., Vickers House, Broadway, Westminster. School: Brooklands. Seven pupils qualified during 1912.

---|---|---+ | Monoplane. | Military | Model and date. | ~1912-13.~ | biplane. | | 2-seater. | ~1913.~ |
---|---|---+ ~Length~ feet (m.)| 25 (7.60) | ... | ~Span~ feet (m.)| 34-1/2 (10.50) | 40 (12.20) | ~Area~ sq. feet (m squared.)| 220 (20) | 385 (35) | {total, lbs. (kgs.)| 730 (331) | ... | ~Weight~ { | | | {useful, lbs. (kgs.)| ... | ... | ~Motor~ h.p.| 80 Gnome | 80 Wolseley | ~Speed~ m.p.h. (km.)| 70 (115) | ... | ~Endurance~ hrs.| 3 | ... | Number built during 1912 | ... | ... |
---|---|---+

Notes.--Steel construction. Landing shock absorbing: 2 wheels and 1 skid. Rectangular enclosed body. _Controls_: warping and rear elevator.

~Monoplane~ climbs 300 feet a minute fully loaded.

~Biplane~ is armed with a Vickers R.C. automatic gun in the bow.

[Illustration: VICKERS. UAS.]

[Illustration: Vickers. Monoplane.]

+---+ | | | | | | +---+ Vickers. Armed biplane.

~W~

WHITE. J. Samuel White & Co., Ltd., shipbuilders and engineers, East Cowes, Isle of Wight. London office: 28, Victoria Street, S.W. This well-known firm of torpedo craft builders, etc., formally opened an aviation department on 1st January, 1913, with Howard T. Wright as general manager and designer.

---|---+ | ~1913.~ | | Navy 'plane. |
---|---+ ~Length~ feet (m.)| 30 (9.15) | ~Span~ feet (m.)| 44 (13.40) | ~Area~ sq. feet (m squared.)| 500 (46-1/2) | {total lbs. (kgs.)| 2000 (907) | ~Weight~ { | | {useful lbs. (kgs.)| 650 (295) | ~Motor~ h.p.| 160 Gnome | {max. m.p.h. (km.)| 70 (115) | ~Speed~ { | | {min. m.p.h. (km.)| 35 (57) | Number built | ... | ---|---+

Remarks.--Hydro-biplane, with Howard T. Wright patent aeroplanes to give wide range of speed. Two patent hydro floats, 21 feet (m.) long, three steps on each.

[Illustration: UAS.]

BRITISH DIRIGIBLES.

~Navy.~ ~Army.~ /----------^----------\ /----------^----------\ ----------+----------+----------+----------+----------+----------+---------- | | | | ~III, & IV & V~ Name and date. | ~II Willows 3.~|~III Astra Torres 2.~| ~IV Parseval 18.~ | ~II BETA.~ | ~GAMMA, DELTA,~ | ~1911.~ | ~1913.~ | ~1913.~ | ~1909 (1910.)~ | ~EPSILON.~ | | | | ~1910, 1912, 1913.~ ----------+----------+----------+----------+----------+---------- ~Volume~ c. feet (m cubed.)| 31,800 (900) | 222,500 (6,500) | 311,000 (8,800) | 21,000 (594) | 70,600 (2,000) ~Length~ feet (m.)| 120 (36.50) | ... | 276 (84) | 104 (31.70) | 152 (46) ~Diameter~ feet (m.)| 40 (12.20) | ... | 49-1/4 (15) | 25 (7.60) | 30 (9.10) {fabric | Spencer | Continental | Metzler | Gold beater skin | Continental ~Gasbags~ {compartments | _nil_ | 3 | _nil_ | _nil_ | _nil_ {ballonets | 1 | 1 | 2 | 1 | 2 {total tons| about 1/2 | about 7 | about 10 | _about_ 3/4 | 2-1/5 ~Lift~ { | | | | | {useful, tons| ... | ... | ... | ... | ... ~Motors~ h.p.| 30 (=30) | 2--120 Chenu (=240) | 2--180 Maybach (=360) | 1--30 Green (=30) | 2--50 Green (=100) {number | 2 (swivel) | 2 | 2 (s.r.) steel | 1 | 2 (swivel) ~Propellers~ {blades | 2 | 2 | 4 | 2 | 2 {diameter feet (m.)| ... | ... | ... | 6 (1.82) | 8-5/6 ~Speed~ max. m.p.h. (km.)| ... | 38 (63) | 42 (68) | 18 (29) | 28 (45) ~Endurance~ full speed| ... | ... | ... | ... | 4 hours ~Max. complement~ | 2 | 15-18 | 10-12 | 3 | 5 ~Station~ | Farnborough | ... | ... | Farnborough | Farnborough ----------+----------+----------+----------+----------+----------+----------

Notes.--All the above are non-rigid. The military ones were all built at the Royal Aircraft Factory.

~Navy Dirigible Pilots.~

Boothby, Lieut. F. L. M. (_F.C._) Everett, Gunner F. Masterman, Comdr. E. A. D. (in command) Usborne, Lieut. N. F. (squad comdr.) Woodcock, Lieut. H. (_F.C._)

Undergoing Naval Aircraft Course:

Crocker, Lieut. W. R. Hicks, Lieut. W. C. Wilson, Lieut. R. A.

~Military Dirigible Pilots.~

~Squadron Commanders.~

Maitland, Capt. E. M.

~Flight Commanders.~

Waterlow, Lieut. C. M.

~Flying Officers.~

Brabazon, Capt. Honble. C. M. P. Fletcher, Lieut. J. N. (R.E.) Hetherington, Lieut. T. G. Mackworth, Lieut. J. D. Pigot, Capt. R.

The following hold dirigible pilot certificates, but are not at present employed:--

Broke Smith, Capt. P. Capper, Col. J. E. Fox, Lieut. A. G.

~Private Dirigibles.~

There are one _Willows_ (1912) (sister to the naval one) and a couple of _Spencers_ about the size of _Beta_.

~Private Dirigible Pilots.~

Willows, E. T. (24-32, Villa Rd. Handsworth, Birmingham).

~BRITISH NAVAL DIRIGIBLES.~

[Illustration: Willows. The naval one is fitted with a boat-shaped car.]

[Illustration: Parseval. (photo of a sister ship.)]

+----------+ | | | ~New Construction.~ | | | | Messrs. Vickers have acquired the Parseval rights for the British | | Empire, and several airships of this type are likely to be put in | | hand by them shortly. | | | | Also reported that a big rigid is projected. | | | +----------+

~BRITISH MILITARY DIRIGIBLES.~

[Illustration: Beta.]

[Illustration: BETA.]

[Illustration: Gamma (Delta the same, but a smaller and enclosed car).]

[Illustration: GAMMA.]

BRITISH COLONIAL AIRCRAFT.

~AUSTRALIAN.~

~Military Aviation.~

In January, 1913, the Australian Flying Corps was instituted, as a part of the citizen forces. During 1913, about L5,600 is to be spent.

The force is to consist ultimately of 4 officers, 7 warrant officers and sergeants, 32 mechanics.

The school is at Duntroon. Course includes--mechanics of the aeroplane, aerial motors, meteorology, aerial navigation by compass, aerial photography, signalling, etc. Mr. Harrison is in command with Lieut. H. Petre as assistant.

~Australian Aviators.~

Banks, R. C. Busteed, H. Duigan, J. R. Hammond, J. J. Harrison +Hart +Lindsay, C. Petre, H. Pickles Watts

+=killed.

~Private Aeroplanes.~

J. R. Duigan has built an aeroplane of his own design.

~NEW ZEALAND.~

Nothing doing worth mention. One _Bleriot,_ 80 h.p., presented 1913 by the _Standard,_ London.

~CANADIAN.~

~Aerial Societies.~

Aeronautical Society of Canada, c/o. M. P. Logan, 99 Gloucester Street, Toronto

McGill Aviation Club, McGill University, Montreal

Oshawa, Ontario Ae. C.

Note.--Owing to the fact that the late Aerial Experiment Association was half Canadian and half U.S.A., it is difficult to draw a very clear dividing line between Canadian and U.S. aviators or machines. Thus, one given here is partly U.S.A., while at least one U.S. machine may be claimed as "partially Canadian."

~AVIATORS.--Private.~

Bell, Dr. Graham McCurdy, J. A. D. (U.S.A. Ae. C. 18) McHardy Symonds, E. F. St. Henry R.

~Canadian Aeroplanes.~

GRAHAM-BELL II. Flights were made by Dr. Graham-Bell in a tetrahedal type, similar to one described in the 1911 edition.

McCURDY-WILLARD. Biplane.

~Maximum length,~ 26-1/4 feet (8 m.) ~maximum breadth,~ 31-1/3 feet (9.50 m.) ~supporting surface,~ ? sq. feet (? m squared.)

~Total weight.~--

~Body.~--Central skid in combination with 4 wheels. Triangular body, base of triangle on top. Fuselage entirely enclosed.

~Planes.~--Maximum span, 31-1/3 feet (9.50 m.) Chord, 3-1/2 feet (1 m.) Gap, 5 feet (1.50 m.) Ailerons at trailing edge of wing tips, 6 feet x 2 feet (1.80x0.60 m.)

~Motor.~--

~Speed.~--

~Tractor.~--Diameter, 7-3/4 feet (2.40 m.) Pitch, 6 feet (1.82 m.)

~Steering.~--Double elevator placed in rear of tail. _Control_, push and pull wheel. Rudder in rear. _Control_, wheel. Ailerons. _Control_, turning steering-wheel left or right.

Remarks.--See _Aeronautics_, U.S.A., August, 1911.

There has been also the _Baddeck_ and other early machines (see 1911 edition), but none of them seem to be in existence at the present time.

~INDIAN.~

~Military Aviation.~--A certain number of officers belonging to the Indian Army have qualified as pilots when home on leave, but there is no organised force. One is, however, proposed.

~Private Aviation.~--In the past two or three home-made machines appeared, and one or two were imported, but most or all are now extinct.

~SOUTH AFRICA.~

~Military Aviation.~--Non-existent.

~Private Aviation.~--J. Weston is a qualified pilot, but at the outside there are not more than two effective machines in the country.

BULGARIAN.

In the Balkan War, 1912-13, Bulgaria hastily organised an aviation corps. This, though necessarily lacking in military organisation, proved very useful on several occasions.

At the end of March, 1913, the aeroplanes effective included 6 _Bristol_ monos.; one 70 h.p. _Bleriot XXI_; 2 _Bleriot XI bis_ (captured from the Turks); also some half-dozen or more miscellaneous machines temporarily hired.

~Military Aviators.~--The principal are Lieuts. Milkoff, Taraxchieff and Petroff.

A number of other officers in various stages of training.

CENTRAL AMERICAN.

~General Note.~--Nicaragua and S. Domingo have both purchased one or two aeroplanes for their military forces: but nothing appears to have been done with them.

CHILIAN.

~Aviator.~--Edwards, Emilio. Sanchez Besa is a Chilian, but resides in Paris (see France).

~Military Aviation.~--In 1912, a commencement was made and one 80 h.p. _Deperdussin_ purchased. Other machines are now on order.

CHINESE.

~AVIATORS.~

Lee, Y. L. (British Ae.C. 148) Tsai Tao Prince

~Military Aviation.~

In March, 1913, orders were placed for six 80 h.p. _Caudrons_, also for six 50 h.p., and a decision arrived at gradually to acquire a force of 700 aeroplanes, but very little has actually been done to date.

DANISH.

~Aerial Societies~:--

Danske Aeronautiske Selskab, 34 Amaliegade, Copenhagen.

~Aerial Journals~:--

None; but Motor (3 Bredgade, Mezz, Copenhagen) deals with aerial matters.

~Flying Grounds~:--

Klampenburg, Copenhagen.

Skandinarisk Aerodrom.

~Army Aeroplanes.~

In 1911 there was an Antoinette. Nothing done since.

~AVIATORS.~

Military.

Ullitkz, Kapt.

Private.

Arntzen, Dr. Christiansen, S. Ellerhammer Folmes, Hansen Maltke, Count Nervoe, A. Svendsen, R. Thorup, K.

DUTCH.

(Revised by I. SCHIERE, Aeronautical Engineer and Librarian of the Dutch Ae. C.)

~Aerial Societies:--~

Haagsche Proefvliegtuig Club (3e V.d. Boschstreet 20, The Hague).

Nederlandsche Vereeniging voor Luchtvaart (Nassau Zuilensteintraat, 10, The Hague). (Ae. C.)

Rotterdamsche Model Aero Club (Rochussenstreet 229b, Rotterdam).

Colonial:

Nederlandsche Indische Vereeniging voor Luchtvaart.

~Aerial Journals:--~

De Luchtvaart (Ged Onde-Gracht, 141, Haarlem). Fortnightly.

Avia, Wynbrugstraat 13, Rotterdam. Fortnightly.

~Flying Grounds:--~

~Breda-Gilske-Rijen.~--6 hangars.

~Soesterberg.~--20 hangars.

~Army Aeroplanes.~

Up to the end of 1911 there were none, though some officers had their own private ones (_H. Farman's_ mostly).

At end of 1912.

2 monos. _Deperdussin_ (for Java). 1 biplane. _De Brouchere_ (for Java).

~AVIATORS.~

(The number against any name is, unless otherwise stated the Ae. C. Nederlandsche pilot certificate number.)

To end of ~1911~.

Military.

Bakker, H. Yandrig Labouchere, Lieut. J. Meel, Lieut. Van Poorton, Lieut. H. ter Versreegh, Lt, W. C. J.

Private.

Bahle, F. K. Boerlage, M. Burgh, Van der Fokker, A. H. G. Hilgers, J. W. E. L. Konings, L. Koolhoven (1) Kueller, G. P. (2) Lutge, F. (4) (323, F.) Mulder, A. Riemsdyk, Van F. (5) Ryk, Madame Bde. Wynmalen, H. (6) (208, F.)

The following Dutch aviator has been killed:

+------------------+ | 1911. | | Van Maasdyck, C. | | (130, Ae. C. F.) | +------------------+

DUTCH AEROPLANES.

DE BROUCKERE. Biplane. _H. Farman_ type. Details, _De Luchtvaart_, No. 8, 1911.

FOKKER. Monoplane. Anthony Fokker, of Haarlem. In early 1912 flew at Breda.

MONNIER-HARPER. Monoplane. (O.P.I.I.) Generally _Bleriot_ type. Built 1911.

VAN DEN BURG. Monoplane. Early in 1912 was flying at Johannisthal, Germany.

VREEDENBURGH. Monoplane. (O.P.I.I.) Blend of _Bleriot_ and _Antoinette_. Motor, 75 h.p. Miesse. Completed December, 1909.

[Illustration]

FOKKER. Monoplane. (See Germany for details.) Firm now established in Germany.

DUTCH DIRIGIBLES.

~Military.~

DUINDIGT. Non-rigid.

(Zodiac make.)

~Length~, 111-1/2 feet (34 m.) ~diameter~, 22-1/2 feet (60.80 m.) ~capacity~, 31,785 c. feet (900 m cubed.)

~Motor~.--18 h.p.

Remarks.--Small edition of _Zodiac III_. (See France.)

FRENCH.

(Special French Editor.)

~Aerial Societies:--~

Aero Club de France.

Academie Aeronautique de France.

Aeronautique Club de France.

Societe des Aeronautes du Siege.

Aero Club du Sud Ouest.

Aero Club du Rhone.

Aero Club du Nord.

La Ligue Aerienne du Sud.

Societe Francaise de Navigation Aerienne.

Societe d'encouragement a l'Aviation.

(_Full list of clubs next page._)

~Aerial Journals:--~

L'Aerophile.

L'Aero.

L'Aeronaute.

Aerostat (Bulletin Aeronautique).

Aerostat (Academie d'Aerostation).

Revue de l'Aerostation.

Le Ballon.

L'Aerostation.

L'Aeronautique.

Bulletin Aeronautique.

Encyclopediede l'Aviation.

La Ligue Nationale Aerienne.

Revue de l'Aviation.

L'Aeromecanique.

~Principal Flying Grounds:~

~Antibes.~--Hanriot school.

~Beauce.~

~Betheny.~--Sommer school. ~Deperdussin School.~

~Buc.~--M. Farman school.

~Buoy.~

~Chalons.~--Sommer school.

~Chalais-Mendon.~ (Military)

~Chatres.~--Savary school.

~Cran~, Marseilles.

~Crotoy.~--Caudron school.

~Croix d'Hins~, Bordeaux (Aer. Lig. du Sud.) Area 6 km. Track. Free sheds.

~Corbeaulieu~ pres. Compregne.--Doutre school.

~Etampes.~--Bleriot school. Farman school.

~Grand Camp, Lyons.~

~Issy les Moulineaux.~--Astra school

~Juan-le-Pias.~--Paulhan aquaplane school.

~Juvissy~, near Paris.--Aerodrome. Caudron school. Goupy school.

~La Brayelle~, Douai.--Breguet school.

~Da motte Brueil dans L'Oise.~

~Le Bourget~, Paris.--100 sheds.

~Le Mans.~

~Moisson.~

~Mourmelon.~--Voisin school.

~Napante.~

~Nice.~--Small and rough surface.

~Pau.~--Bleriot school.

~Reims.~--Aerodrome.

~St. Cyr.~

~Villacoublay~, Paris.--Breguet, Nieuport and Astra schools.

FRENCH AEROPLANES.

~Military Aviation.~

In February, 1912, the then total of 208 effective aeroplanes were divided into "squadrillas" consisting of eight aeroplanes; attached to these eleven or twelve motor cars, one traction car and one fast car, also a repairing car and repairing van.

It was then estimated that at the end of 1912, ~344~ aeroplanes would be available for service.

The estimated _personnel_ was provisionally fixed at 234 officer pilots, 210 scouts, 42 mechanics, 110 officers, 1,600 corporals or sappers and 550 privates.

Approximately L880,000 was spent in aviation during 1912, and L1,000,000 was estimated for future years.

The French military aviation centres are all upon somewhat the same footing as fortresses, and the greater part of the work comes under the head of "confidential." The principal school is at St. Cyr, which was specially selected because the ground is rough and mostly covered with small shrubs: it being held important to train officers from the first to rise and land on ground similar to that most likely to be found in war time. Each station is supplied with large portable wooden-framed hangars covered with canvas. These can be rapidly taken to pieces and re-erected. Each station is supplied with its own special motor transport.

All military machines are provided with a compass and map case in front of the pilot and sketching apparatus in front of the observer.

Although a few non-commissioned officers have been taught flying, the organization only contemplates the employment of commissioned officers as pilots. The age limit is 38.

On April 16th, 1913, the flying corps was modified. The principal features of the corps as now existing are as follows:--

~Establishments.~

1. Schools.

2. Special establishments, dealing with purchase, construction, and big repairs.

3. _Directions._ Administration of _material_.

4. Depots. A species of dockyards dealing with minor repairs, etc.

~Administration.~

There are three main groups, each commanded by a colonel. Each group consists of dirigibles and aeroplane "escadrilles," and is fully equipped with establishments, etc. The three centres are:--

1. Versailles.

2. Reims.

3. Lyon.

~General.~

All squadron units are made up of machines of the same make and power. Pilots are detailed as required to any particular unit, and liable to transfer from one to another, though in practice such transfers are rare.

~Army Aeroplanes.~

During 1912 nearly 500 machines were delivered to the Army, but a great many old machines have been scrapped. At the end of March, 1913, the force stood at 421 effective for war machines, plus an uncertain number of school machines and obsoletes.

About one-third or more of the effective aeroplanes were _Farmans_. The rest consisted of all leading French types, proportionated more or less to the productive capacity of these firms. Also certain other makes experimental.

~Navy Aviation.~

The Navy section of French military aviation is still in the "being formed" process. No data are yet available as to the ultimate force to be provided. At present the number of effective war machines is small. It is made up of hydro-avions of the following types:--_Astra_, _Borel_, _Breguet_, _Caudron_, _Deperdussin_, _Donnet-Leveque_, _Farman_, _Paulhan-Curtiss_, _Sanchez-Besa_, the total at end of March, 1913, being well under 20. There are also two special _Bleriot_ type fitted with floats, which carry 330 lbs of explosive, are fitted with wireless, have a speed of 140 km.p.h. (85 m.p.h.), and a radius of about 600 miles (1,000 km.)

~PRINCIPAL FRENCH ARMY AND NAVY AVIATORS.~

(In each case the number against each name is, unless otherwise stated, the Ae. C. French certificate pilot number.)

Army.

Abadie, Sous Officier Acevedo, Lieut. (740) Acquaviva, Lieut. Paul V. (68) Aiguillon, Lt. R.d' (308) Aubry, Lieut. Balensi, Capt. Albert (173) Bares, Capt. (543) Basset, Lieut. Paul (145) Battini, Lieut. G. (508) Baugnies, Lt. J. B. E. (193) Beatrix, Sous Officier Bellemois, Lieut. G. (546) Bellenger, Capt. M. (45) Berni, Lieut. (760) Biard, Capt. G. M. (261) Bihan, Lieut. Binda, Lieut. Louis (232) Blard, Lieut. (460) Bobillier, Lieut. Boerner, Lieut. Boissonas, Lieut. (443) Bon, Lieut. Boncour, Lieut. (478) Bonnier, Lieut. (478) Bonnier, General (137) Boucher, Lieut. Bousnuet, Lieut. P. (295) Breley, Lieut. Brenot, Capt. Brouchard, Lieut. Brugiere, Lt. Brule, Lieut. (436) Bruncher, Lieut. Burgeat, Capt. M. (44) Camerman, Lieut. F. (33) Camine, Capt. Campagne, Lieut. (782) Casse, Capt. (415) Chabert, Lieut. Charoux, Sous Officier Chavenac, Lieut. E. (551) Cheutin, Lt. E. J. (233) Chevreau, Lieut. R. (132) Clavenad,

Lieut. P. (294) Clerc, Lieut. (465) Clolus, Commdt. G. (97) Couret, Lieut. Coville, Capt. D'Abrantes, Lieut. D'Aquillon, Lieut. De Beruis, Lieut. De Caumont, Capt. De Chanac Lanzac, Capt. De Geyer, Lieut. De Gorge, Lieut. (805) De Goys, Capt. De Lafargue (417) De L'Estrade, Lieut. De Rose, Lieut. P. (477) Destace, Capt. Destouches, Capt. Devarenne, Lieut. Devaulx, Lieut. R. (158) De Ville d'Avray, Lieut. Didier, Sous Officier (765) Do-Ird, Lieut. Drevet, Sous Officier (753) Duparquet, Capt. Duperron, Capt. (196) Dupin, Lieut. Eteve, Capt. A. (89) Erstorac, Capt. Felix, Capt. J. (270) Fequant, Lieut. A. (63) Fequant, Lieut. P. (340) Fierstein, Sous Officier Francezon, E. (410) Foirelline, Lieut. Garnier, Lieut. (305) Garnier, Lt. (826) Gastringer, Lieut. Gaubert, Lieut. E. (313) Germain, Lieut. Girard, Lieut. J. (197) Gironde, Lt. A. de Godefroy, Sous Officier (583) Gouin, Lt. M. E. R. (348) Gourlez, Lieut. (521) Grezaud, S.-Lt. P. (265) Grailly, Lieut. (399) Gronier, Lieut. J. (138) Grandjean, Sapper Guibart, Lieut. Guiton, Sous Officier Hable, Sous-Lt. A. L. (257) Hugoni, Capt. E. (165) Hanouille, Lieut. Henequin, Lieut. Henri, Lieut. (497) Herli, S.-Lt. (257) Hurard, Sous Officier Issartier (531) Jacquet, Lieut. Joly, Lieut. F. (341) Jost, Lieut. R. G. (264) Kass, Capt. Langardt, Lieut. Laurent, Sous Officier (246) Le Beau, Capt. Le Bleu, Lieut. Lelievre, Lieut. E. (522) Lemasson, Lieut. (506) Le Mauget, Capt. Letheux, Lieut. G. (142) Letort, Sapper (170) Letourneur, Lieut. Lucca, Lieut. D. (154) Ludmann, Lieut. G. (255) Lussigny, Lieut. Machin, Lieut. Mailfert, Lieut. F. (146) Maillois, Lieut. J. (131) Malherbe, Lt. de (334) Maneyrol, Lieut. Manoha, Lt. Marc, Lt. Marconnet, Capt. (90) Marie, Capt. Felix (80) Marlin, Lieut. Marmies, Lieut. Marty, Sous Officier (816) Massol, Lieut. Mauger, Lieut. Maurice, Lieut. Mazac, Lieut. (592) Migaud, Lieut. G. (501) Morel, Sous-Lt. P. (262) Morlaye, Lieut. la Mouchard, Lieut. Negre, Capt. Nicaud, Lieut. Nogues, Capt. (114) Normand, Lieut. F. (314) Pelloux, Sous-Lt. M. (346) Peraldi, Lieut. Peretti, Sous Officier Pierre, Lieut. Ponchet, Lieut. Prat, Lieut. Precardin, Lieut. Princetau, Lieut. Postulat, Sergt. Quennehen, Sous Officier Ragot, Lieut. Remy, Lieut. H. C. (143) Reynard, Lieut. (668) Rimbert, Lieut. Rocca-Serra, Lieut. Rochette, Lieut. J. (564) Rolland, Lieut. M. E. (545) Ronin, Lieut. Rougerie, Lieut. Sauleillon, Lt. A. (674) Saunier, Lieut. G. (153) Seguin, Sapper (528) Sevelle, Lieut. (747) Silvestre, Lieut. (599) Sido, Capt. Marie (65) Sourdeau, Lieut. A. (474) Soulielani, Lieut. Thomas, Lieut. (846) Thomas, R. (116) Touzet, E. (485) Tretane, Lieut. Tricornot de Rose, Lt. de (330) Vandamone, Lieut. (535) Van de Vaero, Lt. (491) Vandine, Lieut. Varcin, Lieut. Vaudein, Lieut. Verdier, Sous Officier (538) Vibra, Lieut. Vigne, Lt. Henri (315) Vinda, Lieut. Vitra-Rougerie, Lieut. Vocayeau, Lieut. Vogoya, Capt. Vuilliereme, Lt. L. (174) Watteau, Lieut. Willemenz, Lieut. (759) Yence, Lieut. R, (220)

Naval.

Byasson, Lt. de V. (175) Cayla, Lieut. (458) Conneau, Lieut. (322) "Beaumont" Davelny, Comdt. Delage, Lieut. G. (219) Fournier, Lieut. Hautefille, Lieut. (247) Lafon, Lt. (194) Leve, Lieut. (243) Parasa, Lieut. (179) Reymond, Lieut. (206)

~FRENCH AEROPLANES--PRIVATE.~

~Private Aeroplanes.~

The total number of machines built in France during 1912 has been estimated at about 1,500. This includes military as well as private machines, also machines exported, and appears to be unduly generous even so. The actual total of machines commenced and completed in 1912 is nearer 1,000.

The number of private aeroplanes--excluding demonstration and school machines is small.

~PRIVATE AVIATORS~ (brevets to end of 1911).

(In each case the number against each name is, unless otherwise stated, the Ae. C. French certificate pilot number.)

Algrin, Rene (252) Allard, M. (480) Alincourt (488) Andre, C. (192) Aubrun (21) Bachot, A. (271) Baeder, F. de (107) Bague, E. (337) Balliod, Louis (236) Balaye, A. (275) Balsan, Jacques (22) Baratoux, Marcel (49) Barbotte, Ernest (268) Barra, Franck (171) Barrier, A. (64) Banier, Rene (64) Bathiat, Georges (237) Bathiat, Leon (110) Beard, Pierre (276) Beaud, Edouard (150) Becue, Jean F. (263) Bellier, Albert (297) Bellot, Andre (317) Benoist, Jean (369) Bergognie, Charles (373) Bernard, A. (505) Berlot, Henri J. (450) Biard, Desire J. (460) Bielovucic, Jean (87) Bill, Henri (205) Blanchet, Georges (244) Bleriot, Louis (1) Blondeau, Gustave (101) Bobba, Andre (309) Boillot, Geo. (395) Boissounas, L. (443) Boise de Courcenay, Comte (283) Boivin, Albert (248) Bonzon, Maurice (355) Bouvier, Andre (120) Boyer, Louis (303) Bregi, Henry (26) Breguet, Louis (52) Bresson, Georges (280) Briancon, Lucien (277) Briey, F. de (492) Brindejonc des Moulinais (449) Bruneau de Laborie, E. (67) Bunau-Varilla, E. (16) Busson, Guillaume (121) Caille, Albert (200) Caramanlaki, A. (761) Carles, Fernand (362) Carlin, L. V. (554) Caudron, Rene (180) Cayla, P. (458) Chailliey, Henri (63) Challe, M. J. (523) Champel, Florentin (94) Chanteloup, P. (549) Chapelle, J. (547) Charpentier, Louis (286) Chassagne, Jean (160) Chausse, P. (519) Chaussier, Piere (384) Chatain, Marius L. (267) Chatain, L. M. L. (296) Chateau, Edouard (135) Chaunac-Lenzac de (394) Chemet, Geo. (159) Cheuret, Leon (62) Cherent, L. (62) Chevalier, J. (515) Chevalier, Louis (333) Chevillard, Maurice (385) Chioni, Basile (250) Clerc, Paul A. L. (465) Clement, M. (108) Collardeau, Geo. (393) Collieux, M. (85) Collin, Georges (279) Conard (647) Contard, Paul (351) Contenet, Henri (447) Contour, Ernest (371) Contre (657) Cordonnier, Robert (221) Corso, E. (529) Crochon, Andre (43) Cronier, Andre M. H. (352) Cugnet, Gaston (140) Cure, Gaston M. (242) Daillens, Jean (119) Dancourt, P. H. (520) Debener, M. (562) Deletang, Fernand (42) Delacroix, Maurice (452) Delagrange, Robert (366) De La Roche, Mde. (36) Deloche, R. D. (526) Denis, Auguste (380) Deroy, Francis (374) Derry, Leon (254) Deruissy, Andre (376) Despres, E. M. L. (527) Deschamps de Bois, Hebert (461) Didier, A. (77) Divetain, Pierre (466) Driancourt, M. L. (525) Dubonnet, Emile (47) Ducoweneau (456) Dufour, Jean M. R. (457) Dufour, Jean (96) Dufour, Louis (185) Duval, E. (118) Duval, Emile (118) Echeman, P. M. (466) Esnault-Pelterie, R. (4) Espanet, Dr. G. (532) Farman, Henry (5) Farman, Maurice (6) Fiorellimo, Louis (369) Florencie, Jean (201) Fournie, J. P. S. (502) Frantz, Joseph (363) Francq, Baron de (481) Frey, Alfred (48) Frey, Andre (93) Froussart, Ernest (350) Frugier, Leon (378) Gaget, Joseph (335) Gaillard, J. O. C. (504) Gallie, Fernand (343) Gardey, M. (482) Garros, Roland (147) Garsonnin, L. (555) Gastinger, Edouard M. (455)

Gassnier, Rene (39) Gassier, Marcel (392) Gasnier, Pierre (391) Gaudart, Louis (228) Gaulard, Charles (302) Gautheron, Louis (449) Gaye, Georges (251) Gibert, Louis (92) Gilbert, Eugene (240) Giraud, Etienne (493) Glorieux, Leon (188) Gobe, Armand (102) Gobron, Jean (7) Goffin, Marcel (284) Gouguenheim, P. (388) Goux, Jules (398) Gournay, Henri (186) Goys de Mereyrac, Louis (354) Grandjean, E. C. H. (469) Grandseigne, R. (360) Granel, Marcel (117) Grellet, Alexis (370) Gressard, M. (725) Gue, Albert (216) Guerre, Henri (444) Guidard, V. P. (487) Guilband, C. J. (518) Guillemard, T. (445) Guillaume, C. (651) Hainaux, Marcel R. (239) Hanriot, Marcel R. (239) Hanriot, Rene (368) Herbster, Maurice (41) Herveu, Mlle. Jane (318) Hesne, Paul (113) Houlette, Andre (367) Jacquemart, G. C. (464) Jamblez, Paul A. (266) Janoir, L. (553) Joliot, Andre (202) Joly, C. E. M. (530) Julleriot, Henry (61) Junod, Auguste (253) Kauffman, Paul (198) Kergariou, Engard de (503) Kieffer, C. E. (372) Kummerling, A. (291) Koechlin, Jean P. (203) Kuhling, Paul L. (136) Labouchere, Rene (86) Labouret, Rene (222) Lacombe, P. (534) Ladougne, Emile (81) Lafarge, Henri (278) Lajous, Francois, A. (463) Lambert, Comte de (8) Langhe, Armand de (204) Lastours, H. R. de (552) Larfinty-Tholosan, Marquis Jules (468) Laroche, Mme. Raymonde (36) Latzel, J. (700) Leblanc, Alfred (17) Lecomte, Henri (320) Legagneux, Georges (55) Le Lasseur de Ranzay, G. (479) Lemartin, Theodore (249) Lenfant, Louis (386) Leouet, B. L. (485) Leprince, P. (494) Lesire, Eugene (176) Lesseps, Jacques de (27) Leyat, Marcel (364) Lieutard, H. (497) Liger, A. (573) Lombardi, Henri (241) Loridan, Marcel (224) Magnan, Leon (379) Magneval, Gabriel (359) Mahieu, Georges E. (123) Mallet, J. A. P. (490) Mamet, Julien (18) Marchal, Anselem (328) Maron, P. H. (495) Marquezy, Rene (238) Martin, Edouard (365) Martin, Xavier (162) Martinet, Robert (78) Marvingt, Marie (281) Mauvais, Jean (144) Metrot, Rene (19) Meyer, Jules M. (229) Mignot, Robert (76) Miltgen, Paul (339) Moineau, R. L. (554) Molla, Henri (172) Montalent, O. de (509) Montjou, Guy de (446) Mollien, Elie A. (57) Molon, Leon (25) Molon, Louis (234) Molon, Lucien (235) Montigny, Alfred de (69) Morane, Leon F. (54) Morelle, Edmond (35) Morel, P. F. (524) Morin, Roger (306) Mouthier, Louis (157) Mousnier, Yvon (454) Niel, Albert (104) Niel, Mme. Marthe (226) Nissole, Edouard (383) Noe, A. G. M. (498) Noel, Andre (122) Obre, Emile (148) Ors, Jean (382) Orus, Maurice (256) Osmon, Geo. (361) Paillette, Marcel (99) Paillole, E. C. L. (556) Palade, Antoine (387) Pallier, Mdlle. Parent, Francois (189) Paris-Leclerc, Max (190) Partiot, G. (516) Pascal, Ferdinand (301) Paul, Ernest (91) Paulhan, Louis (10) Pequet, Henri (88) Perin, Albert (161) Perreyon, Edmond (311) Perrigot, J. (499) Picard, Pierre (174) Planchet, Edmond (319) Poillot (182) Pommier, Martin (400) Porcheron, L. A. (471) Pouleriguen, F. (349) Poumet (576) Pourpe, Marc Pourpe, M. M. E. A. (560) Prevost, M. (475) Prevoteau, G. (507) Prier, Pierre (169) Raoblt, Jean (386) Reimbert, Ernest (375) Reichert, Henri (377) Renaux, Eugene (139) Renaud de la Fregeoliere (396) Rey, P. A. P. (517) Reymond, Senator Richet, A. (537) Rigal, Victor (60) Rivolier, Jean (381) Robillard, G. de (184) Robinet, J. (476) Romance, F. de (288) Rougier, Henry (11) Ruby, F. L. (514) Ruchonnet (127) Sallard, H. (794) Sallenave, Henru (66) Savary, Robert (112) Schlumberger, M. (316) See, Raymond (187) Servies, Jules (218) Simon, Rene (177) Sommer, Roger (29) Tabateau, Maurice (128) Taurin, Andre (84) Tetard, Maurice (79) Thieulin, Joseph (459) Tissandier, Paul (13) Tixier, Henri (397) Toussin, Rene (56) Train, Emile Louis (167) Vallier, Edmond P. (269) Vallon, Rene (109) Van Gaver, Paul (338) Vasseur, Narcisse (282) Vedrines,

Jules (312) Vendrines, E. (536) Verliac, Adrien (129) Vergmault, O. (561) Verrier, Pierre (390) Versepuy, Leon (149) Vialard, Charles (342) Vidart, Rene (133) Villeneuve Trans, Louis de (285) Vimard, E. (484) Visseaux, Henri (217) Vittoz-Gallet, G. (500) Wagner, Louis (83) Walleton, Louis (304) Weiss, H. (73) Wintrebert, Henri (300) Zens, Ernest (28)

The following French aviators have been killed:--

+------------------------+ | 1909. | | Ferber, Capt. | | Lefebvre, E. | | | | 1910. | | Blanchard (215) | | Delagrange, Leon (3) | | Le Blon (38) | | Poillot (182) | | | | 1911. | | Byasson, Lt. | | Camine, Capt. | | Caumont, Lieut. (156) | | Carron, Capt. | | Chotard, Lieut. | | De Grailly, Lieut. | | Desparmet, J. (451) | | Dupuis, Lieut. | | Gaubert (59) | | Laffont, A. (111) | | Lautheaume, Lt. | | Level | | Liere, Louis | | Loder, Lt. | | Madiot, Capt. (106) | | Mommlin | | Nieuport, E. (105) | | Noel | | Princeteau, Lt. (331) | | Ruchonnet | | Tarron, Capt. | | Vallon, Rene | | Wachter, C. L. (53) | +------------------------|

~FRENCH PRIVATE AVIATORS, 1912.~

Adam-Gironne (818) Arondel, P. (827) Andenis, C. (788) Badet (622) Balighant, G. (588) Barbarou, M. (702) Basano, F. (828) Baudrin, E. (609) Bedel, R. (668) Beatrix, C. (781) Benoit, O. (771) Benoist, G. (667) Bertin, L. (801) Blaignan (633) Bleu, Le (643) Boiteau, G. (833) Boerlage (666) Bordage, A. (650) Boncour (678) Boucher, F. (600) Borie, A. (803) Brocard, A. (770) Brodin, E. (838) Brouard, E. (807) Bruginere, A. (813) Cailleaux, A. (617) Carreard, G. (779) Castellan, E. (639) Cavalier, M. (764) Caye, M. (672) Cerantes, F. (611) Chabert, V. (631) Chandenier, L. (804) Coblyn, L. (735) Contre (657) Corsini, A. (654) Cornier, R. (605) Coville, F. (594) Couffin, L. (619) Dambricourt, J. (773) De Beausire de Seyssel (756) Debroutelle, P. (806) De Chabot, P. (783) De Gensac, A. (836) De Lareinty Tholozan, H. (822) Delacour, J. (602) Delaunay, P. M. (635) Deleraye, M. (790) De l'Escaille (791) Delmas, M. (837) De Marmies, R. (663) De Mazurkiewicz, W. C. (707) Denhaut, F. (690) Des Pres de la Morlais (636) De Pontac (596) De Reals, R. (686) De Ryk (Mme. B.) (652) De Segonac, R. (669) Desille, L. (581) De Vergnette, C. (792) De Villepin, O. (832) Do Huu, T. (649) Drouhet, F. (727) Dussot, A. (733) Dutertre, C. (748) Ecomand, G. (714) Ehrmann, L. (646) Escot, P. (624) Eymien, S. (726) Fassin, F. (844) Faucompre, L. (814) Fleiche, L. (729) Foudre, R. (808) Foulquier, M. (772) Francois, A. (665) Galon, S. (613) Garros, R. (811) Glaize, F. (845) Godot, J. (815) Grazzioli, A. (687) Grasset, A. (800) Greppo, J. (676) Guerre, P. (730) Guillaux, E. (749) Hanne, A. (681) Helen, E. (586) Hembert (662) Hurard, J. (757) Hustinx, C. (716) Irate, G. (655) Jacquin, A. (582) Jailler, L. (682) Jeannerod, H. (696) Jeansoulin, L. (703) Joachim, H. (610) Jourjon, R. (841) Junquet, P. (621) Kormann (789) Lambert, A. (618) Lanier, P. (684) Lantheaume, C. (616) Latzel, J. (700) Le Bleu, P. (643) Leclerc, P. (593) Lefebvre, L. (691) Lecontellec, H. (810) Lenfant, P. (731) Lemoine, A. (632) Leroy, J. (638) Lesne, M. (796) Levasseur, J. (743) Le Vassor, J. (704) Lewis, J. (642) Loubignac, L. (793) Lumiere, G. (840) Madon, G. (595) Magnin, L. (648) Maicon, A. (695) Mandelli, P. (762) Mauger, D. (750) Malecaze, J. (776) Mancarot (710) Mazier, L. (634) Melin, E. (699) Metairie, A. (689) Mouroux, J. (724) Navarre, A. (584) Noel, L. (656) Nove-Josseraud (825) Olivier, L. (556) Pasquier, Baron R. (728) Penet, H. (809) Pia,

G. (829) Picard, F. (601) Poulet, E. (709) Radisson, V. (834) Raulet, F. (658) Richer, H. (607) Ridont, R. (817) Roussel, L. (659) Roux, H. (715) Saint-Michel Rivet (604) Sallard, H. (794) Sauson de Sausal (812) Schneegaus, C. (712) Senart, J. (661) Sensever, H. (580) Senougue, A. (823) Serant, L. (679) Seyrat, J. (830) Shigeno, K. (744) Soularis, M. (698) Soyer, H. (671) Testulat, P. (821) Thierry de Ville d'Avray (579) Thoret, J. (708) Tierch, M. (645) Tournier, A. (677) Trescartes, L. (842) Vallet, C. (734) Vaudelle, R. (785) Vandinck, A. (787) Vandal, P. (598) Ventre, L. (585) Vidal Soler, E. (686) Vogoyeau, A. (755) Whitehouse, W. (589) Zens, P. (675) Zorra, L. (653)

~Killed.~

+---------------------------------+ | 1912. | | Barillon (307) | | Bedell, R. | | Bernard, Suzanne | | Boerner, Lieut. | | Boncour, Lieut. | | Bressand, Lieut. | | Chanteriers, Lieut. | | Dubois, Capt. | | Ducourneau, Lieut. | | Etienne, Lieut. | | Faure, Capt. | | Lacour | | Madiot, Capt. (106) | | Maguet, Capt. le | | Nieuport, C. | | Olivers, G. | | Peignan, Lieut. A. | | Poutrin, Lieut. | | Sevelle, Lieut. H. P. | | Thiery de Ville d'Avray, Lieut. | | Thomas, Lieut. | | Wagner, A. | | | | 1913. | | Bresson, Lieut. | +---------------------------------+

FRENCH AEROPLANES.

~A~

AERIENNE. L'Aerienne, 25 Quai des Grands Agustins, Paris. Builds to specifications and supplies all parts.

ANTOINETTE. Company has ceased to exist.

ASTRA. "Astra" Soc. de Constructions Aeronautiques, (Anciens Etabs. Surcouf) Soc. An'yme 13 Rue Couchat, Billancourt (Seine). Works: 121-123 Rue de Bellevue, Billancourt. Flying grounds: Issy-les-Molineux Villacoublay (S-&-O). This old established balloon and dirigible firm first took up aviation as French agents for the _Wrights_ in 1909. For a time they built _Wrights_ with certain modifications, but by 1912, little save the Wright system of warping remained. Capacity: about 100 machines a year.

+------------------------------+----------------+----------------+----------------+----------------+----------------+ | Biplane, | Military | Biplane, | Mil. biplane, | Hydro-biplane, | type C. | biplane | Type C. | type C.M. | type C.M. | ~1912-13.~ | type C.M. | ~1913.~ | ~1913.~ | ~1913.~ | Wood. | ~1912-13.~ | Wood & steel. | Wood & steel. | Wood & steel | | Wood | | | |
------------------------------+----------------+----------------+----------------+----------------+----------------+
~Length~ feet (m.)| 34 (10.40) | 36 (10.97) | 34 (10.40) | 36 (10.97) | 32-3/4 (10) ~Span~ feet (m.)| 41 (12.50) | 40-1/2 (12.32) | 41 (12.50) | 40-1/2 (12.32) | 39-1/2 (12) ~Area.~ sq. feet (m squared)| 519 (48.2) | 519 (48.2) | 519 (48.2) | 519 (48.2) | 519 (48.2) {machine lbs. (kgs.)| 1764 (800) | 2365 (1073) | ... | 1411 (640) | 1763 (800) ~Weight~ { | | | | | (unladen) {useful lbs. (kgs.)| 661 (300) | 882 (400) | ... | ... | ... ~Motor~ h.p.| 50 Renault | 75 Renault | 50 Renault | 75 Renault | 100 Renault | | or 75 Chenu | | | {max. m.p.h. (km.)| 56 (90) | 56 (90) | 56 (90) | 56 (90) | 56 (90) ~Speed~ { | | | | | {min. m.p.h. (km.)| ... | ... | ... | ... | ... ~Endurance~ hrs.| ... | ... | ... |

... | ... Number built during 1912 | ... | ... | ... | ... | ...

Remarks.--The 1912-13 and 1913 types differ only in the adoption of metal in the 1913 models, which are consequently considerably lighter.

General features.--Warping wings. Fixed tail planes with two elevators in rear. Single rudder. Single tractor geared down 1 to 2. Type C carries 85 litres petrol; type C.M., 137 litres.

[Illustration: Astra. Military "C.M." 1913.]

[Illustration: Astra. Hydro-avion, 1913.]

~B~

BERTIN. L. Bertin, 23 rue de Rocroy, Paris. About 1908 Bertin began building helicopters. The machine below was exhibited in the 1913 Paris Salon.

[Illustration: Bertin. UAS.]

---+---+ | ~1913.~ | | Monoplane. | | 2-seater. | ---+---+ ~Length~ feet (m.)| 29 (8.80) | ~Span~ feet (m.)| 34 (10.40) | ~Area~ sq. feet (m squared.)| 226 (21) | {machine, lbs. (kgs.)| 770 (350) | ~Weight~ { | | {useful lbs. (kgs.)| ... | ~Motor~ h.p.| 100 Bertin | ~Speed~ {max m.p.h. (km.)| 71 (115) | Number built during 1912 | 1 | ---+---+

Remarks.--Wood and steel construction. On wheels only. _Controls:_ warping and rear elevator.

BESSON. Marcel Besson, 24 rue Marbeuf, Paris. Capacity: small. Besson first appeared in 1911 with a tail-first mono. In the Paris Salon, 1913, he exhibited an improved machine along similar lines.

---+---+ | ~1913~ | | _Canard_ | | 2-seater. | ---+---+ ~Length~ feet (m.)| 22 (6.70) | ~Span~ feet (m.)| 44 (13.40) | ~Area~ sq. feet (m squared.)| 323 (30) | {machine, lbs. (kgs.)| 730 (331.2) | ~Weight~ { | | {useful lbs. (kgs.)| ... | ~Motor~ h.p.| 70 Gnome | ~Speed~ m.p.h.(km.)| 59 (95) | Number built during 1912 | 1 | ---+---+

Remarks.--All steel construction. On wheels and 2 skids. _Control:_ ailerons and front elevator.

BLERIOT Monoplanes. L. Bleriot, "Bleriot-Aeronautique," 39, Route de la Revolte, Paris-Levallois. Flying grounds: Buc Etampes and Pau.

L. Bleriot began to experiment in 1906, along Langley lines. By 1909 he was one of the leading French firms; and the first cross Channel flight was made by him.

Details of standard types:--

---+---+---+

2-seater | 2-seater |(~1911~ onward) | 2-seater mono. | ~1912.~ | ~1913.~ | ~1913.~ | ~1913.~ | | ~1912.~ | | | | ----------+----------+----------+----------+----------+----------+---------- ~Length~ feet (m)| 27-1/3 (8.40) | 27-1/4 (8.24) | 28 (8.50) | 25 (7.60) | 27 (8.20) | ... ~Span~ feet (m)| 36 (11) | 36 (11) | 29-1/2 (9) | 29 (8.80) | 32 (9.75) | 40 (12.25) ~Area~ sq. ft. (m squared.)| 349 (33) | 268 (25) | 129 (12) | 162 (15) | 215 (20) | 270 (25) {unladen, lbs. (kgs)| ... | 727 (330) | 529 (240) | 530 (240) | 660 (300) | 830 (375) ~Weight~ { | | | | | | {useful lbs. (kgs.)| ... | ... | ... | 286 (129) | 550 (250) | ... ~Motor~ h.p.| 50 Gnome | 70 Gnome | 70 Gnome | 50 Gnome | 70 Gnome | 80 Gnome {max. m.p.h. (km.)| 56 (90) | 56 (90) | 78 (125) | 62 (100) | 71 (115) | 75 (120) ~Speed~ { | | | | | | {min. m.p.h. (km.)| ... | ... | ... | ... | ... | ... ~Endurance~ hrs.| ... | ... | ... | ... | ... | ... Number built during 1912 | ... | ... | ... | ... | ... | ... ----------+----------+----------+----------+----------+----------+----------

Note.--The monos., as usual, are of wood construction; wheels only for landing. Rectangular section bodies. Warping wings, elevator in rear. Chauviere propeller. The monocoque has wood, steel and cork construction. Coque body. Skids to landing chassis. Levasseur propeller. Otherwise as the other monos.

Principal _Bleriot_ flyers are or have been:--Aubrun, Balsan, Bleriot, Busson, Chavez, Cordonnier, Delagrange, Drexel, Efimoff, Gibbs, Hubert, Hamel, Moissant, Paulhan, Prevetau, Prevot, Prier, Radley, Thorup, Tyck, Wienzciers, and many others.

[Illustration: Bleriot XI _bis._]

[Illustration: 1913 type of XI _bis._ UAS.]

[Illustration: Bleriot XXVII.]

[Illustration: BLERIOT XXI. UAS.

General standard type of _Bleriot_ 1912 & 1913.]

~Special types of Bleriots.~--In addition to the standard machines, Bleriot from time to time produces special machines, of which the best known is the _Limousine_, built for M. Deutsch de la Meurthe, built 1911 and still existing. One or two Canards have also been built, including an armoured military.

[Illustration: BLERIOT-LIMOUSINE. UAS.]

Early in 1913 a special experimental military machine was produced with considerable secrecy.

[Illustration: BLERIOT MILITARY. Special 1913 military. UAS.]

BOREL. G. Borel & Cie, 25 rue Brunel, Paris. Established 1910. Capacity: about 25 machines a year.

----------+----------+----------+---------- Model. | ~1913.~ | ~1913.~ | ~1913.~ | Monoplane. | Monocoque Racer. | Hydro-mono. | | | 2-seater. ----------+----------+----------+---------- ~Length~ | 22 feet (6.70 m.) | 19 feet (5.80 m.) | 27 feet (8.30 m.) ~Span~ | 30 feet (9.15 m.) | 26 feet (8.00 m.) | 37 feet (11.25 m.) ~Area~ | 152 sq. ft. (14 m squared.) | 116 sq. ft. (11 m squared.) |

237 sq. ft. (22 m squared.) {total | 530 lbs. (240 kgs.) | 608 lbs. (276 kgs.) | 880 lbs. (399 kgs.) ~Weight~ { | | | {useful | 287 lbs. (130 kgs.) | ... | ... ~Motor~ | 50 Gnome | 80 Gnome | 80 Gnome ~Speed~ (p.h.)| 71 m. (115 km.) | 94 m. (150 km.) | 62 m. (100 km.) ----------+----------+----------+----------

Note.--The monocoque is of wood and steel construction, the others wood only. The monocoque has coque body, the others ordinary rectangular section. Floats of the hydro as illustrated. For the rest the ordinary mono. is practically on the same lines as the 1912. The racer is somewhat on _Deperdussin_ lines, but the body is built up inside. No fixed tail. The hydro. is an enlarged edition of the mono. Floats display nothing very original, except that a float under tail is interconnected with the rudder, and that the two front floats are fitted for being rowed. Fitted with a self-starter.

[Illustration: 1913 Borel. Hydro-avion. _By favour of "Flight."_ UAS.]

[Illustration: Hydro-avion.]

There is also a Denhaut design, 1913, about the same as a _Donnet-Leveque_.

[Illustration: Borel. Monocoque. UAS.]

BREGUET. Soc. Anonyme des ateliers d'aviation, Louis Breguet, 16 Boulevard Vauban, Donai (Nord). Capacity: about 200 machines a year. Paris office: 25, Boulevard Jules Sandeau. Schools at La Brayelle, pris Douai, Velisy-Villacoublay, pris Paris.

----------+----------+----------+----------+----------+---------- | | | | Aerhydroplane | ~G2~ bis. | ~G3.~ | ~C-U1.~ | ~C-U2.~ | tandem ~1913 models.~ | 2 or 3-seater | 3-seater | 2-seater | 2-seater | mono. | biplane. | biplane. | biplane. | biplane. | 2-seater, | | | | side by side. ----------+----------+----------+----------+----------+---------- ~Length~ feet (m)| 33 (10) | 29 (8.75) | 29 (8.75) | 29 (8.75) | 29 (8.75) ~Span~ feet (m)| 49 (15) | 45 (13.65) | 45 (13.65) | 45 (13.65) | 42 (12.80) ~Area~ sq. feet (m squared)| 376 (35) | 377 (36) | 387 (36) | 387 (36) | 387 (36) {empty, lbs. (kgs.)| 1323 (600) | 1212 (550) | 1430 (649) | 1160 (522) | 1760 (798) ~Weight~ { | | | | {useful lbs. (kgs.)| 662 (300) | 882 (400) | 662 (300) | 882 (400) | 662 (300) ~Motor~ h.p.| 80 Gnome | 100 Gnome | 80 Canton Unme. | 110 Canton Unme. | 110 Canton Unme. {max. m.p.h. (km.)| 62 (100) | 69 (110) | 62 (100) | 71 (115) | 87 (140) ~Speed~ { | | | | {min. m.p.h. (km.)| ... | ... | ... | ... | 62 (100) ~Endurance~ hrs.| 3-1/2 | 4 | 7 | 7 Number built during 1912 |A total of 41 sold| during 1912 for| military purposes.| | ... ----------+----------+----------+----------+----------+----------

In each case.--

~Construction.~--All steel.

~Landing chassis.~--C consists of three wheels each protected by skids. The two main wheels, placed on either side of the centre of gravity, are fitted with patent "Oleopneumatic" shock absorbers. The steering wheel and the front skid have a spring suspension.

~Military machines.~--The 1912 sales of these were:--32 to France; 5 British; 3 Italian; 1 Swedish.

~Steering.~--The patented control system consists of a wheel mounted on a pivoted lever. The backward and forward movement of the entire system operates the elevator: the sideway movement warps the rear edge of the upper wings, and the rotation of the wheel steers the machine. The latter operation also governs the front wheel of the landing chassis, so that when on the ground the machine can be steered like a motor car.

~Portability.~--The main planes can be folded alongside of the fuselage. The machine can then be towed on any ordinary road, or be housed in places such as farm buildings, stables, &c.

[Illustration: Aerhydroplane, 1913-14.]

[Illustration: BREGUET. Hydro. UAS]

[Illustration: BREGUET. Biplane. UAS]

[Illustration: BREGUET. 1912-13, G3 type 3-seater military. UAS]

C

CAUDRON. Caudron Freres, Rue (Somme). Schools: Crotoy and Juvissy. Capacity: about 100-250 a year.

----------+----------+----------+----------+----------++----------+----------++----------+---------- | ~M2~ | ~N.~ | ~G.D.~ | || ~B.~ | ~E.~ || Monaco type, | Model and Date. | 1912-13 | 1912-13 | 1912-13 | ~1913~ || 1912-13 | 1912-13 || 1912 | ~1913~ | mono. | mono. | mono. | mono. || biplane. | biplane. || hydro-biplane. | hydro-biplane. ----------+----------+----------+----------+----------++----------+----------++----------+---------- ~Length~ feet (m.)| 20 (6.10) | 19-3/4 (6) | 22 (6.75) | 19-1/4 (5.80) || 26-1/4 (8) | 23-1/2 (7.15) || 22 (6.75) | 32-3/4 (10) ~Span~ feet (m.)| 31 (9.40) | 26-1/3 (8) | 34 (10.30) | 27-1/3 (8.50) || 32-3/4 (10) | 35-1/2 (10.80) || 33 (10.10) | 46 (14) ~Area~ sq. feet (m squared.)| 151 (14) | 108 (10) | 268 (25) | 118 (11) || 431 (40) | 301 (28) || 268 (25) | 376 (35) ~Weight~ machine, lbs. (kgs.)| 518 (235) | 496 (225) | 386 (175) | 490 (225) || 683 (310) | 640 (295) || 772 (350) | 882 (400) ~Motor~ h.p.| 50 Anzani or Gnome | 50 Anzani | Anzani or Gnome | 50 Gnome. || Anzani or Gnome | Gnome || Gnome | 70 Gnome ~Speed~ m.p.h. (km.)| 71 (115) | 84 (135) | 75 (120) | 84 || 56 (90) | 56 (90) || 50 (80) | 50 (80) Number built during 1912 | ... | ... | ... | ... || ... | ... || ... | ... ----------+----------+----------+----------++----------+----------++----------+---------- || Lateral control, warping. Wood construction. Notes.--Lateral control, warping. Wood construction. On wheels. Enclosed body. || On wheels as well as || floats. (Special Caudron patent.) ----------++----------

[Illustration: 1912 hydro. _By favour of "Aeronautics," U.S.A._ UAS]

[Illustration: CAUDRON. UAS]

[Illustration: 1913 hydro. UAS]

[Illustration: CAUDRON. Mono. _By favour of "Flight."_ UAS]

CLEMENT-BAYARD. Usines Clement-Bayard, 33 quai Michelet, Levallois-Perret (Seine).

[Illustration]

----------+----------+----------+ | ~1913.~ | ~1913.~ | | Military 3-seater | Military single seater | | biplane. | monoplane. | ----------+----------+----------+ ~Length~ feet (m)| 37 (11.20) | 24-2/3 (7.50) | {upper feet (m)| 52 (16) | 30 (9.20) | ~Span~ { | | | {lower feet (m)| 36 (11) | ... | ~Area~ sq. feet (m squared.)| 533 (50) | 172 (16) | {total lbs. (kgs.)| 2425 (1100) | 1146 (520) | ~Weight~ { | | | {useful lbs. (kgs.)| 1014 (460) | 441 (200) | ~Motor~ h.p.| 100 Gnome | 70 Gnome | {max. m.p.h. (km.)| 53 (85) | 75 (120) | ~Speed~ { | | | {min. m.p.h. (km.)| ... | ... | ~Endurance~ hrs.| ... | ... | ----------+----------+----------+

Notes.--_Control_: lateral, warping.

D

D'ARTOIS. Soc. Anonyme des Anciens Chantiers Tellier, Longuenesse, pres St. Omer. Re-established 1912. Capacity: small.

----------+----------+----------+ | ~1913~ model. | ~1913~ | Model and date. | "Aero torpille" | "Aero torpille" | | hydro-biplane. | biplane. | ----------+----------+----------+ ~Length~ feet (m.)| 23 (7) | 24-3/4 (7.50) | {| 36 (11) | 36 (11) | ~Span~ feet (m.){| | | {| 20 (6) | 20 (6) | ~Area~ sq. feet (m squared.)| 280 (26) | 280 (26) | ~Weight~ empty, lbs. (kgs.)| 772 (350) | 551 (250) | ~Motor~ h.p.| 50 Gnome | 50 Gnome | ~Speed~ m.p.h. (km.)| 56 (90) | 84 (135) | ~Endurance~ hrs.| ... | ... | Number built during 1912 | ... | ... | ----------+----------+----------+

Notes.--Single long boat body, canoe-shape.

[Illustration: _By favour of "Aeronautics," U.S.A._ UAS]

DEPERDUSSIN. Armand Deperdussin, 19 rue des Entrepreneurs, Paris. School: Courey-Betheny (Marne). Established 1910. Capacity: about 150-200 machines a year.

----------+----------+----------+----------+----------+----------+---------- | ~E 1912-13.~ | ~P 1912-13.~ | ~T 1912-13.~ | ~H 1912-13.~ | Monocoque | Mono. | school mono. | single seater | 2-seater | 3-seater | ~1913.~ | ~1913.~ | | mono. | mono. | mono. | 2-seater. | 2-seater. ----------+----------+----------+----------+----------+----------+--------

---------- ~Length~ feet (m)| 24 (7.30) | 24 (7.30) | 24 (7.30) | 29 (8.80) | 19 (5.75) | ... ~Span~ feet (m)| 29 (8.85) | 28 (8.50) | 35 (10.65) | 41 (12.50) | 29-1/2 (8.95) | 36 (11) ~Area~ sq. feet (m squared.)| ... | 162 (15) | ... | 310 (28) | 97 (9) | ... {total lbs. (kgs.)| 661 (300) | 782 (355) | 1212 (550) | 2050 (930) | 882 (400) | ... ~Weight~ { | | | | | | {useful lbs. (kgs.)| ... | ... | ... | ... | ... | ... ~Motor~ h.p.| 30 Anzani | 50 Gnome | 70 Gnome | 100 Gnome | 50 Gnome | 80 Gnome {max. m.p.h. (km.)| 50 (80) | 69 (110) | 65 (105) | 69 (110) | 113 (180) | 105 (170) ~Speed~ { | | | | | | {min. m.p.h. (km.)| ... | ... | ... | ... | 81 (130) | ... ~Endurance~ hrs.| ... | ... | ... | ... | ... | ... Number built during 1912 | 2 | 5 | 27 | 3 | 2 | 1 ----------+----------+----------+----------+----------+----------+----------+----------

Notes.--Wood construction. Lateral control by warping. Mounted on wheels without skids. Fabric: "Aviator" Ramie.

Principal _Deperdussin_ records: 1912 Gordon Bennett (Vedrines) and a number of world records for speed and distance.

Principal pilots include: Busson, Prevost, Vedrines, Vidart.

[Illustration: 50 h.p. monocoque.]

[Illustration: DEPERDUSSIN. 80 h.p. UAS]

[Illustration: The 80 h.p. mounted on floats as a hydro.]

DONNET-LEVEQUE.

----------+----------+----------+----------+---------- | ~A 1912.~ | ~B 1912.~ | ~C 1912.~ | ~1913.~ | 2-seater | 2-seater | 3-seater | 2-seater | hydro-biplane | hydro-biplane | hydro-biplane | hydro-biplane ----------+----------+----------+----------+---------- ~Length~ feet (m)| 26 (7.80) | 27 (8.30) | 27 (8.30) | 34-1/2 (10.50) ~Span~ feet (m)| 29-1/2 (9) | 32-3/4 (10) | 34-1/2 (10.50) | 29-1/2 (9) ~Area~ sq. feet (m squared.)| 194 (18) | 215 (20) | 237 (22) | 194 (18) ~Weight~ lbs. (kgs.)| 683 (310) | 772 (350) | 888 (380) | 888 (380) ~Motor~ h.p.| 50 Gnome | 70 Gnome | 80 Gnome | 50 Gnome ~Speed~ m.p.h. (km.)| 69 (110) | 75 (120) | ... | 50 (80) ~Endurance~ hrs.| ... | ... | ... | ... Number built during 1912 | ... | ... | ... | ... ----------+----------+----------+----------+----------

Notes.--Lateral control by warping ailerons. Motor in gap just below upper plane: propeller in rear, direct driven. Fabric: "Aviator" Ramie.

Floats.--One large central boat 27 feet (8.20 m.) long--two small ones at each extremity of lower plane.

[Illustration: _By favour of "Aeronautics," U.S.A._]

[Illustration: UAS]

DOUTRE. Soc. Anonyme Doutre, 58, rue Talbot, Paris.

----------+----------+----------+ Type. | Biplane 3-seater, | Biplane 2-seater, | | ~1912-13.~ | ~1912-13.~ |

----------+----------+----------+ ~Length~ feet (m.)| 40 (12.25) | ... | {| 53 (16.10) | ... | ~Span~ feet (m.){| | | {| 43 (13) | ... | ~Area~ sq. feet (m squared.)| 533 (50) | ... | {machine lbs. (kgs.)| 1323 (600) | 1323 (600) | ~Weight~ { | | | {useful lbs. (kgs.)| 992 (450) | 992 (450) | ~Motor~ h.p.| 70 Renault | 50 Renault | ~Speed~ max. m.p.h (km.)| 56 (90) | 56 (90) | Number built during 1912 | 1 | ? | ----------+----------+----------+

Notes.--Fabric: "Aviator" Ramie. Both types fitted with the Doutre patent stabiliser, which automatically and instantaneously counteracts troubles due to sudden gusts or partial motor failures. Weight of the 1913 model stabiliser is only 44 lbs. (20 kgs.)

[Illustration: Model 1913 stabiliser.]

[Illustration: DOUTRE. UAS]

F

FARMAN. Henry and Maurice Farman, 167, Rue de Silly, Billancourt (Seine) Aerodromes: Buc, pres Versailles and Etampes. Depots: Camp de Chalons--Reims. Established by H. Farman in 1908. M. Farman established works a little later. In 1912 the two brothers combined. The present works were opened in January, 1912, and had an output capacity of at least 300 machines a year in March, 1913.

----------+----------+----------+----------+----------+----------+---------- | _H. Farman._ | _H. Farman._ | _H. Farman._ | _H. Farman._ | _M. Farman._ | _M. Farman._ | _M. Farman._ | Military. | Single-seater. | 2-seater | 2-seater special | Military biplane. | Big military | Staggered | 2 or 3-seater. | Military. | monoplane. | hydro-biplane. | | biplane. | biplane. | ~1912-13.~ | ~1913.~ | | ~1913.~ | | | | Biplane. | Biplane. | | | | | ----------+----------+----------+----------+----------+----------+----------+---------- ~Length~ feet (m.)| 26-1/4 (8) | 24 (7.35) | 24-1/2 (7.50) | 26 (7.90) | 39-1/3 (12) | 46 (14) | 39 (11.90) ~Span~ feet (m.)| 42-3/4 (13.25) | 31-1/8 (9.50) | 32-3/4 (10) | 45 (13.70) | 50-3/4 (15.50) | 65-3/4 (20) | 36 (11) ~Area~ sq. feet (m squared.)| 376 (35) | 161 (15) | 204 (19) | 344 (32) | 646 (60) | 861 (80) | 323 (30) {total lbs. (kgs.)| 793 (360) | 640 (295) | 628 (285) | 950 (431) | 1102 (500) | 1433 (650) | 882 (400) ~Weight~ { | | | | | | {useful lbs. (kgs.)| 661 (300) | 386 (175) | ... | ... | 617 (280) | 882 (400) | 551 (250) {| | | Designed for | | | | ~Motor~ h.p.{| 70-80 Gnome | 70-80 Gnome | Gnomes from | 50 Gnome | 70 Renault | 70 Renault | 70 Renault {| | | 40 up to 160 h.p. | | | | {max. m.p.h. (km)| 65 (105) | 71 (15) | ... | 52 (100) | 56 (90) | 44 (70) | 69 (110) ~Speed~ { | | | | | | {min. m.p.h. (km)| ... | ... | ... | ... | ... | ... | ... ~Endurance~ hrs.| 3 | ... | ... | ... | ... | ... | ... Number built during 1912 | ... | ... | ... | ... | ... | ... | ... ----------+----------+----------+----------+----------+----------+----------

Remarks.--The whole of the above can easily be converted into hydro-avions--two long narrow floats without steps. _H. Farmans_ are of wood and steel construction; _M. Farman_, wood. In all 1913 biplanes the ailerons are inter-connected. All 1913 machines designed to carry one or in some cases two mitrailleuse, and special attention is paid to facility for taking down for transport and re-assembling. The 1911-12 _H. Farmans_ had elevators forward, were a good deal longer, and had more

surface than 1913 models. Ailerons not inter-connected. The _M. Farmans_ generally as now, except that all planes, etc., had rounded edges. On September 11th, 1912, Foury, in an _M. Farman_ military, made world's endurance record to date, 13 hrs. 22 min., covering 631 miles (1,017 km.) All models of this type, also the "big military," are fitted with the Doutre stabiliser. Fabric: "Aviator" Ramie.

~Latest Hydro.~--In March, 1913, a new hydro was produced experimentally. There is a boat body, without steps, carrying the motor which is chain connected with the propeller. Machine is fitted with wheels and skids as well.

[Illustration: H. Farman. 1912-13 military biplane.]

[Illustration: H. Farman. 1913 latest type military biplane.]

[Illustration: M. Farman. 1912-13 military biplane.]

[Illustration: M. Farman. 1912-13 staggered biplane. This is the type which has done best as a hydro-aeroplane.]

G

GOUPY. A. Goupy, 50, Avenue Marceau, Paris. School: Juvissy (Port Aviation). Capacity: about 30 machines a year.

| | | ~1913.~ Model and date. | ~1913 A.~ | ~1913 B.~ | Hydro-staggered | Staggered biplane. | Staggered biplane. | biplane. ~Length~ feet (m.)| 25 (7.50) | 26-1/4 (8) | 33 (10) ~Span~ feet (m.)| 26-1/4 (8) | 42-3/4 (13) | 42 (12.70) ~Area~ sq. feet (m squared)| ... | ... | 480 (45) {machine lbs. (kgs.)| ... | ... | 992 (450) ~Weight~ { | | | {useful lbs. (kgs.)| ... | ... | 661 (300) ~Motor~ h.p.| 50 Gnome | 80 or 100 Gnome | 80 Gnome {max. m.p.h. (km.)| 62 (100) | 75 (120) | 75 (120) ~Speed~ { | | | {min. m.p.h. (km.)| ... | ... | ... ~Endurance~ hrs.| ... | ... | ... Number built during 1912 | ... | 12 | 1

Fabric: "Aviator" Ramie.

[Illustration: Goupy. Hydro. _From "Flight."_ UAS]

[Illustration: Goupy. Hydro. _By favour of "Aeronautics," U.S.A._ UAS]

H

HANRIOT. Aeroplanes Hanriot & Cie., 145 rue de Neufchatel, Reims. Paris office: 69 boulevard Berthier, Paris. School: Antibes, Reims.

~1913~ models. | ~D I.~ | ~D II.~ | ~D III.~ | ~D IV.~ | ~D VII.~ ~Monoplanes.~ | Single seater. | 2 or 3-seater. | Racer. | Steel. | ~Length~ feet (m.)| 23 (7) | 26-1/3 (8) | 21-3/4 (6.65) | 23 (7) | 23 (7) ~Span~ feet (m.)| 28-1/3 (8.70) | 42-3/4 (13) | 24 (7.30) | 28-1/3 (8.65) | 36 (10.95) ~Area~ sq. feet (m

squared)| 161 (15) | 226 (21) | 91 (8.50) | 161 (15) | 194 (18) {machine lbs. (kgs.)| 661 (300) | 937 (425) | 661 (300) | 661 (300) | 771 (350) ~Weight~ { | | | | {useful lbs. (kgs.)| ... | 616 (280) | ... | 396 (180) | 364 (165) ~Motor~ h.p.| 50 Anzani | 100 Gnome | 100 Gnome | 50 R. Peugeot | 80 Gnome {max. m.p.h. (km.)| 69 (110) | 78 (125) | 106 (170) | 71 (115) | 71 (115) ~Speed~ { | | | | {min. m.p.h. (km.)| ... | ... | ... | ... | ... ~Endurance~ hrs.| ... | ... | ... | ... | ... Number built during 1912 | ... | ... | ... | ... | ... ---

Notes.--There are also two school types 35 and 45 h.p. Records include 1912 world record for speed with passengers.

None of the above machines represent any very particular divergence from recognised _Hanriot_ practice. _D IV_ is all steel construction, the others wood and steel.

[Illustration]

M

MORANE-SAULNIER. Soc. de constructions aeronautiques, Morane-Saulnier. 206 Boulevard Pereire. Capital: 1,500,000 francs. School: Villacoublay. Output capacity: about 50 machines a year.

| ~Military, 1913.~ | ~2 places.~ | | ~Tandem.~ ~Length~ feet (m.)| 21 (6.38) | 21 (6.38) ~Span~ feet (m.)| 30-1/5 (9.20) | 33-1/2 (10.20) ~Surface~ sq. feet (m squared.)| 151 (14) | 172 (16) {total lbs. (kgs.)| 595 (270) | 617 (280) ~Weight~ { | | {useful lbs. (kgs.)| ... | ... ~Motor~ h.p.| 50 h.p. | 80 h.p. ~Speed~ m.p.h. (km.)| 75 (120) | 75 (120) Number built during 1912 | ... | ...

In each case ~body~ is of rectangular section, wood, mounted on wheels only, except for the ~military~ type which has skids also. Fabric: "Aviator" Ramie.

In all there is a rear elevator and a Chauviere tractor.

Note.--Flown in the European Circuit, 1911, by Vedrines, Gajet, Lesire, Morisson, Verept, Frey, Garnier and Dalgier.

[Illustration]

[Illustration: 1913. 100 h.p. Gnome engined.]

MOREAU. Moreau freres, Combs-la-Ville.

| ~1913.~ | Model and date. | 2-seater. | ~Length~ feet (m.)| 31 (9.50) | ~Span~ feet (m.)| 39-1/3 (12) | ~Area~ sq. feet (m squared.)| 258 (24) | {machine lbs. (kgs.)| 992 (450) | ~Weight~ { | | {useful lbs. (kgs.)| ... | ~Motor~ h.p.| 70 Gnome | ~Speed~ max. m.p.h. (km.)| 62 (100) | Number built during 1912 | 2 |

Notes.--Fitted with a special stabilising device.

[Illustration: MOREAU. UAS]

N

NIEUPORT. Etablissements Nieuport, 9 rue de Seine, Suresnes (Seine). Established 1910 by the late Edouard Nieuport. Approximate capacity of works: about 100 machines a year. Chief designer during 1911 was Pagny, who has now joined the Hanriot firm.

----------+----------+----------+----------+----------+----------+----------+----------+---------- Model and date. | ~II N,~ | ~II G,~ | ~IV G, 1912-13.~ | ~IV M, 1912-13.~ | ~1913.~ | ~1913.~ | ~1913.~ | ~1913.~ ~Monoplanes.~ | ~1912.~ | ~1912.~ | 2-seater. | 3-seater. | 2-seater. | 1-seater. | 1-seater. | Hydro 3-seater. ----------+----------+----------+----------+----------+----------+----------+----------+---------- ~Length~ feet (m.)| 23-2/3 (7.20) | 23-2/3 (7.20) | 25-2/3 (7.80) | 25-2/3 (7.80) | 26-1/4 (8) | 21-3/4 (6.60) | 23 (7) | 29 (8.80) ~Span~ feet (m.)| 28-1/3 (8.65) | 28-1/3 (8.65) | 36 (10.90) | 39-1/3 (12.10) | 36 (11) | 28-1/3 (8.70) | 27-2/3 (8.40) | 40 (12.20) ~Area~ sq. feet (m squared.)| ... | ... | ... | ... | 231 (21-1/2) | 140 (13) | 156 (14-1/2) | 242 (22-1/2) {machine lbs. (kgs.)| 529 (240) | 683 (310) | 771 (350) | 1058 (480) | 771 (350) | 573 (260) | 573 (260) | 1230 (558) ~Weight~ { | | | | | | | {useful lbs. (kgs.)| ... | ... | ... | ... | ... | ... | ... | ... ~Motor~ h.p.| 30 Nieuport | Gnome | Gnome | Gnome | Gnome | 50 Gnome | 30 Nieuport | 100 Gnome {max. m.p.h. (km.)| 75 (120) | 87 (140) | 72 (117) | 72 (117) | 69 (110) | 78 (125) | 69 (110) | 72 (117) ~Speed~ { | | | | | | | | {min. m.p.h. (km.)| ... | 75 (120) | 69 (110) | ... | ... | ... | ... | ... Number built during 1912 | ... | ... | ... | ... | ... | ... | ... | ... ----------+----------+----------+----------+----------+----------+----------+----------+----------

Notes.--Early types had a _Hanriot_ style landing carriage; the 1913 models revert to a _Bleriot_ type. Warping wings. Fuselage entirely enclosed, rectilineal with rounded nose.

[Illustration: Nieuport. Hydro. _By favour of "Flight."_ UAS]

P

PAULHAN-CURTISS. Soc. anonyme d'aviation Paulhan, (S.A.P.) 71 boulevard Berthier, Paris. Flying ground: Bois d'Arcy par St. Ayr (S. et O.) Hydro school: Juan-les-Pins, par Antibes (Alpes Maritimes).

Founded by the well-known aviator, L. Paulhan. He first produced biplanes, then triplanes and finally a monoplane type, the _Tatin-Paulhan_ (1911). These are now all abandoned, and the firm devotes itself to building hydro-aeroplanes under Curtiss (U.S.A.) license. Principal type built are:--

----------+----------+----------+ Model and date. | Flying boat. | Flying boat. | ~Biplanes.~ | Single-seater. | 2-seater. | ----------+----------+----------+ ~Length~ feet (m.)| ... | 27

(8.30) | ~Span~ feet (m.)| 35-1/2 (10.80) | 37 (11.30) | ~Area~ sq. feet (m squared.)| ... | 290 (26-3/4) | {machine lbs. (kgs.)| ... | 948 (430) | ~Weight~ { | | | {useful lbs. (kgs.)| ... | ... | ~Motor~ h.p.| 75 Curtiss | 85 Curtiss | ~Speed~ m.p.h. (km.)| ... | ... | Number built during 1912 | 2 | 8 | ----------+----------+----------+

PISCHOFF. Etablissements Autoplan, 4 rue Beranger, Boulogne sur Seine (Seine).

This firm has produced various types in the past, but at present, appears confined to constructing to specifications (See _Pischoff-Werner_ last edition).

[Illustration: Paulhan-Curtiss. Flying boat.]

R

R.E.P. Robert Esnault-Pelterie, Billancourt. School: Bue. One of the earliest established French firms. The first to go in for steel construction. Reported to have amalgamated with _Breguet_ in 1912, but this fell through.

----------+----------+----------+----------+----------+----------+ Model. | ~1912.~ | ~1912.~ | ~1912.~ | ~1913.~ | ~1913.~ | ~Steel monoplanes.~ | 1-seater. | 2-seater. | Military. | 2-seater. | Hydro-mono. | | | 3-seater. | | 2-seater. | ----------+----------+----------+----------+----------+----------+----------+ ~Length~ feet (m.)| 25-1/3 (7.70) | 25-1/3 (7.70) | 25-1/3 (7.70) | 23 (7) | 25 (7.50) | ~Span~ feet (m.)| 35 (10.70) | 38-1/3 (11.70) | 38-1/3 (11.70) | 36 (11) | 38-1/4 (11.60) | ~Area~ sq. feet (m squared.)| 215 (20) | 237 (22) | 323 (30) | 237 (22) | 323 (20) | {machine lbs. (kgs.)| 882 (400) | 661 (300) | 882 (400) | 595 (270) | ... | ~Weight~ { | | | | | {useful lbs. (kgs.)| ... | ... | ... | ... | ... | ~Motor~ make and h.p.| 60 Rep. | 66 Rep. | 90 Rep. | 95 Rep. | 80 Rep. | {max. mph. (km.)| 69 (110) | 69 (110) | 69 (110) | 78 (125) | 78 (125) | ~Speed~ { | | | | | {min. mph. (km.)| ... | ... | ... | 62 (100) | 62 (100) | Number built during 1912 | ... | ... | ... | ... | ... | ----------+----------+----------+----------+----------+----------+

Remarks.--Steel construction. Pentagonal and triangular body. Mounted on wheels and skids. The hydro is on one very large central float.

[Illustration: _Flight._ UAS]

S

SANCHEZ BESA. 2 avenue de Villiers, Paris.

----------+----------+----------+----------+ Model and date. | ~1912.~ | ~1912.~ | ~1913.~ | | Hydro-biplane. | Hydro-biplane. | Hydro-biplane. | | | | (amphibious) | ----------+----------+----------+----------+ ~Length~ feet (m.)| 34 (10.40) | ... | 32-3/4 (10) | ~Span~ feet (m.)| 54 (16.40) | 55-3/4 (17) | 54-3/4 (16.60) | ~Area~ sq. feet (m squared.)| 646 (60) | ... | 646 (60) | {avide lbs. (kgs.)| 1984 (900) | ... | 1102 (500) | ~Weight~ { | | | | {useful lbs. (kgs.)| ... | ... | ... | ~Motor~ h.p.| 100 Renault | 70 Renault | 70 Renault | ~Speed~ max. m.p.h. (km.)| 56 (90) | ... | 50 (80) | ~Endurance~ hrs.| 5 | 5 | 6 | Number built during 1912 | 3 | 1 | 1 | ----------+----------+----------+----------+

Notes.--Wood and steel construction.

Controls.--Ailerons and rear elevators. Floats: The first has two and the second three floats. The 1913 model has a single boat body mounted on wheels.

[Illustration: 1913 hydro.]

SAVARY. Soc. anonyme des aeroplanes. Robert Savary, 31 rue Dunois, Paris. School: Chartres. Output capacity: 100 to 150 machines a year.

----------+----------+----------+---------- Model and date. | ~1912.~ | ~1912.~ | ~1913.~ | Biplane. |Military (3-seater.)| Biplane. ----------+----------+----------+---------- ~Length~ feet (m.)| 36 (11) | 33-1/2 (10.15) | 38-1/2 (11.70) ~Span~ {feet (m.)| 46 (14) | 49 (14.90) | 49-1/4 (15) {feet (m.)| 33 (10) | 37 (11.20) | 33 (10) ~Area~ sq. feet (m squared.)| 510 (48) | 533 (50) | 550 (52) {machine lbs. (kgs.)| 1132 (600) | ... | 1132 (600) ~Weight~ { | | | {useful lbs. (kgs.)| ... | ... | ... ~Motor~ h.p.| various | 70 Labor | 75 Renault | | | (Gnome or Labor) {max m.p.h. (km.)| 56 (90) | ... | 59 (96) ~Speed~ { | | | {min m.p.h. (km.)| 50 (80) | ... | ... Number built during 1912 | ... | 47 | ... ----------+----------+----------+----------

Notes.--Wood and steel construction. _Control_: ailerons and rear elevator. Landing gear: wheels and skids. _Special features_: There are 4 rudders in the gap, and 2 tractors, chain driven. Aeroplatte fabric.

[Illustration: SAVARY. 1913. UAS]

SLOAN. "Bicurve." Sloan & Cie, 17 rue de Louvre, Paris. Works: 9 rue Victor Hugo, Charenton. Flying ground: Port Aviation. Output capacity: small.

----------+----------+----------+ Model and date. | ~1912.~ | ~1913.~ | ----------+----------+----------+ ~Length~ feet (m.)| 31-1/3 (9.50) | 29 (8.70) | ~Span~ feet (m.)| 42-3/4 (13) | 42-1/2 (12.90) | ~Area~ sq. feet (m squared)| 527 (49) | 473 (44) | {machine lbs. (kgs.)| 1100 (500) | 662 (300) | ~Weight~ { | | | {useful lbs. (kgs.)| ... | ... | ~Motor~ h.p.| 100 Gnome | 120 Laviator | ~Speed~ {max. m.p.h. (km.)| 59 (95) | 65 (105) | Number built during 1912 | ... | ... | ----------+----------+----------+

Notes.--Wood construction. Wheels and skids landing gear. _Control_: ailerons and rear elevator.

[Illustration: Sloan.]

SOMMER. Ateliers Roger Sommer, Mouzon, Ardennes. Flying grounds: Douzy, Mourmelon, Vidamme.

~Monoplanes.~ ~Biplanes.~ /----------^----------\ /----------^----------\ ----------+----------+----------+----------+----------+----------+----------+---------- Model and date. | ~E 1912.~ | ~1913.~ | ~K 1912.~ | ~R 1912.~ | ~S 1912.~ | ~L 1912.~ | ~R3 1913.~ | | | Single seater. | 2 or 3-seater | | | 2 or 3-seater -----------+----------+----------+----------+----------+----------+----------

--------+----------+---------- ~Length~ feet (m.)| 22 (6.70) | 23 (7) | 39-1/4 (12) | 36 (11) | 31 (9.50) | 29-1/2 (9) | 38-2/3 (11.70) ~Span~ feet (m.)| 28-1/2 (8.70) | 26-1/4 (8) | 39-1/4 (12) | 51 (15.50) | 42 (12.80) | 39-1/4 (12) | 46 (14) ~Area~ sq. feet (m squared.)| 172 (16) | 172 (16) | 215 (20) | 533 (50) | 350 (32) | ... | 575 (54) {machine lbs. (kgs.)| 595 (270) | 617 (280) | 617 (280) | 992 (450) | 597 (275) | 639 (290) | 882 (400) ~Weight~ { | | | | | | {useful lbs. (kgs.)| ... | ... | ... | ... | ... | ... | ... ~Motor~ h.p.| 50 Anzani | 50 Gnome | Various | Various | Various | Various | 70 Renault | or Gnome | | | | | {max. m.p.h. (km.)| 84 (135) | 84 (135) | 61 (98) | 50 (80) | 57 (92) | 56 (90) | 56 (90) ~Speed~ { | | | | | | | {min. m.p.h. (km.)| 67 (108) | 65 (105) | 53 (85) | ... | 53 (84) | ... | ... ~Endurance~ hrs.| 4 | 4 | ... | ... | ... | ... | ... Number built during 1912 | ... | ... | ... | ... | ... | ... | ... ----------+----------+----------+----------+----------+----------+----------+---------- Wood and steel construction. Landing: carriage |Wood and steel construction. Landing: wheels and skids. wheels. _Control_: warping and rear elevator. |_Control_: ailerons and front rear elevator. Rectangular body. | ----------+----------+----------+----------+----------+----------+----------+----------

[Illustration: SOMMER. UAS]

T

TRAIN. E. Train, Buoy, Camp de Chalons (Marne).

----------+----------+----------+----------+ Model and date. | 1-seater. | 2-seater. | Hydro-mono. | ~Monoplanes.~ | | | | ----------+----------+----------+----------+ ~Length~ feet (m.)| 26-1/4 (8) | 26-1/4 (8) | 26-1/4 (8) | ~Span~ feet (m.)| 30-3/4 (9.30) | 35 (10.66) | 42-1/2 (12.94) | ~Area~ sq. feet (m squared.)| 172 (16) | 215 (20) | ... | {machine lbs. (kgs.)| 573 (260) | 617 (280) | ... | ~Weight~ { | | | {useful lbs. (kgs.)| ... | ... | ... | ~Motor~ h.p.| 30/60 Anzani | 70 Gnome | 80 Gnome | {max. m.p.h. (km.)| 59 (95) | 65 (105) | ... | ~Speed~ { | | | {min. m.p.h. (km.)| 47 (75) | ... | ... | Number built during 1912 | ... | ... | ... | ----------+----------+----------+----------+

Notes.--Steel construction. Landing: carriage wheels and skids. _Control_: warping and rear elevator. The hydro has one very large float which extends a considerable distance ahead of the tractor.

[Illustration: TRAIN. UAS]

TUBAVION. Ponche & Primaud, Long.

----------+----------+ Model and date. | Monoplane. | | ~1913.~ | ----------+----------+ ~Length~ feet (m.)| 29 (8.85) | ~Span~ feet (m.)| 29-1/2 (9) | ~Area~ sq. feet (m squared.)| 194 (18) | {machine lbs. (kgs.)| 772 (350) | ~Weight~ { | | {useful lbs. (kgs.)| ... | ~Motor~ h.p.| 70 Gnome | ~Speed~ max. m.p.h. (km.)| 65 (105) | Number built during 1912 | 1 | ----------+----------+

Notes.--Tubular steel construction. Landing: wheels and 2 very long skids. Propeller: amidships.

[Illustration: TUBAVION. UAS]

V

VINET. Gaston Vinet, 41-47 quai de Seine, Courbevoie: also 2-8 rue Larnac. Established for automobile work, 1893. Aeroplane output capacity: small.

----------+----------+----------+ Model and date. | Type ~D~ | ~1913.~ | | ~1912~ mono. | Mono. | ----------+----------+----------+ ~Length~ feet (m.)| 21-1/2 (6.60) | 21 (6.40) | ~Span~ feet (m.)| 28-1/2 (8.60) | 28 (8.50) | ~Area~ sq. feet (m squared.)| 162 (15) | 162 (15) | {machine lbs. (kgs.)| 550 (250) | 440 (200) | ~Weight~ { | | | {useful lbs. (kgs.)| ... | ... | ~Motor~ h.p.| 50 Gnome | 50 Gnome | ~Speed~ max m.p.h. (km.)| 56 (90) | 60 (95) | Number built during 1912 | 6 | ... | ----------+----------+----------+

Notes.--Wood construction. Landing wheels and skids. _Control_: warping and rear elevator. Rectangular body. The two types are practically identical.

[Illustration: VINET. Type D. UAS]

VOISIN. Voisin Aeroplanes, Boulevard Gambetta, Issy le Molineux, (Seine). School: Mourmelon. Capital 1,000,000 francs. The oldest aeroplane firm in the world, founded by the Brothers Voisin in 1905. (See past editions).

Latest models are:

----------+----------+----------+----------+ | Military | | Military | Model and date. | biplane. | Hydro-biplane. | biplane. | | Model ~1912.~ | Model ~1912.~ | Model ~1913.~ | ----------+----------+----------+----------+ ~Length~ feet (m.)| 37-3/4 (11.50) | 36 (11) | 32-3/4 (10) | ~Span~ feet (m.)| 55-3/4 (17) | 43-1/4 (13.50) | 45-1/3 (13.80) | ~Area~ sq. feet (m squared.)| 387 (36) | 376 (35) | 398 (37) | {total lbs. (kgs.)| 1367 (620) | 1212 (550) | 1102 (500) | ~Weight~ { | | | | {useful lbs. (kgs.)| 772 (350) | 661 (300) | 794 (360) | ~Motor~ h.p.| 70 Renault | 100 Gnome | 80 Gnome | {max. m.p.h. (km.)| 62 (100) | 62 (100) | 65 (105) | ~Speed~ { | | | | {min. m.p.h. (km.)| ... | ... | ... | Number built during 1912 | 47 | 8 | ... | ----------+----------+----------+----------+

[Illustration: Canard with floats. _By favour of "Aeronautics," U.S.A._]

Z

ZODIAC. Societe Zodiac, 10 route du Havre, Puteaux _pres_ Paris (Seine). Aero park: St. Cyr l'Ecole _pres_ Versailles. Established 1896. Capital 850,000 francs.

----------+----------+ Model and date. | ~S2.~ | | ~1913.~ | ----------+----------+ ~Length~ feet (m.)| 38-3/4 (11.75) | {feet (m.)| 49 (15) | ~Span~ { | | {feet (m.)| 36 (11) | ~Area~ sq. feet (m squared.)| 350 (32) | {machine lbs. (kgs.)| 1010 (460) | ~Weight~ { | | {useful lbs. (kgs.)| 551 (250) | ~Motor~ h.p.| 50 Gnome | ~Speed~ max. m.p.h. (km.)| 59 (95) | Number built during

1912 | ... | ----------+----------+

Notes.--Wood construction. _Control_: Ailerons and 1 rear elevator. Upper planes staggered 30 in advance of lower. Quadrilateral fuselage. Piloted passenger side by side. Landing carriage: 2 wheels and 1 skid. Aeroplatte fabric.

The 1912 model was practically the same.

[Illustration: ZODIAC. UAS]

[Illustration]

FRENCH DIRIGIBLES.

~Military.~

---------+----------+---------+-------+---------+------+---------+---------- | | | Capacity | | Speed. | Date. | Name. | Make. | Type. | in m cubed. | H.P. | m.p.h. (K) | Notes.
---------+----------+---------+-------+---------+------+---------+----------
1909 | ~LIBERTE~ | Lebaudy | s.r. | 4800 | 120 | 28 (45) | | | | | | | 1910 | ~COL. RENARD~ | Astra | n.r. | 4100 | 100 | 30 (50) | | | | | | | 1911 | ~ADJUTANT REAU~ | Astra 10 | n.r. | 8950 | 220 | 32 (53) | " | ~LIEUT. CHAURE~ | Astra 11 | n.r. | 8950 | 220 | 32 (53) | " | ~ADJ. VINCENNOT~ | C. Bayard 4 | n.r. | 7500 | 75 | 29 (48) | " | ~SELLE DE BEAUCHAMP~ | Lebaudy | s.r. | 8000 | 75 | 30 (50) | " | ~CAPT. MARECHAL~ | Lebaudy | s.r. | 7500 | 160 | | " | ~LE TEMPS~ | Zodiac 9 | n.r. | 2500 | 75 | 29 (48) | " | ~CAPT. FERBER~ | Zodiac 10 | n.r. | 6000 | 180 | 33 (54) | " | ~COMDT. COUTELLE~ | Zodiac 11 | n.r. | 9000 | 380 | 37 (60) | | | | | | | 1912 | ~SPIESS~ | Zodiac 12 | r. | 11000 | 400 | 40 (65) | " | ~FLEURUS~ | C. Bayard 5 | n.r. | 6500 | 150 | 36 (58) | " | ~ECLAIREUR CONTE~ | Astra 12 | n.r. | 6640 | 75 | 28 (46) | " | ~DUPUY DE LOME~ | C. Bayard 6 | n.r. | 9700 | 244 | 35-1/2 (58) | | | | | | | | _Building_ | _A_ | Astra |} | | | | | _B_ | C. Bayard 7 |} | | | | | _C_ | Lebaudy |} | 17000 | 1000 | 43-1/2 (70) | | _D_ | Zodiac 13 |} ? | | | | | | |} | | | | _Pro._ | _7 new_ 20,000 c.m. | |} | | | | |
---------+----------+---------+-------+---------+------+---------+----------

~Military sheds~ at Belfert, Epinal, Maubenge, Reims, Toul, Verdun (2). --Total 7.

During the year 1912 the principal work done was as follows:--

----------+----------+----------+---------- | Hours out. | Distance travelled. | Gas used. Name. | | m. (km.) | m cubed ----------+----------+----------+---------- _C. Ferber_ | 152 | 3540 (5900) | 45,500 _Adj. Reau_ | 105-1/2 | 2310 (3845) | 81,000 _Dupuy de Lome_ | 100 | 2655 (4424) | 66,500 _Adj. Vincennot_ | 55 | 1340 (2235) | 50,000 _Le Temps_ | 23 | 440 (700) | 9,000 _Fleurus_ | 3-3/4 | 100 (159) | 19,000
----------+----------+----------+----------

~Army Dirigible Pilots.~

Airault, F. Balny D'Avricourt Baudry, A. Bayard de Mendoca Clerget, P. Cohen, A. Herbster, M. Hirschaner, Col. Juchmes, G. Mugnier, Capt. Noe, Martial Perisse, Y. Renard, Col. P. Roussel, A. Schelcher, A.

Note.--There are no dirigibles attached to the Navy.

~Private.~

Capacity		Speed.	Date.	Name.	Make.	Type.	in m cubed.	H.P.	m.p.h. (k.p.h.)	Remarks.
1909	~ASTRA~	Astra 7	n.r.	4475	100	27 (43)				
1909	~ZODIAC III~	Zodiac 3	n.r.	1400	40	28 (45)				
1911	~ASTRA TORRES~	Astra	n.r.	1930	55	34 (56)				
1912	~TRANSAERIENNE II~	Astra 13	n.r.	9000	350	34	(56)			

~Private sheds~ at Chalons-s-Marre, Issy (2), Lamotte-Breuil, Meaux, Melun, Mousson, Reims, Pau, St. Cyr (2).--Total 11.

32 sheds are building or projected by the National Aviation Committee.

~Private Dirigible Pilots.~

Capazza, Louis Godart, Louis Julliott, Henri Kapferer, Henri La Vaulx (de) Compte Santo-Dumont, Albert Surcouf, Edward

=ASTRA CLASS.=

Astra Societe de Constructions Aeronautique, 13, Rue Couchot, and 121, Rue de Bellevue, Billancourt.

This Society was founded by Surcouf for the production of ordinary balloons. The first dirigible work was building part of the old _Lebaudy_ in 1903, followed in 1906 by the _Ville de Paris_. The total number of dirigibles of this type completed by the end of 1912 stood at 14, one very large dirigible in hand for the French Army, and one small one for the British Navy, and another for the Russian Army.

Owing to changes in names, or owing to two names getting supplied to one ship, confusion frequently exists as to the names of the Astra dirigibles. The correct list is as follows:--

1. Part of the LEBAUDY 1903 2. VILLE DE PARIS 1906 3. VILLE DE BORDEAUX 1908 4. VILLE DE NANCY 1909 5. Russian Military dirigible, KOMMISSIONNY,} originally known as CLEMENT-BAYARD I} 1909 6. COLONEL RENARD 1909 7. ASTRA-TRANSAERIENNE-VILLE DE PAU-VILLE} DE LUCERNE[C] } 1909 8. ESPANA (Spanish Military) 1909 9. VILLE DE BRUXELLES 1910 10. LIEUT. CHAURE (French Military) 1911 11. ADJUTANT REAU (French Military) 1911 12. ECLAIREUR CONTE (French Military) 1912 13. TRANSAERIENNE II 1912 14. ASTRA-TORRES I 1911

The general features of the _Astra_ class are: Non-rigid, weights distributed by means of a long girder hung under the gas bags, a long nacelle, and inflated stabilising shapes at the rear end of the balloon.

The _Astra-Torres_ type are also non-rigid, but of trefoil section with a short nacelle.

The Compagnie Generale Transaerienne was first established in 1909 with _Transaerienne I_, and during the summers 1909, 1910 and 1911, this ship made a total of 273 ascents, carried 2590 passengers, and voyaged 7990 kilometres.

The Astra firm has dirigible hangers at Issy, Pau, Meaux, and Reims. Its constructional capacity is sufficient to build six dirigibles at any one time.

"ASTRA I-TRANSAERIEN-VILLE DE PAU-VILLE DE LUCERNE" (1909).

[Illustration]

~Maximum length,~ 197 feet (60 m.) ~maximum diameter,~ 40 feet (12.20 m.) ~volume,~ 158,000 c. feet (4,475 m cubed.)

~Total lift.~--Just over 7 tons=15,763 lbs. (7,150 kgs.) ~Useful lift,~ lbs. (kgs.)

~Gas bags.~--Continental rubbered fabric, yellow.

~Motor.~--One 90-100 C. Bayard.

~Speed.~--27 m.p.h. (43 k.p.h.)

~Propellers.~--One.

[Illustration: SIDE ELEVATION]

COLONEL RENARD. Military (1909).

[Illustration]

~Maximum length,~ 213 feet (65 m.) ~maximum diameter,~ 35 feet (10.50 m.) ~volume,~ 145,000 c. feet (4,200 m cubed.)

~Total lift.~--9,921 lbs. (4,500 kgs.)=about 4-1/2 tons.

~Gas bags.~--Yellow coloured rubber proofed Continental fabric.

~Motor.~--One 110 h.p. 4-cylinder Panhard.

~Speed.~--29 m.p.h.

~Propellers.~--1, at the front end of the car. "Integrale."

~Steering.~--Elevators.

Remarks.--The two side stabilising shapes are duplicated, as they were in the _Ville de Paris_. A webbing stretched on steel tubes is introduced between the inner edges of the 4 main stabilising shapes to provide extra stabilising surface.

[Illustration: COLONEL RENARD. UDS

Note.--An elevator aft has since been added.]

Improved _Col. Renard's_ are:--

LIEUT. CHAURE. Military (1911).

ADJUTANT REAU. Military (1911)

TRANSAERIEN II (1911).

Particulars of these are as follows:--

Lieut. Chaure.	_Adjutant Reau._	_Transaerien II._
~Length~	275-1/2 feet (83.8 m.)	285 feet (86.78 m.)	250 feet (76.25 m.)
~Diameter~	46 feet (14 m.)	46 feet (14 m.)	46 feet (14 m.)
~Volume~	312,550 c. ft. (8,850 m cubed.)	314,000 c. ft. (8950 m cubed.)	318,000 c. ft. (9,000 m cubed.)
~Motors~	2 Panhard, each 110 h.p.	2 Brasier, each 110 h.p.	2 of 175 h.p. each
~Speed~ (p.h.)	32 m. (53 km.)	32 m. (53 km.)	34 m. (56 km.)

Notes.--All have 1 propeller forward of 6 m. diameter, and 2 aft of 3.70 m. The _Lieut. Chaure's_ empeunage is by ballonets; in the other two a cellular system and automatic stabilisation are the special feature.

| Appearance practically the same as for _Colonel Renard_. |

ECLAIREUR CONTE. Military. (1912)

Nominal volume, 6,500 m cubed.

[Illustration]

~Length,~ 213 feet (65 m.) ~diameter,~ 46 feet (14 m.) ~volume,~ 234,500 c. feet (6,640 m cubed.)

~Ballonets.~--Volume, 71,770 c. feet (2,032 m cubed) empeunage: cellular.

~Nacelle.~--Length, 115 feet (35 m.) Breadth, 5-1/2 feet (1.60 m.) Height _about_ 6 feet (2-1.50 m.)

~Motor.~--2 Chenu, 80 h.p. Hele-Shaw clutch.

~Speed.~--_About_ 28 m.p.h. (43-45 km.p.h.)

~Propellers.~--2 central aft, each of 4 m. (13 feet) diameter. 650 r.p.m.

~Empeunage.~--Cellular, Stabilisation automatic.

Notes.--In this type the usual Astra style, rear of gas bag, is entirely done away with. Surface of each elevator is 18m squared, of the rudder 33m squared. There are 2 petrol reservoirs, each of 180 litre capacity.

~Weights.~

lbs. (kgs.) Crew 838 (380) Details 1367 (620) Tools, etc. 220 (100) "Lest d'altitude" 2205 (1000) " securite 661 (300) ---- ------ Total 5291 (2400)

ASTRA-TORRES I.

[Illustration]

~Length,~ 157 feet (47.72 m.) ~diameter,~ 33 feet (10 m.) ~volume,~ 68,150 c. feet (1,930 m cubed.)

~Ballonets.~--Volume, 11,300 c. feet (320 m cubed.)

~Nacelle.~--Length, 18 feet (5.50 m.) Breadth, 5 feet (1.50 m.) Height, 6-1/2 feet (2 m.)

~Useful lift.~--1,219 lbs. (553 kgs.)

~Motor.~--1 Chenu, 55 h.p., at 1,380 r.p.m. Clutch, Ruban.

~Speed.~--31 m.p.h. (50 km.) ~Endurance~ _about_ 5 hours.

~Propeller.~--1 in rear of nacelle. Diameter, 14-3/4 feet (4.50 m.)

Notes.--The special feature of this type is that it is constructed in three lobes, two below and one above. This particular ship is merely experimental, and is known as a "Vedette." Three models of it are to be obtained, (1) this 55 h.p. of 1,930 m cubed. volume. (2) a 75 h.p. of 2,000 m cubed. nominal volume. (3) a 110 h.p. of from 3,000-3,500 m cubed. volume. This latter is designed to have two propellers instead of one.

Larger editions of the type are also projected as follows:--

~"Scouts:"~ 4500-6300 m cubed. of 200 h.p. (2 motors.) ~"Transaeriens:"~ 7,000-8,000 m cubed. of 400 h.p. (2 motors.) ~"Dreadnoughts:"~ 12,000 m cubed. or so, of 750 h.p. (4 motors.)

[Illustration: UDS]

=CLEMENT-BAYARD CLASS.=

Usines Clement-Bayard, 33, quai Michelet, Levallois-Perret (Seine).

These dirigibles closely resemble the _Astra_ class in some main particulars; but (excepting _I_) differ from them in the sharp sterns and absence of stabilisers on stern.

The ships of this class are:--

1 CLEMENT-BAYARD I (Kommissionny) Russian Military 2 " II British Military (wrecked) 3 " 4 " IV (_Adjutant Vincennot_) French Military 5 " V (_Fleurus_) " 6 " VI Private 7 " VII French Military (_building_), To be of 17,000 m cubed.

ADJUTANT VINCENNOT. Military. (1911.) (Clement-Bayard IV.)

[Illustration]

~Maximum length,~ 251 feet (76.50 m.) ~maximum diameter,~ 43 feet (13.22 m.) ~volume,~ 7,500 m cubed.

~Total lift.~--Nearly 8 tons (8,000 kgs.) ~Useful lift,~ 2-3/4 tons (2717 kgs.)

~Gasbags.~--Continental rubbered fabric. Weight, 380 grammes per m squared. Strength 1,000 kg. per metre. Leakage under 10 litres per m squared per 24 hours.

~Motors.~--2 Clement motors, 4-cylinder, of 130 h.p., each placed on either side of the motor space.

~Speed.~--35 m.p.h. (56 km.)

~Propellers.~--2 Chauviere. Diameter, 19-3/4 feet (6 m.) Placed one on either side of the motors, well above the level.

~Steering.~--Vertical steering by means of a treble horizontal rudder over the rear end of the car. Horizontal steering by means of 2 vertical rudders placed one on each side of the rear horizontal rudder.

Remarks.--The feature of this _C.B._ type, which distinguishes it from the Astra ships of about the same size, is the arrangement of the propellers and the use of a 2 speed gear in connection with these. Normally each motor drives its own propeller through two sets of gearing connected by a Cardan shaft. On stopping one motor, the stopped motor is unclutched from its propeller shaft, which is then connected up by chain drive to the opposite shaft. The running motor is then put on to a "low gear," so that it can make the revolutions necessary for obtaining full power, while the propellers run slower than before. The ratio of "low gear" to "high" is 2 to 1, so that a single motor will be running under its best conditions when well throttled down.

A sister, _C. Bayard II_ was sold to the British Army, and wrecked or dismantled, 1911.

LIST OF WEIGHTS.

kgs. Gas bag 1,350 Valves (4) 45 Suspension 195 Girder (complete with fittings) Bow portion (6 m. long.) 128 Engine room (2.5 m.) 1,390 Bridge and passenger space (12 m.) 957 After part (18 m.) 182 Raised tail (4.5 m.) 63 2 Propeller brackets 378 2 Propellers 230 Rudders 150 Water 140 Trail ropes 75 ----- Total 5,283 Lift 8,000 ----- Balance, for ballast fuel, oil, crew 2,717

FLEURUS. Military. (C.B. V.) (1912.)

C. BAYARD VI. (Private.) (1913.)

These two are slightly smaller sisters of the _Adjutant Vincennot_.

=LEBAUDY CLASS.=

Ateliers Lebaudy Freres, Moisson, par La Roche-Guyon (Seine-et-Oise).

~DISTINCTIVE CHARACTERISTICS:~

The cars are short and suspended from a long keel which is suspended close up to the gas bag, and is mostly covered in with fireproof canvas.

The rear end of the keel is expanded into fixed vertical and horizontal fins, and carries a vertical and a horizontal rudder.

The rear end of the gas bag is fitted with thin fixed planes (compare with the pear shaped or tubular fins of the "Astra" class).

The cars are provided underneath with an extraordinarily strong conical structure, which takes the shock of striking the ground and distributes it over the whole car.

Aeroplanes are now fitted, one each side of the keel, well forward.

Ships of this class which have been built:--

~LEBAUDY I~ ~French Military Airship.~ Rebuilt 1909 into _Lebaudy II_. } now 1. ~LEBAUDY II~ " " Original _Lebaudy I_ rebuilt. ~Known as~ _Le Jaune_.} discarded 2. ~PATRIE~ Lost in a storm. 3. ~REPUBLIQUE~ ~French Military Airship.~ Wrecked Autumn, 1909. 4. ~LA RUSSIE~ ~Sold to Russian Government.~ Now _Lebed_. 5. ~LIBERTE~ ~French Military Airship.~ 6. ~CAPITAINE MARECHAL~ " " 7. ~"MORNING POST"~ ~British Military.~ (_Lebaudy III._) Wrecked 1911 8. ~LIEUT. SELLE DE BEAUCHAMP.~ ~French Military Airship.~ 9. New ship of 17,000 m cubed building. " "

To Lebaudy designs:--

~ONE~ ~Austrian Military Airship.~

----------+----------+----------+----------+---------- | | ~CAPITAINE~ | ~SELLE DE~ | New ship. Name | ~LIBERTE~ | ~MARECHAL.~ |~BEAUCHAMP.~ | _Building._ Date | ~1909.~ | ~1911.~ | ~1911.~ | ~1913-14.~ Service | Military. | Military. | Military. | Military. ----------+----------+----------+----------+---------- ~Volume~ c. feet (m cubed)| 4800 | 7500 | 8000 | 17,000 ~Length~ feet (m.)| 220 (67) | 279 (85) | 292 (89) | ~Diameter~ feet (m.)| 35-1/2 (10.80) | 42 (12.80) | 48 (14.00) | {fabric | Lebaudy | Lebaudy | Lebaudy | ~Gasbags~ { | | | | {ballonets | 1 | ... | ... | {total tons| 4-1/2 | ... | 9 | ~Lift~ { | | | | {useful tons| ... | ... | ... | ~Motors~ h.p.| 1--135 Panhard | 2--80 Panhard | 2--80 Panhard | {number | 2 wood | 2 wood | 2 wood | ~Propellers~ {blades | 2 | 2 | 2 | {diam. feet (m.)| ... | 16-1/2 (5) | 16-1/2 (5) | ~Speed~ max. m.p.h. (km.)| 31 (50) | 28 (45) | 28 (45) | ~Endurance~ hrs.| ... | ... | ... | ~Complement~ | ... | ... | 5 | ----------+----------+----------+----------+----------

[Illustration]

[Illustration: LIBERTE.]

[Illustration]

=ZODIAC CLASS.=

Societe francaise de ballons dirigeables et d'aviation, Zodiac, 10 route du Havre, Puteaux (Seine).

These dirigibles were intended primarily for private pleasure purposes. Consequently they are designed to fly when filled with coal gas if necessary.

Every effort is made to render them easily transportable; the long girder frame by which the weight is distributed is made to take to pieces.

It is held in France that numbers of this class of vessel would form an invaluable asset in time of war, as each could be transported in a single cart, filled with a very few bottles of hydrogen, and when so filled could man[oe]uvre for some 6 hours at a speed which compares favourably with that of the standard types.

When the service of reconnaissance was performed, the vessel could be packed up and sent out of harm's way in an hour, whereas this could scarcely be done with a larger vessel on account of the quantity of hydrogen that would be required if it had to be filled afresh for each service. The mooring of an airship in the open during war requires such an amount of preparation and attention as to be a serious drawback to the alternative plan of keeping such vessels unfilled, while the sending of an airship back to its distant shed on each occasion means doubling the work that the ship is called upon to perform.

Ships of class are:--

1. ZODIAC I (_Petit Journal_) 2. " II (_De la Vaulx_) 3 " III 4 " IV Dutch Military 5. " V South American (private) 6. " VI Sold to United States 7 " VII Sold to Russian Army 8. " VIII " " 9. " IX (_Le Temps_) French Army 10. " X (_Capitaine Ferber_) " 11. " XI (_Commandant Coutelle_) " 12. " XII (_Spiess_) " (rigid)

ZODIAC III.

[Illustration]

~Maximum length,~ 134 feet (40.8 m.) ~maximum diameter,~ 28 feet (8.5 m.) ~volume,~ 1,400 m cubed.

~Total lift.~--1-1/2 tons (1,540 kgs.) ~Useful lift,~ lbs. (kgs.)

~Gas bags.~--Light continental rubbered fabric.

~Motor.~--Ballot, 4-cylinder, 40-45 h.p., 1,200 r.p.m.

~Speed.~-- m.p.h. (45 km.p.h.)

~Propellers.~--Driven at 600 r.p.m. Integral type, 12-1/4 feet (3.75 m.) in diameter. Pitch, 6-1/2 feet (2 m.) in rear of the car.

~Steering.~--Vertical balanced rudder in rear of the vertical fin, under the rear of the gas bag. Double elevator above the fore end of the car. Horizontal fins of material spread on iron frames on either side of the rear end of the car.

Remarks.--The car consists of a 130 feet (40 m.) long wooden girder, which can be divided into 4 separate parts of 13 feet (4 m.) each. The suspension is by steel wires fitted with adjusting screws at the lower ends and toggles at the upper ends, by which they connect to the crows' feet which are sewn to the suspension strips.

[Illustration: ZODIAC III.]

DETAILED WEIGHTS OF _ZODIAC III._

kgs. lbs. Gas bag (_including_ ballonet) 330 727-1/2 Valves 12 26-1/2 Suspension wires and gear 15 33 Tail fins 24 53 Horizontal rudder 10 22 Vertical rudder 10 22 Girder car 168 370-1/4 Motor (_including_ pump, magneto, lubricating gear, etc.) 275 606-1/4 Motor bearer and gear 22 48-1/2 Petrol tank 10 22 Radiator 25 55 Reduction gearing 12 26-1/2 Shafting 15 33 Fan 9 20 Steering gear 5 11 Water 8 17-3/4 Petrol 20 44 Miscellaneous: 4 men 300 661 --- --- Total 1,270 2,800 _about_ Ballast 270 595 ----- ----- Total weight 1,540 Total lift 3,395

LE TEMPS. Military. (_Alias ~ZODIAC IX.)

[Illustration]

~Maximum length~, 164 feet (50.25 m.) ~maximum diameter~, 29-1/2 feet (9 m.) ~volume~, 81,250 cubic feet (2,300 m cubed.)

~Total lift.~--

~Gas bag, etc.~--2 ballonets, each of 257 m cubed.

~Motor~.--Dansette-Gillet. 60 h.p.

~Propellers~.--2, chain driven, one on either side of car.

~Speed.~--

~Steering.~--Elevator in _nacelle_ amidships. Rudder aft.

Remarks.--

[Illustration: LE TEMPS. UDS.]

CAPITAINE FERBER. Military. (_Alias ~ZODIAC X.)

~Maximum length,~ 249-1/3 feet (76 m.) ~maximum diameter,~ 40-1/2 feet (12.36 m.) ~volume,~ 6,000 m cubed.

This ship has 2 ballonets of 650 m cubed. each, and a car 35x13x2 m., made up of 5 sections. 2 ~motors~. Dansette-Gillette, 90 h.p., each actuating 2 propellers (4-bladed), geared to 500 r.p.m. Carries petrol for 15 hours work. Completed 1911.

[Illustration: _Photo, Branger._]

[Illustration: CAPITAINE FERBER. UDS.]

COMMANDANT COUTELLE. Military. (ZODIAC XI.)

+------------------------------+ | | | (Enlarged _Captaine Ferber_.) | | _Building._ | | | +------------------------------+

~Maximum length,~ 292 feet (89 m.) ~maximum diameter,~ 46 feet (14 m.) ~volume,~ 9,000 m cubed.

~Gas bags, etc.~--2 ballonnets, each 45,900 c. feet (1,300 m cubed.)

~Nacelle.~--Nickel steel in 5 sections. Length, 131-1/4 (40 m.) Width, 4-1/2 feet (1.30 m.) Hung 16-1/2 feet (5 m.) below the balloon. Pilot in centre. Carries a total crew of six, petrol and oil for 15 hours' continuous work at full power.

~Motors.~--2, each of 190 h.p.=total of 380 h.p. Placed one at either end of the nacelle.

~Propellers.~--4, of 15 feet (4.50 m.) diameter. Two geared to each motor to half engine speed.

~Speed~ (expected).--37 m.p.h. (60 k.p.h.)

ZODIAC XII. Rigid. Military. (SPIESS.)

[Illustration: Spiess. _Building._ _Photo, Branger._]

~Maximum length,~ 341 feet (104 m.) ~maximum diameter,~ 42-3/4 feet (13 m.) ~volume,~ 11,000 m cubed.

~Gas bags.~--Sections 11. Number of cylindrical sections 8. Number of sides to polygon 14.

~Motors.~--Two 6-cylinder 200 h.p. in each nacelle, each driving 2 propellers of 15 feet (4.50 m.) diameter.

~Speed.~ (expected)--40 m.p.h. (65 k.p.h.)

[Illustration]

GERMAN.

(By our special German editor.)

~Aerial Journals:--~

Deutsche Luftfahrer Zeitschrift fur Luftschffahr Berlin, W. (Fortnightly).

Allgemeine Automobil Zeitung, Berlin (Weekly).

Automobil Welt, Berlin (Thrice Weekly).

Das Deutsche Auto, Munich (Weekly).

Die Luftflotte, Berlin (Monthly).

Internationale Revue fuer Autowesen & Aviatik, Leipzig (Fortnightly).

Flugsport, Frankfurt (Fortnightly).

Motor, Berlin (Monthly).

Der Motorwagen, Berlin (Thrice Monthly).

Monatshefte der Reichsfliegerstiftung, Charlottenburg (Monthly).

Zeitschrift fuer Flugtechnik & Motorluftschiffahrt, Berlin (Fortnightly).

~Private Flying Grounds~ (Military see further on):--

~Adlershof,~ Teil des Flugfeldes Johannisthal (_Wright_ School).

~Bork,~ Post Brueck in der Mark (_Mars_ School).

~Burg bei Magdeburg~ (_Schulze_ School).

~Darmstadt~ (Truppenuebungsplatz).

~Dotzheim bei Wiesbaden.~

~Frankfurt a.M.~ (_August Euler_).

~Fuehlungen bei Koeln~ (Koelner Club fuer Flugsport).

~Garching b. Muenchen~ (_Hoffman-Harlan_).

~Griesheim b. Frankfurt a.M.~ (Frankfurter Flugsport-Club & Flugtechn. Verein).

~Habsheim b. Muelhausen i.E.~ (_Aviatik_).

~Hainberg b. Nuernberg~ (Flugtechn. Ges. Nuernberg-Fuerth).

~Hamburg~ (_Grade_).

~Holten. Niederrh. Verein f.L.~ (_Hilsmann_).

~Johannisthal b. Berlin~ Flugschule _Albatros, Dorner, Harlan, Fokker, Luftverkehrs-ges. m.b.H., _Rumpler & Wright_.

~Kitzingen in Bayern,~ 1911 (Hildebrand & Schroth).

~Lindenthal b. Leipzig.~ (School for Deutschen Flugzengwerke).

~Loddenheide b. Muenster.~

~Meerheimb b. Koeln.~

~Milbertshofen b. Muenchen~ (Dr. Wittenstein).

~Neuenlande b. Bremen~ (_Mueller-Aviatik_, Bremer v.t.L.)

~Niederwalluf,~ 1911 (_Goedecker_).

~Oberwiesenfeld bei Muenchen~ (_Gustav Otto_).

~Puchheim b. Muenchen.~

~Reichenberg-Boxdorf b. Dresden.~

~Schneverdingen~ (_Oertz_).

~Schulzendorf b. Berlin~ (_A.E.G._)

~Strassburg i.E.,~ "Polygon" (E.E.C. _Mathis_).

~Suechteln.~

~Teltow bei Berlin.~

~Velten bei Berlin~ (_A.E.G._)

~Wandsbek,~ Exerzierplatz, 1911 (_Rumpler_ und Jordan).

~Weimar~ (_Wright_).

~Wustenbrand b. Chemnitz~ (Flugtechn. Ges. in Chemnitz).

~Zahlbach b. Mainz~ (School for Aut. & Flugtechnik).

~Aerial Societies:--~

Aachener V. f. L. Aix la Chapelle.

Aero Club (Imperial), 3, Nollenderfplatz, Berlin. Sec.: H. Von Frankenberg und Ludwigsdorf.

Akademie fuer Aviatik, Munich.

Allgemeiner Deutscher Automobil Club, Munich.

Anhaltischer V. f. L. (E. U.) M. Antoineatten str. 22a, Dessau.

Augsburger Verein fuer Luftschiffahrt, Augsburg.

Automobil-und Flugtechnische-Gesellschaft (E. V.) Nurnberger Platz 5, Haupyverein Berlin, Hochster Str. 1, Bezirksverein Frankfurt a. M, Neuer Wall 44, II, Hamburg.

Bayerischer A. K. Munich.

Berliner V. f. L., Berlin.

Bilterfeldt V. f. L., Bilterfeld.

Braunschweigische V. f. L.

Breisgau V. f. L., Freiburg.

Bremer V. f. L. (E. V.) N. W., Obernstr, 52/54 I, Bremen.

Bromberger V. f. L. (E. V.) O, Gasanstalt, Bromberg, Stadt.

Chemnitzner V. f. L.

Deutsche Touring Club, Munich.

Deutscher Luftflotten Verein, Mannheim.

Dusseldorfer Luftdahrer-Klub (E. V.) W. Dusseldorf, Breite Str. 25, I.

Erfurter V. f. L. (E. V.) M. Dalversweg 24, Erfurt.

Flugverein Neustadt a. d. Haardt, S. W., Neustadt, I.

Flugzeugkonvention des V. D. M. L., Potsdamer Str. 121 H, III, Berlin W.

Frankfuerter Flugsport-Club (E. V.) Neue Mainzer Str. 76, Frankfurt a. M.

Frankfurter Flugtechn, Verein (E. V.) Bahnhofplatz 8, Frankfurt a. M.

Frankfurter V. f. L. (E. V.) S. W. Kettenhofweg 136, Frankfurt, a. M.

Frankischer V. f. L. (E. V.) S. Kurschnerhof 6, Wursburg.

Hamburger V. f. L. (E. V.) N. W., 36, Colonnaden 17-19, Hamburg.

Hannoverscher V. f. L. (E. V.) N. W., Lortzingstr. 6, Hannover.

Hereforder Verein fur Lufthahrt, Bahnhofplatz, Alfermann, Herford.

Hildesheimer V. f. L., Hilkesheim, Lucienvorder str. 22.

Kaiserlicher Aero-Club, K. Nollendorfplatz 3, Berlin W.

Kaiserlicher Automobil-Club, K, 9 Leipzigerplatz 16, Berlin W.

Karlsruher Luftfahrt-Verein (E. V.) S. W., Bachstr, 28, Karlsruhe.

Kolner Club, f. L. (E. V.) W. Bischofsgartenstr. 22, Koln.

Koniglich Bayerischer Automobil-Club, B. Brienner str. 5 I, Munchen.

Koniglicher Sachsischer V. f. L. (E. V.) Sa, Ferdinandstr. I, Dresden.

Kurhessischer V. f. L. (E. V.) S. W., Physikalisches Institut, Marburg ad Lahn; Cassel Sektion, Kolnische str. 84, Cassel.

Leipziger V. f. L. (E. V.) Sa, Markt 1, Leipzig.

Lubecker V. f. L. (E. V.) N. W., Israeldorfer Allee 13a, Lubeck.

Luftshrverein Gotha (fruther Reichsflugverein Gotha), Gotha, I.

Luftfahrtverein Touring-Club, Pranner str. 24, I. Munchen.

Luftschiffahrt-Verein Munster fur Munster und das Munsterland (E. V.) N. W., Munster i. W, Klosterstr. 31-32.

Magdeburger V. f. L. (E. V.) M, Wetterwarte, Bahnhofstr. 17, Magdeburg.

Mannheimer V. f. L. "Zahringen" (E. V.) S. W., 7-8 Hansa-Haus, Mannheim.

Mecklenburgerischer Aero-Club, Kaiser-Wilhelm-Str. 85, II, I, Schwerin i. M.

Mindener Verein fur Luftfahrt, N. W., Grosser Domhof 1, L, Minden i. W.

Mitterheinischer V. f. L., S. W., Weisenauer, Str. 15, Mainz.

Munchener V. f. L. (E. V.) Residentzstr. 27 III, Munchen.

Niederrheinischer V. f. L. (E. V.) Wilhemstr. 11, Bonn, Wupperthal Sektion, Hauptfeuerwache, Barmen; Essen Sektion, Bachstr. 21, Essen-Ruhr; Bonn Sektion, Wilhelmstr. 11, Bonn.

Niedersachachsischer V. f. L. (E. V.) Hildesheimer Bank, Filiale Gottingen, Gottingen.

Niederschlesisch-Markischer Verein fur Luftfahrt, Grunberg i. Schl, I.

Nordmark-Verein fur Motorluftfahrt (E. V.) Dusternbrooker Weg 38, Kiel.

Obererzgebirgischer V. f. L. (E. V.) Sa, Geschaftsstelle, Schwarzenberg i. S., Erla im Erzgebirge.

Oberrheimischer V. f. L. (E. V.) S. W., Blauwolkengasse 21, Strassburg i. Els.

Oberschwabischer V. f. L. (E. V.) S, Promenade 17, Ulm a. D.

Osnabrucker V. f. L. (E. V.) N. W., Wittekindstr. 4, Osnabruck.

Ostdeutscher V. f. L. (E. V.) O, Courbierestr. 34, II. Graudenz.

Ostpreusischer V. f. L. (E. V.) O, Kneiphofische Langgasse 8 I, Konigsberg i. Pr.

Pfalzischer Luftfahrtverein Speyer, S. W., Speyer a. Rh. I.

Pommerscher V. f. L. (E. V.) Pasewalk (Stettin) F. 65.

Posener Luftfahrer-V. (E. V.) O, Posen, Kronprinzenstr, 101a.

Reichsflugverein (E. V.) Motztrs, 76, Berlin.

Rheinisch-Westfalische Motorluftschaff-Gesellsch. (E. V.) Bachstr. 21, Essen-Ruhr.

Saarbrucker Verein fur Luftfahrt, S. W. Saarbrucken, I.

Sachsisch-Thuringischer V. f. L. Belvedere-Allee 5, Weimar; Halle a. S. Sektion (E. V.) Halle a. Muhlweg 10 und Poststr. 6; Thuringische Sektion, Staaten; Belvederealle, 5, Weimar.

Schlesischer Aero-Club (E. V.) O, Schweidnitzerstr. 16-18 Breslau.

Schlesischer V. f. L. (E. V.) O, Schweidnitzerstr. 16-18 Breslau.

Schleswig-Holstein. Flieger-Club, N. W., Niemannsweg 81b, Kiel.

Seeoffizier-Luftclub (S.L.C.W.) N. W., Peterstr. 80 II, Wilhelmshaven.

Trierer Club. f. L. (E. V.) W., Nagelstr. 10, Trier.

V. D. Luftchiff-Industrieller, Kleiststr. 8, III, Berlin, W.

V. D. Motorfahrzeug-Industrieller, Potsdamerstr. 121b, Berlin, W.

V. f. Flugwesen in Mannheim, S. W., Lange Rotterstr. 106, I. Mannheim.

V. f. L. am Bodensee (E. V.) S. W., Zummsteinstr. 11, Schwedenchanze 3a, Konstanz.

V. f. L. Darmstadt, S. W., Darmstadt, I.

V. f. L. Gieben, S. W., Seltersweg 56, I., Gieben.

V. f. L. Kolmar (Posen), (E. V.) O. Privinzialbank, Kommanditgesellschaft a. A., Kolmar i. Pos.

V. f. L. Limbach (Sa, u. Umgegend) (E. V.) Postr. 5, Limbach (Sachsen).

V. f. L. in Mainz (E. V.) S. W., Grosse Bleiche 48, Mainz.

V. f. L. in Worms, S. W., Worms, I.

V. f. L. und Flugtechnik Nurnberg-Furth, Klaragasse 2 I, Nurnberg.

V. f. L. in Weimar (E. V.) Erfurter Str. 9, Weimar.

Vogtlandischer V. f. L. (E. V.) Sa, Plauen i. V, Furstenstr. 89.

Westfalisch-Lippischer Luftfahrverein (E. V.) N. W., Kavalleriestr, Petri, Bielefeld.

Westfalish-Markischer Luftfahrer-Verein, Herne, I.

Westpreussicher V. f. L. (E. V.) O., Dr. Waldmann, Abte-inng Schiffbau, Technische Hochschule, Danzig-Langfuhr.

Wissenschaftliche Gesellschaft fur Flugtechnik, Nollendorfplatz 3, Berlin W. 30.

Wurttenbergischer Flugsport-Club, S. Hegelstr. 4b, Stuttgart.

Wurtembergischer V. f. L. (E. V.) S. am Salzmannsweg 21, Stuttgart.

Zwickauer V. f. L. (E. V.) Sa, Hauptmarkt 20, Zwickau i. S.

~GERMAN MILITARY AVIATION.~

~Army General.~

The new Army law provides L400,000 (80 million marks) for Army aviation (including dirigibles), in addition to a considerable share of the L4,000,000 which is being spread over a period of five years.

The Army aerial force will be commanded by 2 inspector generals. The aviation force is put at 4 batallions.

Headquarters: Berlin. _Stations_: Aachen, Allenstein, Cologne, Darmstadt, Doebritz, Freiburg, Graudenz, Hannover, Insterburg, Jueterbog, Koenigsberg, Metz, Posen, Strassburg, Zeithain.

The scheme will be complete by the end of the year.

~Army Flying Schools.~

~Diedenhofen.~ ~Doeberitz.~ ~Metz.~ ~Oberwiesenfeld~ bei Muenchen (Bavarian). ~Saarburg.~ ~Sperenberg~ bei Jueterbog.

~Army Aeroplanes.~

At the end of 1912 the aeroplane force was as follows:--

Bought in 1911 10 monoplanes (2 Grade, 1 Schultze, 5 Rumpler). 25 biplanes (3 Albatross, 22 Farman type.) Bought in 1912 91 monoplanes (20 Bristol, 1 Dorner, 2 Etrich Taube, 2 Grade, 6 Harlan, 20 Mars, 40 Rumpler Taube). 144 biplanes (50 Albatros, 12 Aviatik, 30 Euler, 10 Otto, 2 L.V.G., 10 Mars, 6 Wrights). ---- Making a total of ~270~ of which number about 200 were war-effectives.

For ~1913~ there are ~200~ new aeroplanes building or provided for.

Under the new regulations, military machines must comply with the following conditions:--

1. Must be of entirely German manufacture, with ample and comfortable seating accommodation for pilot and passenger.

2. Design must permit of fitting bomb droppers and photographic apparatus.

3. Speed capabilities must not be less than 90 kilometres (56 m.p.h.)

4. Dimensions must not exceed 49 feet span (14.50 m.), 39 feet long (12 m.), 13 feet high (3.50 m.), and the motor not more than 100 h.p.

5. Minimum endurance, 4 hours.

~Army Aviators.~

Ackermann, Lt. K. Albrecht, Ob-Lt. K. Altrichter, Lt. K. v. Apell, Lt. K. Barends, Lt. von Beaulieu, Ob-Lt. W. Berlin, Ob-Lt. E. Blume, Lt. W. Boeder, Lt. O. Braun, Lt. Busch, Lt. H. v. Buttlar, Lt. W. Canter, Lt. Cipa, T. Coerper, W. Lt. von Detten, Lt. G. Demmel, Lt. M. Dransfield, Lt. E. Eich, H. von Eickstedt, Ob-Lt. V. Erhardt, Ob-Lt, R. (119) von Falkenhayn, Lt. F. E. Graf Finck von Finckenstein, Lt. L. Fisch, Lt. W. (107) von Freyberg-Eisenberg-Allmendingen, Lt. F. E. Funck, Lt. W. Geerdtz, F. H. (133) von Gersdorff, Ob-Lt. E. Geyer, Lt. H. Goebel, Ob-Lt. W. Grade, W. H. St. (20) von Hadeln, Lt. F. v. Hammacher, Lt. (49) von Hammerstein Gesmold, Ob-Lt. F.A. Hantelmann, Ob-Lt. M. von Helldorf, Ob-Lt. v. Hiddessen, F. Lt. (47) Hildebrand, Ob-Lt. F. Hofer, Ob-Lt. W. Hoepker, Lt. A. von Jagwitz, Lt. F. Joly, Lt. A. Justi, Lt. K. Kahl, H. Kastner, Lt. H. Keim, Lt. J. (127) Keller, Lt. G. Koch, Lt. W Kohr, Lt. R. Lauer, Lt. R. Lauterbach, Ob-Lt. F. von Lichtenfels, Lt. S. (51) von Liusingen, Lt. L. Ludewig, F. Ob-Lt. Meyer, Lt. W. (136) von Minkwitz, Lt. H. von Mirbach, Lt. K. Mudra, Lt. H. (95) Neumann, Ob-Lt. H. von Obernitz, Ob-Lt. W. Oelsner, Lt. W. von Oertzen, Ob-Lt. J. von Osterroht, Lt. P-H. Petri, Ob-Lt. F. (120) Pfeifer, Lt. L. Pirner, Lt. H. K. von Poser und Gross-Naedlitz, Ob-Lt. F. Pueschel, Ob-Lt. K. Rapmund, Lt. M. Reiche, Lt. A. v. Reichenberg-Wolfskeel, Graf. (93) Reinhardt, Lt. S. Reuss, Lt. W. Ritter, Lt. K. (121) Roser, H. H. (83) Schaefer, Ob-Lt. L. v. Scheele, Lt. A. Schlegel, Lt. O. Schneider, Lt. H. Schreyer, F. Schulz, Lt. J. Schwartzkopff, Lt. H. Serno, Lt. E. Sieber, Lt. H. Solmitz, F. Lt. Sommer, Lt. P. Steindorf, H. Steger, O. Lt. von Stoephasius, Lt. M. Striper, Lt. F. Suren, Lt. E. Suren, Lt. G. Suren, Lt. H. Taeufert, Lt. W. v. Tiedemann, Ob-Lt. R. (17) von Trotha, Ob-Lt. Vogt, Lt. v. Wedemeyer, Ob-Lt. E. Wendler, Lt. W. Weyer, Lt. G. Wiegandt, Lt. W. Wilberg, Ob-Lt. H. (26) Wildt, Lt. K. (43) Wirth, Ob-Lt. W. (92) Wulff, Lt. A. Zwickau, Lt. K.

~Navy General.~

The 1913 expenditure on naval aviation (including dirigibles) is L250,000 (50 million marks), plus a portion of the special expenditure.

~Naval Flying Schools.~

~Holminsel~ b. Danzig. ~Putzig~ b. Danzig.

~Naval Stations.~

~North Sea.~--Cuxhaven (staff to be 5 officers and 192 under officers and men), Emden and Hamburg.

~Baltic.~--Kiel, Putzig and Konigsberg.

~General Headquarters.~--Berlin.

Each station will ultimately consist of one dirigible and a number of hydro-aeroplanes.

~Navy Aeroplanes.~

At the end of 1912 the total effective force was:--

~4 monoplanes~ (_Rumpler_ hydro.) ~10 biplanes~ (4 _Albatros_ hydro., 2 _Curtiss_ hydro., 4 _Euler_). -- Total ~14~

The _Curtiss_ were purchased towards the end of 1911, all the others in 1912.

All are or can be fitted with wireless, range 50 miles.

1913. Others on order, including _Ottos_ on floats (_A.G.O._), of which one was delivered in April.

~Navy Aviators.~

Bertram, Ob-Lt. (123) Coulmann, W. Francke, Ob-Lt. C. (142) Goltz, Kap-Lt. K. von Gorrissen, Lt. (4) Hartmann, R. Ob-Lt. (96) Hering, Kap-Lt. M. Janetzky, Kap-Lt. W. Langfield, Ob-Lt. W. Prinz Heinrich von Preussen (38) Schroeter, Ob-Lt. W. Stemmler, B.

~GERMAN PRIVATE AVIATION.~

To end of March, 1913.

~Private Aeroplanes.~

At end of March, 1913, the number of private aeroplanes in Germany was about 80, of which most were school, etc., machines.

Abelmann, Carl Abramowitch, Wasewolod Albers, Wilhelm Alig, Ernst. Arntzen, Orla, Dr. jur. Schirrmeister, Hans Badowski, Ludwig Baierlein, Anton Basser, Gustav Beck, Otto Becker, Reinhold Beese, Frl. Behrend, Adolf Berliner, Rudolf v. Bieber, Harald, Dr. jur. Birkmaier, August Blattmann, Ernst Bohlig, Edmund Bosenius, Rudolf Bossin, Fritz Boutard, Charles Braselmann, Karl Breton, Raymond Arthur Brociner, Marco Brunnhuber, Simon Buechner, Bruno Charlett, Willi Clauberg, Fritz Cremer, Fritz Curdts, Carl De Waal, Bernard Dick, Fritz Donnevert, Willy Dorner, Hermann Duecker, Werner Eberhardt, Alfred Eckardt, Willy Eckelmann, Frank Engelhard, Paul Erblich, Heinz Euler, August Evers, Heinrich Eyring, Raymund (Dr. Huth) Falderbaum, Heinz Faller, Artur Faller, Otto Flegier, Th. v. Fokker, Anthony Fremery, Hemmann Friedrich, Alfred Gasser, Hermann Geiss, Franz Georgi, Johannes v. Gorrissen, Ellery Grade, Hans Griebel, Otto, Leutnant a D. Grulich, Karl Gruenberg, Arthur Haas, Heinrich Hansen, Hans Hanuschke, Bruno Hartmann, Alfred Hasenkamp, Emil Haeusler, Hugo Heim, Oskar Heirler, Paul Hennig, Alfred Hess, Robert Heydenreich, Fritz Hild, Luc. Hintner, Cornelius Hirrlinger, Albert Hirth, Helmuth Hoff, Wilhelm Hoffmann, Siegfr. Hoos, Josef, Dr. jur. Hormel, Walter Horn, Albin Hoesli, Gordian Ingold, Karl Jablonski, Bruno Jahnow, Reinhold, Leutn. d. Landw. Jaenisch, M. Jeannin, Emil Kahnt, Oswald Kammerer, K. F. Ludwig Kaniss, Gustav Kanitz, Willy Karsten, Otto Kaspar, Referendar Katzian, Artemy Keidel, Fridolin Kern, Willy Kiepert, Rudolf Kleinle, Josef Kober, Theodor Koehler, Erich Kohnert, Herbert Koenig, Benno Koenig, Martin Krastel, Heinz, Reimar Krieg, Friedrich Krieger, Karl Krueger, Leutnant a. D. Krueger, Arthur Krumsiek, Wilhelm Kuehne, Ernst Herbert Kunze, Ernst Kurtscheid, Nicolaus Ladewig, Heinz. Ladewig, Herbert, Leutnant, Inf.-Regt. Lagler, Fraeulein Bozena Laitsch, Felix Laemmlin,

Charles Lange, Paul Langer, Bruno Lecomte, Ingenieur Lenk, Willy Lichte, Carl Lie, Christian Lindpaintner, Otto E. Linnekogel, Otto Lissauer, Walter Lochner, Erich Loew, Karl Luebbe, Fluglehrer Manhardt, Alfred Willy Mente, Willy, Oberleutnant a. D. Meybaum, Theodor Michaelis, G. A. Mischewsky, Bernard Mohns, Karl Moehring, Charlotte v. Mossner, Robert, Oberleutnant a. D. Muegge, Wilhelm, Kapitan d. Handelsmarine Mueller, B. C. Oscar Mueller, Friedrich Mueller, Karl Mueller, Kurt Munkelt, Kurt Muerau, Georg Netzow, Georg Niemela, Edmund, Leutnant a. D. Noelle, Max Oelerich, Heinrick Oster, Franz Ottenbacher, Ernst Otto, Gustav Paul, Alfred Pentz, Hermann Pietschker, Alfred Placzikowski, Udo von Platen, Horst Plochmann, Ernst Pokristev, Penn, Oberleutnant Poulain, Gabriel Reeb, Alfred Reichhardt, Otto Rentzel, Adolf Rode, Franz de le Roi, Wolfram Roempler, Oskar Rosenstein, Willy Roessler, Fritz Rost, Gottlieb v. Rottenburg, Otto Roever, Hans Rupp, Albert Ruetgers, August Schadt, Karl Schaefer, Otto Schakowskoy, Fuerstin Eugenie Schall, Karl Schauenburg, Theodor Schendel, Georg Scherff, Mauricio Schiedeck, Hermann v. Schimpf, Ernst, Dr. jur Schirrmeister, Hans Schlatter, Joseph Schlegel, Ernst Schlueter, Fritz Schmidt, Erich Schmidt, Richard Schmigulski, Hans Schoener, Georg Schultze, Gustav Schuepphaus, Heinrich Ernst Schwandt, Paul Schwarz, Erwin Sedlmayer, Gerhard Senge, Paul Seydler, Frank Siewert, Lotherm Steffen, Bruno Steinbeck, Hans Stiefvater, Otto Stoeffler, Victor Stoldt Stoephasius, Curt von Strack, Karl Strack, Peter Stueber, Joachim, Leutnant d. R. Suvelack, Josef Thelen, Robert Thiele, Erich Toepfer, Otto Trautwein, Max Treitschke, Friedrich Tybelski, Franz Tweer, Gustav Vollmoeller, Hans Wecsler, Rubin Weickert, Julius Artur Weinaug, Ernst Werntgen, Bruno Wertheim, Paul Weyl, Richard Wiencziers, Eugen Wieting, Werner Wirtz, Reinerm Witte, Gustav Wittenstein, Oskar, Dr. Witterstaetter, E. W. Wolter, Richard von Zastrow, Alexanderm

Note.--Abramowitch was a Russian by birth (killed April, 1913).

The following German aviators have been killed:--

+------------------------------+ | 1896. | | Lilienthal | | | | 1910. | | Haas, Lieut. | | Mente, Lieut. | | Plochmann | | Robl, Thaddeus | | | | 1911. | | Bockmueller | | Bournique ("Pierre Marie") | | Choendel | | Dax | | Englehardt, Kapt. | | Eyring, R. | | Frh. v. Freytag-Loringhoven | | Laemmlin | | Lecomte | | Neumann, Lieut. | | Reeb | | Pietschker | | Schendel, G. | | Stein, Lieut. | | Tachs | | Voss | | | | 1912. | | Alig | | Altrichter | | Beissbarth | | Berger | | Birkymayer | | Buchstaetter | | v. Falkenhayn | | Frh. v. Schlichting | | Fischer | | Hamburger | | Hofer | | Hoesli | | Junghans | | Koenig | | Kugler | | Lachmann | | Lang | | Libau | | Meyer | | Preusser | | Pochmeyer | | Rheinle | | Rost | | Schmidt | | Schmigulski | | Stille | | Werntgen | | Witte | | | | 1913 | | Held | | Schlegel | +------------------------------+

GERMAN AEROPLANES.

~A~

ALBATROS. Albatroswerke G.m.b. H, Flugzeugfabr. u. Fliegerschule, Johannisthal bei Berlin. Established 1910. One of the largest constructors in Germany. Capacity: 150 machines a year.

[Illustration]

---------- | ~1911-12.~ | ~1912.~ | ~1912-13.~ | | | 2-seat tractor | Military tractor | Military tractor | Hydro. | Mono. | biplane. | biplane. | | | ---------- ~Length~ feet (m.)| 35-1/2 (10.70) | 34-1/2 (10.5) | 42-1/2 (12.8) | ... | ... ~Span~ feet (m.)| 43-2/3 (13.30) | 52-1/2 (16) | 65-3/4 (20) | ... | ... ~Area~ sq. feet (m squared.)| 430 (40) | 576 (54) | 624 (58.5) | ... | ... {total lbs. (kgs.)| 1058 (480) | 1543 (700) | 1874 (850) | ... | ... ~Weight~ { | | | | {useful lbs. (kgs.)| 661 (300) | ... | ... | ... | ... ~Motor~ h.p.| 100 Argus | 90 Mercedes or | 120 N.A.G. or | ... | ... | | 100 Argus | Aust. Daimler | | {max. m.p.h. (km.)| 56 (90) | 59 (95) | 46 (75) | ... | ... ~Speed~ { | | | | | {min. m.p.h. (km.)| ... | ... | ... | ... | ... ~Endurance~ hrs.| 6 | 6 | 7-5 | ... | ... Number built during 1912 | about 40 | 70 | 30 | 4 | 2 ---------- ---------- ---------- ---------- ---------- ----------

Remarks.--In all the upper plane is slightly staggered. In all the control is duplicated.

[Illustration: ALBATROS. UAS.]

[Illustration: Albatros. Military hydro-biplane.]

[Illustration: Albatros. Monoplane.]

AVIATIK. Autemobil & Aviatik A.G., Muelhausen i.E. Established 1910. Capacity: 100 a year.

[Illustration: 1912 biplane.]

---------- ---------- ---------- ---------- ---------- | ~1912.~ | ~1912.~ | ~1913.~ | ~1912-13.~ | Monoplane. | Biplane. | Racing biplane. | Hydro-biplane. ---------- ---------- ---------- ---------- ---------- ~Length~ feet (m.)| 26-1/2 (8) | 36 (11) | 29-1/2 (9) | 36 (11) ~Span~ feet (m.)| 39 (11.80) | 52-1/2 (16) | 52-1/2 (16) | 62-1/3 (19) ~Area~ sq. feet (m squared.)| 258 (.24) | 517 (48) | 517 (48) | 597 (56) {total lbs. (kgs.)| 1146 (520) | 1323 (600) | 1234 (560) |1653 (750) ~Weight~ { | | | {useful lbs. (kgs.)| 661 (300) | 882 (400) | 882 (400) | 661 (300) ~Motor~ h.p.| 100 Argus | 100 Argus | 100 Argus | 100 Argus {max. m.p.h. (km.)| 68-1/2 (110) | 56 (90) | 62 (100) | 52 (80) ~Speed~ { | | | {min. m.p.h. (km.)| ... | ... | ... | ... ~Endurance~ hrs.| 5 | 6-8 | 7-8 | 4-5 Number built during 1912 | 6 | 20 | 4 | 3 ---------- ---------- ---------- ---------- ----------

Remarks.--The monoplanes are constructed under _Hanriot_ license.

[Illustration: 1913 tractor-biplane (racer).]

[Illustration: Monoplane.]

~D~

DELFOSSE. Ceased to construct.

[Illustration: Dorner]

DORNER III. Monoplane. ~Length.~--34-1/2 feet (10.50 m.) ~Span.~--39-1/3 feet (12 m.) ~Surface.~--280 sq. feet (126 m squared.) ~Weight.~--882 lbs. (400 kgs.)

Type II: ~Length.~--32-3/4 feet (10 m.) ~Span.~--38 feet (11.60 m.) ~Surface.~--268-1/2 sq. feet (25 m squared.) ~Weight.~--661 lbs. (300 kgs.) See _Flugsport_, No. 5, 1911.

~E~

ETRICH. Etrich Fliegerwerke, G.m.b. H, Dittersbach b. Liebau (Schlesien). Capacity: 50 a year.

[Illustration]

----------+---------- | ~1913.~ | ~Etrich~ (original) _Taube._ | monoplane. ----------+---------- ~Length~ feet (m.)| 31 (9.5) ~Span~ feet (m.)| 47-1/2 (14.4) ~Area~ sq. feet (m squared.)| 301 (28) {Total lbs. (kgs.)| 1323 (600) ~Weight~ { | {Useful lbs. (kgs.)| ... ~Motor~ h.p.| 100 Mercedes or Argus { max. m.p.h. (km.)| 71 to 75 (115 to 120) ~Speed~ { | { min. m.p.h. (km.)| ... ~Endurance~ hrs.| 6 ----------+----------

Remarks.--

EULER. August Euler, Frankfurt a.M. In 1908 Euler secured _Voisin_ rights for Germany. In 1910 he took out a patent for a design of his own. In the summer of 1911 he built a successful monoplane, in the autumn of the same year a triplane. Existing models are as follows:--

----------+----------+----------+---------- | ~1912.~ | ~1912.~ | | Triplane. | Monoplane. | Military biplane. ----------+----------+----------+---------- ~Length~ feet (m.)| 23 (7) | | ~Span~ feet (m.)| 23 (7) | | ~Area~ sq. feet (m squared.)| ... | | {total lbs. (kgs.)| ... | no data | no data ~Weight~ { | | | {useful lbs. (kgs.)| ... | | ~Motor~ h.p.| Gnome | | {max. m.p.h. (km.)| 56 (90) | | ~Speed~ { | | | {min. m.p.h. (km.)| ... | | ~Endurance~ hrs.| 3-4 | | Number built during 1912 | _about_ 70 | of various types | ----------+----------+----------+----------

[Illustration: Euler. Triplane.]

[Illustration: Euler. Monoplane, 1912.]

[Illustration: Euler. Military biplane.]

~F~

FOKKER. Monoplanes. Fokker-Aeroplanbau, G. m. b. H., 18 Parkstrasse, Johannisthal bei Berlin. Capacity: 40.

----------+----------+----------+----------+----------+---------- | ~1912.~ | ~1912.~ | ~1912-13.~ | ~1912-13.~ | ~1912-13.~ | ~1913.~ | ~A.~ | ~B.~ | ~A.~ | ~B.~ | ~C.~ | Hydro-aeroplane. ----------+----------+----------+----------+----------+---------- ~Length~ feet (m.)| 26-1/4 (8) | 26-3/4 (8.25) | 29-1/2 (9) | 29-1/2 (9) | 29-1/2 (9) | 31

(9.50) ~Span~ feet (m.)| 37-3/4 (11.50) | 39-1/3 (12) | 42-3/4 (13.20) | 42-3/4 (13.20) | 42-3/4 (13.20) | 52-1/2 (16.20) ~Area~ sq. feet (m squared.)| 226 (21) | 242 (22.50) | 280 (26) | 280 (26) | 280 (26) | ... {total lbs. (kgs.)| 838 (380) | 1036 (470) | 970 (440) | 1146 (520) | 1190 (540) | ... ~Weight~ { | | | | | | {useful lbs. (kgs.)| ... | ... | ... | ... | ... | ... ~Motor~ h.p.{| 70 Argus | 100 Argus | 70 Argus | 100 Argus | 70 Renault | 100 Renault {| | | or Dixi | | | or Mercedes {max. m.p.h. (km.)| 56 (90) | 68 (108) | 52 (83) | 60 (96) | 53 (85) | 59 (95) ~Speed~ { | | | | | | {min. m.p.h. (km.)| ... | ... | 43 (70) | ... | ... | ... ~Endurance~ hrs.| 4-6 | 4-6 | 5-8 | 5-8 | 4-6 | 4 Number built during 1912 | 3 | 2 | 6 | 5 | 2 | ... ----------+----------+----------+----------+----------+----------+----------

Remarks.--The _Fokker_ is a machine of Dutch origin. (See Dutch).

[Illustration: 1912-13 model.]

[Illustration: Hydro.]

FOHN. This firm ceased to exist January, 1913.

~G~

GRADE. Hans Grade Fliegerwerke, Bork, Post Bruck (Mark). Founded 1910 by H. Grade, who was the first man in Germany to fly with a German machine. During 1911 _Grades_ had a considerable vogue, but since then have not been prominent.

[Illustration: 1912 racer.]

----------+----------+----------+---------- Model and date. | Racer, ~1911.~ | Racer, ~1912.~ | Racer, ~1912.~ | ~C.~ | ~D.~ | ~E.~ ----------+----------+----------+---------- ~Length~ feet (m.)| 33 (10) | 21 (6.50) | 26-1/4 (8) ~Span~ feet (m.)| 39-1/4 (12) | 34-1/2 (10.50) | 41 (12.50) ~Area~ sq. feet (m squared.)| 480 (45) | 240 (22) | 360 (33) {machine, lbs. (kgs.)| 375 (170) | 408 (185) | 595 (270) ~Weight~ { | | | {useful lbs. (kgs.)| ... ... | ... ... | ... ... ~Motor~ h.p.| various | ... ... | ... ... ~Speed~ m.p.h. (km.)| 56 (90) | 71 (115) | 71 (115) Number built during 1912 | ? | 1 or 2 | ? ----------+----------+----------+----------

[Illustration: GRADE. UAS.]

GOEDECKER. J. Goedecker, Flugmaschinen-Werke, Niederwalluf a. Rh. Flying School: Flugplatz Grosser Sand bei Mainz.

[Illustration: GOEDECKER.]

----------+----------+---------- | ~1912.~ | ~1911.~ | Monoplane | Monoplane | "Sturmvogel." | "Sturmvogel." ----------+----------+---------- ~Length~ feet (m.)| 32-3/4 (10) | 29-1/2 (9) ~Span~ feet (m.)| 47-3/4 (14.5) | 47-3/4 (14.5) ~Area~ sq. feet (m squared.)| 387 (36) | ... {total lbs. (kgs.)| 992 (459) | 827 (375) ~Weight~ { | | {useful lbs. (kgs.)| ... | ... ~Motor~ h.p.| 100 Dixi | 70 Argus ~Speed~ m.p.h. (km.)| 56 (90) | ... Number built during 1912 | 8 | 2 ----------+----------+----------

~H~

HANSA-TAUBE. Heinrich Heitmann, Aviatik und Konstructions Werkstaetten, Altona.

[Illustration]

----------+----------+---------- | ~1912.~ | ~1913.~ | Monoplane. | Monoplane. ----------+----------+---------- ~Length~ feet (m.)| 24-3/4 (7.5) | 24-3/4 (7.5) ~Span~ feet (m.)| 36-3/4 (11.2) | 36-3/4 (11.2) ~Area~ sq. feet (m squared.)| 237 (22) | 237 (22) {total lbs. (kgs.)| 617 (280) | 573 (260) ~Weight~ { | | {useful lbs. (kgs.)| ... | ... ~Motor~ h.p.| 75 or 100 Argus | 100 Argus ~Speed~ m.p.h. (km.)| 56 to 62 (95 to 100)| 62 (100) Number built during 1912 | 2 | 2 ----------+----------+----------

Remarks.--

HARLAN. Harlan Werke, G. m. b. H., 21 Moltkestrasse, Johannisthal bei Berlin. Established 1909, turned into present Company, 1911. Output capacity about 50 machines a year.

[Illustration]

----------+----------+---------- | ~1912.~ | ~1912-13.~ | Military monoplane. | Military monoplane. ----------+----------+---------- ~Length~ feet (m.)| 26-1/4 (8) | 30 (9.10) ~Span~ feet (m.)| 39-1/3 (12) | 45-1/2 (13.80) ~Area~ sq. feet (m squared.)| 312 (29) | 312 (29) {total lbs. (kgs.)| ... | 1984 (900) ~Weight~{ | | {useful lbs. (kgs.)| ... | 1323 (600) ~Motor~ h.p.| 100 Argus or Mercedes | 100 Argus ~Speed~ m.p.h. (km.)| 69 (110) | 69 (110) ~Endurance~ hrs.| 7-8 | 7-8 Number built during 1912 | 20 | 15 ----------+----------+----------

HANUSCHKE. Bruno Hanuschke, Flugzeugbau, Johannisthal b. Berlin. Capacity: small.

[Illustration]

----------+----------+---------- | ~1912.~ | ~1913.~ | "Typ populaire" | Typ II. ----------+----------+---------- ~Length~ feet (m.)| 24-3/4 (7.50) | 21 (6.50) ~Span~ feet (m.)| 27 (8.25) | 26-1/4 (8) ~Area~ sq. feet (m squared.)| 183 (17) | 172 (16) {total lbs. (kgs.)| 716 (325) | 1102 (500) ~Weight~ { | | {useful lbs. (kgs.)| 385 (175) | 600 (275) ~Motor~ h.p.| 35 Anzani | 50 Gnome ~Speed~ m.p.h. (km.)| 56 (90) | 62 (100) ~Endurance~ hrs.| 2 | 2 Number built during 1912 | 2 | 2 ----------+----------+----------

Remarks.--

~J~

JATHO. Jatho Flugzeugwerke, G. m. b. H., Stader Chaussee 32, Hannover. Karl Jatho built his first aeroplane in 1899, and has produced machines at intervals ever since. Capacity: small.

[Illustration]

----------+---------- | ~1913.~ ----------+---------- ~Length~ feet (m.)| 29-1/2 (9) ~Span~ feet (m.)| 49-1/4 (15) ~Area~ sq. feet (m squared.)| 345 (32) {total lbs. (kgs.)| 2116 (960) ~Weight~ { | {useful lbs. (kgs.)| 992 (450) ~Motor~ h.p.| 100 N.A.G. ~Speed~ m.p.h. (km.)| 75 (120) ~Endurance~ hrs.| 3 Number built during 1912 | 2 ----------+----------

Remarks.--

JEANNIN. Emile Jeannin, Flugzeugbau, G. m. b. H., Stahltauben & Renneindecker Fabrik, Johannisthal b. Berlin. Capacity: small.

[Illustration: 1912 "Taube."]

----------+----------+---------- | ~1912.~ | ~1913.~ | "Taube" monoplane. | Racing monoplane. ----------+----------+---------- ~Length~ feet (m.)| 29-1/2 (9) | ... ~Span~ feet (m.)| 42-3/4 (13) | ... ~Area~ sq. feet (m squared.)| ... | ... {total lbs. (kgs.)| ... | ... ~Weight~ { | | {useful lbs. (kgs.)| ... | ... ~Motor~ h.p.| 100-150 Argus | 150 Argus ~Speed~ m.p.h. (km.)| 68 (110) | 87 (140) ~Endurance~ hrs.| 5-8 | 4-7 Number built during 1912 | 2 | 3 ----------+----------+----------

Remarks.--The 1913 was building only in March.

~K~

KAHNT. Oswald Kahnt, Flugzeugbau, Leipzig. Capacity: small.

[Illustration]

----------+---------- | ~K. F. 1913.~ | "Falke." ----------+---------- ~Length~ feet (m.)| 27-3/4 (8.50) ~Span~ feet (m.)| 42-3/4 (13) ~Area~ sq. feet (m squared.)| 291 (27) {total lbs. (kgs.)| ... ~Weight~ { | {useful lbs. (kgs.)| ... ~Motor~ h.p.| 50-70 ~Speed~ m.p.h. (km.)| 62 (100) Number built during 1912 | new firm ----------+----------

KONDOR. Kondor Flugzeugwerke G. m. b. H., Essen, Ruhr. Fabrik auf dem Flugplatz. Rotthausen. Capacity: 30 or so a year.

[Illustration: 1913 model. (1912 same appearance.)]

----------+----------+---------- | ~1912.~ | ~1913.~ ----------+----------+---------- ~Length~ feet (m.)| 33-3/4 (10.30) | 27 (8.20) ~Span~ feet (m.)| 48-3/4 (14.80) | 46 (14) ~Area~ sq. feet (m squared.)| 258 (24) | 280 (26) {total lbs. (kgs.)| 1543 (700) | 1328 (600) ~Weight~ { | | {useful lbs. (kgs.)| ... | ... ~Motor~ h.p.| 100 Argus | 100 Argus ~Speed~ m.p.h. (km.)| 65 (105) | 70 (112) Number built during 1912 | 2 | ... ----------+----------+----------

Remarks.--Both models torpedo body, on 4 skids. Planes dart ~V~ form. Constructor: J. Suwelack.

KUeHLSTEIN. Kuehlstein Wagenbau, Karosseriefabrik, Salzufer 4, Charlottenburg. This old-established motor car firm commenced to build aeroplanes

in 1911. Capacity: 20 a year.

[Illustration: 96 h.p.]

----------+----------+---------- | ~1912.~ | ~1912.~ | Torpedo monoplane. | Torpedo monoplane. | I. | II. ----------+----------+---------- ~Length~ feet (m.)| 29-3/4 (9.10) | 27 (8.2) ~Span~ feet (m.)| 40-3/4 (12.4) | 35-1/2 (10.8) ~Area~ sq. feet (m squared.)| 291 (27) | 215 (20) {total lbs. (kgs.)| 1984 (900) | 2204 (1000) ~Weight~ { | | {useful lbs. (kgs.)| 1322 (600) | 1543 (700) ~Motor~ h.p.| 100 Argus | 96 Mercedes {max. m.p.h. (km.)| ... | ... ~Speed~ { | | {min. m.p.h. (km.)| 84 (135) | 87 (140) ~Endurance~ hrs.| 3 | 3 Number built during 1912 | 2 | 2 ----------+----------+----------

Remarks.--

~M~

MARS. Deutsche Flugzeugwerke G. m. b. H., Lindenthal bei Leipzig. Established 1911. This is one of the most important and successful aviation works in Germany. Capacity: from 80 to 100 machines a year.

[Illustration: Mars. Monoplane.]

----------+----------+----------+---------- | ~1912-13.~ | ~1912-13.~ | ~1913.~ | Monoplane. | Biplane. | Hydro-aeroplane. ----------+----------+----------+---------- ~Length~ feet (m.)| 31 (9.7) | 31 (9.7) | ~Span~ feet (m.)| 55-1/4 (16.8) | 57 (17.8) | ~Area~ sq. feet (m squared.)| 376 (35) | 495 (46) | {total lbs. (kgs.)| 1234 (560) | 1434 (650) | ~Weight~ { | | | {useful lbs. (kgs.)| 1808 (820) | 2006 (910) | _Building._ ~Motor~ h.p.| 95 N.A.G. | 95 Mercedes | {max. m.p.h. (km.)| 120 (75) | 115 (71) | ~Speed~ { | | | {min. m.p.h. (km.)| ... | ... | ~Endurance~ hrs.| 5-6 | 4-6 | Number built during 1912 | 6 | 16 | ----------+----------+----------+----------

Remarks.--

[Illustration: Mars. Biplane.]

MROZINSKI. Bernard Mrozinski, Berlin-Wilmersdorf. Established 1912.

[Illustration]

~Length~, 23 feet (7 m.) ~span~, 32-3/4 feet (10 m.) ~area~, 215 sq. feet (20 m squared.)

~Weight.~--661 lbs. (300 kgs.)

~Motor.~--20 h.p. Anzani.

~Speed.~--50 m.p.h. (80 km.)

Remarks.--One machine only built in 1912.

~O~

OERTZ. Max Oertz, Yachtwerft, Reiherstieg b. Hamburg. Famous yacht builder. Commenced aeroplane construction in 1911. Existing models as below. Capacity about 25 machines a year.

[Illustration: 1912-13 model.]

----------+---------- | ~M 1911-12.~ | ~M 1912-13.~ | Monoplane. | Monoplane. ----------+----------+---------- ~Length~ feet (m.)| 29-1/2 (9) | 30-1/4 (9.2) ~Span~ feet (m.)| 41-3/4 (12.75) | 41-3/4 (12.75) ~Area~ sq. feet (m squared.)| 247 (23) | 263 (24.5) {total lbs. (kgs.)| 948 (430) | 1212 (550) ~Weight~ { | | {useful lbs. (kgs.)| ... | ... ~Motor~ h.p.| 70 Gnome | 70 Gnome {max. m.p.h. (km.)| 69 (110) | 75 (120) ~Speed~ { | | {min. m.p.h. (km.)| ... | ... ~Endurance~ hrs.| 3 | 4 Number built during 1912 | 3 | 1 ----------+----------+----------

OTTO. Gustav Otto, Flugmaschinenwerke, Schleissheimer Str. 135, Munich. Started building in 1911. Present max. capacity about 30 machines a year.

[Illustration]

----------+---------- | ~M 1912.~ | Biplane. ----------+---------- ~Length~ feet (m.)| ... ~Span~ feet (m.)| ... ~Area~ sq. feet (m squared.)| ... {total lbs. (kgs.)| ... ~Weight~ { | {useful lbs. (kgs.)| ... ~Motor~ h.p.| 100 A. G. Otto. {max. m.p.h. (km.)| 69 (110) ~Speed~ { | {min. m.p.h. (km.)| ... ~Endurance~ hrs.| 6-8 Number built during 1912 | 6 ----------+----------

Remarks.--All 1912 machines purchased for German Army.

~P~

PEGA-EMICH. Flugtechnische und mechanische Werke vorm. Pega & Emich, Falterstrasse 13-15, Griesheim, Frankurt-a-M. Commenced building with a 6-decker in 1910. Capacity: small.

[Illustration]

----------+---------- | ~1913.~ | Buteno monoplane. ----------+---------- ~Length~ feet (m.)| 39-1/4 (12) ~Span~ feet (m.)| 46 (14) ~Area~ sq. feet (m squared.)| 355 (33) {total lbs. (kg.)| 838 (380) ~Weight~ { | {useful lbs. (kg.)| 1102 (500) ~Motor~ h.p.| 70 Argus {max. m.p.h. (km.)| 62 (100) ~Speed~ { | {min. m.p.h. (km.)| ... ~Endurance~ hrs.| 2 Number built during 1912. | ... ----------+----------

PIPPART-NOLL. Pippart-Noll-Flugzeugbau, Mannheim.

[Illustration]

----------+----------+----------+---------- | P. N. 1 | P. N. 2. | P. N. 3. Type. | Sporting. | "Uberland" | Military. | ~1912.~ | ~1912.~ | ~1913.~ ----------+----------+----------+---------- ~Length~ feet (m.)| 31 (9.50) | 28 (8.50) | 28 (8.50 also 7) ~Span~ feet (m.)| 34-1/2 (10.50) | 39-1/3 (12) | 45 (13.70) ~Area~ sq. feet (m squared.)| 215 (20) | 280 (26) | 300 (28) {machine lbs.

(kgs.)| 617 (280) | 838 (380) | 1234 (560) ~Weight~ { | | | {useful lbs. (kgs.)| 330 (150) | 463 (210) | 441 (200) ~Motor~ h.p.| 70 Argus | 70 Argus | 70 Argus {max. m.p.h. (km.)| 59 (95) | 62 (100) | 68 (110) ~Speed~ { | | | {min. m.p.h. (km.)| ... | ... | 50 (80) ~Endurance~ hrs.| ... | ... | ... Number built during 1912 | 1 | 1 | 1 ----------+----------+----------+----------

~R~

RUMPLER. E. Rumpler, Luftfahrzeugbau G. m. b. H., Siegfriedstrasse 202, Berlin-Lichtenberg, also Johannisthal b. Berlin. Established 1909 by E. Rumpler and R. Haessner for the construction in Germany of _Etrich_ (see Austria) monoplanes. These now vary considerably from the original _Etrich_. Capacity at present about 200 to 300 machines a year. Standard models are as follows:--

----------+----------+----------+---------- | ~1912.~ | ~1912.~ | ~1913.~ | Monoplane. | "Taube." | Hydro. ----------+----------+----------+---------- ~Length~ feet (m.)| 29-3/4 (9.50) | 34 (10.30) | 33 (10) ~Span~ feet (m.)| 41-1/2 (12.65) | 46 (14) | 49-1/4 (15) ~Area~ sq. feet (m squared.)| 247 (23) | 336 (32) | 387 (36) {total lbs. (kgs.)| 1398 (630) | 1190 (540) | 1328 (600) ~Weight~ { | | | {useful lbs. (kgs.)| 771 (350) | 551 (230) | 485 (220) ~Motor~ h.p.| 95 Mercedes | 100 Argus | 100 Argus {max. m.p.h. (km.)| 81 (130) | 59 (95) | 56 (90) ~Speed~ { | | | {min. m.p.h. (km.)| ... | ... | ... ~Endurance~ hrs.| 6-7 | 4-6 | ... Number built during 1912 | 1 | 60 | 3 ----------+----------+----------+----------

[Illustration: Rumpler. Hydro-aeroplane.]

[Illustration: Rumpler. Monoplane.]

[Illustration: Rumpler. "Taube." With limousine body.]

RUTH-ROHDE. Ruth-Rohde, Motorgleitflieger, G. m. b. H., Wandsbeck. Established 1912. Capacity: small.

[Illustration]

----------+----------+---------- | ~1912.~ | ~1912.~ | Biplane I. | Biplane II. ----------+----------+---------- ~Length~ feet (m.)| 26-1/4 (8) | 26-1/4 (8) ~Span~ feet (m.)| 36 (11) | 45 (14) ~Area~ sq. feet (m squared.)| 590 (55) | 700 (65) {total lbs. (kgs.)| 1653 (750) | 1764 (800) ~Weight~ { | | {useful lbs. (kgs.)| ... | ... ~Motor~ h.p.| 75 Argus | 75 Argus {max. m.p.h. (km.)| 55 (90) | 55 (90) ~Speed~ { | | {min. m.p.h.(km.)| ... | ... ~Endurance~ hrs.| 3 | 3-4 Number built during 1912. | 1 | 1 ----------+----------+----------

~S~

SCHELIES. Richard Schelies, Conventstrasse 5 und 5b, Hamburg 23. Flying Station, etc.: Dockenhuden a/Elbe.

[Illustration]

----------+---------- | ~1913.~ | Hydro-monoplane. ----------+---------- ~Length~ feet (m.)| 23 (7) ~Span~ feet (m.)|

29-1/2 (9) ~Area~ sq. feet (m squared.)| 323 (30) {total lbs. (kgs.)| 705 (320) ~Weight~ { | {useful lbs. (kgs.)| 220 (100) ~Motor~ h.p.| Rheinische Aero 35 ~Speed~ m.p.h. (km.)| ... ----------+----------

SCHULZE. Gustav Schulze, Flugzeug Werke, Burg b. Magdeburg. Schulze began to build in 1910 light monoplanes, generally along _Santos-Dumont_ lines. Maximum present capacity about 12 machines a year.

[Illustration]

----------+----------+----------+----------+---------- | ~1912.~ | ~1912.~ | ~1912.~ | ~1913.~ | I. | II. | III (2-seater). | I (2-seater). ----------+----------+----------+----------+---------- ~Length~ feet (m.)| 19-3/4 (6) | 26-1/4 (8) | 21-1/3 (6.50) | 23 (7) ~Span~ feet (m.)| 26-1/4 (8) | 34-1/2 (10.50) | 28 (8.50) | 29-1/2 (9) ~Area~ sq. feet (m squared.)| 172 (16) | 215 (20) | 172 (16) | 194 (18) {total lbs. (kgs.)| 330 (150) | 441 (200) | 441 (200) | 551 (250) ~Weight~ { | | | {useful lbs. (kgs.)| | ... | ... | ... ~Motor~ h.p.| 24-30 Hilz | 24-30 Hilz | 35 Haacke | 35 Haacke {max. m.p.h. (km.)| 48 (77) | 53 (85) | 56 (90) | 53 (85) ~Speed~ { | | | {min. m.p.h. (km.)| ... | 43 (70) | 50 (80) | 46 (75) Number built during 1912. | 1 | 3 | 1 | _Building._ ----------+----------+----------+----------+----------

SIGISMUND. Prinz Sigismund von Preussen, Berlin.

[Illustration]

----------+---------- Model and date. | Monoplane. | ----------+---------- ~Length~ feet (m.)| 29-1/2 (9) | ~Span~ feet (m.)| 42-3/4 (13) | ~Area~ sq. feet (m squared.)| 323 (30) | {total lbs. (kgs.)| 950 (430) | ~Weight~ { | | {useful lbs. (kgs.)| 395 (180) | ~Motor~ | Argus, 100 | ~Speed~ max. m.p.h. (km.)| 56 (90) | Number built during 1912 | 2 | ----------+----------

~U~

UNION FLUGZEUGWERKE. Union Flugzeugwerke G. m. b. H. Elsenstrasse 106 & 107, Berlin s. o. 36. Established 1913. Capital 500,000 marks. Capacity of works: 20 machines a year.

[Illustration]

----------+----------+ | ~Bomhard.~ | Model and date. | Pfeilflieger, | | ~1913.~ | ----------+----------+ ~Length~ feet (m.)| 32-3/4 (10) | ~Span~ feet (m.)| 59 (18) | ~Area~ sq. feet (m squared.)| 450 (42) | {total lbs. (kgs.)| 1235 (560) | ~Weight~ { | | {useful lbs. (kgs.)| 617 (280) | ~Motor~ | 100 Argus | {max. m.p.h. (km.)| 69 (110) | ~Speed~ { | | {min. m.p.h. (km.)| 62 (100) | Number built during 1912 | New firm | ----------+----------+

~W~

WRIGHT. Flugmaschine Wright, G. m. b. H., Adlershof, bei Berlin. Company formed to trade in German rights for the Wright Bros.' patents. Considerable

departures have been made from the U.S. pattern, and some have been built with a single propeller only. Capacity of works 100-150 a year.

[Illustration: Armoured war aeroplane.]

----------+----------+----------+----------+---------- | ~1912.~ | ~1913.~ | ~1913.~ | ~1913.~ | Military. | Sporting. | Military. | Military. | | | | | 4-seater. ----------+----------+----------+----------+---------- ~Length~ feet (m.)| 28 (8.50) | 26-1/2 (8.20) | 31-1/2 (9.65) | ... ~Span~ feet (m.)| 39-1/2 (12.20) | 31 (9.60) | 40-1/2 (12.50) | 44-1/4 (13.50) ~Area~ sq. feet (m squared.)| 452 (42) | 323 (30) | 463 (43) | 463 (43) {total lbs. (kgs.)| 992 (450) | 837 (380) | 1433 (650) | 1653 (750) ~Weight~ { | | | | {useful lbs. (kgs.)| ... | ... | ... | 882 (400) ~Motor~ h.p.| 55 N.A.G. | 55 N.A.G. | 100 Argus or | 100 | | | Mercedes | {max. m.p.h. (km.)| 50 (80) | 60 (95) | 60 (95) | 60 (95) ~Speed~ { | | | | {min. m.p.h. (km.)| ... | ... | ... | ... Number built during 1912 | 10 | ? | ... | ... ----------+----------+----------+----------+----------

~Z~

ZIEGLER. Ziegler, Potsdam. Established late in 1912.

[Illustration]

----------+---------- | ~1912-13.~ | Monoplane. ----------+---------- ~Length~ feet (m.)| 31 (9.50) ~Span~ feet (m.)| 39-1/3 (12) ~Area~ sq. feet (m squared.)| 344 (32) {total lbs. (kgs.)| 881 (400) ~Weight~ { | {useful lbs. (kgs.)| 992 (450) ~Motor~ h.p.| 100 N.A.G. {max. m.p.h. (km.)| 60 (90) ~Speed~ { | {min. m.p.h. (km.)| ... ~Endurance~ hrs.| 2 Number built in 1912 | 1 ----------+----------

GERMAN DIRIGIBLES.

(Approximately 1000 m cubed=35,000 c. feet.)

~Military.~

----------+----------+----------+-------+--------+------+----------+---------- Date. | Name. | Make. | Type. |Capacity| Total| Speed. | Remarks. | | | | in m cubed. | H.P. | m.p.h. (km.)| ----------+----------+----------+-------+--------+------+----------+---------- 1908 | ~Z I~ | Zeppelin 3b | r. | 12100 | 190 | 29 (46) | as _rebuilt_ | | | | | | 1910 | ~Z II~ | Zeppelin 9b | r. | 18000 | 345 | 35 (56) | as _rebuilt_ " | ~L. S I~ |Schuette Lanz 1| r. | 20000 | 540 | 40 (62) | | | | | | | 1912 | ~Z III~ | Zeppelin 12 | r. | 17800 | 450 | 49 (79) |was _Schwaben L. Z 10_ " | ~L I~ | Zeppelin 14 | r. | 22000 | 450 | 48 (77-1/2) | Naval: 1 gun | | | | | | 1913 | ~Z IV~ (Z I _Ersatz_) | Zeppelin 15 | r. | 21000 | 450 | 48 (77-1/2) | 4 guns Building. | ~L II~ | Zeppelin 16 | r. | 21000 | 450 | 48 (77-1/2) | Naval: _bldg._ 4 guns | ~S. L II~ |Schuette Lanz 2| r. | 26000 | 450 | 48 (77-1/2) | _Building_ | | | | | | | ----------+----------+----------+-------+--------+------+----------+---------- | | | | | | 1908 | ~P I~ | Parseval 2 | n.r. | 3800 | 85 | 33-1/2 (54) | | | | | | | 1911 | ~P III~ |

Parseval 11 | n.r. | 11000 | 400 | 42-1/2 (67) | | | | | | | 1912 | ~M I~ | Gross-Bas 2 | s.r. | 6000 | 150 | 28 (45) | old 1908 _rebuilt_ " | ~M II~ | Gross-Bas 3 | s.r. | 6000 | 150 | 28 (45) | old 1909 _rebuilt_ " | ~M III~ | Gross-Bas 4 | s.r. | 9000 | 300 | 42-1/2 (67) | old 1910 _rebuilt_ | | | | | | 1913 | ~M IV~ | Gross-Bas 5 | s.r. | 12000 | 400 | 44-1/2 (70) | old 1911 _rebuilt_ " | ~P II~ ersatz | Parseval 8 | n.r. | 8250 | 300 | 41 (66) | _Building_ " | ~P IV~ | Parseval 16 | n.r. | 10000 | 360 | 45 (72) | _Building_
----------+-------------------------+-------------+-------+--------+------+-------------+----------------------

~Private.~[D]

-----------+----------------------+-------------+-------+--------+------+-------------+----------------------
Date. | Name. | Make. | Type. |Capacity| Total| Speed | Remarks. | | | | in m cubed. | H.P. | m.p.h (km.) |
-----------+----------------------+-------------+-------+--------+------+-------------+----------------------
1910 | ~DEUTCHLAND 2~ | Zeppelin 6a | r. | 15000 | 345 | 36 (58) | Deutschland _Ersatz_ | | | | | | Delag | | | | | | 1912 | ~V. LUISE~ | Zeppelin 11 | r. | 17000 | 450 | 40 (62) | Delag " | ~HANSA~ | Zeppelin 13 | r. | 17000 | 450 | 40 (62) | Delag | | | | | | 1913 | ~SACHSEN~ | Zeppelin 17 | r. | 21000 | | 48 (77-1/2) | _Building._ Delag _Bldg._ | | | | | | | | | | | | | |
-----------+----------------------+-------------+-------+--------+------+-------------+---------------------- | | | | | | 1908 | ~P. L 1~ | Parseval 1 | n.r. | 3200 | 185 | 20 (32) | | | | | | | 1910 | ~STOLLWERCK~ | Parseval 6 | n.r. | 9000 | 220 | 31 (50) | | | | | | | 1911 | ~P. L 9~ | Parseval 9 | n.r. | 2200 | 50 | 25 (40) | Sporting[E] " | ~R 2~ | Ruthenberg 2| n.r. | 1700 | | | Experimental | | | | | | 1912 | ~SUCHARD~ |Suchard reb'lt| n.r. | 6730 | 200 | 17 (28) | to be _rebuilt_ 1913 " | ~P. L XII~ | Parseval 12 | n.r. | 8800 | 220 | 33-1/2 (54) | | | | | | | 1913 | ~P. L 10~ | Parseval 10 | n.r. | 2200 | 50 | 25 (40) | _Building_: delayed " | ~R 3~ | Ruthenberg 3 | n.r. | 2700 | | | _Building_
-----------+----------------------+-------------+-------+--------+------+-------------+----------------------

~Dirigible Sheds.~ (_See Note._)

Bickendorf bei Koeln. Biesdorf bei Berlin. *Bitterfeld (_Parseval Co._) Breslau. Cuxhaven. Duesseldorf (_Delag_) Frankfurt a/m. Friedrichshafen (_Zeppelin Co._) Gotha. Hamburg (_Delag_). *Johannisthal (_L.V.G._) Kiel (_private_). Koeln. Koenigsberg i/Pr. Leichlingen. Manzell (_Zeppelin Co._) Metz. *Muenchen (_private_). Oos bei Baden-Baden (_Delag_). Potsdam (_Delag_) Reinickendorf bei Berlin. Rheinau. Strassburg. Stuttgart. Thorn.

Note.--Unless otherwise stated the above are military sheds. All private ones capable of holding _Zeppelins_ are subsidised. *=not large enough for _Zeppelins_.

~Dirigible Pilots.~ For M. dirigibles.

Geerdtz, Oblt. George, Hptm. v. Jena, Hptm. Kirchner, Oberltn. Lohmuller, Hptm. Masius, Oberltn. v. Muller, Hptm. Nichisch v. Rosenegk, Oberltn. Schlutter, Obltn. Sperling, Major. von Zech, Obltn.

~Dirigible Pilots.~

Z=Zeppelin. S=Schutte-Lanz. P=Parseval _pilot_.

(The number after each name is the Imperial Ae. C. certificate number.)

Z Abercron, H. v. Major (1) Z Bassus, K. v. (28) Z Bentheim, Kapt. Lt. a. D. v. (34) Z Blew (25) Clouth, R. (8) P Dinglinger, F. (2) Z Dorr, W. E. (21) Z Durr (9) Z Eckener, Dr. (10) P Forsbeck, Ob. Lt. A. D. (11) Z Glund, F. (23) Z Hacker, (12) P Hackstetter, Reg. B. a. D. (13) Z Hanne, G. (32) Z Heinen, A. (22) Z Holzmann, Ob. Lt. A. (26) S Honold, R. (29) P Hormel, Kap. Lt. (14) P Jordens, W. (19) P Kehler, R. v. (6) P Kiefer, T. (5) Kleist, Hptm. a. D. v. (15) PZ Krogh, Hptm. a. D. v. (16) Z Lange, K. (30) Z Lau (17) Z Lempertz, E. (33) Z Mechlenburg, W. C. (35) Z Meyer, Ob. Lt. E. (27) P Parseval, A. v. (4) Z Stahl, K. (31) P Stelling, A. (3) Z Sticker, J. (24) P Thewaldt, C. H. (20) Z Zeppelin, Graf. v. (7) Z Zeppelin, Graf. F. v. junr. (18)

=GERMAN MILITARY CLASS--GROSS-BASENACH. (Semi-rigid)=

Up to date, these vessels have been designed by Major Gross and Oberingenieur Basenach.

The utmost secrecy is observed as to their details.

The system of employing 2 ballonets has been borrowed from the _Parseval_ type, and presumably the _Parseval_ system of working the automatic valves has also been adopted.

In all other features, these ships appear to resemble the French _Lebaudy_ type, the shape of the hulls being rather better.

List of ships built, re-built and re-building of this type:--

1 = Aeronautical Society. (1,800 m cubed) _non-effective._ 2 (reconstructed) = M 1, Military. (6,000 m cubed) 3 " = M 2 " (6,000 m cubed) 4 " = M 3 " (9,000 m cubed) 5 " = M 4 " (12,000 m cubed)

M I (re-built 1912), & M II (re-built 1912). Military.

[Illustration]

~Length,~ 242-3/4 feet} ~maximum diameter,~ 36 feet (11 m.) ~capacity~, 212,000 c. feet (6,050 kg.) ~total lift~, 13,338 lbs. (6,000 m cubed) _about_ 6 tons

~Useful lift.~--2,756 lbs. (1,250 kgs.)=about 1-1/4 tons.

~Gas bag.~--Continental rubber cloth, diagonal thread. Tapering shape.

~Ballonet.~--One-fifth of total volume.

~Motors.~--2-75 h.p. Daimler. 2 propellers, with 3 aluminium blades.

~Speed.~--About 28 m.p.h. (45 km.).

Remarks.--M I was originally built in 1908, re-built and enlarged 1910 and again in 1912. _M II_ built 1909, re-constructed 1912.

[Illustration]

M III (re-built 1912). Military.

[Illustration]

~Length~, 295-1/4 feet (90 m.) ~diameter~, 39-1/3 feet (12 m.) ~volume~, 317,800 c. feet (9,000 m cubed.)

~Motors.~--4 Koerting of 75 h.p. each = 300 h.p. total.

~Speed.~--19 metres per second = 42 m.p.h. (68-1/2 k.p.h.)

~Propellers.~--2, on outriggers from car, chain-driven.

Remarks.--Built 1910. Burned 13th September, 1911. Rebuilt 1912.

M IV (re-built 1913). Military.

+----------+ | | +----------+

~Maximum length~, 334-3/4 feet (102 m.) ~maximum diameter~, 44-1/2 feet (13.5 m.) ~volume~, 423,800 c. feet (12,000 m cubed.)

~Total lift.~-- lbs. (kgs.) ~Useful lift~, lbs. (kgs.)

~Gas bags.~--Continental.

~Motors.~--2 Koerting, 100 h.p. each = total 200 h.p.

~Speed.~--44-1/2 m.p.h. (70-1/2 k.p.h.)

~Propellers.~--4 (two for each motor.) Carried on outriggers projecting from the car.

Remarks.--Departs from previous practice in having two separate cars, each of which contains one motor. Originally built in 1911 of 7,500 m cubed. Re-built 1912-1913 to details as above.

=PARSEVAL CLASS (Non-rigid).=

Luftfahrzeug-Gesellschaft m.b.H, Berlin, W. 62.

When the "Motorluftschiff Studien Gesellschaft" was formed at the instigation of the German Emperor, a committee was formed to acquire an experimental airship of the most promising type. Major Von Parseval's first airship was selected, and since that time the above company has confined itself to improving this type, and to making exhaustive and costly researches, all of which have been embodied in successive ships.

The characteristic feature of every one of these craft is its unequalled portability. Almost all other so-called Non-rigid vessels distribute the load by means of a long girder which also serves as a car. This girder is awkward to pack up and transport. Parseval uses a comparatively small car, and distributes the weight by hanging it further below the balloon than usual, and also by using 2 ballonets which are placed one near each end of the gas bag.

These 2 ballonets enable the ship to be trimmed by merely pumping air into either at the expense of the other.

Another essential feature of the type is the system by which the valves are worked automatically. At the present time there is no other system of valve working so

reliable as this.

A third essential feature of the class is the use of a swinging car, in such a manner that pitching, due to alterations of propeller thrust, is automatically checked by an alteration of the position of the centre of gravity.

A fourth feature is the use of limp propeller blades. A propeller of this type is very easily packed up.

The shape of these vessels is in accordance with the experiments of Professor Prandtl.

Ships of this class built or building (figures supplied by the Parseval Co.):--

EXPERIMENTAL PARSEVAL 2,300 m cubed. P. L. 1 Kals. Ae. C 3,200 m cubed. MILITARY P I 4,000 m cubed. MILITARY P II 6,600 m cubed. P. L. 4 Austrian Military 2,300 m cubed. P. L. 5 Luftverkehrs Gesellschaft 1,450 m cubed. P. L. 6 " " 9,000 m cubed. P. L. 7 Russian Army 7,600 m cubed. P. L. 8 MILITARY P II Ersatz 8,250 m cubed. P. L. 9 Luftverkehrs Gesellschaft 2,200 m cubed. P. L. 10 Motorluftschiff Studien Gesellschaft 2,200 m cubed. MILITARY P III 11,000 m cubed. P. L. 12 Luftverkehrs Gesellschaft 8,800 m cubed. P. L. 13 Japanese Army 8,500 m cubed. P. L. 14 Russian Army 9,500 m cubed. P. L. 15 Italian Army 10,000 m cubed. P. L. 16 MILITARY P IV, Prussian Army 10,000 m cubed. P. L. 17 Italian Army 10,000 m cubed. P. L. 18 British Navy 8,800 m cubed.

(Of the above, the Experimental is no longer in existence, _P. 2_ is out of service, and _P. L. 3_ has been burned and destroyed).

PARSEVAL (P.L. 1). (1908.) (Belongs to the Kaiserl. Aero Club.) (Parseval class.)

[Illustration]

~Length,~ 197 feet (60 m.) ~max. diam.~ 31 feet (9.4 m) ~capacity,~ 113,000 c. feet (3,200m cubed) ~lifting power,~ 7,800 lbs. (3,583 kgs.)

~Gas bag.~--Cylindrical, with semi-conical front. Of rubber-proofed material in longitudinal strips. Pressure in ballonets and gas bag, 30 mm. of water.

~Motor.~--One 85 h.p. Daimler.

~Fuel.~--700 lbs. (325 kg.) 88 gallons (400 litres)

~Speed.~--20 m.p.h. (32 k.p.h.)

~Propeller.~--One 4-bladed. Semi-rigid Parseval.

This vessel was somewhat altered on being bought by the Society. Her essential principle is that she can be taken to pieces in a few minutes, and carried in a truck. Her main feature is that she has a ballonet at each end. This is described in the case of type A (_P.L. 2_). This class rise with the forward ballonet empty, and inclined up by the bow. The propeller is similar to that of _P.L. 2_. The car also is mounted on wire runners. She was originally 4,000 m cubed. capacity. Built 1908. Station: Bitterfeld. The car is at present in Deutsches Museum, Munich.

PARSEVAL P.L. 2 = P. I. Military. (1908.)

[Illustration]

Built by the "Society for the Study of Motor Air Ships," and taken over by the German War Office.

~Length,~ 197 feet (60 m.) ~maximum diameter,~ 34 feet (10.40 m.) ~capacity,~ 111,270 cubic feet (4,000 cubic m.) ~lifting power,~ 9,200 lbs. (4,180 kgs.)

~Gas bag.~--Front end semi-ellipsoidal with semi-axes 15.4 feet (4.7 m.) and 11.8 feet (3.6 m.), from which it increases to its maximum diameter. This is maintained for about two-thirds of its length, when it begins to taper to a point at the stern. Made of 2 layers of transverse strips of rubber proofed material, crossing each other diagonally. Fitted with a tearing strip.

~Ballonets.~--One at each end, together amounting to about one-quarter of the total capacity. Owing to this disposition, the trim can be altered, and steering effected in the vertical plane by filling either more than the other. Pressure in the ballonets and gas bag, 20 mm. of water pressure.

~Motor.~--4-cylinder 85 h.p. Daimler placed at one side of the car to give more room. 1,000 to 1,200 r.p.m.

~Propellers.~--12-1/3 feet (3.75 m.) diameter, 250 to 300 r.p.m. 4-bladed, the blades being of peculiar construction. When stopped, the fabric of which they are made hangs down limply; when running, these flaps fly out under centrifugal force.

~Speed.~--27 miles per hour. 43 kilometers per hour.

~Car.--Length~, 22-1/3 feet (6.8 m.) Width 4.1 feet (1.22 m.) Made of nickel steel, U bars, screwed together so as to take the pieces rapidly. The sides are lattice girders. The whole is boat shaped and covered with canvas. Contains motor, chart table; trail rope 480 feet (146 m.) long, weighing 220 lbs. (100 kg.)

Wheel for horizontal steering at the bow.

110 gal. (500 litre) cask of petrol on the girders at the after point.

41 feet (12-1/2 m.) below the gas bag. It is capable of swinging horizontally on wires running over rollers.

Whereas without this device a forward swing of the car would lift the nose to a possibly dangerous extent, the free motion of the car shifts the centre of the gravity forward and so preserves stability.

~Steering.~--In vertical plane, by altering the trim. In horizontal plane, by a rudder of 80.7 sq. feet (7-1/2 m squared) immediately behind the vertical plane. Two fixed horizontal planes are placed at the rear end of the gas bag above the central line.

[Illustration: PARSEVAL II.]

~Table of weights~:--

Gas bag 1,653 lbs. Cordage 220.5 " Trail rope 220 " Car and motor 529 " Fuel 770 " Oil 160 " Oil and fuel tanks, instruments, miscellaneous 1,637 " Crew, passengers,

ballast 1,654 " --------- Total 6,834 lbs.

Note.--This remarkably successful ship has performed a continuous flight of 11-1/2 hours. She also remained at a height of 4,800 feet (1,500 m.) for 1 hour. She can be transported in 1 railway truck or 2 pair horse wagons, and be assembled and filled ready for ascent within 3 hours of arrival by train.

Built 1908. Station: Metz.

PARSEVAL P.L. 6. "Stollwerck." (1910.)

[Illustration]

~Length~, 229-3/4 feet (70 m.) ~Diameter~, 49-1/4 feet (15 m.) ~Volume~, 318,000 c. feet (9,000 m cubed)

~Gas bags.~--Riedinger.

~Motors.~--2 N.A.G. of 110 h.p. each = 220 h.p.

~Speed.~--31 m.p.h. (50 k.p.h.)

~Propellers.~--Two 4-bladed. Semi-rigid material.

Remarks.--Station, Johannisthal.

PARSEVAL P.L. 8. = P II. Ersatz. Military. (1913.)

[Illustration]

~Maximum length~, 252-3/4 feet (77 m.) ~maximum diameter~, 50-3/4 feet (15.50 m.) ~volume~, 290,000 c. feet (8,250 m cubed.)

~Total lift.~--5-1/2 tons=12,125 lbs. (5,500 kgs.)

~Gas bags.~--2 ballonets, usual arrangement.

~Motors.~--300 h.p. made up of two 150 h.p. Daimler motors, placed one behind the other.

~Speed.~--41 m.p.h. (66 km.)

~Propellers.~--2 Parseval, 4-bladed, semi-rigid steel.

~Steering.~--As in others.

Remarks.--Station, Cologne, (Coeln).

PARSEVAL P.L. 9 (1910), & 10. (Building 1913.)

[Illustration]

~Maximum length~, 164 feet (50m.) ~maximum diameter~, 26-1/4 feet (8m.) ~volume~, 77,700 c. feet (2,200m cubed.)

~Total lift.~--2,910 lbs. (1,320 kgs.)

~Gas bag.~--Continental fabric. One central ballonet instead of the usual two.

~Motors.~--1 N.A.G. of 50 h.p.

~Speed.~--25 m.p.h. (40 k.p.h.)

~Propellers.~--One 2-bladed, wooden. Diameter, 9-3/4 feet (3 m.)

~Steering.~--Differs from other standard types, in that only one ballonet being fitted, an elevator is introduced under the bow.

Remarks.--Small ships for sporting purposes. A remarkably successful type of small dirigible. A small _P.L. 5_, burned 1912. _P.L. 10_ delayed owing to press of other work.

[Illustration: PARSEVAL TYPE D.]

PARSEVAL P.L. 11. = P. III. Military. (1911.)

[Illustration]

~Maximum length~, 272-1/3 feet (83 m.) ~maximum diameter~, 53 feet (16.20 m.) ~volume~, 388,450 c. feet (11,000 m cubed.)

~Total lift.~--

~Gas bags.~--

~Motors.~--2 Koerting, each of 200 h.p.=400 total.

~Speed.~--42 m.p.h. (67 k.p.h.) (18.3 metres p. sec.)

~Propellers.~--Two 4-bladed Parseval.

Remarks.--Built 1911. Station, Koenigsberg.

PARSEVAL P.L. 12. "Charlotte." (1912.)

[Illustration]

~Maximum length~, 259 feet (79 m.) ~maximum diameter~, 49-3/4 feet (15.20 m.) ~volume~, 300,750 c. feet (8,800 m cubed.)

~Total lift.~--

~Gas bags.~--

~Motors.~--2 N.A.G. of 110 h.p. each=220 total.

~Speed.~--33-1/2 m.p.h.=54 k.p.h. (15 m. per sec.)

~Propellers.~--2 Parseval.

~Steering.~--Usual.

Remarks.--Built 1911. Station: Wanne.

PARSEVAL P.L. 16 = P. IV. Military. (1913.)

+----------+ | | | _Completing._ | | | +----------+

~Maximum length~, 308-1/2 feet (94 m.) ~maximum diameter~, 51-1/2 feet (15.50 m.) ~volume~, 353,000 c. feet (10,000 m cubed.)

~Gas bags.~--Metzler.

~Motors.~--2 Maybach, of 180 each=360 h.p.

~Speed.~--45 m.p.h.=72 k.p.h. (20 m. per sec.)

~Propellers.~--Two 4-bladed, wooden (on trial).

Remarks.--For the Prussian Army. Station: Berlin.

RUTHENBERG II. (1911). H. Ruthenberg, Lehderstrasse 16/19, Weissensee bei Berlin: also Luftfahrzeug-Ges, Ruthenberg, Krefeld.

+----------+ | | | _Small ships on Parseval lines. | | Still existing, but stored away._ | | | +----------+

~Maximum length,~ 151 feet (46 m.) ~diameter~, 24-1/4 feet 7.40 (m.) ~volume~, 60,000 c. feet (1,700 m cubed.)

~Gas bags.~--

~Motor.~--

~Speed.~--

~Propellers.~--2 Ruthenberg.

Remarks.--

RUTHENBERG III. (1913).

+----------+ | | | _Building._ | | | +----------+

~Length~, feet (m.) ~diameter~, feet (m.) ~volume~, 95,000 c. feet (2,700 m cubed)

~Gas bags.~--

~Motor.~--

~Speed.~--

~Propeller.~--Ruthenberg.

Remarks.--

SUCHARD. Non-rigid (Trans-Atlantic). (Re-constructed 1912).

[Illustration]

~Maximum length~, 198-1/2 feet (60/5 m.) ~maximum diameter~, 56-1/4 feet (17.11 m.) ~volume~, 237,681 cubic feet (6,730 m cubed.)

~Total weight.~--About 2 tons (2,130 kgs.)

~Gas bags.~--Metzeler fabric. One ballonet.

~Motors.~--2 of 100 h.p. (one a N.A.G., the other an Escher). Placed one behind the other. A 4 h.p. motor carried for auxiliary purposes. Petrol carried, (1700 kil.) Oil, (300 kil.)

~Speed.~--17 m.p.h. (28 k.p.h).

~Propellers.~--Two 2-bladed Zeise. Diameter, 9-3/4 feet (3 m.) Chain driven.

~Steering.~--Elevation by moving weight slung on cable under body. Rudder aft.

Remarks.--Built March, 1911, with a view to crossing the Atlantic from the Canaries to the Antilles. Re-constructed 1912. Proposed further re-construction in 1913.

SCHUeTTE-LANZ 1. Military. S.L. I. (1911.) H. Heinrich Lanz, Rheinau bei Mannheim.

[Illustration]

~Maximum length,~ 426 feet (130 m.) ~maximum diameter~, 60-1/2 feet (18.40 m.) ~volume~, 706,000 c. feet (20,000 m cubed.)

~Total lift.~--About 20 tons (20,000 kgs.) ~Useful lift.~--About 5 tons (5,000 kgs.)

~Gas bags.~--These are of great strength and of unusual shapes, made to fit the interior, which is encumbered with cross stays. All but two of the bags are always full, and when the gas expands it flows into the remaining two, which are nearly empty at sea level, and full at 6500 feet (2,000 m.) A centrifugal pump is used for distributing the gas. There are 14 gas bags.

~Motors.~--2 Maybach of 270 h.p. each. The propellers are at the ends of the car, driven through 1 set of reduction gear. ~Speed.~--38-43 m.p.h. About 59-64 k.p.h.

~Propellers.~--2 aft. Also 1 with its axis vertical.

~Steering.~--Vertical and horizontal rudders at both ends of the ship. Also see Propellers.

Remarks.--Two of these ships were under construction, and one was to be presented and one sold to the German government. The hull is built of special 3-ply wood made of Russian white fir; this wood is pressed into channel bars, angle bars, and all other requisite shapes. The strength of the hull is such that it can be supported at the ends without damage; its lightness is such that although the ship is nearly half as large again as _Zeppelin II_, yet the hull weighs about 3 tons less. Designed by Prof. Schuette.

In 1910, structural defects were found in _Schuette I_ when the loads were applied. This has necessitated extensive alterations and much delay. In 1911 it was completed, and sold for L25,000 to the German Army.

SCHUeTTE-LANZ 2. Military. S.L. II. (1913.)

+------------------------------+ | | | _Building._ | | Enlarged edition of above. | | 918,000 c. feet (26,000 m cubed.) | | | +------------------------------+

ZEPPELIN type. Rigid. Graf von Zeppelin, Friedrichshafen.

The features of this type are--A rigid framework of aluminium, a number of drum-shaped gas bags, and a thin outer cover.

[Illustration]

At the end of March, 1913, the total of _Zeppelins_, limit and building was 16, including one (number 18) for Austria. Of these several had come to grief in various ways, and the actual total at the date mentioned, was:--

~8 effective~ = 4 Army (of which one _Z4_ was still on trials), 1 naval and 3 private.

3 completing or building = 1 naval, 1 private and 1 for Austria.

Others projected but not actually in hand.

All are on the lines of the above plan, differing only in minor details, such as the provision of a cabin amidships, etc., and in dimensions.

Details see the following pages.

| | | |

[Illustration: Z1 Military. (1908.) _Obsolete._ Shortly to be struck off list.]

[Illustration: ERSATZ DEUTSCHLAND. (Private.) (1910.)]

[Illustration: Z II. Military. (1911.)]

[Illustration: VIKTORIA LUISE. (Private.) (1912).]

[Illustration: Z III. Military. (1912.)]

[Illustration: Zeppelin dirigible. Sachsen.]

[Illustration: HANSA. (Private). (1912.)]

L I. Naval. (1912.) Armament: 1 gun on top.

+----------+ | | | No photo procurable. | | Generally resembles _Z IV._ | | | +----------+

[Illustration: _Photo. Deliius._

Z IV. Military. (1913). Armament: 1 gun on top, 1 in each gondola, 1 can be lowered from central cabin]

GREEK.

~Aerial Societies:~--

None.

~Aerial Journals:~--

None.

~Military Aeroplanes:~--

At end of March, 1913, these included:--1 _Astra_ hydro. (fitted with Scott's bomb dropper), 2 or 3 _Nieuports_, 1--100 h.p. _M. Farman_, and probably some others. Owing to the war, details are unobtainable. 3 _Bleriots_ reported captured from the Turks, and 15 _Farmans_ on order.

~Military Aviators:~--

Adamis (824 Ae. C. F.) Kamberos (744 Ae. C. F.) Montoussis (839 Ae. C. F.) Mutassas, Sub-lt., naval. Savoff, Lt.

~Flying Grounds~:--

~Phaleron.~

+----------+ | Killed 1913. | | Argyropulus, Lt. | | | | +----------+

ITALIAN.

~Aerial Societies:~--

Ae. C. d'Italia (Ae.C.I.), 62 via Colonna, Rome.

Ae. Club di Roma (Ae.C.I.), 183, via del Triton, Rome.

Circolo, Aeronautico Napoletano, 295 v. Roma, Naples.

Lega Aerea Nazionale (L.A.N.), 6 via della Signora, Milan.

Societa Aeronautica Italiana (S.A.I.), 4 via Boccaccio, Milan.

Societa Aeronautica Italiana (S.A.I.), 6, via Cernaia, Turin.

Societa Aviazone, di Torino (S.A.T.), 28 via Roma, Turin.

Societa Ital. di Aviazone, (S.I.A.), 14 v. Monte Napoleone, Milan.

~Aerial Journals:~--

(3 times a week.)

Gazzetta dello Sport, 15 v. della Signora, Milan. L--.05 (=1/2d.)

(Weekly):

Italia Sportiva, Rome. L--.05 per no. (=1/2d.)

Lettura Sportiva, 17 corso Porta Romana, Milan. L--.10 (=1d.)

Sports (Gli). 46 and dei Prefretti, Rome. L--.05 (=1/2d.)

Stampa Sportiva, 3 v. Davide Bertolotti, Turin. L--.10 (=1d.)

Tribuna Sport, 22 via S. Giacomo, Naples. L--.10 (=1d.)

(Monthly):

Rivista della L.A.N. (Lega Aerea Nazionale), Milan.

Rivista del T.C.I. (Touring Club Italiano), 14 v. Monte Napoleone, Milan. L--.40 (=4d.)

La Navigazione Aerea (Bolletino dell' Ae. C. d'Italia). L--1.80 (=1/6.)

(Annual):

Annuario dell' Aeronautica (Touring Club Italiano), 14 v. Monte Napoleone, Milan. L--6.00 (=5/-)

~Flying Grounds~ (Military see next page):--

~Cameri~, Novara.--15 hangars (Thouvenot school).

~Mirafiore~, Turin.--17 hangars (Asteria and Chiribiri schools).

~S. Giusto~, Pisa.--4 hangars (Antoni school).

~Taliedo~, Milan.--26 hangars.

~Vizzola Ticino.~--7 hangars (Caproni school).

~Dirigible Headquarters~ (with hangars, etc., etc.):--

Bracciano. Milan. Rome. Venice. Verona.

~ITALIAN MILITARY AVIATION.~

~ORGANISATION, etc.~

The _Battaglione Aviatori_ has its headquarters at Turin. In July, 1912, it was re-organised along the following lines:--

1 command at Turin. 1 flying work. 1 technical work. 2 troop duty. 6 at the aviation schools, with a certain number of mobile squadrillos.

The recognised grades are:--

_a.a.p._ aspirante allievo (learners). _a.p._ allievo pilota (certificated pilots). _p._ pilota militare (superior military brevet).

In flying work the superior pilots are mostly using _Bleriots_; the ordinary pilots _Bleriot-Caproni_, _Bristol_, _Antonis_, _Deperdussins_ and _Voisins_.

The technical section chiefly supervises the theoretical instruction of the _a.a.p._

The 2 companies on troop duty practical work, preparation for the schools.

~FLYING SCHOOLS.~

The military schools are:--

~Aviano.~--Central school. Size about 5x2 kilometres. Sheltered from all winds except westerly, by banks of trees. Numerous hangers.

~Mirafiori (Turin).~--Mixed military and civil school. Hangers. School machines confined to _Asteria_, _Bleriot_, _Nieuport_ and _Savary_ types.

~Pordenone.~--School for superior brevets. Treeless plain. Principal school machines _Breguets_ and _Farmans_; but some _Bleriots_ and _Caproni_.

~S. Francesco al Campo.~--_M. Farman_ machines. At present for officers trained in France.

~Somma Lombarda.~--Camp school for _Nieuports_.

~Venaria Reale.~--Formed late in 1912. _Bristol_ monos for certificated pilots.

~GENERAL TRAINING.~

50 officers in training during the first quarter of 1913.

Aspirants commence with instruction in the theory of heavier than air machines, resistance of material and particular instruction in the various type of aero motors in use. They are taken for flights as passengers.

All then go to the training camp about 80% for monoplane work, the remainder for biplanes.

Monoplanes. Special attention paid to teaching _Gauchis Dessent_.

Biplanes. Much shorter course. Principal feature: _Vol Plane_.

For the _military brevet_ the examination is most comprehensive, special attention is paid to flying in wind, manoeuvring, climbing, good landings without inconvenience

to passengers, cross country flights, etc. The course is generally modelled on war experiences.

~TOTAL FLYING STRENGTH.~

No particular distinction between naval and military aviators.

Total by end of June, 1913, to be about 225 certificated aviators of whom a fair percentage hold the superior brevet. In addition all the best civilian aviators are held at disposal.

~Military Aeroplanes.~

At the end of 1911 there were about 20 machines, mostly _Bleriots_ and _Farmans_. The majority of these are still in use for school purposes.

At the end of March, 1913, the machines effective for war purposes were roughly as follows:--

Bleriot. } _Bristol_ (mono.) } _Caproni._ } _Deperdussin._ } Total _about_ 50, plus a number _Farman._ } of school machines. _Hanriot._ } _Nieuport._ } _Savary._ }

About 40 machines were on order, including 12 _Bristol-Capronies_.

~Naval Aeroplanes.~

Effective at end of March, 1913.

1 _Calderara_. 1 _Guidoni-Farman_. 4 or 5 others.

~ITALIAN AVIATORS.~

Military.

Agostoni, Capt. V. (45) Bailo, Lieut. (71) *Bolla, Capt. (89) Cannonieri, Lieut. (22) +Cammarotta, Lieut. (15F) De Filippi, Com. (5) *De Rada, Lieut. (38) *Falchi, Capt. (55) Garassini. (29) *Gavotti, Lieut. (25) Gazzera, Lieut. (20) Guidoni, Capt. (58) *Lampugnani, Lieut. (33) +Manazini, Lieut. (98) Moizo, Capt. (40) Neri, Lieut. (345-Ae. C. F.) (106) *Palmadi, Cesnola Lieut. (75) *Piazza, Major (44) Pizzagalli, Capt. (49) Poggi, Lieut. (82) Prandoni, Capt. (69) *Pulvirenti, Lieut. (50) Raffaelli, Lieut. A. (108) Ravelli (453, Ae. C. F.) Roberti, Lieut. (47) Rossi, Capt. (27) +Saghetti, Lieut. (16) Savoia, Lieut. T. U. (4) Surdi, Lieut. (32) *Vece, Lieut. F. (74) +Vivaldi, Lieut. (31)

Brevets in 1912.

Andriani, Capt. O. (137) Antonini, Capt. L. (91) Almerigi, F. (159) Alvisi, Lieut. A. (172) Baglione, Lieut. A. (129) Baracca, F. (167) Bonamici, L. (101) Bongiovanni, Lieut. E. (115) Bongiovanni, c. L. (124) Bonuti, R. (135) Brach, Lieut. F. (146) Buzzi, Lieut. M. (156) Calderara, Lieut. A. (134) Calori, S. (136) Capuzzo (143) Casabella, Lieut, G. (121) Clerici, Lieut. U. (110) Cuzzo, Capt. A. (166) De Giovanni, Lieut. G. (101) De Riso, Lieut. G. (153) Della Chiesaconte, Lieut. A. (109) Ercole, Lieut. E.

(117) Franceschini, Lieut. E. (112) Gallotti, Lieut. A. (150) Garino ing. G. (134) Girotti, Lieut. M. (100) Gordesco, Lieut. M. (151) *Graziani, Lieut, C. (92) Jacoponi, Lieut. A. (171) Kerbaker, Lieut. E. (99) Laureati, Lieut. G. Leffi dott. sott. med. A. (169) *Mareno, M. A. (90) Moreno, Capt. G. (78) Nosari, G. (142) *Novellis di Coarazze, Capt. A. (94) Oddo, A. (147) Pagano, P. (158) Palpacelli, A. (164) Perrucca, D. (162) Poggioli, Q. (107) Pongelli, R. (60) Porta, Capt. E. (145) Prandoni, Capt. E. (69) Resio, Lieut. (120) Rosetti, A. (157) Russi, Lieut. S. (152) Suglia, Lieut. C. (118) Torelli, F. (165) Valdimiro, Lieut. F. (170) Venanzi, U. (155) Zanuso, Lieut. G. (149)

Naval.

(_To end of 1911_).

Calderara, Lieut. (1) Ginnochio, Lieut. (18) *Rossi, Sub. Lieut. (31) Strobin, Lieut. (39)

(_During 1912_).

De Muro, Lieut. (119) Frigerio, Lieut. (154) Scelsi, Capt. difreg. G.

Private.

(_To end of 1911_).

Akachew, C. (61) Amerigo (3) Barigiola, G. (51) Battagli, B. (34) Bianchi, P. (6) Biego, C. (56) Bigliani, A. (63) Borgotti, G. (43) Brilli, D. G. (48) Brociner, M. (87) Cagno, U. (10) Cagliani, A. (23) Cannoniere, Umberto (22) Cattaneo, Bartelomo (2) Cavaglia, Pietro (30) Cavalieri, Alfredo (17) Cei, J. (53, Ae. C. F.) Casaroni, A. (77) +Cirri, Ciro (11) Cobianchi, Mariot (24) Darioli, Ernesto (9) Da Zara, Leonino (7) De Agostina, A. (53) De Antonis, A. (67) Faccioli, Mario (21) Franzoni, R. (62) Garassini, G. G. (29) Gianfelice (59) Ginnochio, T. (18) Graziani, nob. Ettere (28) Lusetti, A. (19) Maffeis, C. (36) Maggiora, C. (72) Manissero, R. (37) +Marra, R. (35) Marro, E. (52) Mogafico, Mario (26) Mosca, Francesco (47) Pasquali, R. (66) Picollo, G. (32) Poggioli, Quinto (117) Porro, A. (113) Ramasotto, M. M. (148) Ravetto, Clemento (12) Re, Umberto (86) Ruggerone, G. (14) Sabelli, G. (93) Santoni, L. (114) Stucchi, Federico (8) Verona, A. (54)

(_Brevets in 1912_).

Amour, ing. E. Arista, A. (131) Ballerini, M. (132) Bergonzi, P. C. (78) Berni, L. (95) +Bertoletti, R. (79) Borsalino, G. M. (102) Brunetta D'Usseaux, G. (125) Carabelli, C. (104) +Caramanlaki, A. (97) Caramanlaki, G. (168) Carminati di, B. N. (163) Colucci, G. (80) Corsini, J. C. (133) Corsini, A. E. (85) Dalla, N. C. (126) Dal Mistro, C. A. (127) De Campo conte, S. (103) Fabri, A. (165) Facchini, E. (141) Garino, G. (134) Gelmetti, A. (83) Grassi, conte A. (88) Leonardi, G. (122) Mandelli, P. (96) Marazzi, E. (140) Nardini, G. (128) Paolucci, G. (144) Piceller, G. (105) Sacerdoti, C. (116) Salengo, R. (138) Vallet, C. (86) Zorra, L. (84)

~Private Aeroplanes.~

At the end of March, 1913, there were about 45 machines in use at the various private schools, and about 6 privately owned aeroplanes.

ITALIAN AEROPLANES

~A~

ANTONI. Soc. di aviazione Antoni, via Vitt. Emanuele, 46, Pisa. School: S. Guisto, Pisa. Output capacity: about 20 machines a year.

----------+----------+----------+ | ~1912-13.~ | ~1912-13.~ | | Single seat mono. | 2-seater military | | | mono. | ----------+----------+----------+ ~Length~ feet (m.)| 33 (10) | 36 (11) | ~Span~ feet (m.)| 28 (8.50) | 28 (8.50) | ~Area~ sq. feet (m squared.)| 172 (16) | 237 (22) | {machine lbs. (kg.)| 660 (300) | 770 (350) | ~Weight~ { | | | {useful lbs. (kg.)| ... | ... | ~Motor~ h.p.| Gnome or Anzani | Gnome and Anzani | {max. m.p.h. (km.)| ... | ... | ~Speed~ { | | | {min. m.p.h. (km.)| ... | ... | ~Endurance~ hrs.| ... | ... | Number built during 1912 | ... | ... | ----------+----------+----------+

ASTERIA. Fabbr. Ital. Aeroplani ing. Darbesio e. C., via Salbertrand, 12, Torino (Turin). School: Mirafiori. Capacity: small.

----------+----------+----------+ | ~1912-13.~ | ~1912-13.~ | | Monoplane. | Biplane. | ----------+----------+----------+ ~Length~ feet (m.)| 21-3/4 (6.50) | 29-1/2 (9) | {| 26-1/2 (8.10) | 44 (13.50) | ~Span~ feet (m.){| | | {| ... | 24-1/2 (7.50) | ~Area~ sq. feet (m squared.)| 162 (15) | 431 (40) | {machine lbs. (kgs.)| 530 (240) | 110 (500) | ~Weight~ { | | | {useful lbs. (kgs.)| ... | ... | ~Motor~ h.p.| 50 Gnome | 70 Renault | {max. m.p.h. (km.)| ... | ... | ~Speed~ { | | | {min. m.p.h. (km.)| ... | ... | ~Endurance~ hrs.| ... | ... | Number built during 1912 | ... | ... | ----------+----------+----------+

~C~

CALDERARA. Navy hydro-monoplane.

----------+---------- Model ~1912-13.~ | "Hydro vol." ----------+---------- ~Length~ feet (m.)| 54 (16.50) ~Span~ feet (m.)| 61 (18.50) ~Area~ sq. feet (m squared.)| 753 (70) {total lbs. (kgs.)| 2644 (1200) ~Weight~ { | {useful lbs. (kgs.)| ... ~Motor~ h.p.{| 150 (formerly 100 {| Gnome) {max. m.p.h. (km.)| 62 (100) ~Speed~ { | {min. m.p.h. (km.)| 50 (80) ~Endurance~ hrs.| 6-1/2 Number Built during 1912 | 1 ----------+----------

Lieut. Calderara's floats consist of a plurality of w.t. compartments with internal lattice frame, well braced. Hull is formed of three skins of wood, sail-cloth between each. Distance between outer floats, 21 feet (6.30 m.) Centre of gravity is only 4-1/2 feet (1.40 m.) above water. If necessary wings can be cut away and the central hull used as a boat with emergency sail.

[Illustration: CALDERARA. UAS.]

CAPRONI. Soc. di Aviazione Ingg, Caproni e Faccanoni, Vizzola Ticino. School: Vizzola Ticino.

----------+----------+----------+----------+---------- Models ~1912-13.~ | Single Seat | Single Seat | 2-seater mono. | 3-seater mono. | mono. A. |

mono. B. | | ----------+----------+----------+----------+---------- ~Length~ feet (m.)| 26-1/4 (8) | 26-1/4 (8) | ... | ... ~Span~ feet (m.)| 29 (8.80) | 29 (8.80) | ... | ... ~Area~ sq. ft. (m squared.)| 162 (15) | 162 (15) | 172 (16) | 226 (21) {machine lbs. (kgs.)| 485 (220) | 660 (300) | 750 (340) | 760 (345) ~Weight~ { | | | | {useful lbs. (kgs.)| ... | ... | ... | ... ~Motor~ h.p.| 35 Anzani | 50 Gnome | 60 Anzani | 80 Gnome {max. m.p.h. (km.)| 56 (90) | 75 (120) | 75 (120) | 87 (140) ~Speed~ { | | | | {min. m.p.h. (km.)| ... | ... | ... | ... ~Endurance~ hrs.| 3-1/2 | ... | ... | 4 Number built during 1912 | ... | ... | ... | ... ----------+----------+----------+----------+----------

Remarks.--At the end of 1912, held Italian record for speed, 200-300 k.m. Flown by Cobioni.

CAPRONI-BRISTOL. Caproni also builds under Bristol license.

CHIRIBIRI. A Chiribiri e. C, via Lamarmora 28, and via Don Bosco 68-73. Torino (Turin).

[Illustration: CHIRIBIRI.]

----------+----------+----------+----------+---------- Models ~1912-13.~ | 45 h.p. mono. | 50 h.p. mono. | Racing mono. | 80 h.p. mono. ----------+----------+----------+----------+---------- ~Length~ feet (m.)| 23 (7) | 23 (7) | 24-3/4 (7.50) | 25-3/4 (7.80) ~Span~ feet (m.)| 29-1/2 (9) | 29-1/2 (9) | 31 (9.30) | 39-2/3 (12.10) ~Area~ sq. ft. (m squared.)| 204 (19) | 204 (19) | 226 (21) | 258 (24) {machine lbs. (kgs.)| 595 (270) | 683 (310) | 772 (350) | 595 (270) ~Weight~{ | | | | {useful lbs. (kgs.)| ... | ... | ... | ... ~Motor~ h.p.| 45 Chiribiri | 50 Chiribiri | 60 Chiribiri | 80 Chiribiri {max. m.p.h. (km.)| 44 (70) | 56 (90) | 103 (165) | 65 (105) ~Speed~ { | | | | {min. m.p.h. (km.)| ... | ... | ... | ... ~Endurance~ hrs.| ... | ... | ... | ... Number built during 1912 | ... | ... | 2 | ... ----------+----------+----------+----------+----------

~F~

FRIULI. E. Pensuti e E. Calligaro, Pordenone. School: Pordenone. A 30-35 h.p. Anzani motor monoplane. Area, 150 sq. feet. (14 m squared). Generally of _Bleriot_ type, but _Hanriot_ type landing carriage.

~G~

GUIDONI. Naval Hydroavions. Either a _Farman_ biplane or a _Nieuport_ mono. is used, mounted on special floats designed by Capitano del Genio navale Guidoni. There are two long floats, each of which is fitted with parallel fins.

~FOREIGN AGENCIES.~

Foreign types of machines are constructed in Italy under licenses as follows:--

BLERIOT. Soc. Ital. Transaerea, corso Peschiera 25, Torino (Turin).

BRISTOL. (British), by Caproni.

DEPERDUSSIN. Soc. Ital. degli Aeroplani, via Giulini 7b, Milan.

NIEUPORT. Carrozzeria Macchi. Varese.

ITALIAN DIRIGIBLES.

~ITALIAN MILITARY DIRIGIBLES.~

~Army.~ ~Navy.~
/----------^----------\ /----------^----------\ ----------+----------+----------+----------+----------+----------+----------+----------+---------- Name and Date. | ~P1.~ | ~P2 & P3.~ | ~P4 & P5.~ | ~Citta di Milano~ | ~Parseval.~ | ~M1.~ | ~M2 & M3.~ | 1909. | 1910 & 1911. | both 1912. | 1912. | (P.L. 17). | 1912. | 1912 & 1913. | | | | | 1912-13. | | ----------+----------+----------+----------+----------+----------+----------+----------+---------- { c. feet| 148,000 | 155,000 | 166,000 | 424,000 | 353,000 | 424,000 | 424,000 ~Volume~ { | | | | | | | { (m cubed.)| (~4200~) | (~4400~) | (~4700~) | (~12000~) | (~10000~) | (~12000~) | (~12000~) ~Length~ feet (m.)| 197 (60) | 207 (63) | 207 (63) | 233 (72) | 279 (85) | 272-1/3 (83) | 272-1/3 (83) ~Diameter~ feet (m.)| 38 (11.60) | 38 (11.60) | 39-1/3 (12) | 59 (18) | 52-1/2 (16) | 56 (17) | 56 (17) {fabric | Silk | Continental | Continental | ... | Riedinger | Metzeler | Metzeler ~Gas bags~ {compartments | 7 | 8 | 8 | ... | 0 | ... | ... {ballonets | 1 | 1 | 1 | ... | 2 | ... | ... {total tons | 3.50 | 3.50 | 3.75 | ... | ... | 9.50 | 9.50 ~Lift~ {useful tons | 1.10 | 1.35 | 1.50 | ... | 3.00 | 3.80 | 3.80 ~Motor~ h.p. {| 1-100 C. Bayard | 1-120 C. Bayard | 2-80 Fiat | 2-85/100 Isotta | 2-170 Maybach | 2-250 Fiat | 4-125 Wolseley {| (=100) | (=120) | (=160) | (=170/200) | (=340) | (=500) | (=500) {number | 2 | 2 | 2 | 2 | 2 Parseval | 2 | 2 ~Propellers~{blades | 2 | 2 | 2 | 3 | 4 | 4 | 4 {diameter feet (m.)| 10 (3) | 10 (3) | 10 (3) | 14 (4.20) | ... | 12-1/2 (3.80) | 12-1/2 (3.80) ~Max. speed~ m.p.h. (km.)| 32 (52) | 35 (56) | 37 (60) | 45 (72) | 40 (65) | 44 (70) | 44 (70) ~Full speed endurance~ hrs.| ... | ... | ... | ... | 20 | 12 | 12 ~Max. complement~ | 5 | 5 | 5 | ... | ... | 14 | 14 ~Station~ | Bracciano | Tripoli | Vigna di Valle | Baggio | Venice | Bracciano | ----------+----------+----------+----------+----------+----------+----------+----------+----------

Notes: All the above are semi-rigid. The _P_ and _M_ are all of the same general type. Principal features of these ships, which were designed by Captains Crocci and Ricaldoni, are the shape of the envelope, (maximum diameter very far forward), keel and the box kite tail. The _Citta di Milano_ (semi-rigid) is an enlarged _Leonardo da Vinci_ (which see). _Special feature_ is the keel built into the envelope. This serves as a nacelle.

Building.--One _Parseval_ (P.L. 15) about the same size as the other (P.L. 17), which was completed first.

~Army Dirigible Pilots.~

Agostoni, Capt. Biffi, Ten. Bosio, Ten. Crocco, Capt. G. Dal Fabbro, Capt. C. Denti di Piraino, March, Capt. Gallotti, Ten. Longo, Ten. Manni, Ten. Menenti, S. Ten. Merzari, Capt. Messina, Ten. Munari, Capt. E. Pastina, Capt. Ricaldoni, Capt. A.

Scelso, Ten L. Seymandi, Capt. G. Stabarin, Ten. Tagliasacchi, Ten.

~Navy Dirigible Pilots.~

Carniglia, Ten. d. vas. Gravina, Ten. d. v. Conte M. Penco, Ten. d. v., A. Ponzio, Ten. d. v., E. Scelsi, Capt. di f., G. Valerio, Sot. V. Valli, Ten. d. v., G.

[Illustration: P.I.]

Elevation of P. I. The later ships only differ in dimensions, and the fact that the middle rudder is done away with.

[Illustration]

[Illustration: Dirigibles M1 & M2.]

[Illustration: Citta di Milano.]

[Illustration: Parseval (P.L. 17). First Italian _Parseval_.]

~ITALIAN PRIVATE DIRIGIBLES.~

----------+----------+----------+----------+----------+----------+ Name and date. | ~Ausonia bis.~ | ~Italia I.~ | ~Italia II.~ | ~Leonardo da Vinci.~ | ~Usuelli.~ | | Reconstructed 1910.| 1905. | 1913. | 1909. | 1909. | ----------+----------+----------+----------+----------+----------+ ~Volume~ (m cubed.)| (~1500~) | (~1500~) | (~2600~) | (~3265~) | (~3870~) | ~Length~ feet (m.)| 121 (37) | 128 (39) | 164 (50) | 131-1/4 (40) | 167-1/3 (51) | ~Diameter~ feet (m.)| 27 (8.25) | 19-3/4 (6) | 32-3/4 (10) | 46 (14) | 32 (9.80) | {fabric | ... | ... | ... | ... | ... | ~Gas bags~ {compartments | nil. | nil. | nil. | 7 | 6 | {ballonets | 1 | nil. | nil. | 1 | 1 | {total tons| ... | 1.35 | 2.20 | 3.00 | ... | ~Lift~ { | | | | | | {useful tons| 0.80 | ... | ... | ... | ... | ~Motor~ h.p.| 1-55 h.p. S.P.A. |1-40/50 Antoinette| 1-50 h.p. | 1-40 Antoinette | 1-80 h.p. S.P.A. | {number | 1 | 1 | 2 | 2 | 2 | ~Propellers~ {blades | 2 | 2 | 2 | 5 | 2 | {diameter, feet (m.)| 10-3/4 (3.20) | 15 (4.50) | 10 (3) | 9 (2.70) | ... | ~Max. speed~ m.p.h (k.m.)| 25 (40) | 25 (40) | ... | ... | 30 (50) | ~Full speed endurance~ hrs.| ... | ... | ... | ... | 6 | ~Max. complement~ | ... | ... | ... | ... | ... | ~Station~ | Bosco Mantico | Schio | building | Laid up at Baggio | Turin | | | | near Milan | | ----------+----------+----------+----------+----------+----------+----------+----------+

Notes:

~Ausonia.~ Nico Piccoli, via Accademia 12, Padova (Padua). Works: Magre, Vicenza (Schio). Semi-rigid.

~Italia.~ Cont Almerico da Schio, Schio. Non-rigid. _Special features_ is a "belly" of Para rubber in lieu of a ballonet.

~Leonardo da Vinci.~ Ing. Enrico Forlanini, via Boccaccio 21, Milan. Works: Baggio. Semi-rigid, keel and nacelle, incorporated in envelope.

~Usuelli.~ Usuelli and Borsalini, Torino (Turin). Non-rigid.

~Private Dirigible Pilots.~

Forlanini, ing. E. Piccoli, D. Usuelli, C.

[Illustration: Italia.]

[Illustration: Usuelli.]

[Illustration: ~FORLANINI.~ UDS.]

[Illustration: Leonardo da Vinci.]

JAPANESE.

(Naval Aviation data. Official).

~Aerial Societies~:--

Tokio, Ae. Co.

Aeroplane Assoc., 1, Yayesu Cho, I-Chome, Kojimachi, Tokio. (Sec.: Dr. Fujioka).

Kikyu Kinkyu Kai (connected with War Office).

~Flying Grounds~:--

Near ~Yokohama~.

~Saitama~, Tokorozawa (Government).--Dirigible shed and hangars.

~Port Arthur~, (Government).

~General Military Aviation.~

This was originally formed as one body without distinction between army and navy. It was subsequently re-modelled on lines somewhat similar to the British Royal Flying Corps with naval and military wings.

~Navy.~

The naval section is superintended by Capt. K. Yamaji, I.J.N.

The naval headquarters are at Oihama (near Yokosuka).

The naval force at the end of 1912 consisted of 4 hydro-aeroplanes (2 _Curtiss_ and 2 _Farman_). The available total of qualified naval aviators was 5.

~Finance.~

The total amount granted for aviation of the navy in 1912 (fiscal year) was 100,000 yen (L10,000).

For the year 1913 the estimates amount was 100,000 yen (but not approved yet).

~Pay of Flying Officers.~

The special pay for officers employed in aerial work is undecided.

~Army.~

The army wing is responsible for the dirigible. Aeroplanes are one or two _Bleriots_, a _Grade_, 2 _Tokogawa_, and a _Farman_.

~AVIATORS.~

Military.

Hino, Major Saigom, Capt. Tokogawa, Capt. Tokogama, Lieut.

Naval.

Narahara, Naval Constr. Kaneko, Lieut. Kono, Lieut. Obama, Eng. Lieut. Umikita, Lieut. Usuioku, Naval Constr.

Private.

Doig, S. Iga, Baron Shigeno, Baron Tsuzuki, Yamada, Isaburo

The following have been killed:--

+----------+ | 1912. | | Aibata, Lieut. | | | 1913. | | Kimura, Lieut. | | Tokuda, Lieut. | | Takeishi. | +----------+

~Private Aviation.~

There are some private aeroplanes being regularly flown in Japan. A number of aeroplanes have from time to time been invented by naval and military officers and private individuals, and some of them are in use. Inventors include Major Hino, naval constructor Narahara and Ushioki, Baron Iga, Baron Shigeno and Mr. Tsuzuki.

JAPANESE AEROPLANES.

[Illustration: Bleriot (since wrecked). Tokogawa. Wright. Grade. Army Flying School ground.]

[Illustration: Narahara.]

[Illustration: Tokogawa II. Type I the same except for minor details.]

JAPANESE DIRIGIBLES.

PARSEVAL type. Military. (P.L. 13.)

[Illustration]

~Length~ 259 feet (19 m.) ~maximum diameter~ 47-3/4 feet (14.50 m.) ~capacity,~ 8,500 m cubed.

~Gas bag.~--2 ballonets. Usual Parseval.

~Motors.~--Total, 300 h.p., made up of two 150 h.p. Maybach.

~Speed.~--42 m.p.h. (65 km.)

~Propellers.~--Two 4-bladed. Parseval.

~Steering.~--Usual Parseval (see German).

Remarks.--Of _Parseval P.L. 12_ type (see German). Built 1911.

YAMADA. Non-rigid. (Private.)

[Illustration: _Photo by favour of M. Samuro Kuki._]

~Maximum length~, feet (m.) ~maximum diameter~, feet (m.) ~volume,~ 700 m cubed. (_about_)

~Gas bag.~--

~Motor.~--American make.

~Speed.~--

~Propeller.~--One.

~Steering.~--Biplane elevator forward. Triangular rudder in rear under gas bag.

Remarks.--Generally of American type.

MEXICAN.

~Army Aeroplanes.~

There are 2 old pattern _H. Farman_; also one or more _Curtiss_ and _Wright_ machines. Nothing seems doing with them.

~AVIATORS.~

Military.

Martinez, N. (Ae. C. F. 462) Mendia, (Ae. C. F. 680)

Private.

Duval, Raoul Lebrija, Miguel Morales Noriega Ramsey, E. L. Saavedra, Alfonso

Probably 2 others

(The above are mostly amateur builders.)

NORWEGIAN.

~Aerial Societies:~--

Aero Club, Norsk Flyveselskad (Christiana). Secretary, D. Barth.

Norsk Luftseilads Forening (Christiana). President, H. Mohn.

~Aerial Journals:~--

None.

~Flying Grounds:~--

~Military Aviation.~

At the end of 1912 the Army possessed two 70 h.p. _M. Farmans_ (Renault motors), and the Navy a 100 h.p. N.A.G. _Rumpler_.

For 1913 the purchase of further machines is contemplated for both arms.

~Private Aeroplanes.~

Total at end of ~1911~ 1

At end of 1912 there were in existence 2--a _Grade_ and a _Deperdussin_.

~AVIATORS.~

~Military.~

Dichi, Lieut. Jacobsen, Lieut.

~Private.~

Hansen. St. Dons.

PERUVIAN.

~Military Aeroplanes.~

The Peruvian Government has made a special grant for aviation students, and war machines are projected. Actual order to end of 1912 was one _Avro_ mono.

~Private Aeroplanes.~

Total at end of ~1910~ 3 " ~1911~ 2 " ~1912~ _none_ probably.

~AVIATORS.~

Bielovucic, J. Chavez, J. Monterc (766 A. C. F.)

Peruvian aviators killed:

+----------+ | 1910. | | Chavez, G. | | | | 1911. | | Tenaud, C. | | | +----------+

PORTUGUESE.

(Revised by J. SCHIERE, Aeronautical Engineer.)

~Aerial Societies:~--

Ae. C. de Portugal (R. Nova docklaemada d. ISL.)

~Aerial Journals:~--

Rivista Aeronautica (Ae. C. Journal.)

~Flying Grounds:~--

Campo do Seigcal.

Mounchavo da Povoa.

~Private Aeroplanes.~

Total at end of ~1910~ 1 " ~1911~ 2 " ~1912~ 2

~Private Aviators.~

De Castro, Sanchez De Silva, Gomez

~Military Aviation.~

In 1912 a military corps was formed. At the end of 1912 it possessed _Avro_ (1--50 h.p.), _Voisin_ (1--80 h.p.), and _M. Farman_ (1--80 h.p.) (since wrecked). 1 _Deperdussin_.

~Private Aviation.~

In 1911 the _Gouveia_ mono. was built, span 9 metres, but it failed to fly. Also the _Avante_ biplane, which also failed. First flight in Portugal by a Portuguese was De Castro in September, 1912, with an old _Bleriot_.

ROUMANIAN.

~Army Aeroplanes.~

At end of March, 1913, there were several 80 h.p. _Bristol_ monos., 2 _Bleriots_, 1 _Nieuport_, 1 _Morane_, 2 _Vlaiclu_, and several _H. Farman_ biplanes.

Government school is at Bucharest.

~AVIATORS.~

Military.

Capsa, Lieut. Negrescu, Lieut. Protpopscu, Lieut. Vacas, Lieut. Poly Zorileann, Lieut. (Ae. F. 587)

Private.

Bibesco, Prince (Ae. C. F. 20) Oznoth

VLAICLU Monoplane. Designed by Ouvret Vlaiclu. First shewn at the Vienna Exhibition, 1911. Modified; it flew very well indeed at Aspern, June, 1912. The 1912 model is of entirely novel type, a tail first monoplane with a propeller either end of the main planes, and a triangular tail aft. Principal details are:--~Length~, 34-2/3 feet (10.50 m.) ~Span~, 30 feet (9.15 m.) ~Height~, 12 feet (3.65 m.) Wing frame in three sections with gap between. ~Motor~, 50 h.p., Gnome chain driven. ~Fuselage~, old style; landing chassis on three wheels only, with a single ash skid in front. Covered in engine driving the 31 foot propeller shaft for the 2 propellers. Rear tail consists of 2 fixed planes, a triangular damping plane and a triangular keel plane. Forward, an elevator and two semi-circular rudders (double faced). From this combination remarkable results are achieved, and all gyrostatic effect from the propellers eliminated. _Control_, horizontal wheel on column. Elevator depressed or otherwise by action on column. Note.--At Vienna, 1912, this machine took first prize for the smallest circle and also for accurate bomb-dropping. The original machine was purchased by the Roumanian Army.

RUSSIAN.

~General Note.~--In the number of military machines and general attention to aviation Russia is only second to France. There are no effective machines of Russian design, but the Aviataka, Dux & Lomatuk firms build at home under foreign license, and there is also the Kennedy school (Anglo-Russian).

~Aerial Societies:~--

(Imperial) Aero Club. 1. Odessa branch. 2. Rostow and Don branch. 3. St. Petersburg "

Finland Ae. C., Helsingfors.

Kieff University Ae. C., Kieff.

Moscow Ae. C., Moscow.

Moscow Imp. Tech. College (Aviation Section).

Riga Ae. C., Riga.

Russian Aeronautical Society, St. Petersburg.

Sevastopol Ae. C.

Students' Aviation Club.

Tomsk Ae. C., Tomsk.

Volunteer Aerial Fleet.

~Aerial Journals:~--

Aeronautical Journal of St. Petersburg.

Aero (6, Liteiny, St. Petersburg). Weekly.

Dans l'Empire des Airs (7, Rota 26, Petersburg). Fortnightly.

Revue de Navigation Aerienne (7 rue Stremmiannaya, Petersburg). Weekly.

Sport (25, Ekaterineska, Odessa).

Wozdookhoplavatel (St. Petersburg). Monthly.

Wosduchoplawanie y Sport (Moscow). Monthly.

~Flying Grounds:~--

~Gatchina Park.~-- Flying here under restrictions. V.F. school.

~Kieff.~--School for pilots.

~Kolomiaggi.~--Racecourse.

~Novo Therkask.~

~Odessa.~

~St. Petersburg.~--Kennedy school.

~Sevastopol.~--Volunteer Fleet school.

~Warsaw.~

~RUSSIAN MILITARY AVIATION.~

~Army Aviation.~

Early in 1912, under the presidency of the Grand Duke Alexander, the special school of the Volunteer Aerial Association was finally formed at Sevastopol for the winter and Gatchina for the summer.

June 1912. Vote for 150 aeroplanes (140 to be built at home). Vote 1,050,000 roubles for new school at Tauride.

November, 1912. Military trials results. (1) Sikorsky in a _Sikorsky_. (2) Haber in a _M. Farman_. (3) Boutmy in a _Nieuport_.

December, 1912. Aeronautical school re-organised. Put under control of one commandant, one assistant, and four juniors. Course made seven months--15 pupils per school at a time. A one month course in aeroplanes, aerial motors, etc. Of the pupils, 10 will be selected for aeroplanes. New flying school established at Taskend in Turkestan.

March, 1913. New schools established at Moscow, Odessa and Omsk.

At the end of 1911 the total number of military aeroplanes was about 100. At the end of March, 1913, the total number was about 250, of which about 150 were modern. Principal types: _Albatross_, _Aviatik_, _Bristol_, _Deperdussin_, _Farman_, _Nieuport_, _Rumpler_, there being an average of 20 of each. The majority built under Russian license in Russia. The number of actual military pilots was 72. There is, however, a special volunteer corps of about 36 private aviators, bringing the available total to 108 or thereabouts.

~Navy Aviation.~

July, 1912. Lieut. Andreadi, 50 h.p. _Nieuport_, did a flight with stops from Sevastopol to St. Petersburg.

September, 1912. Special naval aerodrome for hydro-avions ordered for Golodai Island, near Petersburg, bringing total of military and naval aerodromes to 6. _Sikorsky_ hydro-avion acquired. Also an _M. Farman_ ditto. New naval station projected at Libau.

October, 1912. Naval purchase of several _Curtiss_ hydro-avions after trials at Sevastopol.

At the end of March, 1913, the approximate effective force was as follows (all hydros, or capable of being so fitted): 1 _Astra_, 1 _Breguet_, 2 _Donnet-Leveque_, 1 _Farman_, 4 _Paulhan Curtiss_, 2 _Nieuport_ (50 h.p.), 1 _Sikorsky_. (A number of others on order.)

Early in 1913 experiments were carried out with a combination of floats and skids, invented by M. Lobanoff, of Moscow. This proved equally effective on land or water.

~AVIATORS.~

The following are army, navy or volunteer aviators. The number is the Russian Ae. C., unless otherwise stated. F = French. Prefix + = killed. n = navy.

Abramowitch Wissewold (14) Agababa, N. (668 F.) Agofonoff (20) Aleknovitch, G. (29) Alexandroff, D. (472 F.) n Andreadi, Lt. Artsgouloff (44) Avinass, J. (60) Badowski, L. Bakhmoutoff, N. (6) Berdchenko, V. (7) Bistritsky, V. (8) Boukshevden, Bar. G. (10) Boutmy (de), E. Campo, Scipio (211 F.) Childovski (67) Chioni, B. (250) Chimansky (27) Choudinoff (46) Dmitrieff, J. (9) Dorogouski (125 F.) Dougowezky, A. (1) n Dybovski, V. (12) Efimoff, M. (31 F.) Efimoff, T. Erdeli, G. (45) Eristov, Prince (524 F.) Evsukoff, P. (21) Firstemberg Flegfier, von. Gelgar (33) Glouchenko, S. (48) Godoulsky, A. (59) Gorghkoff, G. (626 F.) Goumberto-Dros, B. (58) Grekoff G. (5.) Grigoraschirilly (577 F.) Houeninsey, A. (227 F.) Husarenko (22) Illin, A. (16) Iougmeister (52) Jankovsky, G. (24) Joukoff (37) Kaidenoff (42) Kamensky, V. (66) Katzian, A. + Kauzminski (228 F.) Kebouroff, V. (210 F.) Kirchstern Kolchin, F. (28) Komaroff, M. (245 F.) Kostine, N. (223 F.) Kauznezoff, P. Kreiner, E. Kroumm, A. Lachtionoff, G. (57) Lambert (de) C. (8 F.) Lebedeff, V. (98 F.) Lerche, M. (25) Lewkowicz, H. (327 F.) Linno, G. (15) Lipowski, H. (330 F.) Kokteff (61) Makaroff, D. (13) Makeef, P. (5) Matyevitch, Matzevitch (152 F.) n+Matyevitch, Capt. (178 F.) Meybaum, T. Miller (35) Monakoff, (565 F.) Naidenoff, G. Naslennikoff, B. Nikiforoff (18) Nikolaieff (49) Nikolsky, P. (17) Oulianine, S. (181 F.) Pehanovsky, B. (401 F.) + Pietrowsky, G. (195 F.) Porcheron, J. (640 F.) Popoff, N. (50 F.) Poliakoff, A. (50) Poplavko (34) Pongolowski, W. (4) Pristchepoff (38) Raevsky, A. (F.) Raygorodsky, A. (207 F.) Rossinsky (68) n Rouaroff, M. (245 F.) Rynin, N. (23) Sakoff, N. de (627 F.) Salesky (41) Samoilo (11) Samouiloff, P. (51) Semeniovitch (226 F.) Semenko-Slavorossoff, H. (40) Semitan (36) Seversky-Prokofieff, N. (47) Sewkowicz, L. Shidloovsky, M. Shimansky, K. Shimkevitch, V. Sikorsky, I. (63) Skarginsky, A. (43) Slusarenko, W. + Smith, V. (231 F.) Sobansky Graf. (3) Soechnikoff, A. Soupnevsky, C. (26) Springuefeld Sredinsky, A. Strelmkoff (71) Tchemiakoff (72) Tkatcheff, V. (64) Tounochensky (32) Tselary, I. (54) Wassilieff, A. (225 F.) Zaikine (191 F.) Zelinsky, Col. (273 F.) + Zolotouchin, M. (31)

~CIVILIAN AVIATORS.~

There are very few purely civilian aviators in Russia. Russians who have obtained brevets include Mdlles Anarta (52), Golantchikova (55), Zvereva (30), Count de Lambert, (8 F.) and Count Malynski (209 F.) and one or two others. Few or none do any flying now.

RUSSIAN AEROPLANES.

~A-Z~

AVIATIK. St. Petersburger Aviatik Gesellschaft, Petersburg. Construct Aviatiks. (See Germany.)

BRONISLAWSKI. Experimental biplane with special stabilising features.

DUX. Fabrica Moscovita Tneerskaja "Dux," Lastawa, Moscow. Construct under license.

GELTOUCHOW. W. G. Geltouchow and A. W. Preiss, 4 Piasnitzkajai, Moscow. Constructs.

GILBERT. C. Gilbert, 195 Twerskaja, Moscow. Constructs.

KENNEDY. Soc. d. Dirigibles and Aeroplanes Kennedy, St. Petersburg.

MOTOR. Riga-Sassenhof.

RODJESTVEISKY. Built a triplane in 1911.

RUSSIAN MILITARY DIRIGIBLES (13).

----------+----------+----------+----------+----------+----------+----------+----------+----------+---------- | (1) | (2) | (3 & 4) | (5, 6, 7) | (8) | (9) | (10) | (11) | (12) | (13) Name | ~LEBEDJ.~ | ~KOMMISSIONY.~ | ~JASTREB~ and | ~ZODIAC VII,~ | ~PARSEVAL.~ | ~FORSZMANN I.~ | ~FORSZMANN II.~ | ~ASTRA 13.~ | ~PARSEVAL 14.~ | ~C. BAYARD 6,~ | | | ~GOLOUBJ.~ | ~VIII & IX.~ | | | | | _bis._ Make | Lebaudy. | C. Bayard I. | Outchebny I & II. | | Parseval 7. | Forszmann. | Forszmann. | | | Date | ~1910.~ | ~1910.~ | ~1910-11.~ | ~1910-11.~ | ~1911.~ | ~1911.~ | ~1912.~ | ~1913.~ | ~1913.~ | ~1913.~ System | Semi-rigid. | Non-rigid. | Semi-rigid. | Non-rigid. | Non-rigid. | Non-rigid. | Non-rigid. | Non-rigid. | Non-rigid. ----------+----------+----------+----------+----------+----------+----------+----------+----------+---------- ~Volume~ c. feet (m cubed.)| ~3700~ | ~3000~ | ~1500~ | ~2140~ | ~7600~ | ~800~ | ~600~ | ~9800~ | ~10,000~ | ~6200~ ~Length~ feet (m.)| 200 (61) | 184 (56.25) | ... | 164 (50) | 236 (72) | 121-1/2 (37) | ... | 259 (77.80) | 279 (85) | 250 (77.60) ~Diameter~ feet (m.)| 35-1/2 (10.80) | 34-3/4 (10.58) | ... | 29-1/2 (9) | 46 (14) | 19-3/4 (6) | ... | 49 (14.90) | 52-1/2 (16) | 42-3/4 (13) {fabric | Continental | Continental | ... | Continental | Continental | ... | ... | Continental | Reidinger | Continental ~Gas Bags~{ballonets | 1 | 1 | ... | 1 | 2 | ... | ... | 2 (3100 m cubed.) | 2 | 2 {compartments| 3 | 2 | 2 | ... | ... | ... | ... | ... | ... | ... ~Lift~ {total tons | 4 | 3-3/4 | ... | 2 | 7 | 1/2 | 1/3 | ... | ... | 7-1/2 {useful tons | 1-1/4 | 1 | ... | ... | ... | ... | ... | nearly 4 | about 3-1/2 | 2-3/4 ~Motor~ h.p.| 1-70 Panhard | 1-105 Clement B. | 1-75 E.N.V. | 1-60 Labor | 2-110 N.A.G. | 1-24 (=24) | ... | 2-150 Chenu | 2-180 Maybach | 2-130 Clement B. | (=70) | (=105) | (=75) | (=60) | (=220) | | | (=300) | (=360) | (=260) ~Propellers~ number| 2 | 1 | 1 | 1 | two 4-bladed | 1 | 1 | 3 | two 4-bladed | 2 ~Speed~ m.p.h. (km.)| 30 (49) | 33-1/2 (54) | 13 (21) | 33-1/2 (54) | 37 (59) | 23 (37) | ... | 36 (60) | 43 (68) | ... ----------+----------+----------+----------+----------+----------+----------+----------+----------+---------- -------- _Notes_ | _ex La Russie._ | _Jastreb_ reported | | | Carries 500 | | One-man | Carries 740 litres | | Special 2 speed | | wrecked, March, | | | litres of petrol. | | dirigible. | petrol. Crew 6. | | gear to propellers. | | 1913. | | | Has done 6-1/3 | | | Weights: | | | | | | hours at 1500 | | |Crew: 1044 lbs. | | | | | | metres, with 9 | | |Tools,

&c. 220 " | | | | | | | on board. | | |Petrol, oil, &c. | | | | | | | | | | 7307 " | | | | | | | | | | ---- | | | | | | | | | | 8541 " | | | | | | | | | | ---- | | | | | | | | | |Forward propeller | | | | | | | | | |6m. diameter; the | | | | | | | | | |two after ones 3 m. | | | | | | | | | | each. | | ----------+----------+----------+----------+----------+----------+----------+----------+----------+----------

Note.--Illustrations see next page.

[Illustration: Lebedj. UDS.]

[Illustration: Jastreb (Outchebny).]

[Illustration: Kommissiony. UDS.]

[Illustration: FURSZMANN.]

Note.--The other dirigibles are of usual type. See France and Germany.

SERVIA.

~Military Aviation.~

At end of March, 1913, there were 7 aeroplanes, and 3 more (_Bleriots_) on order.

SPANISH.

~Aerial Societies:~--

El Real Aero Club de Espana (70 rue Alcala, Madrid).

La Asociacion de Locomocion Aerea (20 Plaza de Cataluna, Barcelona).

Real Aero Club d'Espana.

Cataluna Ae. C.

~Aerial Journals:~--

Boletin Oficial de la Asociacion de Locomocion Aerea, 20, Plaza de Cataluna, Barcelona (monthly).

Espana Automovil, 5, plaza de Isabel II, Madrid. Official organ, Spanish R. Ae. C.

Revista de Locomotion Aerea, 20, Plaza de Cataluna, Barcelona (monthly).

~Flying Grounds:~--

~Carbouchelle~ Military School.

~Army Aeroplanes.~

There are 9 old _Farmans_ (1910-11 model), and one or two more modern monoplanes: but little is doing.

Some hydro-aeroplanes are on order for the Navy.

~AVIATORS.~

Military.

Adaro, Lt. J. Alfaro, Lt. H. Arridaga, Capt. Berron, Lt. E. Echevarria, J. Gonzales, Capt. C. J. Granche Kindelan, Capt. A. Menendez, M. Ortiz, So. Lt. J. Penas, M. de las Pujo, Capt. (467 F)

Private.

Campano Dras, J. F. Jezzi, R. G. L. (British Ae. C. 44)[F] Lailhacar, de Pascal, Ferdinand Pimentel, B. L. Prince Alphonse d'Orleans (1)

The following Spanish aviators have been killed:--

+----------+ | 1909. | | Fernandez, A. | | | | 1911. | | Pola, M. | | Mauvais | | | | 1912. | | Bayo, Capt. | +----------+

~Military Dirigible Pilots.~

Herrera, Lt. E. Kindelan y Duany, Capt. A. Vives y Vich, Col.

SPANISH DIRIGIBLES (Non-rigid).

ESPANA. Military. (ASTRA class.)

[Illustration]

~Maximum length~, 197 feet (60 m.) ~maximum diameter~, 35-1/3 feet (10.75 m.) ~volume,~ 43,057 c. feet4,000 m cubed.

~Total lift.~--9,700 lbs. (4,400 kgs.) ~Useful lift~, ? lbs. (? kgs.)

~Gas bags.~--Yellow coloured rubber proofed Continental fabric.

~Motor.~--One 100 h.p. 4-cylinder Panhard.

~Speed.~--29 m.p.h.

~Propellers.~--1, at the front end of the car, of wood, "Integrale" type.

~Steering.~--As in _Clement Bayard I_ and _Ville de Nancy_.

Remarks.--The two side stabilising shapes are duplicated, as they are in the _Ville de Paris_. A webbing stretched on steel tubes is introduced between the inner edges of the 4 main stabilising shapes to provide extra stabilising surface.

TORRES-QUEVEDO II. Military.

+----------+ | | | | +----------+

~Maximum length~, 147-3/4 feet (45 m.) ~maximum diameter~, 32-3/4 feet (10 m.) ~volume,~ 56,700 c. feet (1,600 m cubed.)

~Total lift.~-- ? lbs. (? kgs.) ~Useful lift~, ? lbs. (? kgs.)

~Gas bags.~--

~Motor.~--60 h.p. Chenu.

~Speed.~--

~Propellers.~--

~Steering.~--

Remarks.--Designed by Captain Kindelan and Engineer Torres Quevedo.

SWEDISH.

~(Revised by Lieut. DAHLBECK, R. Swedish Navy.)~

~Aerial Societies:~--

Svenska Aeronautiska Saellskapet (Stockholm).

Kungl. Automobil klubben: (Fenixpalatset, Stockholm).

Svenska Motor-klubben: Aero sektion (Stockholm).

~Aerial Journals:~--

Svensk Motor-Tidning (Fenixpalatset, Stockholm) Fortnightly.

~Flying Grounds:~--

~Ljungbyhed~ (Skane), sheds.

~Malmsl[~a]tt~, sheds.

~Military Aeroplanes.~

At the end of March, 1913, the Army possessed 1 monoplane, 1 biplane, and 2 biplanes building.

The Navy had 1 _Bleriot_ type monoplane and 3 building.

At the end of 1912 there were 9 privately owned aeroplanes.

~AVIATORS.~

(The number against any name is, unless otherwise stated, the Ae. C. Swedish pilot certificate.)

Military.

von Porat, Lieut. (6) Ljungner, Lieut. (7) Hamilton, Capt. (2)

Naval.

Dahlbeck, Lieut. (3) (British Ae. C. 120) Werner, Lieut. (9)

Private.

Cedarstr[~o]m, Baron C. (1) Fj[~a]llb[~a]ck (4) Angstr[~o]m (5) Sundstedt (8) Thulin, M.A. (10)

SWEDISH AEROPLANES.

ASK. Monoplane.

[Illustration: _Harlan_ type. Built by Ask, 1911.]

NYROP. Naval Monoplane.

[Illustration: _Bleriot_ 2-seater. Built in Sweden by Nyrop, 1911. ~Motor~, 50 h.p. Gnome.]

DAHLBECK.

[Illustration: _Farman_ type. Built by Lieut. Dahlbeck. 1913.]

SWISS.

(By our special Swiss editor.)

~Aerial Societies:~--

Aero Club Suisse (3, Hirschengraben, Berne). Sec.: F. Filliot.

a Ostschweizerischer V. fuer L. (Zuerich).

b Sektion Mittelschwerz (Bern).

c Sektion Westschweiz (Romande) (Lausanne).

d Club Suisse d'Aviation (Geneva).

Club Genevois d'Aviation (Geneva). Sec.: P. Brasier.

Fluegsport Klub (Rorschach). Sec.: A. Zuern.

~Aerial Journals:~--

Bulletin de l'Aero Club Suisse (Berne). Monthly.

La Suisse Sportive (16, Rue de Hesse, Geneva). Weekly.

Sport (35, Boulevard Exterieur, Berne).

Automobil Revue (Berne). Weekly.

Le Sport Suisse (Geneva). Weekly.

L'Auto Sport (Geneva). Weekly.

_A.C.S._ (Swiss Aut. Clubs) (Geneva). Fortnightly.

Das Illustrierte Programm (Zurich). Fortnightly.

Revue Weinfelden. Monthly.

~Flying Grounds:~--

~Avenches.~

~Collex-Versoix.~ (Club Suisse d'Aviation).

~Lucerne.~--60 acre park. Sheds.

~Petit Lancy.~ Geneva (Geneva Club).

~Duebendorf bei Zurich.~

~Dirigible Station~ (with hangars):--

~Lucerne.~

~Army Aeroplanes.~

At the end of March, 1913, there were no army aeroplanes, a _Farman_ bought in 1911 having ceased to exist.

~Private Aeroplanes.~

Total at end of ~1910~ _about_ 10 " ~1911~ " 15

At the end of March, 1913, there were about ~15~ privately owned aeroplanes.

~AVIATORS.~

(The number against any name is, unless otherwise stated, the Ae. C. Suisse pilot certificate number.) + = killed.

Military.

Real, Lieut. T. (4) + Schmidt, Capt. J.

Private.

Audemars, E. (7) Bianchi, P. (6) Bider, O. (32) + Blane, M. (17) Bucher, M. (11) Burkard, H. (20) Burri, E. (24) Casser, E. (28) + Cobioni, E. (15) Domenjoz, J. (10) Durafour, F. (3) Failloubaz, E. (1) Grandjean, R. (2l) Gsell, R. (12) + Hoesli, G. (25) Hug, M. (18) Ingold, K. E. (35) Jucker, A. (13) Kramer, H. (31) Mallei, A. (23) Parmelin, A. (22) + Primavesi, E. (34) Rech, E. (29) Rettig, J. J. (27) Reynold, M. (19) Ruchonnet, E. (5) Rupp, A. (9) Salvioni, C. (16) + Schmid, H. (14) Schumacher, J. (26) Taddoli, E. (2) Trepp, M. (30) Wyss P. A. (8) Zuest, B. (33)

SWISS AEROPLANES.

[Illustration: Grandjean.]

[Illustration: Taddeoli.]

[Illustration: Wetterwald.]

----------+----------+----------+----------+ | ~GRANDJEAN.~ | ~TADDEOLI.~ | ~WETTERWALD.~ | Model and date. | Hydro-monoplane. | Monoplane. | Monoplane. | | ~1911-12.~ | ~1911-12.~ | ~1912.~ | ----------+---------- +----------+----------+ ~Length~ feet (m.)| 33 (10) | 19-3/4 (6) | 24-1/2 (7.50) | ~Span~ feet (m.)| 33 (10) | 29-1/2 (9) | 33 (10) | ~Area~ sq. feet (m squared.)| 191 (18) | 151 (14) | 215 (20) | { total lbs. (kgs.)| 750 (340) | 880 (400) | 705 (320) | ~Weight~ { | | | | {useful, lbs. (kgs.)| 310 (140) | 330 (150) | ... | ~Motor~ h.p.| 50 Oerlikon | 50 Gnome | 40 E.N.V. | {max. m.p.h. (km.)| 62 (100) | 69 (110) | ... | ~Speed~ { | | | | {min. m.p.h. (km.)| 56 (90) | ... | ... | Number built during 1912 | 2 | 1 | 1 | ----------+----------+----------+----------+

TURKISH.

~Army Aeroplanes.~

There is a military aerodrome at S. Stefano, with Amerigo, Renzel and Thanlau as instructors.

In March, 1913, there were about 12 monoplanes (_Harlans & Reps_), and one or two biplanes. Only one seems actually to have been used. Several other aeroplanes were captured during the war--generally in their packing cases unopened. In April, 50 machines were reported ordered in Germany.

~AVIATORS.~

Military.

Fessa, Bey (780, F.) Kienan, Lt. (797, F.) Nouri, Lt. Ratzian Refik, Capt. Sismanoglou, J.

URUGUAY.

~Aeroplanes in the country.~

None.

~AVIATOR:~ Cameo, M. Garcia

U.S.A.

(Edited by E. L. JONES, Editor of "Aeronautics," U.S.A.)

~General Note.~--In the early nineties, Professor Langley and the Bros. Wright were experimenting with heavier-than-air machines, but general interest in the subject is quite recent. Though some small dirigibles exist, American attention is mainly devoted to aeroplanes. Ballooning was quite the thing in 1907-11, but has languished.

It is stated that there are certainly no less than _two thousand_ people in the U. S. A. who have built flying machines. The greater percentage of these have been home-made copies of standard machines. Individual builders of copies and freaks have diminished greatly in numbers, and there remains a few well-established manufacturers.

Although inventors are still prolific in the Patent Office and clubs numerous, the general public takes very slight intelligent interest in aviation. The majority of clubs are inactive.

In the year 1912 commercial development seemed to have great possibilities. The copyists were being weeded out and competent aeronautical constructors financed by adequate means began operations on systematic business lines. The latter half of the year saw a great slump. In the spring of 1913 prospects looked greatly improved, and there was generally increased activity.

~Aerial Journals:~--

~Aeronautics.~--122, East 25th Street, New York. Monthly.

~Aircraft.~--37, East 28th Street, New York. Monthly.

~Fly.~--1701, Chestnut, Philadelphia, Pa. Monthly.

~Aero.~--Chicago, Ills. Weekly.

~Flying Grounds:~--

~Belmont Park, N.Y.~--Old race track. Not very good. Scene of 1910 meet. 30 sheds occupied by few experimenters.

~Dayton.~--_Wright_ school private field.

~Chicago, Ills.~--Two fine fields.

~Fort Myer, Va.~--Government and private sheds.

~Hammondsport, N.Y.~--_Curtiss_ factory. Field (small) and lake for water planes.

~Los Angeles, Calif.~--Several fields in vicinity. Used for _Eaton_ school and private flyers.

~Marblehead, Mass.~--Poor field. Home of Burgess C. Fine for hydro-aeroplanes.

~Mineola, N.Y.~--_Moisant, Sloane_ and another school, and individuals. About 1 by 10 miles level field, without obstructions.

~Oakwood Heights, Staten I., N.Y.~--The Aeronautical Soc. grounds, on bay for use of hydro-aeroplanes.

~San Diego, Calif.~--Winter quarters _Curtiss_ camp; also used by army flyers.

~San Francisco~ (near).--Good.

~St. Louis, Mo.~--Kinloch Park. _Benoist_ school and private owners.

~U. S. A. AERO CLUBS.~

An attempt has been made here to give the name of every aero club that has been formed recently in the United States, or has been in existence for a long time. It is believed this list covers every club in the United States.

Many of these clubs are nothing but a name. They were formed to conduct meets or exhibitions, given by the various aeroplane concerns engaged in this business. Many clubs are not incorporated. Others have no organisation, being run by principals of boys' schools or classes.

Clubs even affiliated with the Aero Club of America have no members' meetings, nor have they in many cases even meeting rooms. There are but a half-dozen live aero clubs worthy the name in America. Three clubs own balloons, which are rented to members for ascensions. Little attempt is made by more than one or two associations to popularise aeronautics, to encourage experimenters, or to indulge in scientific work.

The Aero Club of America, the Aeronautical Society and Aero Club of Illinois, are the principal organisations. The Ae. C. represents the F.A.I., and has a beautiful club house. The Aeronautical Society has rooms in the United Engineering Building,

conducts well-attended lectures twice a month, and has grounds on Straten Island (for hydro-aeroplanes and aeroplanes).

Clubs affiliated with the Ae. C. of America are marked *

~CALIFORNIA.~

New Orleans Aero Club, Wm. Allen, Sec., New Orleans

*Aero Club of California, Prof. H. La V. Twining, Pres., 1308 Calumet St. Los Angeles.

*Pacific Aero Club, Pacific Buildings, 331 Octavia Street, San Francisco

Postal Aero Club, 305 W. Santa Clara Street, San Jose

University of California Aero Club, T. W. Veitch, Sec., Berkeley

Oakland Aero Club, Oakland

*Aero Club of Colorado, 36 West Colfax Avenue, Denver, Col.

Aero Club of Blackstonehill, Oakland, Calif, c/o W. R. Davis, Jr., 474 Prospect Street

Curtiss Amateur Aviation Club, Harold Scott, Secretary, Los Angeles

Santa Clara Valley Aero Club, Chamber of Commerce, San Jose

Aero Club of San Diego, San Diego, Colonel C. C. Collier, Pres.

Aero Club of Pasedena, W. J. Hogan, Pres., 635 Chamber of Commerce, Box 1054

~CONNECTICUT.~

*Aero Club of Connecticut, Pres., A. Holland Forbes, at Fairfield

Yale University Aero Club, New Haven

Aero Club of Hartford, Hiram Percy Maxim, Pres., Hartford

~CUBA.~

*Aero Club de Cuba, Ignario 5, Havana

~DELAWARE.~

Aero Club of Delaware, Wilmington

~DISTRICT OF COLUMBIA.~

Washington Aero Scientific Club, F. L. Rice, Sec., c/o Y.M.C.A., Washington

*Aero Club of Washington, Dr. Albert F. Zahm, Sec., Cosmos Club, Washington

~FLORIDA.~

Aeronautic Society of Florida, Davenport and Kerrison, Secs., 2014 Main Street, Jacksonville

~ILLINOIS.~

*Aero Club of Illinois, F. McCormick, Pres., 240 Michigan Avenue, Chicago

Aeroplane and Kite Club, E. E. Harbert, Pres., 2852 N. Clark Street, Chicago

University of Chicago Aero Club, Chicago

*Aircraft Club of Peoria, c/o Leslie Lord, 505 E. Armstrong Street, Peoria

Aeronautical Society of the University of Illinois, Urbana, R. Watts, Sec., 507 E. John Street, Champaigne

~INDIANA.~

Purdue Aero Club, Purdue University, Lafayette

South Bend Aero Club, South Bend.

*Aero Club of Indiana, Indianapolis

~KANSAS.~

Aero Club of Topeka, Topeka

*Western Aero Association, E. S. Cole, Sec., Topeka

Kansas State Aero Club, C. H. Lyons, Sec., Overland Park

~KENTUCKY.~

Continental Aero Club, Richmond

~LOUISIANA.~

Southern Aero Club, 809 Canal Street, New Orleans

New Orleans Aero Club, Wm. Allen, Sec., New Orleans

~MARYLAND.~

*Aero Club of Baltimore, Col. Jerome H. Joyce, Pres., Baltimore

~MASSACHUSETTS.~

Aero Club of North Adams, North Adams

*Aero Club of New England, A. R. Shrigley, Sec., 26 Trement St., Boston

Amherst Aero Club, Amherst

*Pittsfield Aero Club, L. J. Minahan, Pres., Pittsfield

Springfield Aero Club, c/o Charles T. Shean, Pres., 3 John Street, Springfield

Tufts College Aero Club, Tufts College

*Harvard Aeronautical Society, Prof. A. Lawrence Rotch, Pres., Blue Hill Observatory

Mass. Inst. of Technology Aero Club, John S. Selfridgem, Sec., Inst. of Technology, Boston

Dartmouth Aero Club, Richard F. Paul, Sec., Dartmouth

First Assn. of Licensed Pilots, Chas. J. Glidden, Pres., Hotel Somerset, Boston

Williams Aeronautical Society, Williams College, Robert O. Starret, Sec., Williamstown

~MICHIGAN.~

*Aero Club of Michigan, C. B. du Charme, Sec., Detroit

University of Michigan Aero Club, Ann Arbor

~MINNESOTA.~

Minneapolis Junior Aero Club, Stillman Chase, Sec., 3047 5th Avenue, S., Minneapolis

St. Louis Experimental Ass'n., 5346 Zealand Street, St. Louis

*Kansas City Aero Club, George M. Myers, Pres., Convention Hall, Kansas City

~MISSOURI.~

*Aero Club of St. Louis, 1429 Pine Street, St. Louis

~MONTANA.~

Aero Club of St. Charles College, Helena

~NEBRASKA.~

Aero Club of Nebraska, Col. Wm. H. Glassford, Pres., Fort Omaha

Junior Aero Club or the Y.M.C.A., c/o Y.M.C.A., Omaha

Lincoln Aero Club, Lincoln, c/o G. R. Brownfield, 1234 "O" Street

Aviation Club of Nebraska, Arthur Frenzer, Sec., 2778 California Street, Omaha

~NEW JERSEY.~

Princeton University Aero Club, Princeton

Aeronautic Society of New Jersey, c/o N.J. Automobile & Motor Club, Broad Street, Newark

Aero Club of New Jersey, c/o James K. Duffy, Sec., 315 Madison Avenue, New York

New Jersey Aeronautical League, W. A. Kraus, Sec., Guttenberg

Aero and Motor Club, Asbury Park

Atlantic City Aero Club, Col. Walter E. Edge, Sec., Atlantic City

Model School Aero Club, Trenton, R. G. Teavitt, Sec.

Trenton Aero Club, James Fenton, Sec., Trenton

~NEW YORK.~

*Aero Club of America, 297 Madison Avenue, Chas. Walsh, Sec.

The Aeronautical Society, 250 W. 54th Street, Arnold Kruckman, Gen. Sec.

Aeronautic Alumni Ass'n., c/o West Side Y.M.C.A., West 57th Street

New York Model Aero Club, Adrien Lacroix, Sec., 141 Lexington Avenue

National Model Aero Club, c/o A. Leo Stevens, 282 9th Avenue

Stuyvesant Aeronautic Society, 345 East 15th Street, Percey W. Pierce, Sec.

Columbia Aero Club, Columbia University, 116th Street

Dewitt Clinton High School Aero Club, 58th Street and 10th Avenue

*Aero Club of Buffalo, Lafayette Hotel, Buffalo, N.Y.

Thousand Islands Aero Club, c/o Dr. J. M. Gibbons, 168 Montague Street, Ithaca

Aeronautic Section, Technology Club, Syracuse

Boys' High School Aero Club, Henry St. Pieless, Sec., 815 Avenue, J., Brooklyn

*Rochester Aero Club, c/o L. J. Seely, 10 Culver Road, Rochester

Aero Club of the Y.M.C.A., Harold C. Carpenter, Pres., White Plains

Aero Club, Haliano, U.S.A., 135, West 12th Street, N.Y. Seventy-two members. C. Chiantelli, Sec.

Junior Aero Club, c/o A. E. Horn, Public School, 77 Park Avenue and 84th Street

Aero Club of Long Island, c/o Hohn H. Lisle, Alen Cove

Commerce Aero Club, 65 West 117th Street

*Aero Club of New York, Garden City.

Mechanics Aeronautical Ass'n., c/o H. H. Simms, 304 Cutler Building, Rochester

Aeronautical Research Club of the Y.M.C.A., H. C. Myers, Sec., Buffalo

Aero Club Italiano, Saverio A. Mascia, 403 Park Avenue

Aeronautical Society of Women, Miss Dorothy E. Ball, Sec., 250 West 54th Street

~OHIO.~

*Aero Club of Ohio, Canton

*Aero Club of Dayton, Dayton

International Aeroplane Club, Dayton

Cleveland Aero Club, C. J. Forbes, Sec., Hollanden Hotel, Cleveland

*Aero Club of Cincinnati, c/o P. L. Mitchell, Traction Buildings, Cincinnati

~OREGON.~

Portland Aero Club, E. Henry Wemme, Pres., Portland

~PENNSYLVANIA.~

Aviation Section, Professional Chauffeurs Ass'n. of America, 1933 Spring Gardens, Phil.

*Aero Club of Pennsylvania, Rev. Geo. S. Gassner, Sec., Betz Buildings, Phil.

Ben Franklin Aeronautical Ass'n., c/o Dr. T. Chalmers Fulton, 6th and Diamond Street, Phil.

Philadelphia Aeronautical Recreation Society, Dr. Thos. E. Eldridge, Pres., 1639 N. Broad Street, Phil.

Haverford College Aero Club, Haverford, Pa.

Swartmore College Aero Club, Swartmore, Pa.

Univ. of Penn. Aero Club, Univ. of Penn., Phil., Pa.

Aero Club of Carnegie, Tech. Schools, Pittsburg, Pa.

Intercollegiate Aeronautical Ass'n., Geo. A. Richardson, Pres., Univ. of Penn., Phil.

Pittsburg Aero Club, H. P. Haas, Sec., Magel Buildings, Pittsburg, Pa.

~RHODE ISLAND.~

Pawtucket Aero Club, Pawtucket

Rhode Island Aeronautical Society, Providence, John J. Long, Sec., c/o Brown University

~TENNESSE.~

Nashville Aero Club, Nashville, E. Fisher Coles, Sec.

~TEXAS.~

Dallas Aero Club, c/o Chamber of Commerce

San Antonio Aero Club, c/o Dr. Fred J. Fielding, 423 Hick's Buildings, San Antonio

South Western Aero Club, P.O. Box 821, Fort Worth

Texas Junior Aeronautical Ass'n., Hugh Dumas, Pres., Fort Worth

~UTAH.~

Aero Club of Utah, c/o L. R. Culver, 11 Eagle Block, Salt Lake City

Salt Lake City Aero Club, c/o Mr. Campbell, Walker Bank Buildings, Salt Lake City

~VERMONT.~

Aero Club of Vermont, Chas. T. Fairfield, Pres., c/o Rutland News, Rutland

~VIRGINIA.~

University of Virginia Aero Club, Stanford Swin, Sec., University of Virginia

Virginia-Tennesse Aero Club, Bristol, Va-Tenn, C. W. Morey, Sec.

~WASHINGTON.~

Aero Club of Washington, 415 Union Trust Buildings, Washington, D.C.

Aero Club of Seattle, c/o M. Robert Guggenheim, 511 Lonan Buildings, Seattle

Walla Walla Aero Club, Walla Walla

~WISCONSIN.~

*Milwaukee Aero Club, Milwaukee, c/o Major Henry B. Hersey, Chief of the Weather Bureau, Milwaukee

Milwaukee Aeronautic Society, Pres., Sherman Brown, Manager of Davidson Theatre, Milwaukee

~U.S.A. MILITARY AVIATION.~

~U. S. ARMY AEROPLANE SPECIFICATIONS. (1912).~

~SPEED SCOUT MILITARY AEROPLANE.~

(1) Carry one person with the seat located to permit of the largest possible field of observation. (2) Ascend at the rate of 1500 feet in three minutes, while carrying fuel for one hour's flight. (3) Carry fuel for a three hours' flight. (4) Must be easily transportable by road, rail, etc., and easily and rapidly assembled and adjusted. (5) The starting and landing devices must be part of the machine itself, and it must be able to start without outside assistance. (6) The engine must be capable of throttling. (7) The engine will be subject to endurance test in the air of two hours' continuous flight. (8) Speed in the air of at least 65 miles an hour. (9) Capable of landing on and arising from ploughed fields. (10) The supporting surfaces must be of sufficient size to insure safe gliding in case the engine stops. (11) The efficiency and reliability of the system of control must have been demonstrated before the purchase order is placed. The aeroplane must be capable of executing a figure eight within a rectangle 500 yards by 250 yards, and without decreasing its altitude more than 100 feet at the completion of the figure eight. This test to be made by aviator alone without carrying extra weight. (12) The extreme width of the aeroplane supporting surfaces must not exceed 40 feet.

~SCOUT MILITARY AEROPLANE.~

(1) The aeroplane must carry two persons with seats located to permit of the largest possible field of observation for both. (2) The control must be capable of use by either operator from either seat. (3) The machine must be able to ascend at least 2000 feet in ten minutes while carrying a weight of 600 lbs. including the aviator and passenger, 150 lbs. of gasoline, and extra weight to make 600 lbs. All of the extra weight must be carried on the engine section and not distributed over the wings. (4) The fuel and oil capacity must be sufficient for at least four hours continuous flight. This will be determined by a trial flight of at least one half-hour, measuring the consumption of gasoline while carrying the passenger and weight stated in paragraph 3. (5) Same as No. 4 above. (6) Same as No. 5 above. (7) The engine must be of American manufacture and capable of throttling to run at reduced speed. (8) Same as No. 7 above. This test will be made with aviator and passenger, extra weight and fuel enumerated in paragraphs 3 and 4. (9) The aeroplane must develop a speed in the air of at least forty miles an hour. This test will be made with aviator and passenger, extra weight and fuel enumerated in paragraphs 3 and 4. The maximum speed must not exceed sixty-five miles per hour. (10) Same as No. 9 above. This test will be made with aviator, passenger, extra weight and fuel enumerated in paragraphs 3 and 4. (11) Same as No. 10 above. (12) Same as No. 11 above. (13) Same as No. 12 above.

In ~1913~ additional requirements specified enclosed body, bullet-proof armour, .75 chrome steel, for engine and aviator, provision of necessary instruments and wireless, with, as desirable features, silencer and cut-out, self-starter and an efficient stabilising device.

At end of March, 1913, the effective Army aeroplanes consisted of three 50 h.p. _Wrights_, one _Wright-Burgess_, several old machines.

The Navy had two _Wright-Burgess_ hydros and a few nondescripts. A _Burgess_ flying boat since added.

The estimate for Army effectives at end of the present year (1913) is 21 (5 _Burgess_, 6 _Curtiss_, 10 _Wright_).

~AVIATORS.~

(The numbers after any name is the number of the U. S. Aero Club certificate.)

Army.

Arnold, Lieut. H. H. (29) Beck, P. Capt. (39) Brereton, Lt. L. H. (211) Burge, Corp. V. S. (154) Chandler, C. de F. Capt. (59) Foulois, Lieut. (140) Geiger, Lieut. H. (166) Goodier, Lt. L. E. (200) Graham, Lieut. H. (152) Hennessy, Capt. F. B. (153) Humphreys, Lieut. Kirtland, Lieut. R.C. (45) Lahm, Lieut. F. P. (2) Love, Lieut. M. L. (155) McClaskey, Lieut. J. W. (90) McKay, Capt. G. W. (67) McLeary, Lieut. S. H. (210) McManus, Lieut. Milling, Lieut. (30) Rodgers, J. Lieut. (48) Sherman, Lieut. W. C. (151) Winder, Lieut.-Col. C.B. (130)

Navy.

Herbster, Ens. (103) Ellyson, Lieut. T. G. (28) Rodgers, John, Lieut. Towers, Lieut. J. H. (62)

~U.S.A. PRIVATE AVIATORS (to end of 1911).~

(The number against any name is, unless otherwise stated, the Ae. C. America pilot certificate number. Only a few American aviators have bothered to obtain the Ae. Certificate. America produces a large number of aviators who fly for pleasure or exhibitions only and have not gone into competitions under International Rules. These consequently do not bother about certificates; but most of those recorded could easily obtain them, if they cared to try.)

Adams, Clarence Adams, A. S. (215) Alvarez, F. Ambrose, Charles Andrews, Thornwell Apto, H. J. Arndt, Edw. F. Atwater, Mrs. L. J. Atwater, W. B. (98) Atwood, H. N. (33) Baker, G. H. Baldwin, Ivy Baldwin, Capt. T. S. (7) Barnett, A. E. Barton, Sam Bates, M. F. (66) Beachey, Hillery (89) Beachey, Lincoln (27) Beatty, G. W. (41) Beckly, Wm. A. Beers, W. C. (40) Benoist, T. W. Bergdoll, Louis, J. Betton, Kaid Bishop, Cortland Bleakley, W. H. Boandette, A. B. Bonner, G. T. Bonette, C. C. Bonney, L. W. (47) Brackett, A. J. Brewer, Roy Brindley, O. A. (46) Brinker, H. S. Brodie, O. W. (135) Brookins, W. R. (19) Brown, H. H. (58) Bumbaugh, Capt. G. L. Burgess, W. Starling (136) Burligh, Chas. Bush, J. F. Butler, P. J. Callon, J. L. (102) Champion, Frank (86) Christmas, Wm. Cannon, Jack Cline, W. F. Coffyn, F. C. (26) Cole, R. Coleman, R. F. Cook, W. B. (95) Cooke, Henry C. Cooke, F. G. (26) Cooper, John D. (60) Costello, A. B. Coutourier, C. (79) Crewelson, W. H. Cross, Redmond W. (35) Crosby, R. W. Cummings, J. A. Curtiss, Glenn H. (1) & (Ae. C. F. 1) Curzon, J. W. DeGiers, C. De Hart, D. C. De Kor, F. (72) Dennis, D. L. Dixon, S. D. Dougherty, E. S.

(87) Doyle, H. Drew, A. (50) Drexel, J. A. (8) Durgan, W. E. Dyott, G. M. Eaton, Warren Ecot, Robert G. Eells, Fred. Elton, Albert (75) Engel, A. J. Erickson, Louis, G. Eshoo, D. Evans, W. Ey, G. Fish, Farnam (85) Fortney, Lewis Fowler, R. G. Freeman, A. (84) Fuchs, Joseph Funk, T. B. Gallaudet, E. F. (32) Games, A. B. Gantz, Saxe P. Gardener, Hubbard G. Garner, R. W. Gaskell, Bud Gratz, H. F. Gray, George Green, William, Dr. Gregory, Donald Greider, C. Greider, J. Gressier, Romaine Guey, Fung Joe Hadley, C. O. Hall, Hamilton, C. K. (12) Hamilton, J. W. Hamilton, Thos. W. Hammond, Lee (34) Harper Harkness, H. S. (16) Harmon, C. B. (6) Hartman, A. Haupt, Willie Havens, Beckwith (127) Hendrian, A. Henning, J. C. Henningsen, Fred Heth, Eugene Henry, R. St. Hilliard, W. M. (Brit. Ae. C. 102) Hills, H. V. Hofer, W. Hoff, Wm. H. (91) Hoflake, Charles Holden, J. J. Holt, L. E. (63) Hoover, Fred. (100) Hoover, H. H. Huddleston, E. D. James, Stanley Janicke, W. Jannus, Anthony (80) Jennings, J. C. Jerwan, S. S. (54) Johnson, Frank H. Johnson, Walter E. (164) Jumel, August Kantner, H. (65) Kellrey, H. Kemmerle, Horace Kennedy, F. M. (97) Kiley, J. E. Kimball, Wilbur R. Klein, H. H. Klockles, J. G. Korn, Edward Krasting, Theodore La Chapelle, Duval Lambert, A. B. (61) Lambreath, C. E. Lapadat, N. Laser, G. F. Le Van, Howard Lewis, S. C. (92) Lewkowicz, Ladis Lidstone, Ed. S. Lillie, M. T. (73) Lockwood, Chas. Longfellow, H. W. Loose, Geo. H. Lougheed, A. Longo, T. Ludwig, Vandy Maier Manners, George Mars, J. C. (11) Martin, J. B. Martin, G. L. (56) Martin, J. V. Massar, A. M. Masson Matalach, S. H. Mattingley, O. A. Maynard, Arthur Mayo, Albert (99) McCally, J. B. (94) McCarty, James McClellam McCollum, W. C. McCurdy, J. A. D. (18) McGoey, Thomas McNamara, Geo. E. McManus, L. McMahon, A. J. Medrick, F. H. Meyerhoffer, Orvar Miller, Clinton R. Moisant, Miss M. E. (44) Morok, Chas. B. Mourfield, Carl Murias, De E. F. (38) Murphy, T. Murphy, Wm. Neidmiller, Ed. Nelson, N. B. Nelson, Nels. T. Ovington, E. L. Page, P. W. (68) Paine, N. B. Paridon, Michael Park, Henry Paulding, Dwight Paulhan, L. (3) Pfiel, P. Post, Augustus Powers, H. W. Prince, Norman (55) Prentice, Prospect, Louis Prowse, C. O. Raiche, Mrs. F. Ragot, Louis Reichert, H. D. W. (82) Remington, Earle Reynolds, Dr. Percy, L. Richter, J. (81) Riggs, E. Roat, Arthur R. Robinson, H. (42) Roehrig, B. F. Rowe, F. E. Russell, Geo. Sackett, Harry Sands, H. Hayden (Ae. C. F. 70) Schafer, G. E. Schmidt, G. S. Schneider, Fred, P. Schulz, G. C. Schwartz, A. Schwister, John Scott, Miss B. Seeman, J. R. Seignor, H. A. Seligman, J. (64) Seymour, Joseph Sellers, M. B. Shelton, T. Sherwood, Oliver, B. Shneider, Fred. Shoemaker, Chas. W. (93) Sill, F. Simmonds, O. G. (145) Skinner, S. R. Slaik, E. Slavin, J. J. Smith, A. Smith, Kyle Smith, R. M. Sommerville, W. E. Soreusen, Prof. Sparling, J. N. Steitz, F. M. (88) Stewart, J. G. Stone, A. (Ae. C. F. 15) Summer, Gill. Suppe, Talmage, M. P. Takisow, Tarbox, J. Thomas, W. T. Thomas, O. W. Thompson, George Tickell, Sam Timothy, S. R. Troxey Turpin, J. C. (22) Tuttle, T. T. Vanderbilt, W. K. Vaughan, Stanley Vogt, Jesse S. Walden, Dr. H. W. (74) Walker, Clarence Ward, J. J. (52) Warner, A. P. Webster, C. L. (69) Weeks, F. W. Wells, G. Wetzig, H. H. Weymann, Charles (14) Wilcox Wildman, Dock Willard, C. F. (10) Williams, Beryl (71) Williams, B. J. Willoughby, Capt. Hugh L. Wilson, Edward Wiseman, Fred. E. Witmer, C. C. (53) Worden, John H. (76) Wright, Orville (4) (Ae. C. F. 14) Wright, Wilbur (5) (Ae. C. F. 15) Young, C. M. Yan, J.

The following American aviators have been killed:--

+----------+ | 1908. | | Selfridge, Lt. (Army) | | | | 1910. | | Johnstone, R. (20) | | Moisant, J. B. (13) | | | | 1911. | | Badger, Wm. R. (36) | | Castellane, Tony | | Clark, C. B., Dr. | | Dixon, Cromwell (43) | | Ely, Eugene | | Frisbie, J. J. (24) | | Hoxsey, Arch. (21) | | Johnston, St. Croix | | Kelly, Lieut. (Army) | | Kreamer, Dan. A. | | Miller, F. H. | | Oxley | | Penot, Marcel | | Purvis, Wm. G. | | Rosenbaum, Louis | | Schriver, Tod (9) | +----------+

~U.S.A. PRIVATE AVIATORS. (Brevets, 1912.)~

Aldasoro, J. P. (217) Aldasaro, E. A. (218) Andrews, W. D. (124) Arnold, G. (198) Barlow, F. E. (139) Beckwith, S. F. (137) Beech, A. C. (168) Belcher, O. T. (158) Bell, Dr. F. J. (196) Bell, G. E. (201) Bergdoll, G. C. (169) Berlin, C. A. (109) Bleakley, W. H. (206) Bouldin, W. (157) Boysdorfer, C. (193) Brown, R. M. (185) Bryant, G. M. (208) Burnside, F. H. (212) Carlstrom, O. G. (145) Colovon, P. (160) Crossley, S. J. (187) Dalwigk, G. B. (190) De Hart, D. C. (129) Eaton, W. S. (128) Edelman, D. (191) Elliott, R. (178) Figyelmessy, H. (203) Fritts, E. V. (213) Gilpatric, J. G. (171) Gray, G. A. (142) Gray, J. F. (150) Gunn, T. (131) Hattemer, H. L. (147) Hemstraught, W. H. (146) Hetlick, W. A., jr. (197) Hild, F. C. (216) Hunt, E. N. (163) Holmes, H. (204) Johnson, R. R. (205) Kabitzke, W. (126) Kammski, J. G. (121) Kemper, F. W. (119) Klockler, J. G. (125) Korn, E. (171) Lamkey, W. A. (183) Law, R. B. (188) Maroney, T. T. (106) Masson, D. (202) McMillen, R. E. (111) Meyer, C. (176) Miller, B. A. (173) Niles, C. F. (181) Park, H. (113) Peoli, C. (141) Piceller, W. (116) Prodgers, C. B. (159) Reid, M. E. (114) Reid, P. H. (179) Remer, L. H., de (115) Richardson, R. H. C. (174) Robinson, R. W. C. (162) Ruiz, H. (182) Russell, R. B. (132) Salinas, A. (170) Salinas, G. (172) Schaeffer, J. S. (177) Scholovinck, E. (195) Schuman, F. J. (143) Singh, M. M. (123) Sjolander, C. T. (138) Smith, J. F. (207) Spaulding, J. D. (107) Stark, W. M. (110) Stinson, K. (148) Sverkerson, J. S. (180) Tait, G. M. (184) Takeiski, K. (122) Terrill, F. J. (108) Thomson, C. (112) Thompson, De L. (134) Twombly, W. I. (149) Vought, C. M. (156) Waite, H. R. (186) Weeks, E. O. (214) Weiner, T. (167) Wiggins, C. L. (175) Wood, C. M. (209)

U.S. Aviators killed: _Continued._

+----------+ | In ~1912.~ | | | Blair, R. | | Chambers, W. B. | | Clarke, J. (133) | | Gill, H. W. (31) | | Hazelhurst, Lieut. | | Kearney, H. F. (83) | | Kondo, M. (120) | | Lawrence, C. | | Longstaffe, J. L. | | Mitchell, L. (51) | | Page, R. (96) | | Parmelee (25) | | Peck, P. (57) | | Quimby, Miss H. (37) | | Rodgers, C. P. (49) | | Rockwell, Lt. L. C. (165) | | Scott, Corp. F. | | Southard, F. J. | | Stevenson, J. | | Turner, H. | | Underwood, G. | | Walsh, C. F. | | Welsh, A. L. (23) | | | In ~1913.~ | | | Boland, F. E. | | Chandler, Lieut. R. | | Park, Lieut. T. D. (223) | +----------+

U.S.A. AEROPLANES.

~A~

AERIAL EXHIBITION Co. (Biplane), 1777, Broadway, New York. Built a _Curtiss_ type with Kirkham motor, 1911. 2 skids, with wheel between, and usual _Farman_ rubber shock absorbers.

AERIAL YACHT CO., San Francisco. Inc. 1913. Capital, $25,000.

AERONAUTICAL SUPPLY CO. See _Cordeaux-Etter_.

AMERICAN AEROPLANE SUPPLY HOUSE (Monoplane), 266, Main Street, Hempstead, N.Y. Builders of monoplanes after the _Bleriot_ type. Half-a-dozen machines were built and sold during 1911. Fitted with Gnomes or American engines.

~B~

BALDWIN Biplanes. Captain Thos. S. Baldwin, PO Box, 78, Madison Square, N.Y. About half-a-dozen steel biplanes have been produced in 1911 by Captain Baldwin, and he and other aviators, Badger, Hammond, Miss Scott Mass, etc., have flown these at various exhibitions and meets, and are classed with well-known successful American biplanes.

[Illustration: _Photo, Edwin Levick, N.Y._]

Details of _Baldwin_ ("Red Devil").

~Length.~--28-1/4 feet (8.60 m.) ~Span.~--28-3/4 feet (8.75 m.)

~Motor.~--50-60 h.p. Hall-Scott ~Propeller.~--One Requa-Gibson in rear of main planes. Diameter, 7 feet (2.13 m.) Pitch, 6 feet (1.82 m.)

~Speed.~--60 m.p.h. (97 k.p.h.)

[Illustration: BALDWIN. RED DEVIL. UAS.]

BENOIST. Benoist Aircraft Co., 6628, Delmar Boulevard, St. Louis, Mo. (formerly Aeronautic Supply Co.)

----------+----------+----------+ | ~1912-13.~ | ~1913.~ | Model and date. | "Headless." | Flying boat. | | | Tandem biplane. | ----------+----------+----------+ ~Length~ feet (m.)| 22-1/2 (6.85) | 27 | ~Span~ feet (m.)| 30 (9.15) | 42-1/6 (12.80) | ~Area~ sq. feet (m squared.)| ... | ... | {total lbs. (kgs.)| ... | 1004 (455) | ~Weight~ { | | | {useful lbs. (kgs.)| ... | ... | ~Motor~ h.p.| ... | 75 Roberts | {max. m.p.h. (km.)| 68 (110) | ... | ~Speed~ { | | | {min. m.p.h. (km.)| 31 (50) | ... | ~Endurance~ hrs.| 3 | | ----------+----------+----------+

Notes.--The boat of the flying boat is 23-5/6 feet long, by 2 feet 2-1/2 inches wide. Shipable wheels. See _Aeronautics_, January, 1913.

[Illustration: BENOIST. Flying boat. UAS.]

BOLAND. Boland Aeroplane & Motor Co., 1821, Broadway, New York. Works: Ft. Center St. Newark, N.J.

[Illustration]

----------+----------+ Model and date. | ~1913.~ | | "Tailless." | ----------+----------+ ~Length~ feet (m.)| 21-1/6 (6.45) | ~Span~ feet (m.)| 35-1/2 (10.80) | ~Area~ sq. feet (m squared.)| ... | {total lbs. (kgs.)| 900 (408) | ~Weight~ { | | {useful lbs. (kgs.)| ... | ~Motor~ h.p.| 60 Boland | ~Speed~ m.p.h. (km.)| 60 (95) | Number built during 1912 | 1 | ----------+----------+

A refinement of the original machine of the late F. E. Boland, which first flew in 1911. _Control_ by two special jibs which work inward. Designed to be used also as a hydro, with three step floats. No rudder or ailerons. Full details, etc., see _Aeronautics_, U.S.A., May, 1913, and _Aircraft_, U.S.A., May, 1913.

BURGESS. Burgess Co. & Curtis, Marblehead, Mass. Built _Wright_ types under license, also machines of their own.

Model and date.	Military tractor.	Coast defence hydro.	Naval flying boat.
~1912-13.~	~1913.~	~1913.~	
~Length~ feet (m.)	37-3/4 (8.50)	33-1/3 (9.55)	31 (9.45)
~Span~ feet (m.)	34-1/2 (10.50)	37-3/4 (12)	43 (13.10)
~Area~ sq. feet (m squared.)	...	...	397 (37)
{total lbs. (kgs.)	...	...	...
~Weight~ {			
{useful lbs. (kgs.)	...	775 (352)	...
~Motor~ h.p.	70 Renault	60 Sturtevant	70 Renault
	muffled		
~Speed~ m.p.h. (km.)	45 (70)	59 (95)	...
~Endurance~ hrs.	4-1/2	4-1/2	...
Number built during 1912	...	...	...
Remarks.--	Lumina fabric.	Special clear view for observation.	Boat 29-1/2 feet long.
	Petrol, 48 gallons.	(U.S.A.), May-June, 1912.	2--2 step floats. Details, _Aeronautics_, (U.S.A.), May, 1913.
		Useful weight includes floats.	2--1 step mahogany and copper floats. Details, _Aeronautics_, (U.S.A.), Feb., 1913.

[Illustration: Burgess-Wright.]

[Illustration: Burgess-Wright as a hydro (the U.S. Navy has two of these).]

[Illustration: Military tractor. _By favour of "Aeronautics," U.S.A._ UAS.]

[Illustration: "Coast defense" hydro. _From "Aeronautics."_]

[Illustration: Burgess-Curtis. 1913 Naval flying boat.]

~C~

CHRISTMAS. Durham Christmas Aeroplane Sales & Exhibition Corporation, Inc. 1913. Capital: $10,000 to $50,000. Claims for it are that it is "automatically balanced." This is attained by the shape of the machine, not through the agency of any auxiliary apparatus.

[Illustration]

CORDEAUX-ETTER. Cordeaux-Etter Mfg. Corporation, Brooklyn, N.Y. Capital: $10,000. Took over, 1913, the Aeronautical Supply Co., of N.Y.

COOKE. Weldon B. Cooke Aeroplane Co., Sandusky, Ohio. Founded 1913 by the well-known aviator, W. B. Cooke.

[Illustration]

---------------------------+------------------+ Model and date. | ~1913.~ | ---------------------------+------------------+ ~Length~ feet (m.)| 25 (7.60) | ~Span~ feet (m.)| 24 (7.30) | ~Area~ sq. feet (m squared.)| 240 (22) | {total lbs. (kgs.)| 750 (340) | ~Weight~ { | | {usefullbs. (kgs.)| ... | ~Motor~ h.p.{|75 Roberts 2 cycle| {| _upside down_ | ~Speed~ m.p.h. (km.)| ... | Number built during 1912 | new firm | ---------------------------+------------------+

Details, _Aeronautics_, U.S.A., February, 1913.

[Illustration: COOKE. UAS]

CURTISS. Curtiss Aeroplane Co., Hammondsport, N.Y. Glenn H. Curtiss in 1907 and 1908 was a member of the Aerial Experiment Association, formed by Dr. and Mrs. Alexander Graham Bell. This Association built four machines, each along the lines of one of the four engineers belong to the Association, F. W. Baldwin, Lieut. T. E. Selfridge, G. H. Curtiss and J. A. D. McCurdy. The last built was the _June Bug_, designed by Curtiss and was the most successful. In the spring of 1908, the Association was disbanded and The Aeronautical Society gave Curtiss an order for an aeroplane with _carte blanche_ as to design. He produced a 4 cyl. machine, Curtiss engine, and flew it. A duplicate was hurriedly built, 8 cyl. engine installed, and taken to Europe for the first Gordon Bennett, which he won. Returning, the same type was continued with minor improvements. Later the front elevator was brought closer in, finally discarded, and the fan tail adopted and this remains the standard land machine to-day. In April, a military tractor was built and flown.

On January 26th, 1911, first successful flights were made with a hydroaeroplane, at the Winter camp at San Diego, Calif. This had two floats tandem. One was finally adopted and great success was achieved, and remains standard at the present time. With this machine various experiments were made. It was altered in a tractor for one occasion, it was lifted on board warships; made into triplane, etc.

In 1912 he brought out his present type of flying boat. This is being rapidly developed and minor changes in details are made in practically every machine put out.

In May, 1913, he produced a special 4-passenger flying boat for a customer on special order.

Note.--In addition to those tabulated, special small racing machines have been built, as well as similar machines with extra sections simply added either side for Army use.

---------------------------+--------------------+---------------------+--------------------- Model and date. | ~Type D.~ | ~Type E.~ | ~Type F.~ | ~1913.~ | ~1913.~ | ~1913.~ ---------------------------+--------------------+---------------------+--------------------- ~Length~ feet (m.)| 26-2/3 (8.10) | 27-1/3 (8.33) | 27-1/3 (8.33) ~Span~ feet (m.)| 26-1/4 (8) | 31-1/4 (9.50) | 38-1/3 (11.70) ~Overall~ feet (m.)| 33-1/12 (10) | 36-1/4 (11) | 41-2/3 (12.70) ~Area~ sq. feet (m squared.)| 214 (19-1/2) | 288 (26-1/4) | 421-1/2 (39) {total lbs. (kgs)| ... | 1700 (771) | ... ~Weight~ { | | | {useful lbs. (kgs)| ... | 500 (227) | ... ~Motor~ h.p.| Curtiss | 80 Curtiss | Curtiss ~Speed~ m.p.h. (km.)| ... | 59 (95) | ...

---------- Remarks.-- | Land service, but | Fitted either with |Used to date only |is also made fitted | wheels, pontons, or |as military tractor | with floats. | boat. |or heavy flying boat. | Panels. | _Vilas boat._ | _McCormick boat._ | | Boat 24 ft. long. |Boat 25 ft. long 4 ft. | |Beam 54-1/2 ft. long. |wide. Freeboard 46 | |Height 41 ins. long. |ins. Cockpit 84 ins. | |Cockpit 3 ft. long by |long by 46 ins. wide. | | 4 ft. 2 ins. wide. |Length of tail, incl. | | | elevator 12 feet. ----------

For full details of the tractor (F) see _Aeronautics_, U.S.A., February, 1913.

[Illustration: 1913 Tractor. Type F.]

[Illustration: 1912 flying boat. _By favor of "Aeronautics," U.S.A._]

[Illustration: Curtiss. 1913 flying boat. UAS.]

~G~

GALLAUDET. Gallaudet Eng. Co., Norwich Ct.

[Illustration]

In 1912 produced a special racer as above. ~Span~, 32 feet (9.75 m.) ~Area~, 200 sq. feet (18-1/2 m squared.) ~Speed~, 100 m.p.h. (160 k.p.h.) ~Motor~, 100 Gnome.

~K~

KIRKHAM Biplanes. Chas. B. Kirkham, Motor Manufacturers, Savona, N.Y. Began to manufacture aeroplanes in 1912, after previous experiments and flights near his factory.

~Length~, ? feet (? m.) ~span~, 34 feet (10.40 m.) ~surface~, ? sq. feet (? m squared.)

~Weight.~--Complete, _without pilot_, 980 lbs. (445 kgs.)

~Motor.~--50 h.p. Kirkham, located in front under bonnet. 70 h.p. also fitted.

~Speed.~--56-62 m.p.h. (90-100 k.p.h.)

Remarks.--Rises easily at under 35 m.p.h., and has a full speed radius of 5-1/2 hours. Full details in _Aeronautics_, U.S.A., January, 1912. 1913, no changes.

[Illustration]

~L-S~

LOENING. Monoplane aero boat, with one very deep step. See _Aeronautics_, U.S.A., May-June, 1912.

SELLERS. Quadruplane. Matthew B. Sellers, R.F., D2, Norwood, Ga. Has been successfully experimenting for a number of years with a staggered quadruplane, and has given the aviation world a number of valuable papers. His aim is to fly successfully with the least possible horse power. For several years he has been making flights with various engines delivering from 5 to 6 h.p. on careful test. The

actual thrust has been measured and recorded in late 1911 experiments. Details in _Aeronautics_, June, 1909; October, 1909; November, 1910; January, 1911; January, 1912. No actual details of the machine are available, but it follows closely the patent drawings (see references). He is one of the few real scientific flying men in the U.S.A. The original machine with slight changes was still flying at end of 1912 with only 5 h.p. B.H.P. The flying speed is 20 m.p.h.

[Illustration]

SLOANE. Sloane Aeroplane Co., 1733, Broadway, New York. Established 1911. Agents for _Caudrons_ and _Deperdussins_. Run a school for these.

~T-V~

THOMAS Biplanes. Thomas Bros., Bath, N.Y., O.W., and W.T. Thomas began experimenting and flying in 1908 with a machine on the order of a _Curtiss_. In the winter of 1909-10, a type of their own was produced and was flown during 1911 by Walter Johnson in exhibitions. In 1912 they continued the same type, with refinements. In 1913 they adopted the overhanging top plane type, but of the same general high order of construction.

---------- | ~1912.~ | | ~1913.~ | ~1913.~ | ~1913.~ Model and date. | Tractor | ~1913.~ | Standard | Special | Flying boat. | biplane. | Monoplane. | biplane. | biplane. | ---------- ~Length~ ft.(m.)| ... | 30 (9.15) | ... | 25 (7.62) | ... ~Span~ ft.(m.)| 37 (11.27) | 32 (9.75) | 37 (11.27) | 33 (10) | 33 (10) | 27 (8.23) | ... | 27 (8.23) | 23 (7) | 23 (7) ~Area~ sq. ft.(m squared.)| ... | ... | ... | ... | ... {total lbs. (kgs.)| 900 (408) | 750 (340) | 900 (408) | 850 (385) | ... ~Weight~ { | | | | | {useful lbs. (kgs.)| ... | ... | ... | 400 (181) | ... ~Motor~ h.p.| 65 Kirkham | 70 Kirkham | 65 Kirkham | 65 Kirkham | 100 | | _muffled_ | | | Maximotor ~Speed~ m.p.h.(km)| 58 (94) | ... | 58 (94) | 60 (97) | ... ~Endurance~ hrs.| 2 | ... | 2 | 2-1/4 | ... Number built during 1912 | 1 | _building_ | ... | ... | _building_ ---

Remarks. Control in all: Ailerons, 4 rudders. Elevator operated by rocking post on which wheel is mounted. The 1912 tractor was given up as less efficient than the Standard 1913. Special: full description _Aeronautics_, U.S.A., May, 1913.

The move was evolved 1912, but not built till well into 1913.

[Illustration: 1913 Standard biplane. UAS.]

~W~

WASHINGTON. Washington Aeroplane Co., Washington, D.C. In 1913 built a flying boat to private order. ~Length~, 29 feet (8.83 m.) ~Maximum span~, 38 feet (11.85 m.) ~Motor~, 80 h.p. Gyro. Boat with eight compartments and one 3 inch step.

[Illustration: Miss Columbia. UAS.]

[Illustration: Latest Thomas.]

WITTEMAN. Witteman Bros., 17, Ocean Terrace and Little Clare Road, Staton Island, N.Y. These people do a considerable business building Curtiss type machines or machines to special designs for others. They built the _Baldwin_ biplanes for Captain Baldwin, to his design, using steel tubing throughout. See _Aeronautics_, December, 1911, for a _Witteman_ of special design shown by them at the Aero Show.

[Illustration: Witteman. 1912-13.]

WRIGHT BROS. Biplanes. The Wright Co., Dayton, Ohio. The original type of _Wright_ machine was mounted on skids only, and started along a rail. Its special features were a biplane elevator forward, main planes with warpable tips to trailing edge, small keel in gap, 2 propellers, chain driven in rear of planes, double rudder in rear and no tail. Wilbur Wright flew a machine of this type for 2 h. 20 m. 23-1/2 s. in 1908. (Details of early _Wrights_ see previous editions of this book.)

Model and date.	~B.~	~C.~	~EX.~	~E.~
~Length~ feet (m.)	31 (9.45)	29-3/4 (9)	...	...
~Span~ feet (m.)	39 (11.90)	38 (11.58)	32 (9.75)	32 (9.75)
~Area~ sq. feet (m squared.)	500 (47)	500 (47)	...	...
{total lbs. (kgs.)	1250 (567)	...	...	...
~Weight~ {				
{useful lbs. (kgs.)	...	...	...	...
~Motor~ h.p.	30-35 Wright	30-35 Wright	30 or 50 Wright	30 or 50 Wright
~Speed~ m.p.h. (km.)	45 (75)	45 (75)	...	...
	1913 standard.	For exhibition	1913	
	This machine as	work only.	for exhibition	
	a hydro is fitted	Single seater	work only.	
	with two 3 step	small duplicate	Single seater	
	floats.	of B.	duplicate of EX	
	Mea magneto.	except fitted with		
	a single propeller			only.

[Illustration: Wright. Model B. UAS]

[Illustration: Wright. Model C. _From "Aeronautics," U.S.A._ UAS.]

U.S.A. DIRIGIBLES.

There are a few small dirigibles in the U.S.A., but they are in no way to be compared to French and German productions. Up-to-date, they have only been used as attractions at fairs about the country. In the past several larger ones of poor design have been built and found failures.

~Military.~

BALDWIN (1908) 20,000 c. feet (560 m cubed)

Part B.

HISTORICAL AIRCRAFT.

In the following pages an attempt has been made to include photographs of all aeroplanes of the past six years, which, for one reason or another, "made history" in their own day.

While many are merely freak machines, which in the light of present knowledge seem ridiculous, the germ of modern practice is to be found in many other aircraft illustrated in this cemetery of dead ideals; and it is worth noting that at least one constructor, who is one of the first in the field to-day, commenced operations with machines which were entirely "freaks."

~AUSTRIAN.~

[Illustration: WELS & ETRICH (1908). Original form of the modern _Etrich_ (q.v.).]

[Illustration: HIPSSICH (1908). Tandem mono. with one propeller before and another in rear of rear plane.]

[Illustration: NEMETHY (1908). The first "Aviette."]

[Illustration: SOLTAU (1910). An ornithopter based on the earlier ideas of _Adehmar de la Hault_ (see Belgium).]

~BELGIAN.~

[Illustration: DE LA HAULT (1907). One of the earliest attempts at an ornithopter. No flights.]

[Illustration: DE LA HAULT II. (1910-11). The ornithopter principle applied to a monoplane. No success met with.]

[Illustration: D'HESPEL (1909-10). Single plane and suspended body. Early example of enclosed body. No flights.]

~BRAZIL.~

[Illustration: SANTOS-DUMONT XIX. This little machine, surface only 9 m squared. made an extraordinary sensation in France in 1909. It flew at the then incredible speed of 65 m.p.h. (100 k.p.h.) Santos-Dumont presented all rights to the world soon afterwards, and a large number were built before it was realised that only an extremely light weight pilot could fly in one. Few of the copies ever left the ground.]

~BRITISH.~

[Illustration: AVRO (1906). This 24 h.p. biplane, designed by A. V. Roe, was the first British machine to leave the ground.]

[Illustration: AVRO (1907). Tractor triplane of only 9 h.p. This flew in Lea Marshes--the lowest horse power yet flown in Europe to the present day.]

[Illustration: CODY (1909). Development of a much earlier machine. This one was a general laughing stock for a long time; but it was the direct predecessor of the machine (not very materially different) which was an easy first in the British Army aeroplane trials, 1912.]

[Illustration: DE HAVILLAND (1909). The performances of this machine secured a Government appointment for its aviator-designer.]

[Illustration: HOWARD WRIGHT (1908-09). The first machine in the world in which special attention was paid to securing a stream line body and minimised wind resistance.]

[Illustration: HUMPHREY (1908-09). Earliest British attempt at a hydro-aeroplane; possibly the earliest design ever produced anywhere.]

[Illustration: HUNTINGDON (DUNNE II) (1910). One of the earliest aeroplanes in existence--designed by Captain Dunne about 1905-06, previous to the secret experiments of the British War Office in Scotland, on the Duke of Atholl's estate. Assigned to Prof. Huntingdon in 1910. Made a few short flights.]

[Illustration: PORTE (1908). Designed by Lieut. Porte, R.N., in conjunction with Lieut. Pirrie, R.N. This machine, on which the former well known aviator commenced his flying career, was smashed up in preliminary trials as a glider on Portsdown Hill, Portsmouth. Its design apparently preceded the _Goupy_ in the use of staggered planes.]

[Illustration: "SAFETY" (1909-10).]

[Illustration: SEDDON. (1910). Designed by Lieut. Seddon, R.N.]

[Illustration: SHORT (1910). The first machine to Short's own design. (The tail here shown is a specially large one fitted by Moore-Brabazon).]

[Illustration: VALKYRIE (1910). This was one of the first "tail first" machines to be designed. The experimental machine (also known as the _A.S.L._), was completed in Feb., 1910.]

~DANISH.~

[Illustration: ELLEHAMMER (1905). On 12th September, 1906, this machine made the first free flight in Europe. On 28th June, 1908, it won the prize at Kiel for the first flight in Germany (distance, 47 m.) It was a tractor biplane with a revolving Ellehammer motor. It also had a pendulum seat as a stabilising device.]

~FRENCH.~

[Illustration: ANTOINETTE IV (1909). In this machine Latham made the first attempt to fly the Channel, 19th July, 1909.]

[Illustration: BLERIOT IV (1907-8).]

In 1909 the famous _Bleriot XI_ was built. This did very well at Reims, 1909. On 25th July, 1909, Bleriot made the first Cross-Channel flight in the machine illustrated below.

[Illustration: Bleriot XI.]

This machine had ~length~, 23 feet (7 m.) ~Span~, 25-3/4 feet (7.80 m.) ~Area~, 167 sq. feet (15-1/2 m squared.) Aspect ratio 4-1/2 to 1. ~Motor~, 22-25, 3 cylinder Anzani. ~Speed~, _about_ 45 m.p.h. (73 k.m.) _Special features_: Fixed wings with rounded edges. Twin elevator and fixed surface tail.

[Illustration: BOUSSON-BORGNIS (1907-08).]

[Illustration: BREGUET (1906). The first Breguet, known as _Breguet Gyroplane I_. Made a flight in October, 1906, being the first helicopter to leave the ground.]

[Illustration: BREGUET-RICHET II bis. (1909). A large and unsuccessful development of the gyroplane.]

[Illustration: BREGUET IV (1910). On its appearance, this machine was generally laughed at and nicknamed the "Coffee Pot," till in Aug., 1910, it made a world's record by carrying six, and later proved itself superior in stability to anything then existing.]

[Illustration: CHAUVIERE (1909-10). Attempt to develop a monoplane with propellers in rear. The idea has been resuscitated for some 1913 military monoplanes.]

[Illustration: COLLOMB (1907-09). Ornithopter, from which great things were once expected.]

[Illustration: CORNU (1908). An early helicopter for which flights were claimed, but have also been denied.]

[Illustration: D'EQUIVELLY (1907-08). Interesting example of the strange machines devised by pioneers.]

[Illustration: H. FARMAN (1907). This famous machine is the first _Voisin_, and the one on which H. Farman taught himself to fly. It was the first machine to make a turn in the air. Won the Deutsh-Archdeacon Grand Prix, 13th January, 1908, with a flight of 1 minute, 28 seconds. The extra third plane was added later. An Austrian Syndicate subsequently bought the machine.]

[Illustration: H. FARMAN (1908). Farman's first idea of a monoplane. It proved too heavy to fly with the power provided. Was eventually sold to a German officer. Three sets of wings and entirely enclosed body.]

[Illustration: GABARDINI (1909-10). Very early hydro-aeroplane, antedating the _Fabre_.]

[Illustration: GIVAUDIN (1908-09). Built by the Vermorel Co. The first conception of an idea which has since attracted a certain class of inventor in Germany, Italy and the U.S.A.]

[Illustration: MILITARY (1909). The first special military aeroplane ever built. It was specially designed by Capt. Dorand, for what were then held to be the aerial necessities of the French Army. The planes were placed well above the body, giving the pilot a very clear uninterrupted view.]

[Illustration: PISCHOFF-KOECHLIN (1906 or earlier). Dates from the days when a box-kite was the elementary idea in design, and the accepted position of the aviator lying prone.]

[Illustration: PISCHOFF-KOECHLIN (1908). Very early example of a tractor biplane. The extra span of the upper plane is also of interest. The machine had twin mono-elevators aft and also twin rudders.]

[Illustration: R.E.P. (1908). Early example of enclosed stream line body. Apparently the first machine in which steel construction appeared.]

[Illustration: VOISIN (1908). The first European aeroplanes to fly with any real success.]

[Illustration: VUITTON-HUBER (1908). Early helicopter.]

[Illustration: VUIA (1908). Earliest known machine with folding wings.]

[Illustration: WITZIG-LIORE-DUTILLUEL (1908-09). First or one of the first appearances of the idea of a series of staggered planes, with which Sellers has ever since experimented in the U.S.A.]

~GERMAN.~

[Illustration: BEILHARZ. (1909). First design in which a completely closed in body figured.]

[Illustration: GEISLER (1908).]

[Illustration: GRADE (1908). The first German built machine to fly.]

[Illustration: LORENZEN (1908-09).]

[Illustration: PARSEVAL (1909). Early hydro-aeroplane. Specially designed for military purposes by Major Parseval.]

[Illustration: SCHOLTZ (1908). Never left the ground.]

~ITALIAN.~

[Illustration: MILLER (1908-09). First aeroplane to be designed and constructed by Italians.]

~SWISS.~

[Illustration: DUFAUX (1908-09). First Swiss machine.]

~U.S.A.~

[Illustration: BOKOR (1909). The third American machine to leave the ground; the second purely U.S. one.]

[Illustration: CALL II (1909).]

[Illustration: CYGNET II (1908). Designed by Dr. Graham Bell, of the Aeronautical Society of America. Bell (Canadian), Glen Curtis (U.S.), Herring (U.S.), and Burgess (Canadian). It made short flights.]

[Illustration: ENGLISH (1909). In 1909 extraordinary claims were made for this machine and great things expected. On a full power trial in its shed it broke loose, and smashed itself against the roof. No recorded outdoor results.]

[Illustration: HERRING-BURGESS (1910).]

[Illustration: HULBERT (1910). This strange machine built in Switzerland by Dr. Dane Hulbert, achieved several flights. The planes were placed longitudinally instead of in the usual way.]

[Illustration: JUNE BUG (1908-09). Famous machine of its era. Built by the Aeronautical Society of America (see _Cygnet II_). Second machine to fly in the U.S.A. Did 2000 miles before being broken up.]

[Illustration: KIMBALL (1909). First machine in which a large number of propellers was attempted. Failed.]

[Illustration: LOOSE (1910).]

[Illustration: LUYTIES OTTO (1908).]

[Illustration: MOISSANT (1910). Built entirely of aluminium. Designed by the late John Moissant. Failed.]

[Illustration: RICKMAN (1908).]

[Illustration: ROSHON (1908).]

[Illustration: WILLIAMS (1908).]

[Illustration: ZERBE (1909).]

[Illustration]

[Illustration: WRIGHT (1908). Two views of the machine with which Wilbur Wright startled all Europe from August, 1908 to April, 1909. First U.S. machine to fly.]

Part C.

AERO ENGINES,

ALPHABETICALLY ARRANGED IN ORDER OF COUNTRY OF ORIGIN.

~AUSTRIAN, BELGIAN, BRITISH, FRENCH, GERMAN, ITALIAN, U.S.A.~

~Note.~--So far as possible this is a complete list of all the aero engines of any importance.

Data are confined to what is now being made or actually in use; untested "show novelties" are ignored.

In the case of some engines it has for various reasons proved impossible to obtain full data in time for inclusion in this edition.

In a general way these lists are confined to aeroplane engines.

~AUSTRIAN.~

~Revised by Herr Ing. W. Isendahl.~

~AUSTRO-DAIMLER.~	~KOeRTING.~
35-40 h.p., 4 cyl., 100x120 (1450 r.p.m.) 165 lbs.	[Illustration]
65-70 h.p., 4 cyl., 120x140 (1350 r.p.m.) 232 lbs.	
120 h.p., 6 cyl., 130x175 (1200 r.p.m.) 419 lbs.	Note.--This engine is no longer made, but it is to
be found still in some dirigibles.	Vertical water-cooled.
H.T. Magneto.	
All Valves overhead. Rocking levers and piston	
rods.	
Forced lubrication.	
Pressed steel pistons.	
Nickel-chrome crank shaft, hollow and closed.	+ White-metal bearings.
Cast-iron single cylinders (copper jackets).	
Single camshaft.	
[Illustration: 120 h.p.]	
The 120 has 2 carburetters and 2 H.T. magnetos.	

~BELGIAN.~

~METALLURGIQUE.~	~PIPE.~
40 h.p., 4 cyl., 85x130 (1850 r.p.m.)	50-70 h.p., 8 cyl., 100x100 (1950 r.p.m.) 239 lbs.
60 h.p., 4 cyl., 100x150 (1850 r.p.m.) 300 lbs.	110 h.p., 8 cyl.
90 h.p., 4 cyl., 125x150 (1600 r.p.m.) 550 lbs.	
Vertical, air-cooled (fan).	Vertical, water-cooled.
H.T. magneto.	H.T. magneto.
Mechanical inlets.	Mechanical inlets.
Pump lubrication.	Pump lubrication.

MIESSE.	VIVINUS.
50-60 h.p., 4 cyl. lbs.	50 h.p., 4 cyl., 106x120 (1600 r.p.m.) 205 lbs.
100 h.p., 8 cyl., 130x140 245 lbs.	60 h.p., 4 cyl., 112x130 (1600 r.p.m.) 236 lbs
70 h.p., 4 cyl., 115x130 (1800 r.p.m.) 280 lbs.	Horizontal opposed, air-cooled (fan).
Vertical, water-cooled.	H.T. magneto
Mechanical inlets.	H.T. magneto.
Pump lubrication.	Mechanical inlets.
Pump lubrication.	Features.--
Air cooling is carried out by fans which drive	

|air through air jackets on cylinders. | |All valves in cylinder heads, rocker operated. | |Vertical crank shaft. |
+----------+----------

~BRITISH.~

----------+----------+---------- ----------+---------- ~A.B.C.~ | ~GREEN.~ | ~N.E.C.~ | ~WOLSELEY.~ All British Engine Co., Ltd., Brooklands, Surrey.| Green Engine Co., Ltd., 455, Berners Street, | New Engine (Motor) Co. Ltd., 9, Grafton Street, | Wolseley Tool & Motor Car Co., Ltd., | London, W. | Bond Street, London, W. | Adderley Park, Birmingham. ----------+---------- ----------+----------+---------- -------- 30 h.p., 4 cyl., 95x80 (1450 r.p.m.) 155 lbs. | 30-35 h.p., 4 cyl., 105x120 158 lbs. | 50 h.p., 4 cyl., 95x115 (1250 r.p.m.) 205 lbs. | 60-80 h.p., 8 cyl., 95x140, type A 325 lbs. 45 h.p., 6 cyl., 95x80 (1450 r.p.m.) 225 lbs. | 50-60 h.p., 4 cyl., 140x146 263 lbs. | 90 h.p., 6 cyl., 96x115 (1250 r.p.m.) 405 lbs. | " " " " B 345 lbs. 60 h.p., 8 cyl., 95x80 (1450 r.p.m.) 231 lbs. | 90-100 h.p., 6 cyl., 140x152 298 lbs. | | " " " " C 315 lbs. 85 h.p., 6 cyl., 125x105 (1700 r.p.m.) 290 lbs. | Vertical, water-cooled (pump). H.T. magneto. | Two stroke vertical for the 90 h.p. ~V~ for the 50. | " " " " D 335 lbs. 115 h.p., 8 cyl., 125x105 (1400 r.p.m.) 380 lbs. | Mechanical inlets. Forced lubrication.| | 120 h.p., 8 cyl., 125x175 (1150 r.p.m.) 630 lbs. 170 h.p., 12 cyl., 125x105 (1400 r.p.m.) 520 lbs. | | H.T. magneto. | ~V~ type. Types A and B of the 60-80 air-cooled 225 h.p., 16 cyl., 125x105 (1400 r.p.m.) 640 lbs. | _Features._-- | Valveless. | (water-cooled exhausts). The others water-cooled. | Cast-iron cylinders. Overhead cam shaft. | Forced lubrication. | Types A and C of the 60-80 are for direct coupling ~V~ type, water-cooled (pump). | Copper jackets. Nickel chrome crank shaft.| | of propeller, with double thrust ball bearings. | Overhead valves. White metal bearings. | _Features._-- | Types B and D geared to half crank shaft speed. H.T. magneto. | | | Bosch dual ignition. Mechanical inlets. | [Illustration] | Pistons after uncovering exhaust ports open | Mechanical inlets. Forced lubrication. | | the inlet ports. Air from blowers | Forced lubrication. | | scavenges. Strong mixture enters | _Features._-- _Features._-- | | immediately on compression stroke. | Steel cylinders (single). | | This is effected by a central mechanism. | Overhead valves (removable seats). Steel cylinders (steel and copper jackets). | | | Carburettor between cylinders. Overhead vertical valves. | | (N.B. Older types see previous editions.) | Nickel chrome crank shaft, on 3 bearings. Cast-steel crank case. | | | Nickel chrome crank shaft, white metal | | [Illustration] | [Illustration] bearings. | | | ----------+ ----------+----------+----------

~FRENCH.~

----------+----------+---------- ----------+---------- ~ANZANI.~ | ~BERTIN.~ | ~CANTON-UNNE (SALMSON).~ | ~CLEMENT BAYARD.~ 71, _bis_

Quai d'Asinieres (Seine). | 8, rue Garancier, Paris. | E. Salmson, 55, rue Grange aux Belles, Paris. | Usineo Clement Bayard, 33 quai Michelet, | | | Levallois-Perret, (Seine). ----------+---------- +---------- 30 h.p., 3 cyl., 105x130 (1575 r.p.m.) 154 lbs. | 50 h.p., 4 cyl., 116x150 (1100 r.p.m,) 132 lbs. | 60 h.p., 7 cyl., 75x260 (1300 r.p.m.) 220 lbs. | 40 h.p., 4 cyl., 100x120 242 lbs. 30 h.p., 3 cyl., 105x120 (1300 r.p.m.) 121 lbs. | 100 h.p., 8 cyl., 116x150 (1100 r.p.m.) 209 lbs. | 80 h.p., 7 cyl., 120x140 (1250 r.p.m.) 298 lbs. | 100 h.p., 4 cyl., 135x160 (1500 rp.m.) 463 lbs. 40-45 h.p., 6 cyl., 90x120 (1300 r.p.m.) 154 lbs. | | 110 h.p., 9 cyl., 120x140 (1300 r.p.m.) 353 lbs. | 130 h.p., 4 cyl., 155x185 50-60 h.p., 6 cyl., 105x120 (1300 r.p.m.) 200 lbs. | ~X~ type air cooled. | | 180 h.p., 6 cyl., 155x185 (1200 r.p.m.) 80 h.p., 10 cyl., 90x130 (1250 r.p.m.) 238 lbs. | | The 60 h.p. has parallel a.c. cylinders, the other | 200 h.p., 4 cyl., 190x230 (1200 r.p.m.) 1100 lbs. 100 h.p., 10 cyl., 105x140 (1100 r.p.m.) 308 lbs. | | two are radial w.c. | | | | H.T. magneto. Radial type, air-cooled (but water-cooling is occasionally | | There is also a horizontal radial engine (w.c.) | G.A. carburetter. Forced lubrication. fitted). | | 300 h.p., 9 cyl., 150x210 (1200 r.p.m.) 990 lbs. | | | | Overhead valves worked by two cams only. H.T. magneto. | | H.T. magneto. | Exhausts opened and closed by spring on tappet. Mechanical inlets. | | Mechanical inlets. | Forced lubrication. | | Forced lubrication. | _40 h.p._, vertical, _en bloc_, water-cooled, copper, | | Steel cylinders, copper jackets. | jacket, all valves same side, single cam shaft, _Features._-- | | Overhead Valves. | splash lubrication. Special carburetter, jet in | | Single special steel crank shaft on ball bearings. | centre of float chamber. Extremely simple construction. | | Aluminium alloy or steel crank case. | Mainshaft single crank. | | | _130 and 180 h.p._ (for dirigibles), cylinders in pairs, Flywheel specially balanced to compensate. | | [Illustration] | water-cooled. Overhead valves, single over-head Zenith carburetter. +----------+ | cam shaft. Two ignitions. Expanding | ~BURLAT.~ | | clutch. | 289 Avenue de Saxe, Lyon (Rhone). | | | | | [Illustration: Dirigible engine.] | 35 h.p., 8 cyl., 95x120 (956 r.p.m.) 187 lbs. | | | 60 h.p., 8 cyl., 120x120 (940 r.p.m.) 264 lbs. | | | 75 h.p., 8 cyl., 120x170 (940 r.p.m.) 308 lbs. | | | 120 h.p., 16 cyl., 120x120 (900 r.p.m.) 495 lbs. | | | | | Rotary, air-cooled. | | ----------+---------- +---------- ~CHENU.~ | ~CLERGET.~ | ~DANSETTE GILLET (LAVIATOR).~ | ~DE DION.~ Chenu, 10 Rue Fontaine-Saint-Georges, Paris. | Clerget & Cie, 11 rue Leon-Cogniet, Paris. | Dansette Gillet & Cie., 36 quai de. | Etablissements de Dion-Bouton, 52 avenue des | | Suresnes, Suresnes (Seine). | Champs-Elysees, Paris. ----------+---------- +---------- 50 h.p., 4 cyl., 110x130 (1300 r.p.m.) 253-1/2 lbs. | 43 h.p., 4 cyl., 100x120 (1600 r.p.m.) | 80 h.p., 8 cyl., 100x130 (1200 r.p.m,) 418 lbs. | 80 h.p., 8 cyl., 100x120 (1700 r.p.m.) 484 lbs. 75 h.p., 6 cyl., 110x130 (1300 r.p.m.) 375 lbs. | 50 h.p., 4 cyl., 110x120 (1500 r.p.m.) 172 lbs. | 110 h.p., 6 cyl., 130x160 (1100 r.p.m.) 616 lbs. | 150 h.p., 8 cyl., 125x150 (1600 r.p.m.) 968 lbs. 200 h.p., 6 cyl., 150x200 860 lbs. | 100 h.p., 4 cyl., 140x160 (1250 r.p.m.) 342 lbs. | 120 h.p., 4 cyl., 145x175 (1200 r.p.m.) 484 lbs. | ~V~ type, air-cooled for the 80; water-cooled for the | 50-60 h.p., 7 cyl., 120x120 (1200 r.p.m.) 198 lbs. | 120 h.p., 8 cyl., 114x160 (1200

r.p.m.) 418 lbs. | 150. Vertical, water-cooled (thermo syphon). | 200 h.p., 8 cyl., 140x160 (1275 r.p.m.) 495 lbs. | 200 h.p., 8 cyl., 147x175 (1100 r.p.m.) 715 lbs. | | | 250 h.p., 6 cyl., 180x200 (1050 r.p.m.) 1210 lbs. | H.T. magneto. H.T. magneto. | The 43 h.p. 50 and 100 vertical engines, w.c. | | Pump lubrication. Automatic lubrication. | | The 110, 120 (4 cyl.) and 250 are vertical, the | Forced lubrication. | The 50-60 h.p. is a radial, rotary. | others are ~V~ type. | Cylinders in pairs. | | | [Illustration: De Dion.] | The 200 h.p. has 2 carburetters and 2 magnetos, | | | and is ~V~ type. | | | | | | [Illustration: 200 h.p. Clerget.] | | | | | ----------+----------+----------+---------- ~DUTHEIL CHALMERS (EOLE).~ | ~GNOME~ | ~LABOR AVIATION.~ | ~PANHARD.~ Dutheil Chalmers & Cie., 81-83 avenue d'italie, Paris.| Societe des moleurs Gnome, 3 rue La Boetie, Paris. | Soc. anonyme des moteurs Labor Aviation, 29 rue de | Societe Panhard & Levassor, avenue d'Ivry, Paris. | | la Revolte, Levallois Perret (Seine). | ----------+----------+----------+---------- 40 h.p., 4 cyl., 125x120 250 lbs. | 50 h.p., 7 cyl., 11$1x$220 (1200 r.p.m.) 165 lbs. | 42 h p., 4 cyl., 90x150 (1200 r.p.m.) 221 lbs. | 35-40 h.p., 4 cyl., 110x140 210 lbs. 60 h.p., 6 cyl., 125x120 350 lbs. | 70 h.p., 7 cyl., 13$1x$220 (1300 r.p.m.) 183 lbs. | 72 h.p., 4 cyl., 100x210 (1200 r.p.m.) 353 lbs. | 55 h.p., 6 cyl., 110x140 341 lbs. | 80 h.p., 7 cyl., 124x140 (1200 r.p.m.) 191 lbs. | 120 h.p., 4 cyl., 120x250 419 lbs. | 100 h.p., 8 cyl., 110x140 (1500 r.p.m.) 440 lbs. Opposed horizontal, water-cooled. | 100 h.p., 14 cyl., 110x120 (1200 r.p.m.) 220 lbs. | | | 140 h.p., 14 cyl., 130x120 (1200 r.p.m.) 286 lbs. | Vertical water-cooled (pump). | Vertical, water-cooled. H.T. magneto. | 160 h.p., 14 cyl., 124x140 (1200 r.p.m.) 308 lbs. | | Automatic inlets. | | H.T. magneto. | H.T. magneto. Pump lubrication. | Radial rotary, air-cooled. | Mechanical inlets. | Mechanical inlets. | | Forced lubrication. | Pump lubrication. also | H.T. magneto. | | | Automatic inlets. | _Features:_-- | _Features._-- ~EOLE.~ (Dutheil Chalmers.) | Forced lubrication. | | | | Automatic carburetter. | Cast-iron cylinders, jackets in casting. 40 h.p., 4 cyl., 130x130 198 lbs. | _Features._-- | | Valves at side. | | [Illustration] | Cam shaft in crank case. Horizontal w.c., with central crank shaft over | Single crank pin +----------+ Nickel chrome crank shaft; white metal head valves. | Steel cylinders turned from solid. | ~LA RHONE~ | bearings. | Single ignition point. | | 100 h.p., 8 cyl. | Gas admitted through hollow crank shaft to | 50 h.p., 7 cyl., 105x140 176 lbs. | [Illustration] | crank case, thence to pistons; oil enters | 80 h.p., 9 cyl., 105x140 242 lbs. | Crank shaft at either end, all valves in centre. | in a similar way. | 100 h.p., 14 cyl., 105x140 308 lbs. | There is also a 120 h.p. 4 cylinder for dirigibles. | Nickel chrome crank shaft, ball bearings. | 160 h.p., 18 cyl., 105x140 374 lbs. | [Illustration] | Steel crank case. | (1200 r.p.m. in all.) | | The 100 h.p. has seven cylinders behind seven | | | others. Larger sizes ditto. | Rotary, air-cooled. | | Older engines of 50-100 h.p. do not differ in | | | general details. | | | | | [Illustration: 50 h.p. Gnome.] | | | | | ----------+----------+---------- ~RENAULT~ | ~R.E.P.~ | ~ROSSEL-PEUGEOT.~ | ~VIALE.~ Automobiles Louis Renault, 15 rue Gustav-Sandoz, | Etablissement Robert Esnault Pelterie, 149 rue de |Soc. anonyme de constructions aerienne Rossel-Peugeot,| Viale & Cie. 19 rue de la Mairie, Boulogne-sur-Seine Billancourt (Seine). | Silly, Billancourt

(Seine). | rue de Longchamp, a Suresnes (Seine). | (Seine). ----------
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+---------- 25 h.p., 4 cyl., 90x120 243 lbs. | 45 h.p., 5 cyl., 100x140 243 lbs. | 30-40 h.p., 7 cyl., 105x110 (1100 r.p.m.) 165 lbs. | 30 h.p., 3 cyl., 105x130 (1250 r.p.m.) 165 lbs. 35 h.p., 8 cyl., 75x120 243 lbs. | 60 h.p., 5 cyl., 110x160 (1100 r.p.m.) 330 lbs. | 40-50 h.p., 7 cyl., 110x110 (1100 r.p.m.) 172 lbs. | 50 h.p., 5 cyl., 105x130 (1250 r.p.m.) 199 lbs. 50 h.p., 8 cyl., 90x120 375 lbs. | 90 h.p., 7 cyl., 110x160 (1100 r.p.m.) 463 lbs. | 50-55 h.p., 7 cyl., 110x110 (1150 r.p.m.) 165 lbs. | 70 h.p., 7 cyl., 105x130 (1250 r.p.m.) 254 lbs. 70 h.p., 8 cyl., 96x140 397 lbs. | | | 100 h.p., 10 cyl., 105x130 (1250 r.p.m.) 320 lbs. 90 h.p., 12 cyl., 96x140 640 lbs. | Radial, air-cooled. | Rotary, air-cooled. | (All at 1800 r.p.m.) | | | Radial, air-cooled. | H.T. magneto and accumulators. | H.T. magneto. | Cylinders at 90 deg. | Mechanical inlets. | Mechanical inlets. | [Illustration: VIALE.] ~V~ type, air-cooled. | Forced lubrication. | Forced lubrication. | | | +---------- H.T. magneto. | _Features._-- | [Illustration: ROSSEL-PEUGEOT.] | Mechanical inlets. | | | ~VERDET.~ Pump lubrication. | In the 7 cyl. the cylinders are in two planes, | There is also a vertical water-cooled motor (1913). | | four being in front of the others. | 100 h.p., 140x140 (1300 r.p.m.) 352 lbs. | 55 h.p., 7 cyl., 112x140 (1100 r.p.m.) 176 lbs. _Features._-- | | | The 5 cyl. engines are fan shape in one plane. | | Rotary, air-cooled. Two to one shaft, made specially strong to | | | admit of the direct coupling of a propeller. | [Illustration: 7 cylinder.] | | | | | Inlet valves operated from below, exhausts | | | placed above them at the side. | | | | | | Plain bearings. | | | | | Special cooling. | | | | | | [Illustration] | | | | | | ----------+----------
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~GERMAN.~

~Revised by Herr Ing. W. Isendahl.~

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----------+---------- ~ARGUS.~ | ~BENZ.~ | | Argus-Motoren G.m.b.H., Flottenstrasse 39 and 40, | Benz & Cie, Mannheim. | ~DELFOSSE (radial.)~ | ~DELFOSSE (rotary.)~ Reinickendorf bei Berlin. Established 1900. | | | ----------+----------
----------+----------+---------- 70 h.p., 4 cyl., 124x130 (1400 r.p.m.) 254 lbs. | 100 h.p., 4 cyl., 130x180 (1250 r.p.m.) 337 lbs. | 24-30 h.p., 3 cyl., 110x130 (1500 r.p.m.) 100 lbs. | 30 h.p., 3 cyl. (1500 r.p.m.) 121 lbs. 100 h.p., 4 cyl., 140x140 (1250 r.p.m.) 290 lbs. | Vertical, water-cooled (pump). | 30-40 h.p., 3 cyl., 120x140 (1400 r.p.m.) 120 lbs. | 50 h.p., 5 cyl., 110x130 (1400 r.p.m.) 176 lbs. 150 h.p., 6 cyl., 140x140 (1250 r.p.m.) 353 lbs. | 2 H.T. magneto (Bosch). | 35-45 h.p., 4 cyl., 110x130 (1500 r.p.m.) | 70 h.p., 7 cyl., 110x138 (1200 r.p.m.) | Mechanical inlets. | 50-70 h.p., 4 cyl., 120x140 (1500 r.p.m.) | Vertical, water-cooled (pump). | Forced lubrication. | 50-60 h.p., 6 cyl., 110x130 (1500 r.p.m.) | Rotary air-cooled. | _Features._-- |80-100 h.p., 6 cyl., 120x140 (1500 r.p.m.) | H.T. magneto (Bosch). H.T. magneto (Bosch). | | | Overhead valves.

Mechanical inlets. | Cast-iron cylinder, steel jackets. | Radial air-cooled. | Automatic inlets. _Features._-- | Single cylinders. | H.T. magneto (or 6 volt accumulator). | Steel cylinders. | All valves overhead (single cam shaft). | Automatic inlets. | Crank shaft on ball bearings. Cast-iron cylinders. | Crank shaft hollow oil 5 metal bearings. | Forced lubrication. | Cylinders in pairs. | | | [Illustration: DELFOSSE] Valves one side (single cam shaft). | [Illustration] | _Features._-- | Crank shaft on ball bearings, closed and hollow. | | | | This engine won the Kaiser's prize of 50,000 marks.| Special metal cylinders. | [Illustration: 100 h.p.] | | Very large valve chambers. | | | Chrome nickel crankshaft and big ends. | | | Water-cooling fitted if required at a 10% increase | | | of weight. | | | | | [Illustration: DELFOSSE RADIAL.] | ----------+----------+----------+---------- ---------- ~DIXI.~ | ~HILZ.~ | ~MERCEDES-DAIMLER.~ | Dixi Luftfahrt-u-Bootsmotoren-Verkaufsgesellschaft | Hilz Motorenfabrik G.m.b.H., Fuerstenwallstr. 189, | Daimler Motoren G.m.b.H., Stuttgart-Unterturkheim. | m.b.H., Bulowstr. 11, Berlin W. 25. Established 1911.| Duesseldorf. | | ----------+----------+---------- +---------- 50 h.p., 4 cyl., 100x140 (1400 r.p.m.) 198 lbs. | 25-30 h.p., 3 cyl., 105x130 (1400 r.p.m.) ? | 70 h.p., 4 cyl., 120x140 (1400 r.p.m.) 276 lbs. | 75 h.p., 4 cyl., 120x170 (1300 r.p.m.) 308 lbs. | 50 h.p., 5 cyl., 105x130 (1400 r.p.m.) ? | 70 h.p. (as above, but _inverted_) 298 lbs. | [Illustration: 70 h.p. Mercedes-Daimler.] 100 h.p., 4 cyl., 140x200 (1200 r.p.m.) 452 lbs. | | 90 h.p., 6 cyl., 105x140 (1350 r.p.m.) 309 lbs. | | Radial, air-cooled. | | Vertical, water-cooled (pump). | H.T. magneto (Bosch). | Vertical, water-cooled (pump). | | Automatic inlets. | H.T. magneto (Eismann in the 70, two Bosch in | H.T. magneto (Bosch). | Splash and forced lubrication. | the 90). | Mechanical inlets. | Steel cylinders. | Mechanical inlets. | Forced lubrication. | Crank shaft, hollow, on white metal bearings. | Forced lubrication. | | | | _Features._-- | [Illustration] | _Features_ of the 70's-- | | | | Cast-iron cylinders, copper jackets. | | Cast-iron cylinders (in pairs). | Single cylinders. | | Overhead valves. | Overhead inlets (single cam shaft). | | Single cam shaft. | Crank shaft, hollow, on 3 metal bearings. | | Crank shaft, hollow, on metal bearings. | | | | [Illustration: 100 h.p.] | | _Features_ of the 90.-- | | | | | Steel cylinders, with steel jackets. | | | 2 carburetters (Mercedes-Daimler), _otherwise | | | as the 70's_. | | | | | | [Illustration: 90 h.p.] | ----------+----------+---------- ~N.A.G.~ | ~OTTO ("A.G.O.")~ | ~ROTOR.~ | ~SYLPHE.~ Neue Automobile Ges. m.b.H., Berlin-Oberschoneweide. | Gustav Otto, G.m.b.H., Karlstrasse 72, Munich. | | ----------+----------+---------- ---------- 60 h.p., 4 cyl., 118x100 (1400 r.p.m.) 254 lbs. | 50 h.p., 4 cyl., 110x150 (1400 r.p.m.) 199 lbs. | 70 h.p., 7 cyl., 110x150 (1100 r.p.m.) 199 lbs. | 40 h.p., 5 cyl., 110x130 (1200 r.p.m.) ? lbs. 95 h.p., 4 cyl., 135x165 (1350 r.p.m.) 353 lbs. | 70 h.p., 6 cyl., 110x150 (1400 r.p.m.) 287 lbs. | 90 h.p., 9 cyl., 110x150 (1100 r.p.m.) 243 lbs. | | 100 h.p., 4 cyl., 140x150 (1300 r.p.m.) 353 lbs. | | Rotary, air-cooled. Vertical, water-cooled (pump). | | Rotary, air-cooled. | | Vertical, water-cooled (pump). | | H.T. magneto (Eismann). H.T. magneto (Bosch), 2 in the 95 h.p. | | H.T. magneto. | Automatic inlets. Mechanical inlets. | H.T. magneto (Bosch). | Automatic inlets. | Forced lubrication (fresh oil).

Forced lubrication. | Mechanical inlets. | Forced lubrication (fresh oil). | | Forced lubrication. | | _Features._-- _Features._-- | | _Features._-- | | _Features._-- | | Chrome nickel steel cylinders. Cast-iron cylinders, copper jackets. | | Steel cylinders. | Single cylinders. Cylinders in pairs. | Cast-iron cylinders. | Single cylinders. | Overhead valves. Single cam shaft. | Single cylinders, all connected by long bolts | Overhead valves. | Crank shaft, hollow, on metal bearings. Overhead valves. | and nuts. | Crank shaft, hollow, on ball bearings. | Crank shaft, hollow, on 5 metal bearings. | Overhead valves in the 100 h.p. Side valves | | [Illustration: 30-40 h.p.] | in the 50 and 70. | | [Illustration: 95 h.p.] | Single cam shaft. | | | Crank shaft hollow, on metal bearings. | | | | | | [Illustration] | | ----------+----------+----------+----------

~ITALIAN.~

----------+----------+---------- ~CAPRONI & FACCANONI.~ | (_Dirigibles only._) ~FIAT.~ | (_Dirigibles only._) ~ISOTTA-FRASCHINI.~ | ~ITALA.~ Soc. di Aviazione Ing^{ri} Caproni & Faccanoni, | Fabbrica Italiano Automobile Torino, 30-35 | Fabbrica Automobili Isotta-Fraschini, 79 Via | Itala Fabbrica Automobili, Barriera Orbassano, Vizzola Ticino. | Corso Dante, Turin (Torino). | Monte Rosa, Milan. | Turin (Torino). ----------+----------+---------- 60 h.p., 6 cyl., 105x130 176 lbs | 60 h.p., 4 cyl., 150x200 (1200 r.p.m.) 220 lbs. | 100 h.p., 4 cyl., 130x180 (r.p.m.) 662 lbs. | 50-55 h.p., 4 cyl., 115x140 (1500 r.p.m.) 397 lbs. 120 h.p., 12 cyl., 105x130 lbs. | 200 h.p., 4 cyl., 170x250 (1200 r.p.m.) 1443 lbs. | 500 h.p., 8 cyl., 150x200 (r.p.m.) 1543 lbs. | | | Vertical, water-cooled (pump). Radial, air-cooled. | Vertical. | Vertical. | | | H.T. magneto. H.T. magneto. | H.T. magneto and accumulators. | H.T. magneto. | Mechanical inlets. Mechanical inlets. | Mechanical inlets. | Mechanical inlets. | Forced lubrication. Forced lubrication. | Forced lubrication. | Forced lubrication (pump). | | | | _Features._-- | _Features._-- | _Features._-- | | | | Cylinders in pairs. | Enclosed valves. | Overhead inlets. | Overhead inlets. | Single cast-iron cylinders. | Horizontal exhausts. | Automatic carburetter. | | Special radiation. | +----------+ Zenith carburetter. | [Illustration] | ~GNOME.~ | | | Fab. Italiana Mot. Gnome, 73 Strada Venaria, | | | Turin (Torino). | | +----------+ | | Works of the Italian built Gnome engines. | | | | | ----------+----------+ ~L. U. C. T.~ | ~S. P. A.~ | Ladetto-Ubertalli & Cavalchini, Via Cavalli.-Angolo| Societa Ligure Piemontese Automobili, Barriera | Via Circonvallazione Turin (Torino). | Crocetta, Turin. | ----------+----------+ 50 h.p., 7 cyl., 110x120 lbs. | 40-50 h.p., 4 cyl., 95x150 (1200 r.p.m.) 199 lbs. | 80 h.p., 9 cyl., 110x120 lbs. | | 100 h.p., 9 cyl., 122x150 lbs. | Horizontal, water-cooled (pump). | | | Rotary, air-cooled. | H.T. magneto and accumulators. | | Mechanical inlets. | H.T. magneto. | Forced lubrication. | Mechanical inlets. | | Forced lubrication. | _Features._-- | | | | Two pistons per cylinder. | | Ball bearings throughout. | | | |

[Illustration: _Dirigible engine._] | | There is also a vertical 160 h.p. dirigible engine.| | | | | | | | | ----------+----------+

~SWISS.~

+---------- | OERLIKON. |Societe Oerlikon Suisse de Machines Outils, Oerlikon. | +---------- | 55 h.p., 4 cyl., 100x200 (1200 r.p.m.) 176 lbs. | | Horizontal opposed, water-cooled (pump). | | H.T. magneto (2 circuits for 2 sets of plugs). | Mechanical inlets. | Forced lubrication. | | Features.-- | | Steel cylinders (copper jackets). | Single cylinders. | Overhead valves. | Single cam shaft. | 2 carburetters (one for each pair of cylinders). | Crank shaft, solid, on ball bearings. | Open crank case. | | [Illustration] | +----------

~U.S.A.~

----------+----------+---------- ----------+---------- ~ADAMS-FARWELL.~ | ~ALBATROSS.~ | ~CALL.~ | ~CURTISS.~ 21, Athol Street, Dubuque, Iowa. | Albatross Co., Detroit, Mich. | Aerial Navigation Co. of America, Girard, Kansas. | Curtis Aeroplane Co., Hammondsport, N.Y. ----------+------ ----------+----------+---------- ---------- 36 h.p., 5 cyl., 102x 88 (1200 r.p.m.) 97 lbs. | 50 h.p., 6 cyl., 113x125 (1230 r.p.m.) 250 lbs. | 50 h.p., 2 cyl., 150x131 185 lbs. | 40 h.p., 4 cyl., (1100 r.p.m.) lbs. 63 h.p., 5 cyl., 142x127 (1200 r.p.m.) 250 lbs. | 100 h.p., 6 cyl., 137x125 275 lbs. | 100 h.p., 4 cyl. 325 lbs. | 75 h.p., 8 cyl., 100x100 (1100 r.p.m.) 250 lbs. 72 h.p., 5 cyl., 152x152 285 lbs. | | | also | Radial. The 50 is air-cooled, the 100 water-cooled.| | 60 h.p., 6 cyl., (1350 r.p.m.) lbs. Rotary horizontal. | | Horizontal opposed, water-cooled. | | | | 40 and 75, ~V~ shape, water-cooled (pump). H.T. magneto. | | | 60, vertical water-cooled (pump). Special valves. | | Mechanical inlets. | | | Magneto ignition (Bosch). | H.T. magneto (Bosch dual). _Features._-- | | Special silencer. | Mechanical inlets. | | Vanadium iron cylinders. | Splash and forced lubrication. No flywheel. | | Forced lubrication. | All valves in cylinder head, actuated by a | | | _Features._-- single push and pull lever worked by a | | | single cam. Valves close outwardly and | | _Features._-- | Single cylinders, copper jackets. are held shut by centrifugal force. | | | All valves in cylinder heads, actuated by rocking Variable lift. | | [Illustration] | levers from single cam shaft. Exhaust ports. | | | Mechanical oil feed. | | | [Illustration] | | | Engine weights are "fully complete." | | | | | [Illustration] | | | ---- ----------+----------+---------- ----------+---------- ~DETROIT AEROPLANE CO.~ | ~ELBRIDGE.~ | ~HALL-SCOTT.~ | ~KEMP (GREY EAGLE).~ Detroit Aeroplane Co., Detroit, Mich. | Elbridge Engine Co., 10, Culver Road, Rochester, | Hall-Scott Motor Car Co., San Francisco, Cal. | Kemp Machine Works, Muncie, Ind. | N.Y. | | ----------+----------+------- ----------+---------- 30-40 h.p., 2 cyl., 127x127 (1200 r.p.m.) 110 lbs. | 40 h.p., 4 cyl., 123x114 198 lbs. | 30 h.p., 4 cyl.,

100x100 142 lbs. | 1912 _models_: | 60 h.p., 6 cyl., 123x114 257 lbs. | 40 h.p., 4 cyl., 100x125 150 lbs. | 2 cycle horizontal, air-cooled. | | 60 h.p., 8 cyl., 100x100 235 lbs. | 35 h.p., D 4 cyl., 100x113 lbs. | Vertical, water-cooled. Valveless. Oil in gas. | 80 h.p., 8 cyl., 100x125 270 lbs. | 50 h.p., E 6 cyl., 100x113 260 lbs. H.T. magneto. | | 100 h.p. lbs. | Automatic inlets. | _Features._--Extra large bearings. | | 1913 _models_: Splash lubrication. | | First two are vertical, the others | | [Illustration] | V type, water-cooled (pump). | 16 h.p., G 2 cyl. lbs. _Features._-- | | H.T. magneto (Bosch). | 35 h.p., I 4 cyl. lbs. | | Mechanical inlets. | 55 h.p., H 6 cyl. lbs. All valves in cylinder heads operated by a | | Pump lubrication. | 75 h.p., J 8 cyl. lbs. single cam. | | | Valves easily detached. | | Cast-iron cylinders. | Vertical, air-cooled. Very large valves. | | All valves overhead. | Schebler carburetter | | Copper jacketted. | H.T. magneto. | | Special Stromberg carburetter. | Mechanical inlets. | | Special radiators. | Pump lubrication. | | | [Illustration: Over-all length of the 30/40 is 19 in.]| | [Illustration] | _Features._-- | | | | | Overhead valves. | | | Extra large exhausts in centre of cylinders. | | | Special semi-steel (grey iron) cylinders. | | | Designed to work at 350 deg.-400 deg. Faht. | | | | | | [Illustration] ----------+----------+----------+----------

~KIRKHAM.~ | ~MAXIMOTOR.~ | ~ROBERTS.~ | ~STURTEVANT.~ C. Kirkham, Savona, N.Y. | Maximotor Makers, Detroit, Mich. | Roberts Motor Co., Sandusky, Ohio. | B. F. Sturtevant Co., Hyde Park, Boston, Mass. ----------

45 h.p., 4 cyl., 105x120 (1400 r.p.m.) 180 lbs. | 50 h.p., 4 cyl., 113x127 (r.p.m.) 200 lbs. | 50 h.p., 4 cyl., 113x125 165 lbs. | 40 h.p., 4 cyl., 113x113 (1300 r.p.m.) 200 lbs. 65 h.p., 6 cyl., 105x120 (1300 r.p.m.) 235 lbs. | 60-70 h.p., 4 cyl., 127x127 (r.p.m.) lbs. | 75 h.p., 6 cyl., 113x125 (1100 r.p.m.) 240 lbs. | 60 h.p., 6 cyl., 113x113 (r.p.m.) 285 lbs. 75 h.p., 6 cyl., x (1300 r.p.m.) 255 lbs. | 70-80 h.p., 6 cyl., 157x127 (r.p.m.) lbs. | | 110 h.p., 8 cyl., 105x120 (1200 r.p.m.) 310 lbs. | 80-100 h.p., 6 cyl., (r.p.m.) lbs. | Vertical, 2 cycle, water-cooled (pump). | H.T. magneto (Mea). | 100 h.p., 4 cyl., 150x150 (r.p.m.) lbs. | | Mechanical inlets. Vertical, water-cooled (pump). | 150 h.p., 6 cyl., 150x150 (r.p.m.) lbs. | H.T. magneto. | Pressure feed lubrication. | | Rotary inlets. | H.T. magneto (Bosch, 2 spark). | Vertical, water-cooled (pump). | Forced lubrication. | _Features._-- Forced lubrication. | | | | H.T. magneto (Bosch or Mea). | _Features._-- | Semi-steel cylinders (jackets cast with them). _Features._-- | Mechanical inlets (automatic in the 50 and 70.) | | Single cylinders. | Forced lubrication. | 2 carburetters. | Single cam shaft. Cast iron cylinders and pistons. | | Special magneto advance. | No overhead valves. Patent poppet-sleeve valves. | _Features._-- | Babbit bearings. | Exhaust valve lifters. | | Very large hollow crank shaft. | Nickel steel hollow crank shaft (5 bearings in [Illustration] | Started from aviator's seat. | Special metal cylinders (aerolite). | the 4 cyl., 7 in the 6 cyl.) | Double plugs. | Special by-pass. | | Half compression fitted. | Rotary inlets. | | Crank shaft, hollow, on 3 ball bearings. | | | | [Illustration] | ----------

~WELLES & ADAMS.~ | ~WRIGHT.~ | Wells & Adams, Bath, N.Y. | The Wright Co., Dayton, Ohio. | ----------+ 50 h.p., 4 cyl. 200 lbs. | 30 h.p., 4 cyl., 112x100 (1650 r.p.m.) 190 lbs. | | 50 h.p., 6 cyl., 112x100

(1150 r.p.m.) 230 lbs. | Vertical, water-cooled (pump). | | | Vertical, water-cooled (pump). | H.T. magneto. | | Mechanical inlets (overhead). | H.T. magneto. | Forced lubrication. | Rotary valves. | | Pump lubrication. | ~Note.~ _Features._-- | Silencer fitted. | | | There are a good many other U.S. engines of Single cylinders (large brass jackets). | | little or no account. The majority of these are Double plugs. | | merely more or less accurate copies of well-known No valve cages. | | European engines, and none of them have any Chrome nickel crank shaft, on 5 bearings. | | vogue. | | ----------

~Part D.~

AERIAL "WHO'S WHO," DIRECTORY & INDICES.

~Note.~--So far as possible the directory lists are exhaustive for the entire world. Anyone accidentally omitted is requested to communicate with the Manager, _All the World's Air-craft_, 5, Queen Victoria Street, London, E.C.

SUB-HEADS OF THIS SECTION.

"WHO'S WHO" IN AVIATION.

~DIRECTORY:~

CARBURETTERS. FABRICS. GARMENTS FOR AVIATION. HANGAR AND SHED BUILDERS. INSURANCE. LUBRICANTS. MAGNETOS. MISCELLANEOUS ACCESSORIES. PACKERS AND SHIPPERS. PATENT AGENTS. PETROL. PROPELLERS. RADIATORS.

~INDICES:~

ALPHABETICAL AEROPLANES. " DIRIGIBLE TYPES.

"WHO'S WHO" IN AVIATION.

ADER (Clement), Chateau de Ribonnet, Beaumont-sur-Leze (Haute-Garonne, France). Born 1841. Officer Leg. d'Hon. Experimented from 1892. His _Avion_ flew 300 metres at Satory, 12th October, 1897. This was the first flight ever made by a power machine in Europe. One of his early machines is in the _Arts et Metiers_ Museum, Paris.

ALEXANDER (H. I. H. Grand Duke), Michailovitch of Russia, Xenia Palace, St. Petersburg. Born 1866. Admiral of the Russian Navy. Prime mover in anything having to do with aviation in Russia.

ALEXANDER (Patrick Y.), 2 Whitehall Court, London, S.W. Donor of the L1000 Patrick Alexander prize for British Aerial engines. Founder and supporter of various aerial clubs and societies.

ANDRE (Ing. A.), 82 Rue d'Amsterdam, Paris. Editor _Revue Francaise de Construction Automobile et Aeronautique_. Writer on aviation. Experimenter.

ARBUTHNOT (C. B.) (Major General H. T.) Chairman of the Aerial League of the British Empire.

ARCHDEACON (Ernest), 77 Rue de Prony, Paris. Born 1863. Chev. Legion d'honneur. Vice-President _Ligue Nat. Aerienne_. Barrister. From 1884 made balloon ascents. In 1904 made glider experiments with Gabriel Voisin. Giver of the Archdeacon Cup, won by Santos Dumont, 29th October, 1906. Also part giver with Deutsch of the Deutsch-Archdeacon prize, for a Kilometre flight, closed circuit, won by Henry Farman, 13th January, 1908.

ARNOUX (Rene), 45 Rue du Ranelagh, Paris. Born 1858. Vice-President Tech. Com. A.C.F. Designer of the electric motor of the _Tissandier_ dirigible, 1882. Member Soc. Civil Eng. and Soc. Internat. Elec. Contributor to _Omnia_, etc. Inventor of the _Arnoux_ biplane.

ATTWOOD (Harry). Well-known American aviator. In August, 1911, covered 1,435 miles in eight days.

AUFFM-ORDT. (Swiss). 2 Avenue Hoche, Paris. Pioneer Aviator.

AVERY, American pioneer aviator, associated with Herring, Chanute, and others.

BACON (_late_ Rev.), British prominent aeronaut and lecturer.

BACON (Miss), daughter of the above. Aeronaut and lecturer.

BADEN-POWELL (Major, B.), F.R.A.S., F.R. Met. Soc., late Scots Guards. 32 Prince's Gate, London, S.W. Inventor of the Baden-Powell Box Kite. President of the Aeronautical Society from 1902 to 1909. Early experimenter and investigator with aeroplanes. Lecturer. Editor of _Aeronautics_.

BALDWIN (Capt. Thomas S.), 78 Maddison Square, New York. Well known U.S. Aeronaut of many years standing. Invented the _Baldwin_ dirigible.

BALSAN (Jacques), 52 Quai Debilly, Paris. Born 1868. Aeronaut from 1905. Made an altitude record of 8,558 metres. In 1906 went by balloon, Paris to England. Vice-President _Aero Club de France_.

BANNERMAN (Major Sir Alexander, Bart.) In command of the British Army Air Battalion, 1911.

BARBER. British. Aeronautical Syndicate, 1909-12. _Valkyrie_ type, etc., etc.

BARNWELL. British. Instructor 1912, Vicker's School.

BARRA. Well known French aviator.

BASENACH. German. Associated with Major Gross in producing the _M_ type German dirigibles.

BARTON (Dr.) Built the first British dirigible in 1904. Capacity, 235,000 c.f.

BATHIAT (Georges). Frenchman. After only one hour's tuition at the Hanriot School, secured his certificate at Rheims, October, 1910. Brother of the Bathiat who flew a _Breguet_.

BAUMANN (Otto), Berlin. Second German to fly.

BAUMANN. French. Instructor at the Ewen School in 1912.

BEACHEY (Lincoln). American subject. In August, 1911, reached 11,578 feet (3,527 m.)--world's record to that date. Flew Niagara, 27th June, 1911, in a _Curtiss_.

BEATTY (George W.) American aviator. At the Chicago meet, August, 1911, he made the world's passenger flight duration trip in 3 hours, 42 minutes, 22 seconds. Also made American records for 2 man altitude 3080 feet and 3 man duration on a _Wright_.

BECKE (Captain). British Army. In December, 1912, made a (to that date) record flight--Flamborough to Plymouth, and later back again--4-1/2 hours out; 2 hours back, excluding landings en route.

BEESE (Nellie). First German lady to get her pilot certificate, which she did on a _Rumpler_.

BELL (Dr. Alexander Graham Bell). Canadian. One of the founders of the Aerial Exp. Assoc. in U.S.A. Began experiments 1894. Inventor of the Tetratedal, etc.

BENDALL. British. Instructor Bristol School at Brooklands, 1912.

BERGET (Alphonse). French subject. Professor Inst. Oceanographique. Past President Soc. Francoise de Nav. Ae. Author of _La Conquete de L'Air_.

BERNARD. French. Tester for _Farmans_ 1912-13.

BERRIMAN (A. E.) British. 44, St. Martin's Lane, London, W.C. Technical editor of _Flight_. Author of _Principles of Flight_, etc., etc.

BERSON (Prof. Arthur), Haupstrasse 9, Lehlendorp, Germany. Born 1859. Austrian. Well-known author on meteorological and similar subjects affecting aviation.

BESANCON (Georges), 35 Rue Francois I., Paris. Born 1866. Chev. Leg. d'Hon. Editor of _L'Aerophile_. Secretary Ae. C.F. Experimented with balloons from 1886 onward.

BESSONNEAU (J. B. Lieut. de reserve), 29 rue de Louvre, Paris. French. Born 1880. Pioneer aviation helper. Produced special steel cables of high resistance; also the well known Bessonneau _hangars demontables_. Organised the first town-to-town flights, 1910; also first Grand Prix, 1912.

BEZOLD (Professor Wilhelm Von), Director of the Meteorological Institute, Berlin, etc. Author of several works on aeronautics, etc.

BISS (Gerald), British, 1, Melina Place, Grove End Road, London, N.W. Automobile correspondent of the _Standard_. Aviation expert.

BLANCHARD. Frenchman. 1753-1809. First man to cross the English Channel in a balloon (1781).

BLAND (Lillian E., Miss), Carnmoney, Belfast, Ireland. First woman aviator to design and build her own machine, _The Mayfly_. Has since ceased.

BLERIOT (Louis), 56 Boulevard Maillot, Paris. Chev. Legion d'Honneur. Inventor of the _Bleriot_ monoplane. Pioneer. Experimented from 1906. Has had more falls than any other aviators. First man to fly the Channel, which he did in _Bleriot XI._, 25th July, 1909. Member, Com. d'Aviation of the Ae. C.F.

BIELOVUCIC. Peruvian. Flew the Alps, 1912. Well known aviator.

BISS (Gerald). 1, Melina Place, Grove End Road, London, N.W. Well-known writer on automobile and aviation.

BOCKLIN. Swiss. 1827-1901. In 1850 became interested in aviation. In 1881 built gliders and a model aeroplane. Triplane, 1881. Biplanes, 1882-1887. In 1888 a monoplane with electric motor.

BOLOTOFF (Prince), Reigate Priory, Reigate, England. Russian subject. Pioneer aviator.

BOOM (J. A.) Editor of the _De Luchtvaart_, Ged. Aude gracht 144, Haarlem.

BORGNIS (Achille), 48 Rue d'Universite, Paris. Early experimenter and inventor. Vice-President of the Com. d'Aviation Aeronautiques Club de France. Member Ae. C.F. (See aviators.)

BOOTHBY (Lieut. F. L. M.) British Navy. Served on board the _Hermione_ when she was mother ship for aviation and aeronautics. Holds certificates for both aeroplanes and dirigibles.

BOSQUET (Chev. du), 8 Place de la Concorde, Paris. Chev. de l'Ordre de Leopold. Sec. Commission auto-aerienne.

BOUTTIEAUX (Col.) In command French military aviation, 1911-13.

BRACKE (Albert), 11 Chemin de Saint-Denis, Casteau-Mons, Belgium. Engineer. Editor _L'Aero Mecanique_. Inventor of the _Bracke_ and _Misson_ monoplane. Author on aerial matters.

BREGUET (Louis Charles), 31 Rue Morel, Donai (Nord), France. Born 1880. Began experimenting June, 1906, and in July, 1908, his gyroplane flew 20 yards at a height of 14 feet. President de la Section du Nord de la Ligue Nat. Aerienne.

BRERETON (J.). British. Instructor at British Deperdussin School, 1912.

BREWER (W.). Author of a standard technical work, _The Art of Aviation_. Formerly manager of Grahame-White & Co. Writer on aerial matters.

BRINDLEY (Oscar). American citizen. In August, 1911, he made 11,726 feet, at Chicago. These figures were afterwards stated to be incorrect, so the record was not allowed.

BROOKINS, U.S.A. Up to August, 1910, held world's height record, 6,338 feet (1,922 m.), made in a _Wright_ at Atlantic City, U.S.A. Badly injured in an accident, August, 1910.

BUIST (A. Massac). Well-known British writer on aviation subjects. Technical contributor to _Morning Post_, _Country Life_, etc., etc.

BURGEAT (Captain). The first French Officer after Captain Ferber to take up flying. He purchased _Antoinette VI_.--the first _Antoinette_ sold to the public.

BUSTEED (Harry). Australian. _Bristol_ pilot in the British Military Competition, 1912.

BUTLER (Frank Hedges). F.R.G.S., 155 Regent Street, London, W. Founder of the R. Ae. C. Crossed the Channel in a Balloon 1905. Member Ae. C. F.

BUTTENSTEDT (Carl). 95a, Friedrichshaven Str, Berlin. Born 1845. Author of aviation works and early experimenter over many years. Designer of aeroplanes, etc.

CAILLETET (Louis Paul), 75 Boulevard S. Michel, Paris. Officer Leg. d'Hon. Doctor. President Ae. C. F.

CALDERARA (Lieut.). Italian Navy. Sent to France to study aviation in 1908. Has made many good flights ever since. In 1912-13 produced a naval hydro-aeroplane of his own design.

CAPAZZA (Louis). Frenchman. Born 1862. Head of the Clement-Bayard Works.

CAPPER (Col.). Formerly in command of British Army aviation headquarters, Farnborough, 1909-10.

CASSINONE (Alexander), Nordpolstr. 2, Vienna. Leading Figure in Austrian aeronautical circles.

CASTAGNIERIS (Capt. Guido), 70 via della Muratte, Rome. Founder and secretary leading Italian aero clubs, etc.

CASTILLON DE SAINT-VICTOR (Comte G. de), 74 Avenue Marceau, Paris. Born 1870. Aeronaut since 1898. Did a trip, Paris to Sweden. Treasurer Ae. C. F., 1911.

CATERS (Baron de), Berchem-les-Anvers, Belgium. Born 1875. Motorist of renown in the early days. Early aviator pioneer.

CATTANEO. Italian. Well-known aviator since 1910.

CARDEN (Capt.). Experimental officer, appointed 1911, to British Army Air Battalion.

CAUMONT (_late_ Lieut.). French aviator. Killed in a _Nieuport_ monoplane, December 30th, 1910.

CAYLEY (George, Sir). Experimented about a hundred years ago with models and man-carrying gliders. Also wrote on Aviation, and is known as "the Father of Aviation."

CHANDLER (Capt. C. de F.) Commanding Signal Aviation School, U.S.A.

CHANUTE (Octave), U.S.A. Frequently alluded to as "the father of aviation." In company with Herring he joined Langley in 1905. He did much work with gliders. He propounded the theory that little was to be learned from studying birds. Discovered

that the greatest lift was obtained from a plane flat in front and arched from the side. Died November, 1910. Aged 78.

CHATLEY (Professor H.), B.Sc, Imperial Eng. Col. Tientsin, China. Britisher. Author of _The Force of the Wind_ (Griffin & Co.), and an authority on aviation matters in general.

CHAVEZ (Georges). Peruvian aviator, resident in France. Maker of many records. First aviator to fly the Alps, 22nd September, 1910. Fatally injured on that occasion.

CHEREAU. Frenchman. London manager of the Bleriot Co. and Bleriot School at Hendon.

CHOENDEL (_late_). German aviator, who made an altitude record of 1680 metres with a passenger. Killed on alighting.

CLEMENT (Gustave Adolphus), 33 Quai Michelet, Levallois-Perret (Seine), France. Born 1855. Officer Leg. d'Hon. Creator of the _Clement-Bayard_ dirigibles, etc.

COCKBURN (Geo. B.), Gloucester, England. One of the first Englishmen to take up aviation.

CODY. American; naturalised British, 1909. Inventor of the Cody kite. Employed by the British War Office for aviation work, 1905-1909. Inventor of Cody biplanes. Won Michelin prize 1910 and 1911. One of the best-known British aviators. In August 1912, made a biplane speed record of 72.4 m.p.h. Constructor.

COLLOMB. Frenchman. Early experimenter with flappers, etc.

COLMORE (Cyril). British. Ae.C. Pilot 15. Flying partner with the late Cecil Grace. Now given up flying.

COLSMAN (Alfred), Friedrichshaven, Germany. Director of the Zeppelin Co., etc.

CONNEAU (Lieut.) French Navy. Winner of the _Daily Mail_ L10,000 prize, 1911, with a _Bleriot_. Winner of the Paris to Rome and the Circuit of Europe races, 1911. Flies under the name of "Beaumont."

CORNU (Paul), 24 Rue de la Gare, Lisieux, France. Pioneer experimenter with helicopters. In 1908 one of his inventions rose 16 inches.

CROCCO (Lieut.) Italian. Had a good deal to do with the designing of the _Ricaldoni_ dirigible.

CROOKSHANK (Major C. de W.), R.E. Prominent supporter of aviation. Member of the R. Ae. C. Committee, 1910-11.

CURTISS (Glen. H.), Hammondsport, N.Y., U.S.A. Won the Gordon Bennett in 1909 on the _Curtiss_. Formerly a member of the Aerial Experiment Association, out of which the _Curtiss_ was evolved. Is Ae. C.F. Pilot 2. Head of the Curtiss Aeroplane Co.

DAHLBECK (Lieut.). First Swedish naval aviator. Trained in England.

DAVELNY. Commandant French Navy. Appointed 1911, to take command of French naval aviation.

DAUCOURT. Frenchman. First pilot to fly from Paris to Berlin, 16th April, 1913. Average speed 100 k.p.h. Time 12 hours, 32 minutes, including two stops.

DE BAEDAR (F.), 7 Rue Rameau, Paris. Editor _Revue Sportive de l'Aviation et de l'Automobile_.

DE DION (Marqus), 104 Avenue des Champs Elysees, Paris. Born 1856. Principal founder and Hon. President Ae. C. F.

DE HAVILAND (G.) British aviator. Designer of a biplane and a motor purchased by the War Office, in December, 1910. He was subsequently engaged by the Government for work on Salisbury Plain. In August, 1912, made the British altitude record to date of 9,500 feet with a passenger.

DELAGRANGE (the _late_ Leon). Born 1872. French sculptor. Took up aviation early in 1907. He purchased _Voisin No. I._, which made its first trials 28th February, 1907. Subsequently engaged in experiments with Archdeacon. In 1908 bought a _Voisin No. III._ Later on got a _Bleriot_. Killed 4th January, 1910, at Croix d'Hins, Bordeaux, in a _Bleriot_. Was Ae. C. F. pilot 3.

DEMANEST (Rene). French. 25, rue d'Orleans, Neuilly sur Seine. Began flying an _Antoinette_ in 1909. Won the Ae. C. F. prize.

DEPERDUSSIN. (See machines).

DEPREZ (Marcel). Frenchman. Writer on Aerial subjects.

DESBLEDS (L. Bein). Lecturer on Aeronautical Engineering, Polytechnic, London.

DEUTSCH (Henri de la Meurth), 4 Place des Etats-Unis, Paris. Officer Leg. d'Hon. Founder member of the Ae. C.F. Donor of the prize of 100,000 francs won by Santos Dumont, 19th October, 1901. Owner of the dirigible _Ville de Paris_. Vice-Pres. Legue Nat. Aerienne. Donor in part of the Deutsch Archdeacon prize. Offered 1909 to found a Technical Institute of Aviation, Paris University.

DICKSON (Captain). Ex-British Army officer. The first British aviator to distinguish himself at an International flying meet.

DOUTRE. French lawyer, interested in aviation. Invented a stabilising device in which Maurice Farman was interested.

DREXEL (A.) Scotland. American citizen. Made world's record at Lanark, 12th August, 1910, in a _Bleriot_, 6,750 feet (2,057 m.), beating previous record of Brookins.

DRIVER. British aviator. Flew in first aerial post, 1911.

DRZEWIECKI (Stefan), 62 Rue Boileau, Paris. Russian. Born 1844. Chev. Leg. d'Hon. In 1885 investigated aviation in connection with bird flight. Well known otherwise as an inventor of submarines, torpedo tubes, etc.

DU CROS (Harvey), M.P., 14 Regent Street, London, S.W. Born 1876. Takes considerable interest in aviation. Member of the Parliamentary Committee thereon.

DUFAUX (Armand). Swiss. He and his brother Henry were interested in aviation in 1903, and in 1904 built an helicopter. In 1909 the first Swiss aeroplane built by them appeared.

DUNNE (Lieut.), Eastchurch, Sheppey, Kent, England. Ex-British Army officer. Engaged by British War Office to carry out heavier than air experiments immediately after the aeroplane had been demonstrated a possibility. (See _Dunne_ in part I.)

DUPUY DE LOME. Frenchman. Made a hand-propelled dirigible in 1870-72.

DUeRR (Ludwig). German. Born 1878. Chief engineer Zeppelin works.

DUTRIEU (Mdlle. Helene). Belgian. Second woman to take up aviation.

EFIMOFF (Michael). Russian. Made his first appearance in France early in 1910. (Ae. C. F. pilot 31). Distinguished himself on _H. Farmans_ and _Sommers_. On his return to Russia he was made chief instructor of the special school of the Volunteer Aerial Association.

ELLEHAMMER (J. C. H.), Istedgade 119, Copenhagen. Commenced aviation studies in 1905. On 12th Sept., 1906, he made a flight--the first in Europe since Ader.

ELLYSON (Lieut. T. G.) U.S. Navy. In company with Lieut. Towers made the first flights ever made in a hydro-aeroplane.

ELY (Eugene B.) American. Was the first to fly successfully off a warship, which he did in a _Curtiss_ biplane on January 19th, 1911, from the U.S. cruiser _Pennsylvania_. Killed 1911.

ENGLEHARDT (Kapitan). Prominent figure in German aeronautical and aviation circles. Writer on aerial subjects. Began flying in 1910, in which year he won several prizes. Killed 1911.

EQUIVELLY (Marquis d'), 2 Place Wagram, Paris. Pioneer aviator, with a queer multiplane, 1907.

ERBSLOCH (the _late_ Oscar). Well known aeronaut. Inventor of a German dirigible, the _R. M. W. G._, afterwards named after him. He was killed in it with four others, July, 1910.

ESDAILE. British. Pioneered aviation displays in India, 1912.

ESNAULT-PELTERIE (Robert), 149 Rue de Silly, Billancourt (Seine), France. Early experimenter with aeroplanes. Flew the first _R.E.P._, October, 1907. Designer of the _R.E.P._ engine.

ESPITALLIER (Georges), 25 Rue St. Petersburg, Paris. Associated with the late Col. Renard in early dirigible experiments. Author of many aeronautical works.

ETRICH (Igo), Rotunde, Vienna II. Pioneer aviator with Wels. Designer of the _Etrich_ monoplane--the first Austrian machine to fly.

EVANS (William Evans), 1428, Charlotte Street, Kansas City, Mo. Purchased a biplane built by Dr. William Greene, who has since given up aeroplane building. Evans made a number of exhibitions in the middle West, but had given up flying by summer of 1911.

EWEN (W. H.) British. Head of the School for British _Caudron_.

"F. A. I." Federation of the leading Aero Clubs of all countries, for control of International Aviation Meets, Pilot certificates, etc., etc. The bulk of certificates were first obtained in France, but in 1910 they were made obtainable in any country from its own Ae. C., under identical rules. No aviator may compete in any International event without a certificate. Aviation has now more or less outgrown the F.A.I, on account of the virtual disappearance of private aviation events before military interests; but it did excellent service in its time and is still of considerable indirect value.

FARMAN (Henri), 22 Avenue de la Grande Armee, Paris. Born in Paris, 1874, but is of English descent. Chev. Leg. d'Hon. First a racing cyclist, then racing motorist. Took up aviation. Bought _Voison No. II_ (known as "_Farman I._"). On January 13th, 1908, he won the Deutsch-Archdeacon prize for covering a triangular course of one kilometre. In 1909 designed and built his own machine. Won the Michelin cup in 1909, making the record of 4 h. 17 min. 35 2.5s. in the air. Ae. C. F. pilot 5. In 1910, did 8 h. 12 mins. in the air, covering 288-3/4 m. (463 km.)

FARMAN (Maurice), 3 Rue Villaret de Joyeuse, Paris. Brother of above. Went in for aeronautics and motor racing. Bought _Voisin No. IV_ at an early stage of aviation. He fitted this with alterations of his own, and subsequently evolved the _M. Farman_ biplane. Ae. C. F. pilot 6.

FAURE (Jacques), 32 Rue Washington, Paris. Born 1873. Has long been prominent in aeronautical circles. Has crossed the Channel five times in gas bag balloons. Owns the _Faure_ dirigible. Member of Committee Ae. C. F.

FELIX (Capt.) In 1911 in charge of the Bleriot Military School at Etampes.

FERBER (the _late_ Capitane), flew as "De Rue." Born 1862 at Lyon. Commenced experiments with gliders in 1899 on Lilieuthal lines. In 1903 he built a power-driven machine. He taught Gabriel Voisin how to fly gliders. In 1908 was very active and flew several machines. Killed in a _Voisin_, 22nd September, 1909.

FERNANDEZ (the _late_). A Spanish tailor, resident in Paris, killed in 1909 in a machine of his own design.

FISHER (E. U. B.) First flew on a _Hanriot_, early in 1911. August, 1911, engaged as pilot by Messrs. Vickers.

FOKKER (Antony), Haarlem, Holland. Born in Java, 1890. Designed a monoplane in 1911 with special stabilising device. He flew this at Johannisthal. Subsequently started a company.

FOURNY. French. On September 11th, 1912, broke all previous distance and duration records by flying, non-stop, 13 hours, 22 minutes in _M. Farman_. Renault

motor. Distance 1,017 km. (631 miles) at Etampes, France.

FRISBIE (J. J.) American aviator. Killed in a _Curtiss_ at Norton, Kan., having been driven to fly in unsuitable weather by the jeers of a hostile crowd.

FUeRSTENBERG (Prince). Austrian. President of the Centre Aeronautical Committee, formed in June, 1912.

GALANSCHIKOFF (Mdlle.) Russian. On November 22nd at Johannisthal, made world's altitude record for lady fliers with 2,400 meteres.

GARROS. French aviator. Came in second in Paris to Rome and the Circuit of Europe races, 1911. Up to November, 1911, held world's height record (13,000 feet). Made in a _Bleriot_.

GASNIER (Rene), 1 Rue Scribe, Paris. Winner of many prizes in balloon events. French champion for the 1907 Gordon-Bennett. On Committee of Ae. C. F. Hon. President Ae. C. d'l'Ouest. Inventor of an aeroplane, 1908.

GAST (Madame C. Crespin du), 12 Rue Levoux, Paris. Well-known in aeronautical circles.

GASTAMBIDE (Robert), 27 Boulevard de Courcelles, Paris. Born 1882. Civil engineer. Took great interest in aviation at the start. Designed the _Gastambide-Mengin_ from which _Antoinettes_ were evolved. This was the first monoplane to carry a passenger (September, 1908.)

GELEYNS (C.) Editor of the _Avia_, Wynbrugstreet 13, Rotterdam.

GERRARD (Lieut.), R.M.L.I. British. August 17th, 1911, made world's passenger record to date, 4 hours, 13 minutes, on _Short_ No. 34.

GIFFARD (H.) Britisher, resident in France. In 1850 built the first practical dirigible. It had a steam motor. In 1852 it made a controlled speed of about 5 m.p.h.

GILBERT. French. On March 28th, 1913, flew from Lyons to Villacoublay in 3 hours, 10 minutes, a world's record to date from town to town non-stop.

GILL (Howard). U.S.A. aviator. In October, 1911, flew for 4 hours 16 minutes 35 seconds in a _Wright_. American record to that date.

GILMOUR (Graham). British. Pilot Ae. C.F., April, 1910. In 1911, flying a _Bristol_: with which many of his flights have been directly or indirectly of a highly sensational nature. These have included a flight alleged to be over London (reported to R. Ae. C.--case dismissed), flying low over Henley Regatta (certificate suspended, with subsequent litigation). Won second prize in the Brooklands-Brighton Race, May, 1911. Killed February, 1912.

GIBERT. French aviator who made records, 1911.

GLAZEBROOK (Dr. R. T.), C.B., F.R.S. Director of the British National Physical Laboratory.

GLIDDEN (Charles J.) The well-known American motorist. Founder of many of the U.S.A. Ae. clubs.

GODARD (Louis), 170 Rue Legendre, Paris. Builder of the _America_ Wellman Arctic Airship; inventor of the Godard Kite-Balloon; designer and builder of the _La Belgique_, etc., etc.

GORDON-BENNETT (James), 104 Avenue des Champs Elysees, Paris. American citizen. Owner of the _New York Herald_. Giver of the Gordon-Bennet aviation Prize. Previous to this he had instituted a similar event for motor cars, and few, if any, have done so much to advance the International sporting side of automobilism.

GOUPY (Ambrose), 59 Avenue Marceau, Paris. An early pioneer in aviation experiments--had the first triplane built for him by _Voisins_. Now a well known constructor.

GRACE (_late_ Cecil). Naturalised British subject, ex-Chilian. Lost at sea while competing for the Baron de Forest prize, December, 1910.

GRADE (H.), Magdeburg, Germany. First man to fly in Germany. He did this on a Grade triplane early in 1909. Now a well known German constructor.

GRAHAME-WHITE (Claude), 1 Albemarle Street, Piccadilly, London. Pilot 30 Ae. C. F. on a _H. Farman_. Attempted to win the _Daily Mail_ L10,000 London-Manchester prize, 1910. Gordon Bennett, 1911. Now constructor.

GREENE (Dr. W.), Treasurer, Aeronautic Society, U.S.A. Has done a great deal to advance aviation in the U.S.A. Designer of the _Greene_ biplane--a machine which in no way infringed the Wright patents.

GRESWELL. British aviator. Flew in first aerial post, 1911.

GREY (Chas. G.), 166 Piccadilly, London, W. Editor of the _Aeroplane_. Well-known writer on aerial matters, formerly as "Aero-Amateur," later under his own name. By 1912, had come to occupy a unique position of his own by an uncompromising statement of facts without regard to other circumstances.

GROSS (Major). In command of the German war dirigibles. Designer of the _M_ type. (_Gross_).

GRUBB (Capt. A. H. W.) D.S.O., R.E. Prominent supporter of aviation. Member of R. Ae. C. Committee, 1910-11.

GUILLEMEAU (R.), 82 Rue d'Amsterdam, Paris. Editor _Revue Francaise, de Const. Autble et Aeronautique._

HAENLEIN (Paul). German, 1835-1905. Early experimenter with dirigibles. Inventor of the "semi-rigid" system.

HAMEL (Gustav). British. Well-known aviator. Winner of Brooklands-Brighton Race, May, 1911. Flew the first British aerial mail, 1911. In April, 1913, on behalf of the London _Standard_, made a non-stop flight with a passenger London to Cologne in a _Bleriot_. Many other famous flights.

HAMMOND (J. J.) Australian. Pilot 258, Ae. C. F., on a _Sanchis Besa_, 4th October, 1910. In 1911, visited Australia with a _Bristol_, when he made many

sensational flights.

HARGRAVE (Lawrence), Sydney, N.S.W., Australia. A pioneer in aviation, 1890-95. Experimenter with and inventor of box kites.

HARKNESS (Harry). American aviator. Has made various records.

HARMON, (Clifford B.) One of the best-known U.S.A. amateur aviators. Made U.S.A. time record (2h. 3m.), 2nd July, 1910.

HARRISON (Eric). Australian subject. Instructor of the Bristol school at Lark Hill, Salisbury Plain, 1912.

HARRISON (Lieut. L. C. R.) British R. F. C. Killed 28th April, 1913, in the famous _Cody_ which won the British Military Aeroplane competition.

HAULT (Adhemar de la), 214 Rue Royale, Brussels. Editor of _La Conquete de l'air_. Well-known aviation pioneer. Interested in Ornithopters.

HAWKER (H. G.) Australian subject. On October 24th made British duration record to date--8 hours 23 mins. in _Sopwith_ biplane. Awarded the Michelin Cup, 1912.

HEKKING (R.) Frenchman. In September, 1909, carried out experiments with a biplane glider of 7 m. span and 25 m squared. surface. He rose to a height of 25 m., and is stated to have remained stationary for 5 minutes. Not confirmed, however.

HELEN. French aviator. Has appeared in various competitions since early in 1911.

HENDERSON (Brig. Gen.) British Army. First general to obtain British R. Ae. C. aviator certificate. Flying under the name of "Davidson" he obtained his certificate on a _Bristol_, at Brooklands, after seven days' training.

HENRY, Prince of Prussia. Well known for practical interest in motoring and aviation. Has driven his own car in races, and is a certificated aerial pilot for Germany.

HENSON. Died 1842. Projected a steam-driven monoplane early in the XIX century.

HERRING (A. M.), Freeport, Long Island, U.S.A. Started the study of aviation 1894. With Langley, 1895. With Chanute, 1896. Joined Ae. Exp. Assoc. and associated with Curtiss in the _Herring-Curtiss_. Subsequently (1910) with Burgess in the _Herring-Burgess_.

HERVE (Henri), 1 Rue Hautefeuille, Paris. Well-known authority on matters aeronautical. Author, etc.

HEWLETT (Mrs. Maurice) ("Madame Franck"). First lady aviator to obtain a British R. Ae. C. certificate.

HILDEBRAND (Kapitan Alfred), 10 Martin-Lutherstrasse, Berlin W. 30. Retired from German Army. A very well-known aeronaut. Owner of a Baldwin dirigible. Author of many works on aeronautics and aviation--the best known German writer on these subjects.

HINTERSTOISSER (Hauptmann Franz), Luisenstrasse 35, Vienna V. Commanding Austro-Hungarian Aeronautical service. 1911-12.

HIRTH (Helmuth). German. Made German passenger altitude record to date, at Johannistal, September-October, 1911. Height 2475 metres. Many other records. The best known of all German aviators.

HOFFMAN (Joseph). German. Built a steam-driven aeroplane in 1906.

HOLDEN (Col. H. C. L.) R.A.F.R.S. Prominent supporter of aviation. Member of the R. Ae. C. Committee, 1910-11.

HOWARD-FLANDERS. See British Aeroplanes, Part A.

HOWARD-WRIGHT. British. Early designer (_see_ Part B). In January, 1913, became manager of S. White & Co., of Cowes.

HOUDINI (Harry). British. The famous "Handcuff King" flew a _Voisin_ so long ago as November 2nd, 1909. He took it to Australia and won the first aeroplane flight prize there. He is the first to have taken out a "third party" insurance, which he did with the Albingia-Versicherungs-Aktien-gesellschaft, Hamburg, Germany, on November 29th, 1909. The policy was for 150,000 marks.

HUBERT. French aviator. Flew in the first British aerial post and was badly injured.

HUCKS (B. B.) British aviator. Has made several fine exhibition flights on a _Blackburn_. First man to make the double journey across the Bristol Channel, also to carry out wireless telephone experiments with aeroplane.

HUNTINGDON (Prof. A. K.), 14 Buckingham St., Charing Cross, London, W.C. Born 1856. Balloon expert, 1906-1908. Connected with the _Dunne_ machines. Member of the R. Ae. C. Committee, 1910-11.

HURLBERT (Dr. Dane), Vermont, Lucerne, Switzerland. U.S.A. citizen. Experimenter in original types of aeroplanes, 1909-11

ILLNER. First man to fly in Austria, which he did on an _Etrich_.

ISSATIER. French private soldier who obtained three weeks' leave and secured his flying certificate at Betheny after fourteen days, in a _Deperdussin_.

ISENDAHL (Walther). German. Holsteinstrasse 21, Berlin-Wilmersdorf. Leading authority on aerial and boat motors.

JANE (Fred T.), The Hill, Bedhampton, Hants, England. Naval author, founder and editor of _All the World's Air-craft_.

JANNUS (Antony). American. Well-known aviator.

JATHO (Karl), Stader Chausse 22, Hanover, Germany. Born 1873. Pioneer aviator from 1893. Has built various machines--none very satisfactory.

JEANNIN (Emil), Berlin. Prominent German aviator.

JEFFERIES (Dr. John). 1760-1820 _about_. American. Accompanied Blanchard in the first balloon voyage across the English Channel, 1784.

JENKINS (F. Conway). In May, 1911, obtained his certificate (74) after only four flights on a _Roe_ biplane.

JOHNSTONE (St. Croix). American aviator. 27th July, 1911, beat American duration records in a flight of 4 hours, 1 minute, 54 seconds. Distance 176 miles. Killed 1911.

JONES (Ernest L.), 250 West 54 Street, New York. Editor of _Aeronautics_ (U.S.A.)

JOSEF FERDINAND (Grand Duke of Austria). Enthusiastic aeronaut and moving spirit in aviatory matters in Austria.

JOYNSON-HICKS. British M.P. who has specialised in endeavouring to advance aviation.

JULLIOT (Henri), 3 Rue de Flandre, Paris. Born 1855. Chev. Leg. d'Hon. Technical director of the _Lebaudy_ works. Originator of the _Lebaudy_ type of dirigibles. Designer of _Lebaudy_ aeroplane. Member of Committee Ae. C. F.

KAPFERER (Henry), 26 Rue de Clichy, Paris. Chev. Leg. d'Hon. Director of the _Astra_ Cie, and the Cie Gen. Transaerienne. Part designer of the Clement-Bayard dirigibles. Took an early interest in the aeroplane movement, and had a biplane built to his own design by Voisins in 1907. Also had an early monoplane about the same date or a little later. On Committee Ae. C. F.

KASSNER (Carl), Wilhelmstr. 10, Berlin. Professor, German writer on technical aviation matters.

KENNEDY (Rankin), British authority on aviation subjects.

KENNEDY. St. Petersburg. British subject. Engineer who has studied aviation for many years. In 1911, was an honorary aerial adviser to the Russian Government on matters aerial.

KINDELAN (Captain), Guadalajara, Spain. Born 1879. Interested in balloons since 1906. Designer of the Spanish military airship _Torres Quevedos_.

KNIGHT. British. Instructor 1912, Vickers School.

KOENIG. German aviator. Won the 1st prize given by the Berliner _Zeitung am Mittag_. 1,182-1/2 kilos.

KRAUSS. Well-known German aviation engineer. Author of many articles.

KRESS (Wilhelm), Waaggasse 13, Vienna. Born 1836. Flew a model ornithopter in 1888. Author.

KRIEGER (Hans). German. Formerly chauffeur to the Kaiser. Built a monoplane of his own design, and on September 5th, 1911, obtained his certificate on it.

LAFFONT (_late_ A.) Killed in an _Antoinette_, December 28th, 1910.

LAHM (Frank), Washington D.C., U.S.A. Well-known aeronaut.

LAMBERT (Albert B.) President of Ae. C. of St. Louis, U.S.A. Flies a _Wright_.

LAMBERT (Count Charles), 74 Rue Charles-Lafitte, Neuilly-sur-seine, Paris. Russian subject. Born 1865. Interested in aviation 1893 onward. First pupil of Wilbur Wright.

LAMMLIN. German. Killed at Strasburg, May 23rd, 1911.

LANA (Francisco), (1631-1687), (Italian). Jesuit who projected flying machines.

LANCHESTER. Author of well-known aerial classic.

LANE (Howard), 50, Parliament Street, Westminster, London, S.W. British citizen. Mechanical and Chemical Engineer. Born 1852 at Warwick. Government Contractor; Birmingham City Councillor, 1895-1900. Honours, South Kensington, 1873. Inventions, the Seamless Steel Gas Cylinder, 1882; Multiple Stage Gas Compressor, 1884; Roller method of Skin Balloon Construction, 1887; Regenerative Hydrogen Producing Plant, 1903; Turbine Aero-Motor, 1909, etc., etc.

LANGLEY (Samuel Pierpont). Born 1834. Died 1906. American pioneer from 1887. Commenced work in 1893 with Dr. Graham Bell, and later, Herring and Chanute. In May, 1896, he flew a large steam-driven model tandem monoplane--the _Langley_ type. Author of _Experiments in Aero Dynamics_ and other aerial classics.

LANZ (Karl), Lachnerstrasse 18, Mannheim, Germany. Wealthy patron of aviation in Germany. Giver of the L2000 Lanz prize won by Grade. Financed the _Schuette_ dirigible.

LAROCHE (Madame la Baronne Raymonde de). The first lady aviator. Pilot Ae. C. F. 36. Purchased a _Voisin_ in the summer of 1909 and entered for International events. Badly injured in an accident at Reims, July, 1910. In 1913, took up flying again.

LATHAM (Hubert), 7 Rue Rembrandt, Paris. Of English descent one side. Pilot 9 Ae. C. F. Director of the _Antoinette Cie_. Attempted to fly the Channel, 1909. (1) in _Antoinette IV_. (2) in _Antoinette VII_. (3) in August, 1910. Maker of many records. Killed by a buffalo, 1912.

LA VAULX (Comte Henri de), 2 Rue Gaston de St. Paul, Paris. Born 1870. Chev. Leg. d'Hon. Vice-President Ae. C. F., and one of its promoters. Founder and Vice-Pres. of the F.A.I. Took up aeronautics in 1900, since when he had made over 250 ascents. Record holder for "gas bags." Owner of a _Zodiac_ dirigible in 1909-10.

LEBAUDY (Robert), 12 Rue de Lubeck, Paris. Sugar refiner. Member Ae. C. F. Founder of the _Lebaudy Dirigible Cie_.

LE BLANC (Alfred), 17 Rue Lakanal, Paris. Born 1869. Aeronaut in 1904. Winner of the _Circuit d' l'Est._, Aug., 1910.

LE BLON (_late_). Frenchman. Born 1875. Killed in a _Bleriot_ at S. Sebastien, 2nd April, 1910.

LEFEBVRE (Eugene). French aviator. Killed on a _Wright_, 7th September, 1909, at Juvissy.

LEGAGNEUX. In December, 1910, made a flight of nearly 6 hours, at Pau, (322 mile--53 m.p.h.) average, in a _Bleriot_.

LESSEPS (Comte Jacques de), 11 Avenue Montaigne, Paris. Well-known aviator in the early days.

LEVAVASSEUR. Known in France as "Pere Levavasseur." Chief engineer of the Antoinette Works and _deus ex machina_ of the type. He severed his connection early in 1910, but rejoined in June, 1910, and remained as long as the company existed.

LEVE (Pierre), 17 Rue Cassette, Paris. Editor of _La Revue Aerienne_, official organ of _La Ligue Nat. Aerienne_.

LILIEUTHAL (Gustav), 5 Marthastrasse, Gross-Lichterfelde, Germany. Brother of the late Otto Lilieuthal, whose work he has carried on. Author.

LILIEUTHAL (Otto, the _late_). German subject. Began his interest in aviation when 15 years old. In 1889 published his _Bird Flight as a Basis of the Flying Art_, the result of 25 years observation of sea-gulls and storks. In 1891 he made glider flights. In 1895 he produced a biplane glider. On 12th Aug., 1896, he was killed while experimenting. Lilieuthal was the fountain head of modern aviation.

LINKE (Dr. Franz), Kettenhofweg 181, Frankfurt, Germany, Scientist. Born 1878. Author of _Moderne Luftschiffahrt_ and other works.

LIORE (F.), 4 bis Rue de Corneille, Levallois-Perret, France. Early pioneer with the _Witzig-Liore-Duthileuil_. Since then evolved a monoplane.

LOME (Dupuy de). See DUPUY DE LOME.

LORIDAN. In July, 1910, in a _H. Farman_ racer broke the existing altitude record by making 3,280 m. (10,758ft). Did 702 km., July, 1911.

MALONE (Lieut. Cecil J. L'Estrange). R.N. Navy Wing of British R. F. C. Assistant to director of Flying at Admiralty, end of 1912.

MAHIEU. In September, 1911, made the world's passenger record of 2460 metres (7981 feet) in a _Voisin_, at Issy. Duration of flight 3-1/2 hours.

MANNING (H.) British. Aeroplane designer.

MAREY (Professor). Inventor of the Whirling table, 1870.

MARIE (Capitaine). French Army. On staff of Inspector General of Aeronautics.

MARIE (Pierre). Alsatian. Real name was Bournique. He made his name on a _R.E.P.'s_. In May, 1911, he was trying a 100 h.p. _Deperdussin_ when the machine capsized and fell. He was taken to hospital and died a few hours later. His passenger, Lieut. Depuis, was burned to death.

MARS ("Bud"). Well-known American aviator. Has more than once been reported killed; but always appears again.

MARTIN (Glen L.) Santa Ana, California, U.S.A. Flying _Curtiss_ types. Obtained a considerable reputation, and local amateur record at Los Angeles meet, end of 1910.

MASSAC BUIST. (See BUIST).

MATSIEVITCH (Kapitan). Russian Army. Was Instructor of the Military Aviation School at Sevastopol. Killed at Sevastopol, 1911.

MAXIM (Sir Hiram), Baldwyn's Park, Kent. Inventor of the Maxim gun, etc. American by birth, naturalised British subject. Began experiments with propellers, etc., in 1889. In 1890 to 1893 he experimented with a full-sized aeroplane, steam propelled. Abandoned the experiments after spending L20,000 on them. Resumed work 1909, without success. Author of _Artificial and Natural Flight_.

McCLEAN. British aviator. Towards the end of 1910, he loaned two _Short_ biplanes to Eastchurch flying ground for the training of naval officer in aviation, and himself acted as instructor. These were the first machines used by the British naval officers, consequently Mr. McClean may be regarded as the founder of the British Naval Aeroplane Division. Member of the R. Ae. C. Committee.

MENGIN (L.), 2 Rue Debrousse, Paris. Born 1881. Early experimenter. Flew in 1908 in the _Gastambide-Mengin_, from which the _Antoinette_ was evolved. Director of the late _Antoinette Cie_.

MERRIMAN. British. Expert Bristol flyer at Brooklands, etc., 1912. Instructor.

MESSNER (Haupt. E.), Claridenstr. 36, Zurich. In command Swiss military aviation section, 1911-12.

MICHELIN (A. J.), 105 Boulevard Periere, Paris. Born 1853. Chev. Leg. d'Hon. Director of the well-known tyre manufacturers. Donor of the Michelin prize for aviation. Founder member Ae. C. F.

MOEDEBECK (Hermann W. L.) Born 1857. Died 1910. German author on aerial matters.

MOEDEBECK (Lieut. Col.). German subject. Author of _Fliegen de Menschen_ (Salle), a very useful work on aviation. Also of a _Pocket Book of Aeronautics_, etc.

MOINEAU. Frenchman. In August, 1911, with two passengers made a record on a _Breguet_, of reaching 900 metres in twenty minutes at Douai.

MOISANT (Miss Matilda). Sister of the late J. M. Moisant. Second American lady to obtain certificate. Used a _Moisant_.

MOISANT (John). Architect. American citizen, resident in Paris. Invented two monoplanes. In Aug., 1910, flew the Channel with a passenger in a _Bleriot_. This was the first cross-Channel passenger trip. Killed 1911.

MONTAGU (Lord), of Beaulieu. Editor of _The Car Illustrated_. Prominent in arousing British interest in aviation.

MONTGOLFIER (Joseph Michael and Jacques Etienne). Frenchmen, who about the year 1780 invented hot air balloons. In 1783, one such, of 35 feet diameter, rose to a

height of about 1,500 feet.

MONTGOMERY (John Professor), U.S.A. citizen. Began experiments with gliders in 1884, which he continued till his death by accident with one, on 31st October, 1911, at Evergreen, Santa Clare, California.

MOORE BRABAZON (J. T. C.), 29 Chesham St., London, S.W. Born 1884. Originally sporting motorist; winner Circuit des Ardennes, 1907. Took up aviation at an early stage. Bought an early _Voisin_ which he named _Bird of Passage_. This machine was later sold to A. George, who had a smash in it, and sold it later to Grace. Moore Brabazon was the first Britisher to fly. Pilot I, R. Ae C.

MOORHOUSE (W. B. R.), Portholme Aerodrome, Huntingdon. British. Has done a good deal of cross country flying, 1911. Part inventor of the _Radley-Moorhouse_ (R. M.) monoplane, 1911.

MORANE (Leon). Frenchman. Well-known _Bleriot_ pilot. Subsequently built the _Morane_ monoplane. Very badly injured in an accident, Autumn, 1910.

MOREAU. French amateur. Inventor of a special stabilised aeroplane.

MORIS (Colonel). Commanding Italian air battalion, 1911-13.

NEMETHY (Emil), Arad, Hungary. Born 1867. Built his first effort, a helicopter, in 1899. Has experimented ever since, but without much success. Inventor of the _Aviette_.

NEUMANN. Germany. Author of various very reliable works on dirigibles.

NICKEL (Hugo Ludwig), Kahlenbergerstrasse 97, Vienna. Born 1867. Aerial author and journalist.

NIMFUeHR (Dr. Raimund), Lerchengasse 15, Vienna. Born 1874. Experimentalist 1900 onward. Author.

NORTHCLIFFE (Alfred Charles Harmsworth) Lord. British subject. Founder and proprietor of the "Daily Mail." Donor of many important aviation prizes, including the L10,000 London-to-Manchester prize.

OERTZ (Max), Holzdamm 40, Hamburg, Germany. Interested in gliders. Connected with German North Pole Dirigible Expedition. Designer of various aeroplanes.

OGILVIE (A.) Represented Great Britain in both the 1910 and 1911 Gordon Bennetts. Took fourth place in 1911. Average speed, 51 miles per hour. Flew a _Wright_. In December, 1910, flew for nearly 4 hours on a _Wright_ over the Camber sands; distance being 139-3/4 miles. Associated with the Wright Brothers experiments at Kitty Hawk, October, 1911.

O'GORMAN (Mervyn). Well-known authority on aviation matters, and Superintendent of the Royal Aircraft Factory.

OSMONT. Frenchman. Formerly racing cyclist. Did some fine flights at Chalons in 1910. In February, 1911, appointed chief aviation instructor to the Spanish Army.

OTTO (Fried), Hohenstaufeurstrasse 35, Berlin. W. 30. Aerial journalist, etc.

OVINGTON (Earle). U.S.A. aviator. Carried first U.S.A. aerial post, Sept 1911. Winner of many prizes in America.

OXLEY. Instructor to the Blackburn School, at Filey, 1911.

PAINE (Capt. G. M.) M.V.O., R.N. Commandant of the British Central Flying School at Upavon, Salisbury Plain. Appointed early in 1912.

PARKE (Lieut. Wilfred, R. N.) Started flying in 1910, and made a large number of meritorious performances. Killed in a monoplane at Wembley, December 15th, 1912.

PARSEVAL (Major Von), _late_ German Army. Inventor of the _Parseval_ type dirigibles and the _Parseval_ monoplane. Leading figure in all aerial matters in Germany. (See Part A.)

PATERSON (Compton). British aviator. Liverpool Motor House, Ltd., Liverpool. Designed a successful machine in 1909. Also flies _Farmans_. Designed new machine 1911.

PATIALA (Maharajah of). In December, 1910, purchased a _Bleriot_ and a _Voisin_. Member of the R. Ae. C.

PAGNY. French. Designer of _Hanriots_, 1913.

PAULHAN (Louis). Frenchman. Born 1883. Served afloat as a boy. Later served with the _late_ Col. Renard; also with the _late_ Captain Ferber. In 1907 was with Surcouf. In his spare time he made models. In 1909 he won a _Voisin_ biplane, given as prize for the best model in France, and rapidly came to the front. In 1910 he won the _Daily Mail_ L10,000 prize for the London to Manchester flight. Numerous other prizes have been won by him. Took up construction in 1911 without much success till in 1912 he took over French, etc., agency for _Curtiss_ hydros.

PEQUET (H.) Frenchman. Certificated June, 1910. Flew a _Humber-Sommer_ in India, where he conveyed the first officially recognised aerial post at Allahabad.

PERRIN (H.) British. Secretary of the R. Aero Club.

PERRY (Ida), Metropol Theater, Berlin. German actress, who has gone in for aviation.

PFITZNER (_late_ Lieut. Alexander L.) Hungarian. Born 1875. Served in the Austro-Hungarian Artillery. Leaving the Army he went to the U.S.A. and became connected with the Herring-Curtiss work, designing the novel Pfitzner monoplane. He met many mishaps with this, which depressed him. He returned to Hungary early in 1910, but meeting no success there came back to America. Drowned in Marblehead Harbour, 12th July, 1910.

PHILLIPS (Horatio F.), Wealdstone, Harrow. Pioneer experimenter. Discoverer of the "dipping front edge," patented by him 1884 and 1891. ("Philips' entry.") Leading authority on aviation subjects.

PICKLES (Sydney). Australian. Chief pilot at the Ewen school at Hendon, 1912.

PICHAN (Court). Early French experimenter. Flew a flapper model 1889.

PICOLLO (_late_ Jules). Brazilian aviator. Killed December 28th, 1910.

PIERRE (Petit). Frenchman. The _late_ secretary of Bleriot School at Hendon. Assassinated at Hendon, August, 1911, by a Swiss pupil named Hanot, who went insane at not learning to fly so quickly as he had expected.

PILCHER (_late_ Percy S.) Born 1866. British naval engineer. Commenced glider experiments, 1895, on Lilieuthal lines. Designed a power machine in 1899, but was killed in glider experiments before it was completed.

PIXTON (H.) British R. Ae. C. pilot 50. Qualified at Brooklands, January, 1911, on a triplane. Has since done some very fine flights on an _Avro_, taking various prizes.

PISCHOFF (Alfred de), 12 Rue Amiral de Joinville, Paris. In conjunction with Koechlin was a pioneer of French aviation. In December, 1907, he flew a kilometre on a biplane. His earliest machine was practically a large box kite with a motor fitted. In 1910 produced a monoplane of his own design. He is an Austrian resident in France.

POPPER (Josef). Austrian. Concerned with aviation, etc., ever since 1872.

POLLOCK (C. F.) Prominent supporter of aviation. Member of the R. Ae. C. Committee, 1910-11.

PONNIER. Frenchman. Director of the Hanriot Company.

PRANDTL (Dr. Ludwig). Prinz Albertstrasse 20, Goettinger, Germany. Born 1875. Leading Figure in German aerial circles. Connected with the _Parseval_ design.

PREVOST (M.) French. Created world's record on December 2nd, 1911, by reaching a height of 9,800 feet at Rheims.

PRIER (Pierre). Made London-Paris in 3 hours, 56 minutes, 12th April, 1911. Designer to the _Bristol_ Co., 1911.

QUEROZ (the _late_). Brazilian. Killed at S. Paulo, June, 1911, in a monoplane of his own design.

QUIMBY (Miss Harriet). Mineola, U.S.A. First American lady to qualify for aviator certificate, 1st August, 1911. Used a _Moisant_. Killed 1912.

QUOIKA (Haupt. Emanuel), Margarethenstrasse 16, Vienna. Aeronaut from 1904. Now aviator and writer on subject.

RADLEY (James). Well-known British aviator, flying a _Bleriot_. Patented a special wing. Represented Great Britain in the 1910 Gordon Bennett. At Lanark, 1910, broke the world's then speed record, and did 75 miles per hour. Pilot R. Ae. C. 12, June 14th, 1910. August, 1911, flew the Channel in 22 minutes; Calais to Folkestone. Subsequently embarked on construction.

RAYNHAM (F. R.) British subject. Flew 7 hours, 30 mins. competing for the Michelin Cup. Used an _Avro_ fitted with a 60 horse Green.

REISSNER (Dr. Ing. Hans), Lutticherstrasse 166, Aachen. Born 1874. Professor on matters aerial.

RELTICH. French. Cyclist who succeeded in getting an avietter to fly one metre, October, 1912. Won the Dubos prize.

RENARD (_late_ Colonel). In association with Krebs built a dirigible in 1884, with electric motor. Killed.

RENARD (Commandant Paul), 41 Rue Madame, Paris. Born 1854. Officer Leg. d'Hon. Brother of late Col. Renard, with whom he worked. Vice-President, _Ligue Nat. Aerienne_. Professor _Ecole Sup. d'Aeronautique_. Has written a good deal on aerial subjects.

RENAUX. Did 12 hours 12 minutes on a _M. Farman_, 7th August, 1911. (690 k.m.) Won the Quentin Bauchart Prize, 1911.

RENAUX. French aviator. Winner of the Grand Prix Michelin, March, 1911, Paris, to top of the Puy de Dome. Machine, _Maurice Farman_.

RICHET. French patron of early aviation experiments, 1896. Tatin built a large model machine for him in those days, which after a 150 yard flight fell into the sea and was lost.

RIDGE (Theodore). Assistant Superintendent of the Army Aircraft Factory. Killed on August 21st, 1911.

ROBINSON (Hugh). Well-known U.S.A. aviator.

ROBL (_late_ Thaddeus). German aviator. Killed on a _Farman_, 1910, through attempting to fly in unpropitious weather in order to allay the complaints of sightseers. Has been designated the "first martyr of aviation"--not without some cause.

RODGERS (C. P.) U.S.A. aviator. _Wright._ In September-October, 1911, he flew across America, distance 4,321 miles. He started to win the Hirst prize of L10,000, but having taken longer than 30 days was disqualified.

ROE (A. V.) Clifton St., Miles Platting, Manchester. Was the first man to fly in England, and also the first to fly an all-British machine. Is a persistent experimenter on original lines. Has flown with as little as 9 h.p. in one of his triplanes. Now builds mono. and biplanes (_Avro_).

ROEHRIG (B. F.) U.S.A. aviator. Obtained wide reputation with _Curtiss_ types on Pacific Coast.

ROGER, 8 Rue Grange-Bateliere, Paris. Founder and editor of _Revue de l'Aviation_.

ROGUES (General). French Army. Inspector General Military Aeronautics, 1911.

ROLLS (_late_ Hon. C.) Well-known British sportsman, motorist, and aviator. First Englishman to order an aeroplane--a _Wright_. Flew the Channel both ways early in 1910 (first record). Killed at Bournemouth, July, 1910, in a _Wright_.

RUCK (Major-General), C.B., R.E. Chairman of the Aeronautical Society of Great Britain.

RUSSIJAN. Austrian aviator. Killed January 9th, 1911.

SALMET (Henri). French. Born 1878. Made British height record, 8,070 feet, November, 1911. Made record London-Paris flight, March, 1912. Time: 3 hours, 14 minutes.

SAMPSON (Lieut.) British Navy. On August 17th, 1911, made British flight duration record to date, 4 hours 58-1/2 minutes, at Eastchurch on a _Short_ 38. Now Acting-Commander. Employed by Naval Wing, R.F.C.

SAMUELSON (Arnold), Hamburg Waterworks, Germany. Born 1837. Writer on aerial matters.

SANTOS-DUMONT (Alberto), 150 Avenue des Champs-Elysees, Paris. Brazilian, of French descent. Born 1873. Officer Leg. d'Hon. Took up ballooning at an early age. He was the first to use a petrol motor in a balloon. In 1900 the fifth dirigible constructed by him crossed the Seine. On Oct. 19th, 1901, in No. 6, he circled the Eiffel Tower and won the 100,000 franc Deutsch prize. In 1906 he became interested in heavier than air machines, and began on a helicopter. Abandoning this he built a box kite type of aeroplane, and on October 23rd, 1906, won the Archdeacon prize for a heavier than air flight of not less than 25 metres. Thereafter, comparatively little was heard of him, except that he was experimenting with the _Demoiselle_, till in 1909 he made a record on this type--the designs of which he presented to the world. Has not been prominent since.

SCHABSKY (Athanasius Ivanovitch). Russian. Builder of the _Outchebny_ type dirigible.

SCHIERE, J. Aeronautical engineer. Stephonsonstraat 41, The Hague, Holland. Librarian Dutch Ae. C.

SCHUeTTE (Prof. Johann), Jaeschkenthal 47b, Danzig-Langfukr, Danzig, Germany. Born 1873. Designer of the _Schuette_ dirigible.

SCHWANN (Commander Oliver). British Navy. Navy Air Dept., 1912-13. In 1911, conducted a number of hydro-aeroplane experiments.

SCRAGG (Geo. H.), American citizen, 19-21, Great Queen Street, Kingsway, London, W.C. European correspondent of American _Aeronautics_.

SELLERS (M.B.) (See U.S. aeroplanes)

SELLS (Chas. de Grave), La Colombara, Cornigliano-Ligure, Italy. British. A leading authority on all matters having to do with engineering. Also a writer on these subjects. Authority on matters having to do with aviation in Italy.

SHAFFER (Cleve T.) American citizen. West Coast correspondent to _Aeronautics_ (U.S.A.) Writer on aerial subjects generally.

SIMON (Rene). August 18th, 1911, tied with Sopwith for the world's Climbing speed at Chicago; 500 metres in 3' 35".

SMITH (H. White). British. Secretary to the Bristol Co.

SOMMER (Roger) Mouzon, Ardennes, France. Born 1877. Early interested in aviation. In 1908 built a machine of his own design. This was a failure. He then bought one of the first _Farman's_, on which he rapidly achieved success. Towards the end of 1909 he produced the _Sommer_ biplane.

SOPWITH (T.) British. Won the Baron de Forest prize on a _Howard Wright_, 1910. Also won many other prizes in England and America. 19th August, 1911, tied with Simon, world's climbing speed--500 metres in 3' 35"--at Chicago. Now a constructor.

SPENCER (Stanley). Early British dirigible builder (1902). Died 1913.

SPOONER (Stanley), 41 St. Martin's Lane, W.C. Editor of _Flight_. Prominent supporter of aviation. Member of R. Ae. C. Committee.

STEIN (Lieut.) German aviator. Killed at Doerlitz, February 6th, 1911.

STRINGFELLOW. British. A very early experimenter. In 1868 he evolved a triplane model.

SUETER (Capt. R. N.) British. In command of British Navy dirigible section, 1911. Admiralty Air Dept., 1912-13.

SURCOUF (Edward Louis), 33 Boulevard Lannes, Paris. Born 1862. Chev. Leg. d'Hon. Secretary Com. Sport Ae. C. F. Sec. Com. Aerienne Mixte. Director of the _Astra_ Societe. Constructor of the majority of French dirigibles.

SWANN (Rev. Sydney), The Vicarage, Crosby Ravensworth, Westmoreland, England. First clerical aviator. Ceased.

SYKES (Major F. H.) Officer Commandant in Charge of Records, Royal Flying Corps, Military Wing.

TABUTEAU. French aviator. Winner of the Michelin Trophy.

TADDEOLI. Swiss. First Swiss to obtain an aviator's certificate, which he did on a _Dufaux_, October, 1910. Badly injured at Lausanne, June, 1911, during exhibition flights. 1912, built a hydro-aeroplane.

TATIN (Victor), 14 Rue de la Folie-Reynault, Paris. Chev Leg. d'Hon. Born 1843. Commenced heavier than air experiments so long ago as 1879, when he made an aeroplane driven by compressed air. Designed the _Ville de Paris_. Had a good deal to do with the _Bleriot_ in its early days. In 1909 designed the _Clement-Bayard_ monoplane. Associated with Paulhan in 1911. Writes on all aerial subjects.

TAYLOR (Vincent P.) Australian subject. Well-known aeronaut, using the _nom de plume_ of Capt. Penfold. In 1912 went in for aeroplaning, using a _Bristol_.

TISSANDIER (Gaston). French Pioneer aeronaut. Made an electrically-propelled dirigible in 1881. Born 1843. Died 1899.

TISSANDIER (Paul), 17 Avenue Victor Hugo, Paris. Son of Gaston Tissandier. Born 1881. Instructor of aviation. Taught many of the best known aviators.

TURNBULL (W. R.) American Engineer. In the year 1906 commenced to experiment with hydro-aeroplanes; and may be regarded as the originator of all

experiments in this direction. The French _Gabardine_ of much later date did not differ materially from his early models, while the more recent _Fabre_ and the successful _Curtiss Triad_ embodied similar ideas.

TURNER (Charles E.) Authority on aviation matters, special aerial correspondent of the _Observer_, etc.

TURNER (Lewis W. F.) British. Chief pilot of the Ewen School, 1912.

TWINING (S. Frisco). Cal. U.S.A. Experimenter with flappers, man propelled, from 1910 onward.

USBORNE (Lieut. Neville F.), R.N. First British naval officer detailed for aerial work. Was appointed to _Clement-Bayard II_ in 1909, and subsequently to the first Naval Dirigible. 1912, Naval Wing, R.F.C.

VANNIMAN (Melvin). Built the gondola of the first _Wellman_ airship, and intimately concerned with _Wellman II_. Also designed a triplane 1908. Designed _Akron_, 1911. Killed 1911.

VEDRINES. French. Second in the _Daily Mail_ L10,000 prize, 1911, in a _Morane_. Won Paris-Madrid, 1911. One of the best known aviators. Began life as a mechanic.

VIVALDI (_late_ Lieut.) Italian naval officer. Killed in a _M. Farman_, August, 1910.

VOISIN (Charles), 34 Quai du Point du Tour, Billancourt (Seine), France. Born 1882. Director of _Voisin Freres_. Flew the _Delagrange I._ in 1906. Induced H. Farman to be interested in aviation.

VOISIN (Gabriel). Brother of above. Born 1880. Chev. Leg. d'Hon. Director of _Voisin Freres_. Commenced to study aviation in 1902 with Archdeacon. Experimented with gliders. Founded _Voisin Freres_ in 1903. Designer of the _Voisin_ biplane. Killed 1912 in a motor accident.

VUIA. French pioneer, who with a machine somewhat like a _Demoiselle_, flew 6 yards in 1906 and 60 yards in 1907.

WALDEN (Dr.). U.S. citizen. Badly hurt, 1910, in a machine of his own design, but not killed as reported. (See U.S. aeroplanes.)

WALSH (C.F.) American aviator. Winner of various trophies on a _Curtiss_.

WARCHOLOWSKY. Austrian aviator. On October 30th, 1911, made world's record to date by flying 45 minutes with three passengers.

WEILLER (Lazare), 36 Rue de la Bienfaisance, Paris. Officer Leg. d'Hon. Head of the syndicate which in 1908 was responsible for Wilbur Wright coming to France.

WEISS (Jose). British subject. Pioneer experimenter in aviation. The starting stage used by him for early glider experiments is still to be seen near Arundel Castle, Sussex. Much of our knowledge as to the distribution of weights is due to him.

WELLMAN. An American who hoped to reach the North Pole by dirigible. His first ship came to grief at Spitzbergen. In Oct., he attempted a cross-Atlantic voyage, but

failed. (See Vanniman).

WEYMANN (C). American. Won the 1911 Gordon Bennett on a _Nieuport_. Average speed, 78 miles per hour.

WHEELER (R.F.) British Navy. As naval cadet at the age of 15 he obtained his pilot certificate at the Bristol School.

WHITE (Sir George, Bart. LL.D. J.P.) Founder and Chairman of the British and Colonial Aeroplane Co., Ltd. President of the Bristol and West of England Ae. C.

WIDMER. Austrian aviator. In October, 1911, made a flight over the Adriatic, Venice to Triest.

WILLOWS (E.T.) Cardiff, Wales. British Airship pilot 4. Inventor of the _Willows_ airship. Patentee swivelling propellers. Started a dirigible school, 1913.

WISEMAN (Fred T.) American aviator. Flies his own type machine. Has made sensational flights delivering newspapers at farmhouses, April, 1911, also letters.

WRIGHT (Howard). See HOWARD WRIGHT.

WRIGHTS (the) (Orville and Wilbur), 7 Hawthorn Street, Dayton, Ohio, U.S.A. Chevs. Leg. d'Hon. In 1896 the Brothers Wright began to study aerial flight. In 1900 they were making glides. In 1903 they first fitted a motor, and on December 17th of that year made a power flight of about 250 yards. Reports of this were received with incredulity, and right up to July, 1908, when Wilbur Wright appeared in France, many people still regarded the Wrights as a myth. Wilbur Wright easily beat the French machines in circling, etc. He won the Michelin Cup, being up 2 h. 20 m. 23-1/3 sec. Distance 76-1/2 miles official record. Actual, estimated at 93 miles. The exploits of Wilbur Wright put aviation on quite a new footing. Since 1908 the _Wright_ type has been surpassed by others; but to the Wrights will always belong the credit of having made a decided step in the science. Wilbur died of typhoid, 1911.

WYNMALEN (Henri). Dutch. Reached 9,121 feet in a _Farman_ in 1910, and was then compelled to descend because after 8,000 feet blood oozed from his finger nails and lips. Ae.C.F. pilot 208, 27th August, 1910. Has made many famous flights.

ZENS (Ernest), 3 Rue la Boetie, Paris. Born 1878. Pioneer aviator. On committee of Ae. C. F. First passenger in an aeroplane (carried by Wilbur Wright, 6th September, 1908). Built a monoplane, 1912.

ZEPPELIN (Count). The first Zeppelin dirigible was tried in 1900 on Lake Constance. It made a small speed against a 12-16 m.p.h. wind. It also circled. The experiments exhausted the Count's resources until 1905. Details of this and later _Zeppelins_ will be found on the German dirigible pages in Part A.

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CHOTIN (G.), 34 Rue des Archives, Paris

CIRET (F.) & Cie, 140 Rue Rivoli, Paris

COOK & Cie, 23 Rue Auber, Paris

CRABETTE, 54 Faubourg Saint-Honore, Paris

DAMERVAL (A.), 9 Rue Reamur, Paris

DAROLES-VINCENT, 22 Rue de Faubourg-du-Temple, Paris

DAY, 162 Rue du Faubourg-Saint-Martin, Paris

DEITZ (E.), 56 Rue d'Aboukir, Paris

DENIAU & Cie, 86 _bis_, Rue de Rome, Paris

DEWACHTER, 53 Boulevard Voltaire, Paris

DUGAS, Freres, 10 Boulevard Sebastopol, Paris

DUROT & LERY, 25 Rue des Trois-Cailloux, Amiens

DUBESSY (J.), Villefranche

DUBREUIL & PARMENTIER, 34 Rue Montorqueil, Paris

EGGER & Cie, 2 Rue de la Vrilliere, Paris

ESDERS (Maison Henri), 115 Rue Montmartre, Paris

FASHIONABLE HOUSE, 16 Boulevard Montmartre, Paris

FELDSTEIN, 91 Rue des Marais, Paris

FRAENKEL (H.), 28 Rue du Quartre-Septembre, Paris

GALERIES LAFAYETTE, 40 Boulevard Haussmann, Paris

GRANDE MAISON (A La), 7 Rue Croix-des-Petits-Champs, Paris

HALIMBOURG-AKAR (Etablissements), 1 Places des Victoires, Paris

HENRY-TREILLE, Marcigny

HIGH-LIFE, 112 Rue de Richelieu, Paris

HUTCHINSON (Etablissements), 60 Rue Saint-Lazare, Paris

KRIEGCK & Co., 23 Rue Royale, Paris

LACHASSAGNE (E.), Saint-Etienne

LAMBLIN (A.), 15 Rue Tiquetonne, Paris

LAMARTINE, 24 Rue des Bons-Enfants, Paris

LECONGE & WILLMANN, 2 Rue du Renard, Paris

LEON, 21 Rue Daunou, Paris

LOUVRE (Grande Magazines du), 164 Rue de Rivoli, Paris

LYON (Grand Bazaar de), 31 Rue de la Republique, Lyon

MAGNANT & Cie, 117 Rue Reaumur

MAGNE (A.), Moulins, France

MANBY, _les_ 19 Rue Auber, Paris

MARCHAL (M.), 30 Rue le Peletier, Paris

MARECHAL (A.), Nevers

MAX-AUSPITZ, 374 Rue Saint-Honore, Paris

MICHEL JACKSON (A.), 92 Rue Richelieu, Paris

MICHEL JACKSON (E.), Halluin

MENAGERE (A la), 20 Boulevard Bonne-Nouvelle, Paris

METTEZ (Maison), 5 Place de l'Hotel de Ville, Paris

MOLAY (Jacques), 181 Rue du Temple, Paris

MATHAN (G.), 27 Rue Saint-Sabin, Paris

NICOLLE, 29 Rue Tronchet, Paris

OLD ENGLAND-- 12 Boulevard des Capucines, Paris 114 Via Nazionale, Milan, Italy

OLIVIERI & Co., 101 Rue Claude-Decaen, Paris

PAGUIN (J.) BERTHOLLE & Cie, 43 Boulevard des Capucines, Paris

PARIS-TAILLEUR, 3 Rue du Louvre, Paris

PAYEN (Maison G.), 7 Rue de la Republique, Lyon

PETIT MATELOT (Au), 41 quai d'Anjou, Paris

PFEIFFR-BRUNET, 17 Rue de l'Ancienne-Comedie, Paris

PRINTEMPS (Magasius du), 70 Boulevard Haussmann, Paris

RAGEUNEAU, 25 Avenue de la Grande-Armee

REAUMUR (A.), 82 Rue Reaumur, Paris

REVILLON, Freres, 77 Rue de Rivoli, Paris

RIBBY, 16 Boulevard Poissoniere, Paris

RICOUR, 26 Rue du Bouloi, Paris

RODDY, 2 Boulevard des Italiens, Paris

ROFFY, 2 bis, Rue du Bouloi, Paris

ROUSSEAU, 61 Passage du Havre, Paris

ROYAL TAYLOR, 41 Avenue de Wagram, Paris

RUSSIAN AMERICAN INDIA RUBBER Co., 47 Rue Saint Ferdinand, Paris

SAINT, Freres, 34 Rue du Louvre, Paris

SAMARITAINE, Rue du Pont-Neuf, Paris

SEYNOHA (F.), 249 Rue Saint-Honore, Paris

"SIEG," 19 Avenue de la Grande-Armee, Paris

SORIN & MARZETTIER, 2 Rue Haudaudine, Nantes, Paris

SPORT (The), 17 Boulevard Montmartre, Paris

STEINMETZ, Freres, 16 Rue Cambronne, Paris

STROM (D. SCHNEIDER & Cie)-- 16 Rue de la Chaussee-d'Antin, Paris 33 Avenue de la Gare, Nice

TELEPHONE (Societe Industrielle des), 25 Rue du Quartre Septembre, Paris

THIERY & SIGRAND, 18 Boulevard Sebastopol, Paris

TORRILHON (J. B.), Chamalieres

TROIS-QUARTIERS (Aux), 17 Boulevard de la Madeleine, Paris

TUNMER (A.) & Co., 27 Rue du Quartre-Septembre, Paris

VELOCE-CLUB (Au), 21 Avenue de la Grande-Armee, Paris

VINCENE, 148 Rue du Temple, Paris

VOLLANT (A.), 34 Boulevard Sebastopol, Paris

WEST END TAILORS, 10 Rue Auber, Paris

WILLIAMS & Cie, 1 Rue Caumartin, Paris

~GERMAN.~

ANWANDER (A.), 22 Sonnenstrasse, Munich

HERTZOG (R.), 15 Breiterstrasse, Berlin

~ITALIAN.~

MARTINY (Manufacture), 5 Via Pietro Micca, Turin

SANGUINETTI (Frat), 8 Corso Vittorie Emanuele, Milan

~SPANISH~

SANCHA (M.), 12 Calle de la Cruz, Madrid

~SWISS.~

GEISTDORFER & Co., 4 Paradeplatz, Zurich

~U.S.A.~

SCANDINAVIAN FUR & LEATHER Co., 16 West 33rd Street, New York

~HANGAR AND SHED BUILDERS.~

~BRITISH.~

AEROPLANE SUPPLY Co., Ltd., 111 Piccadilly, London, W.

HARBROW (W.), South Bermondsey Station, London, S.E.

HARRISON, SMITH Buildings, Ltd., Vauxhall Works, Dollinan Street, Birmingham.

HUMPHREYS Ltd., Knightsbridge, London, W.

MORTON, FRANCIS & Co., Ltd., Hamilton Ironworks, Garston, Liverpool.

~PIGGOTT, Bros. & Co., Ltd.~, 220, 222, 224, Bishopsgate, London, E.C.

SMITH (F.) & Co., Carpenters Road, Stratford, London, E.

WIRE-WOVE ROOFING Co. & PORTABLE BUILDINGS Co., 108 Queen Victoria St., London, E.C.

~FRENCH.~

~BESSONNEAU~-- 29 Rue du Louvre, Paris 21 Rue Louis Gain, Angers

COMPAGNIE AERIENNE, 63, Avenue des Champs, Elysees, Paris

CONSTRUCTIONS DEMONTABLES (Compagnie des), 54 Rue Lafayette, Paris

CONSTRUCTIONS ECONOMIQUES (Societe de), 11 Avenue de l'Opera, Paris

DUBOIS et Cie, 7 Rue Saint-Amand, Paris

LAPEYRERE (L.), 44 Rue de l'Eglise, Paris

OFFICE d'AVIATION, 3 Avenue de l'Opera, Paris

RUBEROID (Societe du), 82 Boulevard Beaumarchais, Paris

SAINTE-BEUVE (A.), 196 Quai Jemmapes, Paris

~GERMAN.~

MULLER (A.), 27 Fritcherstrasse, Berlin-Charlottenburg

~HYDROGEN SUPPLIES.~

~BRITISH.~

BRITISH HYDROGEN Co. (Lane's System), 49-50 Parliament Street, London, S.W.

BRITISH OXYGEN Co., Ltd.:-- Elverton St., Westminster, London, S.W. Saltley Works, Birmingham Great Marlborough St., Manchester Boyd St., Newcastle-on-Tyne Rosehill Works, Polmadis, Glasgow

~KNOWLES' OXYGEN Co., Ltd.~, Wolverhampton.

WOLF (J.), 15 Seething Lane, London, E.C.

~FRENCH.~

ELECTROLYSE FRANCAISE (L'), 4 Rue des Ecluses, Saint Martin, Paris

HYDROGENE pour l'Aerostation et l'Industrie (Soc. Francaise de l') (Lane's System), Boulevard Senart, St. Cloud (Seine et Soise).

HYDROXGENE PUR (L') 22 Rue de Douai, Paris Marais de Lomme, Lille (Nord)

OXYDRIQUE FRANCAISE (L'), 2 Rue Nouvelle, Paris

~INSURANCE (AVIATION).~

~BELGIAN.~

MONET (Alfred), 3 Avenue de Cortambert, Bruxelles, Belgium

~BRITISH.~

AEROPLANE SUPPLY Co., Ltd., 111 Piccadilly, London, W.

~BRAY, GIBB & Co., Ltd.~, 14 Sherborne Lane, King William Street, London, E.C.

~CAR & GENERAL INSURANCE CORPORATION, Ltd.~, 1 Queen Victoria Street, London, E.C.

DOLAMORE (W. T.), AVIATION INSURANCE BROKER, 199 Piccadilly, W.

FORBES (M. W.) & Co., 15 Queen Street, London, E.C.

GLASGOW ASSURANCE CORPORATION, Ltd., 10 Queen Street, Cheapside, London, E.C.

GOLD (Guy), 1 Cornhill, London, E.C.

KINLOCH (D. A.), 13 Leadenhall Street, London, E.C.

PLANCHE, HEARN & Co., 12 Newgate Street, London, E.C.

WHITE CROSS INSURANCE ASSOC., 1 Cornhill, London, E.C.

~FRENCH.~

ASSURANCE SPECIALES d'AUTOMOBILES (Les) 20 Rue Taitbout (Seine), Paris

BANDU DE CHANTPIE (Ch.), 8 Rue Blanche, Paris (Seine)

CAPRON & HAREL, 10 Rue Viollet-le-Duc, Paris

CASANIVA ET GRIBAUMONT, 50 Boulevard Maesherbes, Paris

CAUBERT ET GARNIA (E.), 5 Rue Moreau, Paris

FASTINGER (L.), 8 Rue du Sentier, Paris

HANCIAN (G.), Omnium des Assurance Terrestries, 59 Rue de Chateaudun

HURET (G.), 56 Rue d'Amsterdam, Paris

LAURIERS (Des) et DUMONT, 43 Rue Lafitte, Paris

LAW-CAR, 42 Rue Pergotese, Paris

LE CHARTIR ET DARDONVILLE, 12 Avenue Moatespan, Paris

LEFEVRE (P.), 7 Rue Villaret-de-Joyeuse, Paris

LLOYD (Continental), 17 Rue Druout, Paris

MULLER & DESPIERRES (G.), 26 Rue Etienne-Marcel, France

NICOLLEAU (Auguste), 36 Rue de la Chapelle, Paris

PIEFR (G.), 92 Boulevard Richard-Lenoir, Paris

STEVENS (Pierre), 26 Rue Bergere, Paris

TERRIER (V.), Courtier d'Assurances, 81 Boulevard Sebastopol, Paris

TROLLET (H.), 131 Rue de Rome, Paris

~LUBRICANTS.~

~AUSTRIAN.~

GERSON BOEHN & ROSENTHAL, 20 Donaueschingenstrasse, Vienna XX.

~BELGIAN.~

BENZO-BELGE (la), 11 Boulevard du Regent, Brussels

GUELETTE & Cie, Hug. (_Diamond-Running Oil._)

~BRITISH.~

ADAMS BRITISH OIL Co., Ltd., Plough Bridge, Deptford, London, S.E.

~ANGLO-AMERICAN OIL CO., Ltd.~, 22 Billiter Street, London, E.C.

ANGLO-BOSPHORUS OIL Co., Ltd., Bristol

BOWRING PETROLEUM Co., Ltd., Finsbury Court, London, E.C.

BRITISH MONOGRAM OIL Co., Ltd., 177 The Vale, Acton, London, W.

BUTTERWORTHS, Ltd., 5 Roscoe Chambers, Liverpool

BRITISH PETROLEUM Co., Ltd., 22 Fenchurch Street, London, E.C.

CARLESS, CAPEL & LEONARD, Hope Chemical Works, Hackney Wick, London, N.E.

COUNTY CHEMICAL Co., Ltd., Chemico Works, Bradford Street, Birmingham

DICK & Co,, Ltd., 33 Eastcheap, London, E.C.

ENGLEBERT & Co., 119 Finsbury Pavement, London, E.C.

GRINDLEY & Co., Ltd., Poplar, London, E.

KAYE (J.) & Sons, Ltd., 93 High Holborn, London, W.C.

MONOVO Co., Mono Works, Stewart's Road, London, S.W.

O'BRIEN (H.F.) & Co., Broadheath Oil Works, Manchester

PETROLEUM Co., Ltd. (The British), 22 Fenchurch Street, London, E.C.

PRICE'S PATENT CANDLE Co., Ltd., Belmont Works, Battersea, London, S.W.

ROSE (Sir W. & Co.), 66 Upper Thames Street, London, E.C.

STERN-SONNEBORN (A. G.), Royal London House, Finsbury Square, London, E.C.

VACUUM OIL Co., Ltd., Caxton House, Westminster, London, S.W.

WAKEFIELD (C. C.) & Co., 27 Cannon Street, London, E.C.

WHITE, 47 Curtain Road, London, E.C.

WILCOX & Co., Ltd., 23 Southwark Street, London, S.E.

~DANISH.~

BEAUVAL (de) Saxlund, 18 Kobmagergade, Copenhagen

MEYER & HENCKEL, 60 Kobmagergade, Copenhagen

~FRENCH.~

ACKER, 7 Rue de Bac, Ivry Port (_Auto Victoire._)

ANDREE (A.) Fils (Societe Anonyme), 8 Rue de la Tour-des-Dames, Paris (_Volgaline & Spidoleine._)

AMELIN & RENAUD, 37 Rue Jean-Jacques-Rosseau, Paris

AMERICAN OIL Co., 42 Rue Lepeletier, Paris

BADIN, 3 Rue de la Mare, Paris

BAILLY, 8 Rue de la Michodiere, Paris

BANTEGNIE & NEVU, 10 Rue Bateau, Aubervilliers

BARBAT (C.), Charenton

BAUD, 24 Rue Saint-Roch, Paris

BAUDOUIN, 32 Quai Saint-Vincent, Lyon

BEDFORD PETROLEUM Co., 67 Boulevard Hausmann, Paris

BESANCON (E.), Saint-Denis

BONIFACE, Freres, Sotteville-L-Rouen

BONNEVILLE, ROUILLY & Cie, 27 Rue du Landy, Saint-Denis

BORREL & Fils, 58 Rue de Vincennes, Bagnolet

BOUCHON & BERTRAND, 17 Rue des Bateliers, Clichy

BOUGAULT & Cie, 32 Boulevard Ornano, Paris

BOURGEOIS-OUDRY, 18 Rue de la Paix, Vincennes

BUISINE & Cie, 35 Rue de Viarmes, Paris

BURCKHARDT, 18 Rue Poliveau, Paris (_Auto-Gazoline._) (_Auto-Moto._)

CABANNE-NIROUET, 124 Route de Joinville, Champigny-s-Marne

CALISCH-ORESTE, 4 Avenue du Cog, Paris

CAMUS, 5 _bis_, Rue des Rosiers, Paris

CAPET, 61 Rue de la Verrerie, Paris

CATHALIFAUD, 120 Boulevard Magenta, Paris

CAUET, 18 Boulevard Pagel, Saint-Denis

CAYEUX, Place de Marche-aux-Herbes, Compiegne

CHAILLY, 15 Rue Catulienne, Saint-Denis

CHATELET, 30 Rue de Fontenay, Nogent-sur-Marne

CHAUDIN & Cie, 132 Faubourg, Saint-Denis, Paris

CHEMET, 143 Route de Versailles, Boulogne

CHEMIN (A.), 10 Rue Gresset, Amiens (_Lubrifa._)

CHICHIGNAUD Au CORNILLON, Saint-Denis

CHOUILLOU, 14 Rue Duphot, Paris

CLAUDY, 92 Rue Neuve-des-Charpennes, Lyon

COLMET & Cie, 70 Rue de Rivoli, Paris

COLUMBRIA (Soc. des Prod. & Pub.), 48 Rue de Paris, Saint-Denis

COSTADAU, 13 Rue Vendome, Lyon (_Golden Oil._)

DANIEL, 4. Rue Villedo, Paris.

DEGREMONT, 21 Rue Gudot-de Mauroi, Paris (_Lion_.)

DEGUEANT, Avenue Lagache, Villemonble

DELAGE, Quai d'Issy, 37 Issy-les-Moulineaux

DELETTREZ. 7 Rue Gide, Levallois-Perret (_G.D._)

DELIGNY, 3 Rue de Buisson-Saint, Louis, Paris

DESCROIX (P.) & LESAGE, 18 Rue de Normande Asnieres

DESSALLE, 39 Rue de Paradis, Paris

DEUTSCHE (Les Fils de), 50 Rue de Chateaudun, Paris (_A.D._) (_Jupiter._) (_Viscositas._)

DION BOUTON (De), 36 Quai National, Puteaux

DOMONT, 36 Boulevard Ornano, Paris

DROUOT, 172 Faubourg Saint-Martin, Paris

FAUCHER, 106 Boulevard Sebastopol, Paris

FEIGEL, 14 Rae Barbette, Paris

FERRANDON, 164 Avenue de Valmy, Paris

FERRON, 59 Boulevard Saint-Denis, Courbevoie

FIRBACH, 16 Rue Violet, Paris

FLOQUET, 36 Rue de la Haie-Cog., Paris

FOURNIER, Freres, 12 Rue Casteres, Clichy

FRANCO-RUSSE, Cie, 10 Rue Thimonier, Paris (_Newoleine._)

GAGNEPIAN, GONNOT & Cie, 109 Rue Victor-Hugo, Levallois-Perret

GALENA OIL Co., Paris

GAMARD & LAFLECHE, 8 Rue de Thorigny, Paris

GARDAIR, 71 Rue de Vaugirard, Paris

GAUBERT, 40 Avenue de la Grande-Armee, Paris

GENERAL INDUSTRIELLE (La), 5 Boulevard Voltaire, Paris

GEORGIER (A.), 8 Route de Flandre, Bourget

GIRARD, 102 Rue du Gazometre, Montreuil (_La Becanine_)

GONNOT, 33 Boulevard de la Chapelle, Paris

GUILLAUD & VALLAT, 36 Chemin, Saint-Matthieu, Lyon

GUILLET-PUSARD, Fils et Cie, 4 Rue Poccard, Levallois-Perret (_Royal Oil._)

GUYENOT (J.), 1 Rue du Printemps, Paris (_Motoleine._)

HACHARD, 43 Boulevard, Richard-Lenoir, Paris

HAMELIN, 65 Rue Rivay, Levallois-Perret

HAMELLE, 21 Quai de Valmy, Paris (_Valvoline_).

HARMIGNIES, 105 Rue de Paris, Ivry Port

HERZEMBERG, 60 Rue Saint-Mande, Saint-Ouen

HUILES & GRAISSES INDUSTRIELLES DE, 18 Rue Gambetta, Nice (_Omnia._)

HUILES-VITESSE (Soc. An. des.), Rue des Minimes, Courbevoie

INDUSTRIELLE GENERALE (L'), 27 Rue la Bruyere, Paris

LACARRIERE & GRAVELIN, 11 Rue de Neuilly, Clichy (_La Preferee._)

LAGET, 181 Rue Lafayette, Paris

LAMPE, Freres, 32 Rue Saint-Lazarre, Paris

LA SELVE & BOURGEON, 54 Chemin des Cures, Lyon (_Auto Oil_)

LAVOIX, Le Bourget

LEBRASSEUR & Cie, 155 Rue de Paris, Saint-Denis

LEBRASSEUR, 11 Rue de la Vega, Paris

LECLERC (C.), 33 Rue Auger, Pantin

LENOIR, 24 Rue Michelet, Pantin

LENORMAND, 18 Avenue Saint-Germain, Puteaux

LEONHARD, 14 Rue Coypel, Paris

L'HERITIER & Cie, 86 Rue de Paris, Saint-Denis

LILLE & BONNIERES, 10 Rue des Pyramides, Paris

LUBIN, 47 Rue du Liegat, Ivry-Port

LUBRICATING Oil Co., Route de Sartrouville, Pecg.

LYNDALI & Cie, 80 Rue Taitbout, Paris

MACKAY, 2 Cite Trevise, Paris

MAILLET, 9 Rue Alfred Condre, Abbeville

MALICET & BLIN, 103 Avenue de la Republique, Aubervilliers (_Mab._)

MANCEAU, 60 Rue de Flanders, Pantin

MARECHAL, 75 Avenue du Chemin-de-Fer, Le Vestinet

MARTIN (V.), 50 Boulevard de Strasbourg, Paris

MARVILLE & Cie, Rueil

MAUPRE, 112 Rue de la Chapelle, Paris

MICHEL, 15 Rue Ferragus, Aubervilliers

MORIN, 48 Rue de l'Aqueduc, Paris

NANTERRE, 18 Rue Gambetta, Nice (_Omnia._)

NASSOY & RIBAUD, 78 Rue Charles-Nodier, Pantin (_Colzarine_)

NICKMILDER, 82 Rue Daquerre, Paris

NOBLET, 1 Rue Pastuer, Ivry-Port

NORTZ, 29 Boulevard Sebastopol, Paris

OLEO, 30 Rue Perrier, Levallois-Perret (_Oleomoto_)

OLEONNAPHTES (Societe Anoyme), 164 Avenue de Paris, Saint-Denis

OLEONNAPHTES EMULSIONNES (Societe Anonyme), 3 Avenue Victor-Hugo

ORANGE & Cie, 432 Avenue de Paris, Saint-Denis

PELON, 76 Avenue de la Republique, Paris

PENNSYLVANIA OIL Co., 39 Rue Sainte-Cecile, Marseilles

PETROLES OIL Co., 2 Rue Fongate, Marseilles (_Onctua._)

PEUGEOT, Freres, 71 Avenue de la Grande-Armee, Paris Valentigney

PIETRATERRA (A.), 10 Rue des Augustins, Argenteuil

POURCHEIROUX, 41 Rue Saint-Ferdinand, Paris

POULET & TAYART, 108 Avenue de la Republique, Aubervilliers

PRADERE & Cie, 16 Rue du 14-Juillet, Pre-Saint-Gervais (_Virginia_)

QUERVEL, 35 Rue du Port, Aubervilliers (_Kervoline_)

RASTIT (H.), 38 Rue Bicolas, Marseilles

RECORD, 27 Quai Gailleton, Lyon

REGNIER, Fils & RODDE, 11 Rue Etienne-Dolet, Paris

RENAUD-LEVEQUE & Cie, 37 Rue Jean-Jacques-Rosseau, Paris

RENAULT (V.), 145 Avenue, Parmentier, Paris

REVAUX, 63 Boulevard Thiers, Amiens

RICBOURG, 19 Quai aux, Fleurs, Paris

RINCK, Fils, 66 Rue de Rivoli, Paris

ROBERT, 25 Rue Drouot, Paris

RONDEL, 101 Rue Marceau, Montreuil

RONDEL (Ch.), 57 Rue de Saint-Mande, Montreuil

SAUTET, Freres, 99 Route d'Orleans, Montrouge

SIMONET (L.), 45 Rue Gambetta, Nancy

SIMON-ROCHE, 17 bis, Avenue du Mans, Tours (_Auto Sims_)

SIVAN, 8 Place de l'Evesche Marques, Frejus (_Record, Aeroline, Motord_)

STANDARD OIL Works, 69 Rue d'Hauteville, Paris

STORACE (B.), 15 Rue de Paris, Nice

SYLVESTER (E.), 6 Rue Nationale, Rouen (_W.S._)

TESSE, 15 Rue de Surene, Paris

TORRE & Cie, 112 bis, Rue de Paris, Vincennes

TOURNEL, 18 Avenue d'Italie, Paris

TRABET (L.), 1 Rue Amelot, Paris (_Trabeoline_)

VACUUM OIL Co., Ltd., 34 Rue de Louvre, Paris

VILLENEUVE (A.), 1 Boulevard Saint-Jacques, Paris

WALLACH & Cie, 60 Avenue de la Republique, Aubervilliers

WALLET, 12 Rue Rennequin, Paris

WILSNER (G.), 29 Rue de Neuilly, Clichy

ZEMMER, 91 Rue Petit, Paris

~GERMAN.~

DEPAUW & Cie, 6 Rue de la Linere, Brussels

DEUTSCHE [OE]LVERKE, 1 Prinz-Louis, Ferdinandstrasse, Berlin

PETROLEUM RAFFINERIE, Breme (_Veloscol_)

SPILCKE, 94 Chausseestrasse, Berlin

STERN-SONNEBORN (A. G.), 21 Ritterstrasse, Berlin S. 42

SUeDDEUTSCHE OELWERKE, Fribourg-en-Brisgau

VALVOLINE OIL Co., 7 Hobzbruecke, Hambourg

VOGT & Cie, Goerlitz (_Vostol_)

~ITALIAN.~

ARNOLDI & Cie, 37 Via Paolo do Cannobio, Milan

CECCARELLI, TEDESCHI & Cie, Corso XXII., Marso, 34, Milan (_Teuff_)

CORLIE RE, 8 Via Santa-Azata, Boulogne

FOLTZER (E.), Rivarolo-Lugure, Genes

KOCH (O.), 50 Via Abbadesse, Milan

MIRAGOLI & PETSATORI, 67 Foro Bonaparte, Milan

OLEUM, Galleria Nazionala, Turin

PETROLIO, 76 Piazza Cinque Lampade, Genes

REINACH & Cie, 90 Via Lario, Milan (_Oleoblitz_)

VOLPATO & Cie, 11 Via Santa-Maria-Fulcornia, Milan

~ROUMANIAN.~

TRAJON, Bucharest, Roumania

~RUSSIAN.~

CHABANIAN (R.), Batoum-Bakou

KAISER (R.), Baku

MALLARD, Caucase, Batoum

NOBEL, Freres, St. Petersburg

PITOEFF & Cie, Tiflis

SCHIBAEFF & Cie, Bakau

TER AKOPOFF, 3 Place Isaac, St. Petersburg

~SPANISH.~

FONTAGUD, 6 Fuentes, Madrid

OLEON Co., 13 Asalto, Saragossa

USERA (De), 47 Legdnitos, Madrid

VACUUM OIL Co., 598 Cortes, Barcelona

~SWISS.~

GRISARD (G.), 302 Route de Greuzach, Bale

HALLER, 8 Splugenstrasse, Zurich

HEUMANN (A.) & Cie, Winterthur

HUILES MINERALES, Route de Frontenex, Geneva

LAMBERCIER (J.) & Cie, Geneva

LUMINA (S. A.), Geneva-Vollandes

MOEBIUS (H.) & Fils, Bale

OMNIA (Maison), Chene-Bourg, Geneva

SCHMID, 133 Murtenstrasse, Berne

~U.S.A.~

DIXON (J.) CRUCIBLE Co., Jersey-City, New York, (_Graphite_)

KEYSTONE LUBRICATING Co., Philadelphia

WHITE & BAGLEY Co., Worcester (_Oilzum_)

~MAGNETOS.~

~AUSTRIAN.~

DENES & DRIEDMAN, 11 Mitterbergasse, Vienna XVIII.

ERBEN (S.) & ARNOLD FRIEDMANN, 14 Stubenring, Vienna I.

~BELGIAN.~

BOSCH MAGNETOS, 121 Rue de l'Instruction, Brussels

PERNSTEIN (Ateliers), 8 Rue Laporte, Liege-Nord

~BRITISH.~

BOSCH MAGNETOS-- 40-42 Newman Street, London, W.C. 28 Store Street, Tottenham Court Road, London, W.C.

BRITISH TELLIER Co., 10 Coburg Place, Hyde Park, London, W.

~EISEMANN MAGNETO Co.~, 43 Berners Street, London, W.

FULLER (J. C.) & Son, Woodland Works, Wick Lane, Bow, London, E.

~MEA MAGNETO Co.~, Gresse Buildings, Stephen Street, Tottenham Court Road, London, W.

NILMELIOR (Societe d'Electricite), 36-37 Alfred Place, Tottenham Court Road, London, W.C.

RICHES (G. T.) & Co., 19 Store Street, Tottenham Court Road, London, W.C.

SIMMS MAGNETO Co., Ltd., Welbeck Works, Kilburn, London, N.W.

VAN RADEN & Co., Ltd., Great Heath, Coventry.

~DUTCH.~

BOSCH MAGNETOS, Willem Van Rijm, Keizergracht 181, Amsterdam

~FRENCH.~

BARDON (L.), 61 Boulevard National, Clichy

BAUDOT ET PAZ, 22 Avenue de la Grande-Armee, Paris (_Simms._)

BOIN, 33 Rue du Four, Paris

BOSCH MAGNETOS-- Depot 295 Avenue de Saxe, Lyon 17 Rue Theophile-Gautier

BREGUET (Maison), 19 Rue Didot, Paris

DEBEAUVE, 68 Rue de Sevres, Paris (_Vestale_)

~EISEMANN & Co.~, Lavalette & Cie., 175 Avenue le Choisy, Paris

EXTRA. (_See_ Giffard.)

GIANOLI, 28 Boulevard Magenta, Paris

GIBAUD, 309 Rue de Faubourg, Saint-Antoine, Paris

GIFFARD, 283 Rue des Pyrenees, Paris (_L'Extra_)

GIRARDEAU (A.), 7 Rue Scribe, Paris

GUENET. 5 Rue Montmorency, Paris

GUILLOU, 41 Rue de Bagneux, Montrouge

HENRIQUE, 54 Quai de Courbevois, Courbevois

HERDTLE & BRUNEAU, 93 Rue Pelleport, Paris

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FOOTNOTES:

[Footnote A: Twelve of these _B.E._ were held up pending some special steel wire strainers which had been specified.]

[Footnote B: Killed in Russia.]

[Footnote C: This ship has frequently figured as four different dirigibles.]

[Footnote D: P.L 9 reported sold to Turkey, April, 1913.]

[Footnote E: Three other dirigibles, _Unger_, _Veeh_, and _Siemens-Schuckert_, are generally credited to Germany. Of these _Unger_ is merely a project. _Veeh_ has been talked about for four years, but has never reached completion. _Siemens-S._ has ceased to exist.]

[Footnote F: Jezzi lives in England where he is a well known amateur constructor.]

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